Prefix Sum, Two pointer, Sliding Window

Prefix AP

Difficulty: **Medium**Accuracy: **31.46%**Submissions: **854+**Points: **4**

Given an array of N integers and Q queries. Initially, all the elements of the array have value 0.

In each query 4 integers A, D, L, R is given, which means perform AP addition

Arr[L]+=A

Arr[L+1]+=A+D

Arr[L+2]+=A+2D

Arr[L+3]+=A+3D

.

.

Arr[R]+=A+(R-L)*D.

After performing all the operations. Print the final array % 10^9+7.

Input:

The first line contains T, the number of test cases. 1<=T<100.

- The first line of each test case contains two space-separated integers N, Q where 1<=N<=10^6, 1<=Q<=10^6.
- Next, Q lines contain four space-separated integers A, D, L, R where 0<=A,D<=10^4, 1<=L<=R<=N.
- The sum of T over all N and Q is less than 10^6.

Output:

Print the N space-separated integer representing the final array %10^9+7.

Example

Input:

1

5 5

1 3 1 5

2 4 2 2

5 6 3 5

6 9 5 5

1 4 2 4

Output:

1 7 17 30 36

Library Visits

Difficulty: **Medium**Accuracy: **35.81%**Submissions: **271+**Points: **4**

There are **n** students in a school. Each student has a unique ID from 1 to **n**. Every time a student accesses the library facility, the software records the ID of the student and the software also displays the list of other students who had visited the library premises, sorted by decreasing times of their visit (so the member who visited most recently is the first of the list). However, the times of last seen are not displayed.

The principal of the school can access this list anytime. She accesses this list once in the morning at 9:00 hrs and once in the evening at 17:00 hrs. Both times she downloaded the list on her computer.

She gives the lists to you in two arrays and asks you to find the **minimum number of students** that must have visited the library premises at least **once** between 9:00 and 17:00.

As per the rules, no two students can visit the library at the same time. You can assume that no one visits the library at 9:00 hrs and 17:00 hrs.

Input:

The first line contains T ($1<=T<=10^4$) , the number of test cases. Each test case is as follows:

- The first line of each test case contains an integer **n**, the number of students in the school.
- The second line contains n integers $a_1,a_2,,a_n$ the list of IDs of the members, sorted by decreasing times of last visit at 9:00.
- The third line contains n integers $b_1,b_2,,b_n$ ($1 \le b_i \le n$) the list of IDs of the members, sorted by decreasing times of the last visit at 17:00.
- For all $1 \le i < j \le N$, it is guaranteed that $a_i \ne a_j$ and $b_i \ne b_j$. It is also guaranteed that the sum of the values of n over all test cases does not exceed 10^5.

Output:

For each test case, print a single integer as described.

Constraints:

$1<=n<=10^5$

$1 \le a_i \le n$

$1 \le b_i \le n$

Example:

Input:

```
3
5
1 4 2 5 3
4 5 1 2 3
6
1 2 3 4 5 6
1 2 3 4 5 6
8
```

8 2 4 7 1 6 5 3
5 6 1 4 8 2 7 3

Output:

2
0
4

Explanation:

In the first test case, students with ID 4,5 must have been online between 9:00 and 17:00.
In the second test case, it is possible that nobody has been online between 9:00 and 17:00.

Number Theory Basics

Find GCD of two numbers

Difficulty: **Easy**Accuracy: **59.53%**Submissions: **746+**Points: **2**

Given two positive integers A and B, find the GCD of A and B.

Example 1:

Input: A = 3, B = 6
Output: 3
Explanation: GCD of 3 and 6 is 3

Example 2:

Input: A = 1, B = 1
Output: 1
Explanation: GCD of 1 and 1 is 1

Your Task:

You don't need to read input or print anything. Complete the function **gcd()** which takes two positive integers as input parameters and returns an integer.

Expected Time Complexity: O(log(min(A, B)))
Expected Auxiliary Space: O(1)

Constraints:
$1 \leq A, B \leq 10^9$

BASIC Practise Problems

Second Lecture

Difficulty: **Easy** Accuracy: **34.57%** Submissions: **2K+** Points: **2**

Geek teaches a maths class in Geekland state university in which you have enrolled. Today is your second lecture. In the previous lecture Geek has taught integer numbers so before starting the next topic he wants to test your understanding of integers for which he gave you a problem - Given an integer **N** find the ratio of sum of squares of first **N** positive integers by sum of first **N** positive integers. Sometimes the ratio is not an integer, then you have to print "NO" (without quotes) as you know only about integer numbers currently. Otherwise print "YES" (without quotes) and the ratio separated by a space.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer on a new line.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq N \leq 10^9$

Example:

Input:

2

2

4

Output:

NO

YES 3

Explanation:

Test Case 1: sum of first N natural numbers = 1 + 2 = 3 and sum of squares of first N natural numbers = 1 + 4 = 5. Since 5 is not divisible by 3, you have to print "NO".

Test Case 2: sum of first N natural numbers = 1 + 2 + 3 + 4 = 10 and sum of squares of first N natural numbers = 1 + 4 + 9 + 16 = 30. Since 30 is divisible by 10, you have to print "YES 3".

Triplet

Difficulty: **Easy** Accuracy: **43.21%** Submissions: **2K+** Points: **2**

Geek is hosting a coding contest in which he wants to set a problem based on array, he loves number three so much, so he wants some positive number of triplets with all same elements in the array. But Geek does not have time to check

whether his array have triplets or not so he asks you to tell him whether he can use his array or not. Given an array **arr** of size **N**, the task is to check if there exists any triplet with all same elements.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**
3. The next line contains **N** space-separated integers which represents **arr[]**.

Output: For each test case, print "**yes**" if some triplet exists, otherwise print "**no**".

Constraints:

1. $1 \leq T \leq 10$
2. $1 \leq N \leq 10^5$
3. $1 \leq arr[i] \leq 10^5$

Example:

Input:

2
5
1 2 1 2 1
3
1 2 3

Output:

yes
no

Explanation:

Test Case 1: There is one triplet of 1's.
Test Case 2: There is no triplet.

Dividing Chocolate

Difficulty: **Easy** Accuracy: **52.69%** Submissions: **1K+** Points: **2**

Sasha has two children which initially have **a** and **b** grams of chocolate respectively. Sasha has **c** grams of chocolate which she wants to divide between both of her children. She will divide her chocolate into two parts (one part can be possibly empty) and will give exactly one part to each child. She wants to divide her chocolate in such a way that difference between amount of chocolates both children have after Sasha has given them her chocolate is minimum. For dividing chocolate Sasha uses a weighing machine which has precision upto six decimal places. You have to print how many grams of chocolate each child has after Sasha has given them her chocolate separated by space. You should use the same weighing machine for finding out the answer.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains three-space separated integers **a, b** and **c**.

Output: For each test case, print the answer. Precision of the answer should be exactly upto six decimal places.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq a, b, c \leq 10^9$

Example:

Input:
2
10 1 10
10 6 3
Output:
10.500000 10.500000
10.000000 9.000000
Explanation:
Test Case 1 : Sasha gives 0.500000 grams
of her chocolate to first child and rest
of the chocolate to other.
Test Case 2 : Sasha gives all of her chocolate
to second child.

Shopping Sale

Difficulty: **Easy**Accuracy: **42.76%**Submissions: **1K+**Points: **2**

Geek went to the DS-mall as there is a shopping sale going on in the graph shop. There are **N** graphs in the shop and the cost of **i^{th}** graph is **a[i]**. Geek wants to buy all the graphs which are in the shop and for every **K** graphs he buys he gets one extra graph free due to the sale. He wants to spend as less as possible. You have to find out minimum amount of money he has to spend to buy all the graphs.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two-space separated integers **N** and **K.**
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 1000$
2. $1 \leq K \leq N \leq 10^5$
3. $1 \leq a[i] \leq 10^9$
4. It is guaranteed that sum of N over all test cases does not exceed 10^5.

Example:

Input:

2
3 2
3 1 2
2 1
1 1000000000

Output:

3
1

Explanation:

Test Case 1 : Geek buys second and third graph and he gets the first graph for free.
Test Case 2 : Geek buys the first graph and he gets the second graph for free.

Good Numbers

Difficulty: **Easy** Accuracy: **83.37%** Submissions: **681+** Points: **2**

Geek admires good numbers. According to his definition good numbers consists of only one digit. Geek has a number and he wants to transform it into a good number using some number (possibly zero) of a particular operation. The

operation is defined as update x to F(x) where F(x) is a function which returns the sum of digits of a number which is given to it as a parameter. You are given N - number which Geek has. Find the good number which is obtained when we apply some number (possibly zero) of operations to N.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^9$

Example:

Input:
2
23
3749833
Output:
5
1
Explanation:
Test Case 1 : After applying one operation we
will get a good number. F(23) = 2 + 3 = 5.

Kill the Enemy

Difficulty: **Easy**Accuracy: **50.59%**Submissions: **878+**Points: **2**

Geek is playing a video game in which he has to kill the enemy before the enemy kills his character. Rules of the video game are as follows :

1. Geek's character does damage to enemy every second. Whenever Geek does damage to enemy, enemy's **XP** decreases by that much amount. Enemy is assumed to be killed if his **XP** becomes zero or negative.
2. Damage depends on two things - strength and anger. Initially i.e. in the first second strength of Geek's character is **a** and his anger is **b**. Damage done at any second is product of strength and anger at that time i.e. in the first second damage done is **a * b**.
3. Every second strength decreases by **d** i.e. if strength in $\mathbf{i^{th}}$ second is **x** then in $\mathbf{(i+1)^{th}}$ second strength becomes **x - d**.
4. Every second anger increases by **r** times i.e. if anger in $\mathbf{i^{th}}$ second is **x** then in $\mathbf{(i+1)^{th}}$ second anger becomes **x * r**.
5. If enemy is not killed within 100 seconds then enemy will kill the Geek's character.

You are given **a**, **b**, **d**, **r** and **XP** of the enemy before start of the game. You have to find out minimum time in seconds, Geek's character will take to kill the enemy or print **"-1"** if it is impossible to kill the enemy.

It is guaranteed that damage done is positive at any point of time before enemy is killed.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains five space-separated integers **a**, **b**, **d**, **r** and **XP**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq a, b, d \leq 10^9$
3. $1 \leq r \leq 9$
4. $1 \leq XP \leq 10^{15}$

Example:

Input:

2

10 2 1 1 40

100 1 1 1 1000000

Output:

3

-1

Explanation:

Test Case 1 : Damage done in first three seconds is :

$(10 - 0 * 1) * (2 * 1^0) = 20$,

$(10 - 1 * 1) * (2 * 1^1) = 18$

and $(10 - 2 * 1) * (2 * 1^2) = 16$ respectively.

Since 20 + 18 + 16 = 54 > 40, enemy is killed in the third second.

Test Case 2 : Total damage done in 100 seconds is less than the XP of the enemy.

Mean and Median

Difficulty: **Easy**Accuracy: **61.69%**Submissions: **786+**Points: **2**

While learning statistics Geek came across two terms - Mean and Median.
Given a **sorted** array **a[]** of size **N** Geek wants to find out the absolute difference between Mean and Median of the array.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer. Answer should be precise upto exactly six decimal places.

Constraints:
1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^5$
3. $1 \leq a[i] \leq 10^9$

Example:

Input:
2
3
1 2 4
2
5 6
Output:
0.333333

0.000000

Explanation:

Test Case 1 : Mean = (1 + 2 + 4) / 3 = 2.333333

Median = Middle element = 2

Test Case 2 : Mean = (5 + 6) / 2 = 5.5

Median = Average of two middle elements =

(5 + 6) / 2 = 5.5

Mean and Median

Difficulty: **Easy**Accuracy: **61.69%**Submissions: **786+**Points: **2**

While learning statistics Geek came across two terms - Mean and Median.

Given a **sorted** array **a[]** of size **N** Geek wants to find out the absolute difference between Mean and Median of the array.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer. Answer should be precise upto exactly six decimal places.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^5$
3. $1 \leq a[i] \leq 10^9$

Example:

Input:
2
3
1 2 4
2
5 6
Output:
0.333333
0.000000
Explanation:
Test Case 1 : Mean = (1 + 2 + 4) / 3 = 2.333333
Median = Middle element = 2
Test Case 2 : Mean = (5 + 6) / 2 = 5.5
Median = Average of two middle elements =
(5 + 6) / 2 = 5.5

Easy Sequence

Difficulty: **Easy**Accuracy: **67.49%**Submissions: **758+**Points: **2**

Geek recently started learning about sequences. He came across an easy sequence where **N^{th}** term is defined as :
a(N) = a(N - 1) * a(N - 2) * a(N - 3) * ... * a(2) * a(1) * a(0) + 1 where **a(0) = 2**.
As you are his firend, he challenged you to find the **N^{th}** term of this sequence. If you do so, Geek will consider that you are an expert in mathematics.
As answer can be very large, find it modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.

2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^5$

Example:

Input:

2

2

3749

Output:

7

317982199

Explanation:

Test Case 1 : a(0) = 2,

a(1) = a(0) + 1 = 2 + 1 = 3

a(2) = a(1) * a(0) + 1 = 3 * 2 + 1 = 7

Tribonacci Sequence

Difficulty: **Easy**Accuracy: **76.94%**Submissions: **603+**Points: **2**

N^{th} Fibonacci number is defined by **F(n) = F(n-1) + F(n-2)**.

Similarly, **N^{th}** Tribonacci number is defined by **T(n) = T(n-1) + T(n-2) + T(n-3)**.

Given first three numbers of Tribonacci sequence, find the **N^{th}** Tribonacci number modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains an integer **N**.
3. The second line of each test case contains three space-separated integers **T[1], T[2]** and **T[3]**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 100$
2. $4 \leq N \leq 10^5$
3. $1 \leq T[1] \leq T[2] \leq T[3] \leq 10^9$

Example:

Input:

2
4
1 2 3
38746
1 23 678

Output:

6
332683912

Explanation:

Test Case 1 : T[4] = T[3] + T[2] + T[1]
= 3 + 2 + 1 = 4

Finding Progression

Difficulty: **Easy**Accuracy: **56.72%**Submissions: **827+**Points: **2**

Geek has been given three successive members of a sequence **A ,B, C** and his homework is to find the next term. The problem is that he forgot if his homework was on arithmetic progression or on geometric progression. Help him to determine the type of the progression (**arithmetic or geometric**) and hence the **next successive member.**

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains three space-separated integers **A**, **B** and **C**.

Output:

For each test case, print the 4th term.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq A, B, C \leq 10^9$

Example:

Input:

2
2 4 8
1 2 3

Output:

16
4

Explanation:

Test Case 1 : It is a GP with common ratio 2.

Counting Numbers

Difficulty: **Medium**Accuracy: **66.57%**Submissions: **814+**Points: **4**

Given an integer **N**, Geek wants to count all the numbers less than 10^5 such that their sum of digits is **N**. Help him as he is your true friend and always helps you when needed.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^5$

Example:

Input:

2

1

20

Output:

5

5631

Explanation:

Test Case 1 : There are five numbers less than 10^5 such that their sum of digits is 1 : 1, 10, 100, 1000 and 10000.

Sweet Tooth

Difficulty: **Easy**Accuracy: **76.63%**Submissions: **566+**Points: **2**

Santa gave little Jack lots of sweets but his mother told him that he will not be allowed to eat all of them as they can be bad for his teeth. She told him that he had to divide the number **N** which is the number of sweets he got into two non-empty parts for example N = 1234 can be divided into 1 and 234 or 12 and 34 and so on and he could have as many sweets as the product of the sum of digits of both parts. Help him eat as many as he can.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains an integer **N**, the numbers of sweets.

Output:

For each test case print, the maximum number of sweets Jack can have.

Constraints:

1. $1 \leq T \leq 10^4$
2. $10 \leq N \leq 10^{16}$

Example:

Input:

2

1234

100

Output:

24

0

Explanation:

Test Case 1 : (1+2+3)*4 = 24 which is maximum

among all possibilities.

The Necklace

Difficulty: **Easy**Accuracy: **57.66%**Submissions: **796+**Points: **2**

Geek had **N** beads, all of **different colours** and he made a necklace out of them. He decided to sell it for as much money as was the number of **trailing zeroes** in the number **X** which was the number of ways he could have made the necklace using those beads. Clockwise and Anti-clockwise orders of necklace are considered same. Print how much he would get for the necklace.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $3 \leq N \leq 10^9$

Example:

Input:

2

3

41

Output:

0

9

Explanation:

Test Case 1 : There was only one way the necklace could have been made so he would

have to sell it for free.

BIT MASKING

Geek Learns Bitwise Operations

Difficulty: **Easy**Accuracy: **71.15%**Submissions: **687+**Points: **2**

Geek recently learned bitwise AND and OR operations. He has two numbers **x1** and **x2**, now he wonders whether there exist two numbers such that their bitwise AND is **x1** and bitwise OR is **x2**. You have to help Geek as always. You have to output "**YES**" (without quotes) if it is possible, otherwise output "**NO**" (without quotes).

Input:

1. First line of the test case contains a single integer T, denoting the number of test cases.
2. First line of each test case contains two space-separated integers x1 and x2.

Output:

1. For each test case output "YES" or "NO" in a single line.

Constraints:

$1 <= T <= 10^4$

$1 <= x1, x2 <= 10^9$

Example :

Input:

2

1 7

2 5

Output:

YES

NO

Explanation:
Testcase 1: Two numbers can be 3 and 5 such that their bitwise AND is 1 and bitwise OR is 7.
Testcase 2: There are no two numbers exist whose bitwise AND is 2 and bitwise OR is 5.

XOR on Array

Difficulty: **Easy**Accuracy: **84.0%**Submissions: **581+**Points: **2**

Geek has an array of **N** elements **arr[]**.
He wants to do following operations **N-1** times on array -
1. Remove two elements from the array .
2. Take Bitwise XOR of the removed elements and insert them into the array.
In the end only one element will remain in the array. Geek wants this element to be minimum possible. You have to find the minimum possible element which will be left in the array.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **arr[]**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 1000$
2. $2 \leq N \leq 10^6$

3. $1 \leq arr[i] \leq 10^9$
4. The sum of the values of N over all test cases does not exceed 10^6

Example:

Input:
2
3
3 1 2
2
1 1000000000
Output:
0
1000000001
Explanation:
Test Case 1 : At first Geek will remove 3 and 2 and insert 3 ^ 2 = 1 into the array. Array will become {1, 1}, then Geek will remove 1 and 1 and insert 1 ^ 1 = 0 into the array, which is minimum possible.
Test Case 2 : Geek has only one option Geek will remove 1 and 1000000000 and insert 1 ^ 1000000000 = 1000000001 into thearray which is minimum possible.

Bit Manipulation

Difficulty: **Medium**Accuracy: **95.27%**Submissions: **394+**Points: **4**

Given an integer **N**, find the minimum number of rounds required to convert its binary representation to zero.
In one round you can perform any one of the following operations :
1. Change **i^{th}** bit if **$(i+1)^{th}$** bit is ON, and every bit from **(i+2)** to the right end are OFF.
2. Change the rightmost bit.
Note: Rightmost bit is the least significant bit *i.e.* binary representation of 6 would be 110.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq N \leq 10^9$

Example:

Input:

3
3
4
786437641

Output:

2
7
884990990

Explanation:

Test Case 1 : We will do following
sequence of operations : 11 -> 01 -> 00.

Test Case 2 : We will do following
sequence of operations : 100 -> 101 -> 111
-> 110 -> 10 -> 11 -> 01 -> 00.

Interesting Ternary Numeral System

Difficulty: **Easy**Accuracy: **73.16%**Submissions: **436+**Points: **2**

The Ternary numeral system is a numeral system in which base is 3 and value of every digit is either 0, 1 or 2.

According to Geek Interesting Ternary numeral system is a numeral system in which base is 3 and every digit is represented by either '**z**', '**n**' or '**p**' where '**z**' corresponds to value 0, '**p**' corresponds to value 1 and '**n**' corresponds to value -1. You are given a normal ternary number in the form of string **s** as input, you have to find its corresponding interesting ternary number.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains **s** denoting ternary number.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq |s| \leq 100$

Example:

Input:

2

11

102

Output:

pp

ppn

Explanation:

Test Case 1 : Value of $11_3 = 4_{10}$ as
1*3 + 1*1 = 4 and value of $pp_3 = 4_{10}$ as
1*3 + 1*1 = 4.

Test Case 2 : Value of $102_3 = 11_{10}$ as
1*9 + 0*3 + 2*1 = 11 and value of ppn_3 =
11_{10} as 1*9 + 1*3 + (-1)*1 =11.

Yet Another Bitmask Problem

Difficulty: **Easy**Accuracy: **53.45%**Submissions: **636+**Points: **2**

Given an integer **N**, find the absolute difference between bitwise **AND** of every integer from **1** to **N** and bitwise **OR** of every integer from **1** to **N**.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq N \leq 10^{18}$

Example:

Input:

2
4

786437641273623

Output:

7

1125899906842623

Explanation:

Test Case 1 : Bitwise AND of 1, 2, 3 and 4 is 0 and Bitwise OR of 1, 2, 3 and 4 is 7. Absolute difference between both values is 7.

Array Conversion

Difficulty: **Easy**Accuracy: **95.54%**Submissions: **397+**Points: **2**

Geek has an array **a[]**, he wants to convert this array to **b[]** using only XOR operation. He will choose an index **i** and a value **x** such that **a[i] ^ x = b[i]**.
Help him find suitable value of **x** for every index **i** from **1** to **N**.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.
4. The third line of each test case contains **N** space-separated integers which represents **b[]**.

Output: For each test case, print the **N** space-separated integers which represents value of **x** for **i^{th}** index from **1** to **N** on a new line.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^5$

3. $1 \leq a[i], b[i] \leq 10^9$

Example:

Input:
2
3
7 2 2
3 4 2
1
1000000000
3762893
Output:
4 6 0
1000579277
Explanation:
Test Case 1 : 7^4 = 3, 2^6 = 4 and 2^0 = 2.
Test Case 2 : 1000000000^1000579277 = 3762893.

XOR Sort

Difficulty: **Hard**Accuracy: **50.67%**Submissions: **563+**Points: **8**

Given three integers **A**, **B** and **C**, you have to sort them in the order **A ≤ B ≤ C**.
For sorting you should choose an integer **X** and take bitwise XOR of **A**, **B** and **C** with **X**. If there are multiple possibilities of choosing **X**, you should choose the smallest one.
If it is impossible to sort them using above operation, print "**-1**" (without quotes).

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.

2. The first line of each test case contains three space-separated integers **A**, **B** and **C**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq A, B, C \leq 10^9$

Example 1:

Input:

2

3 6 4

1 3 3

Output:

2

0

Explanation:

Test Case 1 : 3^2 = 1, 6^2 = 4 and 4^2 = 6.

Test Case 2 : A, B and C are already sorted.

Example 1:

Input:

1

14 12 8

Output:

6

Explanation:

Test Case 1 : 14^6 = 8, 14^12 = 10 and 14^8 = 14.

Test Case 2 : A, B and C are already sorted.

Switching Cost

Difficulty: **Easy**Accuracy: **82.96%**Submissions: **375+**Points: **2**

Geek made a robot while attending a robotics workshop. The Robot is ON when circuit is at state **m** and OFF when circuit is at state **n** where **m** and **n** are integers. While switching Robot from ON to OFF or OFF to ON, state has to be changed from m to n or n to m.

Geek has to encode these integers in the circuit in binary representation. He wants that while switching, number of bits changed is least possible.

You have to find whether he should use normal binary representations of **m** and **n** or gray code representations of **m** and **n** to encode them into the circuit. If number of bits changed is same in both the cases, he will use normal binary representation.12

Print "**GRAY**" (without quotes) if he should use gray code representation, otherwise print "**BINARY**" (without quotes).

NOTE: Find more about Grey Code here.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **m** and **n**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$

2. $1 \leq m, n \leq 10^9$

Example 1:

Input:
2
3 4
32467 987897

Output:
GRAY
BINARY

Explanation:
Test Case 1 : Number of bits changed when
normal binary representation is used
11 -> 100 = 3. Number of bits changed when
gray code representation is used 10 -> 110 = 1.
Hence, Geek will use gray code representation.

Test Case 2: Number of bits changed when using normal binary representation is 11. While number of bits changed in
Gray Code representation is 12. Hence, Geek would use
Normal Binary Representation.

Example 2:

Input:
2
8 10

156 157

Output:
BINARY
BINARY

Explanation:
Test Case 1 : Number of bits changed when
normal binary representation is used
1. Number of bits changed when
gray code representation is greater than 1.
Hence, Geek will use Binary Representation.

Test Case 2: Number of bits changed when using normal binary representation is 1. While number of bits changed in
Gray Code representation is also 1.It is a tie So, Geek would use
Normal Binary Representation.

Simple Calculator

Difficulty: **Easy**Accuracy: **87.77%**Submissions: **318+**Points: **2**

Balanced ternary is a numeral system with base 3 where every digit has value 0, 1 or -1. To represent -1 we use 'z'.
Given two positive Balanced ternary numbers **A** and **B**, implement a simple calculator to add them. The output should also be a Balanced ternary number.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a Balanced ternary number denoting **A**.
3. The second line of each test case contains a Balanced ternary

number denoting **B**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq$ Integer equivalent of A and B $\leq 10^9$

Example:

Input:

2

1z

11

1z0010z

1zzzzzz

Output:

1z0

11zzz11

Explanation:

Test Case 1 : $(1z = 2_{10}) + (11 = 4_{10}) = (1z0 = 6_{10})$.

Queries on Binary Number

Difficulty: **Easy**Accuracy: **53.12%**Submissions: **718+**Points: **2**

Given a long binary number in a string **S** (decimal value of S is greater than 0), you have to perform **Q** queries on it. In each query, you have to multiply the binary number by 2 if " * " is given as a character or divide it by 2 if " / " (integer division) is given as a character.

You have to print the binary number after performing all the operations without any leading zeroes.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains **S**.
3. The second line of each test case contains one integer **Q**.
4. The next **Q** lines contains a character - either " * " or " / ".

Output: For each test case, print the answer on a new line.

Constraints:

1. $1 \leq T \leq 10$
2. $1 \leq |S|, Q \leq 10^5$

Example:

Input:

2
11
2
*
*
1101
2
/
*

Output:

1100
1100

Kth Bit

Difficulty: **Easy** Accuracy: **69.06%** Submissions: **433+** Points: **2**

You are given a **M-bit** nonnegative integer **N**. You can perform a complement operation on this integer any arbitrary number of times. You have to ON

the **K**th bit in the integer from the right end. Print the integer after turning ON the **K**th bit.

Input:
1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains three space-separated integers integers **N, M,** and **K**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^3$
2. $1 \leq K \leq M \leq 30$
3. $0 \leq N < 2^M$

Example:

Input:
2
3 2 2
1 2 2
Output:
3
2
Explanation:
Test Case 1 : Binary representation of 3 is
11, where 2nd bit is already ON.
Test Case 2 : Binary representation of 1 is
01, we will take compliment of this integer
to turn on the 2nd bit.

XOR Operation

Difficulty: **Easy**Accuracy: **63.53%**Submissions: **477+**Points: **2**

Geek loves XOR operation. He wants to solve a problem related to XOR operation but his algorithm is taking exponential time, so he needs your help.

Given an array, **a[]** of **N** elements, you have to partition the array into two non-empty parts *i.e.* partition at **i** will divide the array into two arrays - one consisting of elements from index **1** to **i** and the other consisting of elements from the index **(i + 1)** to **N**. You want to divide the array in such a way that the XOR of the sum of elements of the first array and the sum of elements of the second array is minimum.

You have to find the minimum XOR value.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$
2. $2 \leq N \leq 10^5$
3. $1 \leq a[i] \leq 10^9$

The sum of N overall test case does not exceed 10^6.

Example:

Input:

2

5

1 2 5 9 3

4

1 2 4 8

Output:

4

15

Explanation:

Test Case 1 : We will partition the array at index 3 which will give us two arrays - {1, 2, 5} and {9, 3} and minimum value of XOR of sum of elements of first array and sum of elements of second array = (1 + 2 + 5) ^ (9 + 3) = 8 ^ 12 = 4. Any other partition will give a larger XOR value.

Test Case 2 : We can partition the array at any index. Every partition will give us XOR value of 15.

Bit Magic Query

Difficulty: **Hard** Accuracy: **87.81%** Submissions: **295+** Points: **8**

Given an array of N integers and Q queries. In each query, you are given two parameters L and R; you have to find the following:

- Smallest integer X1 such that the value of (A_L XOR X1) + (A_{L+1} XOR X1) + ... + (A_R XOR X1) is the maximum possible.
- Smallest integer X2 such that the value of (A_L OR X2) + (A_{L+1} OR X2) + ... + (A_R OR X2) is the maximum possible.
- Smallest integer X3 such that the value of (A_L AND X3) + (A_{L+1} AND X3) + ... + (A_R AND X3) is the maximum possible.

Range of X1, X2, X3 allowed → 0 <= X1, X2, X3 < 2^31.

Print the sum of X1, X2, and X3 for each query in a newline.

Input:

The first line of the input contains a single integer T denoting the number of test cases, $1<=T<=10^3$

- The first line of each test case contains a single integer N, ($1<=N<=10^5$).
- The second line contains N space-separated integers A_1, A_2, A_N, ($1<=Ai<=10^9$).
- The third line contains an integer Q, ($1<=Q<=10^5$).
- The next Q lines contain two space-separated integers L, R (1<=L<=R<=N).
- The sum of Q over all test cases does not exceed 10^6, the same for N.

Output:

For each query print the value of X1 + X2 + X3 in a new line.

Example:

Input:

```
5

5

1 2 3 4 5

5

1 5

2 5

3 4

2 4

3 3

3

1 2 2

3
```

1 2

2 3

1 3

4

1 2 2 4

5

1 2

2 3

3 4

2 4

3 4

Output:

4294967300

4294967294

4294967294

4294967299

4294967291

4294967294

4294967292

4294967295

4294967294

4294967292

4294967294

4294967298

4294967294

Fibonacci Numbers

Interesting Names

Difficulty: **Easy**Accuracy: **49.78%**Submissions: **994+**Points: **2**

Geek wants to name his new-born nephew. Geek lives in Geekland where exists only one consonant and one vowel.

Geek finds a name interesting if the length of the name is not more than **N** and there are no consecutive consonants in the name.

Given **N**, find number of distinct interesting names Geek can choose from to name his nephew.

As the number can be pretty large, find it modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^6$

Example:

Input:

3

1

2
3

Output:

2
5
10

Explanation:

Test Case 1 : Two names are possible: One containing the consonant and other containing the vowel.

Test Case 2 : if 'v' denotes vowel and 'c' denotes consonant then there are 5 names possible whose length are not more than 2 like- c, v, cv, vc, vv

Yet Another Chocolates Distribution

Difficulty: **Medium**Accuracy: **70.89%**Submissions: **429+**Points: **4**

Geek has **N** chocolates and he wants to give equal number of some chocolates to his two friends.

He follows following procedure while giving chocolates:

1. He chooses an integer **X** in range **[0, floor(N / 2)]** (both inclusive).
2. He first treats each chocolate identical and gives **X** chocolates to his first friend.
3. He then treats remaining **N - X** chocolates distinct and gives **X** chocolates to his second friend.

Your task is to find number of ways Geek can give chocolates to his two friends. As the answer can be very large, find it modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^{18}$

Example:

Input:

2

2

2837091546

Output:

2

391845697

Explanation:

Test Case 1 : There are two ways to give chocolates:

1. X = 0 and Geek gives 0 chocolates to both his friends.
2. X = 1 and Geek gives first chocolate to his first friend treating all chocolates identical and remaining one to the second friend treating remaining chocolates distinct.

Primality of Sum of Squares of Fibonacci Numbers

Difficulty: **Medium**Accuracy: **49.46%**Submissions: **595+**Points: **4**

Given two integers **N** and **X**, find whether sum of squares of fibonacci numbers from **1** to **N** (both inclusive) is co-prime with **X**th fibonacci number or not.
Print "**Yes**" if it is divisible or "**No**" otherwise (without quotes).

Input:

1. The first line of the input contains a single integer **T** denoting the number of

test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **X**.

Output: For each test case, print the answer.

Constraints:
1. $1 \le T \le 10^5$
2. $1 \le N, X \le 10^9$

Example:

Input:
2
4 2
23673332 12
Output:
Yes
No
Explanation:
Test Case 1 : $F_1^2 + F_2^2 + F_3^2 + F_4^2 = 1^2 + 1^2 + 2^2 + 3^2 = 1 + 1 + 4 + 9 = 15$ and 15 is co-prime with F_2 which is 1.

Computation on Fibonacci Numbers

Difficulty: **Easy**Accuracy: **81.3%**Submissions: **419+**Points: **2**

Geek loves doing problems related to fibonacci numbers as he finds them pretty easy but today he came across a problem in which he got confused. Given two numbers N and M, the task is to find GCD (Greatest Common Divisor) of F_N and F_M, where F_X is the Xth fibonacci number.

$F_1 = 1$

$F_2 = 1$

$F_x = F_{x-1} + F_{x-2}$, if x > 2
You have to help Geek solve the problem. As the answer can be very large, find it modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **M**.

Output: For each test case, print the answer.

Constraints:
1. $1 \le T \le 10^5$
2. $1 \le N, M \le 2 * 10^6$

Example:

Input:
2
8 12
36049 4230
Output:
3
971215059
Explanation:
Test Case 1 : $F_8 = 21$, $F_{12} = 144$ and
GCD(21, 144) = 3.

Fibonacci Coding
Difficulty: **Easy**Accuracy: **65.09%**Submissions: **421+**Points: **2**
Geek lives on Geekland where all computers run on Fibonacci-coded numbers as compared to our planet where computers run on base-2 binary-coded numbers. Given the decimal number **N**, you have to find its corresponding Fibonacci

coding and the absolute difference in a number of bits used to represent **N** in Fibonacci coding and base-2 binary representation.

Input:
1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print two lines - the first containing the Fibonacci coding of **N** and the second containing the absolute difference in a number of bits used to represent **N** in Fibonacci coding and binary representation.

Constraints:
1. $1 \leq T \leq 10^3$
2. $1 \leq N \leq 10^9$

Example:

Input:
2
1
2
Output:
11
1
011
1
Explanation:
Test Case 1 : Fibonacci coding of 1 is "11".
Number of bits used to represent 1 in base-2
binary form is 1, hence difference between
number of bits is 1.

Test Case 2 : Fibonacci coding of 2 is "011".
Number of bits used to represent 2 in base-2
binary form is 2, hence difference between
number of bits is 1.

Nth Fibonacci Number

Difficulty: **Easy**Accuracy: **75.37%**Submissions: **390+**Points: **2**

Given a positive integer n, find the nth Fibonacci number. Since the answer can be very large, return the answer modulo 1000000007.

Input Format:

The first line of input consists of a single integer T, denoting the total number of test cases, and each test case contains a positive integer n.

Output Format:

For each test case, print a single integer mod $10^9 + 7$ representing the nth Fibonacci number.

Constraints:

$1 <= t <= 10^3$

$1 <= n <= 10^5$

The sum of n overall test case does not exceed 10^6.

Example:

Input:

2

2

5

Output:

1

5

Explanation:

Test case 1, 1 is the 2nd number of the Fibonacci series.
Test case 2, 5 is the 5th number of the Fibonacci series.

DIVISORS

Largest Factor

Difficulty: **Easy**Accuracy: **76.33%**Submissions: **502+**Points: **2**

Shawn likes to find out factors of numbers.So given an integer N, he wanted to find out the largest number which divides N, except N ,help him find out!

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the answer

Constraints:

1. $1 \leq T \leq 1000$
2. $2 \leq N \leq 10^9$

Example:

Input:

2
8
7

Output:

4
1

Expalnation:

Testcase1: divisors of 8 are 1, 2, 4, 8 and we can see that largest divisor is 4 which

is not equal to 8 itself.
Testcase2: divisors of 7 are 1, 7 and largest divisor is 1 which is not equal to 7.

Unusual Gift

Difficulty: **Medium**Accuracy: **70.03%**Submissions: **440+**Points: **4**

Santa gave Geek a number **N** and led him to his pile of infinite gifts where each gift had a number written on it. Geek could take those gifts but the conditions were:

1. He could take only those gifts that had a **composite number** written on them. Assume that for each number, there were infinite gifts that had it written on them i.e there were infinite gifts with 1 written on them, infinite gifts with 2 written on them and so on upto infinity.

2. Sum of numbers on the top of the gifts he took should **exactly be equal to N**.

Tell Geek the **maximum possible number of gifts** he could take or print **-1** if he would have to return empty handed.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of the **T** test cases follows.
2. The first line of each test case contains the integer **N.**

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^9$

Example:

Input:
2
12

1

Output:

3

-1

Explanation:

Test Case 1 : 12 = 4 + 4 + 4 = 4 + 8 = 6 + 6 = 12,
but the first splitting has the maximum possible
number of summands.

Total Chocolates I

Difficulty: **Easy**Accuracy: **70.23%**Submissions: **487+**Points: **2**

Geek's maths teacher wanted to teach her class of **N** children about factors of numbers so she gave them all a number each and told them to find out the **sum of all factors** of their number and they would get as many chocolates as the answer. Find out the total number of chocolates the teacher has to buy to be able to give all the children as promised.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains the integer **N,** the number of children in the class.

3. The second line of each test case contains **N space-separated integers**, the i'th integer denoting the number given to the i'th student.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$
2. $1 \leq N \leq 10^5$
3. $1 \leq a[\ i\ ] \leq 10^7$

The sum of N overall test case does not exceed 10^6.

Example:

Input:
2
2
1 6
1
10000000
Output:
13
24902280
Explanation:
Test Case 1 : Sum of all factors of 1 = 1 and
that of 6 = 1 + 2 + 3 + 6 = 12 , so the answer
is 1 + 12 = 13.

Class Test

Difficulty: **Easy**Accuracy: **60.28%**Submissions: **647+**Points: **2**

Geek teaches maths in Geekland state university in which **Q** students have enrolled. Geek took a class test of all the students in which he asked them to find any number and their score would be the sum of all the elements in its prime factorisation.
Now Geek has to check the submissions of all **Q** students but he does not have so much time, so he asked your help as you are his only friend. You have to find the score of each student.

Input:
1. The first line of the input contains a single integer **Q** denoting number of students.
2. Next **Q** lines contains one integer **N**.

Output: For each student, print his score.

Constraints:

1. $1 \leq Q \leq 1000$
2. $2 \leq N \leq 10^9$

Example:

Input:

2

12

8275

Output:

7

341

Explanation:

Test Case 1 : Prime factorisation of 12 is 2 * 2 * 3. Hence, score of the student would be 2 + 2 + 3 = 7.

Least Common Prime Divisor

Difficulty: **Easy**Accuracy: **70.16%**Submissions: **509+**Points: **2**

You recently got a decent rank in problem solving contest at peeksforpeeks.com. As Geek is your friend you were boasting about your problem solving skills in front of him, so he challenged you to solve a problem.

Given two integers **A** and **B**, you have to find their least common prime divisor if it exists. Least common prime divisor is a **prime number** which is the least possible prime number to divide both **A** and **B.**

If least common prime divisor does not exist you have to output **"-1"** (without quotes).

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **A** and **B**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 2*10^5$
2. $1 \leq A, B \leq 10^7$

Note : It is highly recommended to use fast input/output.

Example:

Input:

3
4 8
10 25
1 10000000

Output:

2
-1
-1

Explanation:

Test Case 1 : Least possible prime number to divide 4 and 8 is 2. Hence, 2 is the least common prime divisor.

Test Case 2 : Least possible prime number to divide 10 is 2 whereas least possible prime number to divide 25 is 5.

Hence, least common prime divisor of
10 and 25 does not exist.

Make them Co-prime

Difficulty: **Easy** Accuracy: **59.89%** Submissions: **421+** Points: **2**

Given two integers **A** and **B**. Your task is to make them co-prime using some number (possibly zero) of following operation -
Choose the largest of **A** and **B** and divide it by its smallest prime factor.
You have to find minimum number of operations required to make them co-prime.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **A** and **B**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 5*10^5$
2. $1 \leq A, B \leq 10^7$

Note : It is highly recommended to use fast input/output.

Example:

Input:

3
2 6
4 5

10000000 10000000

Output:

1

0

27

Explanation:

Test Case 1 : After applying operation one time,
A and B becomes 2 and 3 which are co-prime.

Test Case 2 : A and B are already co-prime.

Geek and LPF

Difficulty: **Easy**Accuracy: **86.79%**Submissions: **384+**Points: **2**

Geek is standing at point P(N, 0) and his home at point Q(1, 0). At each step, he can move from point P1(x,0) to P2(x/lpf(x), 0). The task is to find the number of steps required for the geek to reach his home.

Note: lpf(x) is the least prime factor of x.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the answer

Constraints:

1. $1 \leq T \leq 10^3$
2. $1 \leq N \leq 10^5$

The sum of N overall test case does not exceed 10^6.

Example :

Input:

2
8
1

Output:
3
0

Explanation:
Test Case 1: (8, 0) -> (4, 0), -> (2, 0) -> (1, 0).

Total Chocolates II

Difficulty: **Easy**Accuracy: **37.22%**Submissions: **733+**Points: **2**

This time Geek's maths teacher wanted to teach her class of **N** children about **prime factors** of numbers so she gave them all a number each and told them to find out the sum of all distinct prime factors of the number (including itself if it is prime) and they would get as many chocolates as the answer. Find out the total number of chocolates the teacher has to buy to be able to give all the children as promised.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains integer **N,** the number of children in the class.

3. The second line of each test case contains **N space-separated integers**, the ith integer denoting the number given to the i'th student.

Output: For each test case, print the answer.

Constraints:
1. 1 ≤ T ≤ 1000

2. $1 \leq N \leq 10^5$
3. $1 \leq a[i] \leq 10^7$

Example:

Input:
2
10
1 2 3 4 5 6 7 8 9 10
2
1 15
Output:
36
8
Explanation:
Test Case 2 : Sum of prime divisors of 1 = 0
and that of 15 = 3 + 5 = 8 , so the answer
is 0 + 8 = 8.

Prime Factorization

Class Test

Difficulty: **Easy**Accuracy: **60.28%**Submissions: **647+**Points: **2**

Geek teaches maths in Geekland state university in which **Q** students have enrolled. Geek took a class test of all the students in which he asked them to find any number and their score would be the sum of all the elements in its prime factorisation.
Now Geek has to check the submissions of all **Q** students but he does not have so much time, so he asked your help as you are his only friend. You have to find the score of each student.

Input:

1. The first line of the input contains a single integer **Q** denoting number of students.
2. Next **Q** lines contains one integer **N**.

Output: For each student, print his score.

Constraints:

1. $1 \leq Q \leq 1000$
2. $2 \leq N \leq 10^9$

Example:

Input:

2

12

8275

Output:

7

341

Explanation:

Test Case 1 : Prime factorisation of 12 is 2 * 2 * 3. Hence, score of the student would be 2 + 2 + 3 = 7.

Least Common Prime Divisor

Difficulty: **Easy**Accuracy: **70.16%**Submissions: **509+**Points: **2**

You recently got a decent rank in problem solving contest at peeksforpeeks.com. As Geek is your friend you were boasting about your problem solving skills in front of him, so he challenged you to solve a problem.

Given two integers **A** and **B**, you have to find their least common prime divisor if

it exists. Least common prime divisor is a **prime number** which is the least possible prime number to divide both **A** and **B.**
If least common prime divisor does not exist you have to output **"-1"** (without quotes).

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **A** and **B**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 2*10^5$
2. $1 \leq A, B \leq 10^7$

Note : It is highly recommended to use fast input/output.

Example:

Input:
3
4 8
10 25
1 10000000

Output:
2
-1
-1

Explanation:

Test Case 1 : Least possible prime number to divide 4 and 8 is 2. Hence, 2 is the least common prime divisor.

Test Case 2 : Least possible prime number to divide 10 is 2 whereas least possible prime number to divide 25 is 5. Hence, least common prime divisor of 10 and 25 does not exist.

Make them Co-prime

Difficulty: **Easy** Accuracy: **59.89%** Submissions: **421+** Points: **2**

Given two integers **A** and **B**. Your task is to make them co-prime using some number (possibly zero) of following operation -

Choose the largest of **A** and **B** and divide it by its smallest prime factor.

You have to find minimum number of operations required to make them co-prime.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **A** and **B**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 5*10^5$
2. $1 \leq A, B \leq 10^7$

Note : It is highly recommended to use fast input/output.

Example:

Input:
3
2 6
4 5
10000000 10000000
Output:
1
0
27
Explanation:
Test Case 1 : After applying operation one time,
A and B becomes 2 and 3 which are co-prime.
Test Case 2 : A and B are already co-prime.

Geek and LPF

Difficulty: **Easy**Accuracy: **86.79%**Submissions: **384+**Points: **2**

Geek is standing at point P(N, 0) and his home at point Q(1, 0). At each step, he can move from point P1(x,0) to P2(x/lpf(x), 0). The task is to find the number of steps required for the geek to reach his home.

Note: lpf(x) is the least prime factor of x.

Input:
1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the answer

Constraints:
1. $1 \leq T \leq 10^3$

2. $1 \leq N \leq 10^5$

The sum of N overall test case does not exceed 10^6.

Example :

Input:

2

8

1

Output:

3

0

Explanation:

Test Case 1: (8, 0) -> (4, 0), -> (2, 0) -> (1, 0).

Total Chocolates II

Difficulty: **Easy**Accuracy: **37.22%**Submissions: **733+**Points: **2**

This time Geek's maths teacher wanted to teach her class of **N** children about **prime factors** of numbers so she gave them all a number each and told them to find out the sum of all distinct prime factors of the number (including itself if it is prime) and they would get as many chocolates as the answer. Find out the total number of chocolates the teacher has to buy to be able to give all the children as promised.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.

2. The first line of each test case contains integer **N,** the number of children in the class.

3. The second line of each test case contains **N space-separated integers**, the ith integer denoting the number given to the i'th student.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 1000$
2. $1 \leq N \leq 10^5$
3. $1 \leq a[i] \leq 10^7$

Example:

Input:

2

10

1 2 3 4 5 6 7 8 9 10

2

1 15

Output:

36

8

Explanation:

Test Case 2 : Sum of prime divisors of 1 = 0

and that of 15 = 3 + 5 = 8 , so the answer

is 0 + 8 = 8.

Prime Numbers

Is It Broken?

Difficulty: **Easy**Accuracy: **77.4%**Submissions: **542+**Points: **2**

Geek has the most powerful device with him on the planet which is popularly known as "primality checker". If an array of **N** elements **a[]** is supplied to the primality checker, it returns a binary string **s** of length **N** such that **s[i] = '1'** if **a[i]** is

a prime number, otherwise **s[i] = '0'**.
Geek suspects that his device is broken, so he wants your help to make sure if it is broken or not and for that Geek gave you the array of elements which was supplied to the primality checker and the string which it returned.
Output **"YES"** (without quotes) if you are sure that the device is broken, otherwise print **"NO"** (without quotes).

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.
4. The third line of each test case contains **s** - binary string of length **N**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10$
2. $1 \leq N \leq 10^3$
3. $1 \leq a[i] \leq 10^6$
4. $|s| = N$

Example:

Input:
2
2
3 5

10
3
1 2 3
011

Output:

YES
NO

Explanation:

Test Case 1 : 5 is a proven prime number but according to the string returned by primality checker it is not a prime number. Hence, the device is broken.

Test Case 2 : According to the string returned by primality checker 1 is not a prime number and 2 & 3 are prime numbers, which is true. Hence, the device is not broken.

Maths Assignment

Difficulty: **Easy**Accuracy: **48.28%**Submissions: **1K+**Points: **2**

Mike got a maths assignment in which he is given two integers **m** and **n**. He has to calculate sum of all prime numbers which exist in the range **[m, n]**. Like most of the students in the world Mike does not like maths, so he came to you for help. You have to find sum of all prime numbers which exist in the range **[m, n]**.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two-space separated integers **m** and **n**.

Output: For each test case, print sum of all prime numbers which exist in the range **[m, n]**.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq m \leq n \leq 10^6$

Example:

Input:
2
2 10
35 123
Output:
17
1433
Explanation:
In the first test case, there are four prime numbers in the range : 2, 3, 5 & 7 and 2 + 3 + 5 + 7 = 17.

Primes in Range

Difficulty: **Easy**Accuracy: **47.25%**Submissions: **311+**Points: **2**

Geek was arguing with his friend upon how many prime numbers are there in the universe. Geek was stating that for every 1000 numbers in the universe there exist one prime number but his friend was not ready to believe him as he thinks that for every 200 numbers in the universe there exist one prime number.
To prove his point wrong, his friend gave him **Q** queries. Each query consist of two integers **L** and **R**. For each query Geek has to find out number of prime numbers in the range **[L, R]**. As Geek is not that much good at maths as he is at fighting. So he asked your help to find out the answer to each query.

Input:
1. The first line of the input contains a single integer **Q** denoting the number of queries. The description of **Q** queries follows.
2. Each query contains two space-separated integers **L** and **R**.

Output: For each query, print the answer.

Constraints:

1. $1 \le T \le 10$
2. $1 \le L, R \le 10^{10}$
3. $R - L \le 2*10^5$ *i.e.* length of range does not exceed $2*10^5$.

Example:

Input:

2

4 12

9786362811 9786562811

Output:

3

8606

Explanation:

Test Case 1 : There are three prime numbers

5, 7 and 11 in the range [4, 12].

Geek and Function III

Difficulty: **Easy**Accuracy: **76.81%**Submissions: **427+**Points: **2**

Given an integer N, the task is to find fun(N).

```
int fun(int x)
{
  int flag = 1;
  for(int i=2; i<x ; i++)
  {
     if(x%i==0)
     {
```

```
        int y = x/i;
        flag = x - y*i;
      }
    }
    return flag*x;
}
```

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of T test cases follows.
2. The first line of each test case contains a single integer N.

Output:

For each test case, print the answer.

Example:

Input:

2
19
1

Output:

19
1

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^5$

Prime Subarray

Difficulty: **Easy**Accuracy: **70.0%**Submissions: **476+**Points: **2**

Geek likes prime numbers and his little sister likes to play with arrays. She needs your help as he gave her an array arr of integers of size N, and the task to find the number of subarrays with prime sum.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**
3. The next line of each test case contains N space-separated integers.

Output: For each test case, print the answer

Constraints:

1. $1 \leq T \leq 10$
2. $1 \leq N \leq 10^3$
3. $1 \leq arr[i] \leq 10^4$

Example:

Input:

2
3
1 5 8
1
131

Output:

2
1

Explanation:

Test Case 1: {5}, {5, 8} are subarray

Geek and Non-Primes

Difficulty: **Easy**Accuracy: **60.15%**Submissions: **542+**Points: **2**

Geek, Jessy, and Walter participated in a coding contest, and they solved all the questions except for this question. They called you and asked for help. The question contains two integers **l** and **r** and the task is to find the sum of squares of non-prime numbers in the range **l** to **r**. Can you help Geek's team to win the contest?

Note: The answer can be very large. So, output answer modulo 10^9+7.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **l** and **r.**

Output: For each test case, print the answer

Constraints:

1. $1 \le T \le 10^3$
2. $1 \le l \le r \le 10^5$

Example:

Input:

2
1 5
7 7

Output:

17
0

Explanation:

Test Case 1: Non- Prime numbers in the range 1 to 5 are 1 and 4.

Prime and Palindrome

Difficulty: **Easy**Accuracy: **83.27%**Submissions: **405+**Points: **2**

Vanya's teacher gave her an integer N and told her to find out if N is palindrome or not and she could do it. Now she asked her to check if N was both prime and palindrome or not and now Vanya needs your help.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the **"yes"** if N is prime and palindrome. Otherwise, print **"no"**

Constraints:

1. $1 \leq T \leq 1000$
2. $1 \leq N \leq 10^9$

Example:

Input:

3
3
131
22

Output:

yes
yes
no

Make It Prime !

Difficulty: **Easy**Accuracy: **79.39%**Submissions: **403+**Points: **2**

Geek was gifted a composite number N by his father on his birthday. But Geek is not satisfied with his gift as he only likes prime numbers. So he wants to make this number a positive prime number by adding some integer value x. He wants you to find what could be the minimum value of the magnitude of **x** *i.e.* **|x|**.
It is guaranteed that **N** is a positive composite number.

Input:
1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the minimum magnitude of **x** *i.e.* **|x|**.

Constraints:
1. $1 \leq T \leq 1000$
2. $1 \leq N \leq 10^9$

Example:

Input:
2
4
3445345
Output:
1
12
Explanation:
Test Case 1: Geek can add -1 to 4 which
will make it 3 which is a prime number.

GCD and LCM

Maximum LCM

Difficulty: **Easy**Accuracy: **75.02%**Submissions: **400+**Points: **2**

While reading a maths book, Geek came across an interesting problem, so he decided to share it with you so that you can also enjoy solving it.

Given a number **G**, you have to find a pair of two numbers such that their Greatest Common Divisor is **G** and their Least Common Multiple is maximum possible. Both numbers should be **less than or equal to 10^9**.

You have to output the Least Common Multiple of the found pair.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **G**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq G \leq 10^9$

Example:

Input:

2

2

94773678

Output:

499999999000000000

8529631020

Explanation:

Test Case 1 : 1000000000 and 999999998 are the two numbers less than 10^9 such that their GCD is 2 and LCM is largest possible which is 499999999000000000.

Test Case 2 : 947736780 and 852963102 are the two numbers less than 10^9 such that their GCD is 94773678 and LCM is largest possible which is 8529631020.

Geek and Line

Difficulty: **Easy**Accuracy: **48.9%**Submissions: **548+**Points: **2**

Given a line equation **AX + BY = C**, where **C** is the greatest common divisor of **A** and **B**. The task is to find an integer point **p(x, y)** which lies on the given line. Find **x** and **y** such that **|x| + |y|** is the minimum possible. If there exist several **x** and **y** satisfying the minimal criteria, output the pair for which **x ≤ y**.

Note: |x| represents the absolute value of **x.**

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **A,** and **B.**

Output: For each test case, print two space-separated values **x** and **y.**

Constraints:

1. $1 \le T \le 1000$
2. $1 \le A, B, C \le 10^9$

Example:

Input:

2

2 6

15 12

Output:

1 0

1 -1

Explanation:

Test Case 1: gcd of 2 and 6 is 2 and 2*(1)+6*(0) = 2 which satisfy the given equation and |x| + |y| = |1|+|0| = 1 which is minimum.

Test Case 2: gcd of 15 and 12 is 3 and 15*(1)+12*(-1) = 3 which satisfy the given equation and |x| + |y| = |1|+|-1| = 2 which is minimum.

Jessy and Walter

Difficulty: **Easy**Accuracy: **73.78%**Submissions: **286+**Points: **2**

Geek has two friends named Jessy and Walter. Jessy can write **X** lines of code and Walter can write **Y** lines of code. The task is to find the least value of **Z** such that geek can generate any code of size (number of lines of the code) greater than or equals to **Z** using Jessy and Walter code. Geek can use Jessy's code **arbitrary** times and the same goes with Walter's code. If no such **Z** exist then print "**-1**".

Note: It is not necessary to use both the codes an equal number of times.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **X** and **Y.**

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$

2. $1 \leq X, Y \leq 10^9$

Example:

Input:
3
2 3
1 4
5 10
Output:
2
0
-1
Explanation:
Test case 1: Geek can genrate any number of lines of code greater than or equal to 2 like wise -
2 lines code -> use Jessy's code one time.
3 lines code -> use Waltar's code one time.
4 lines code -> use Jessy's code two times.
5 lines code -> use Jessy's code one time and Walter code's one time.
and so on ....
Test case 2: Geek can genrate any number of lines of code greater than or equal to 0 like wise -
0 lines code -> not use anyone's code.
1 lines code -> use Jessy's code one time.
2 lines code -> use Jessy's code two times.
3 lines code -> use Jessy's code three times.
and so on ...

Geek and GCD

Difficulty: **Easy**Accuracy: **53.39%**Submissions: **502+**Points: **2**

Geek has an array **arr[]** of positive integers of size **N** and he asked you to calculate the maximum value of GCD of any subarray of size greater than one.
Formally, find the maximum value of **gcd(arr[i], arr[i+1],.....arr[j]))** for all **1 <= i < j <= N**

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N**.
3. The next line contains **N** space-separated positive integers represents array **arr[]**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10$
2. $2 \leq N \leq 10^5$
3. $1 \leq arr[i] \leq 10^5$

Example:

Input:
2
5
1 2 4 6 8
2
1 1
Output:
2

1

Explanation:

Test Case 1: We can take subarray from index 2 to 3 i.e. {2, 4} to get maximum value of gcd which is 2.

Test Case 2: There is only one option of choosing the subarray which is the whole array.

One Number

Difficulty: **Easy**Accuracy: **50.42%**Submissions: **571+**Points: **2**

Geek has an array **a[]** of size **N**. Geek wants to add one more element in the array such that LCM of all the elements of the array does not change after adding this element. Geek wants your help to find the maximum number such that he can insert it into the array and LCM of all the elements will not change.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer. It is guaranteed that the answer is less than 10^{18}.

Constraints:

1. $1 \leq T \leq 10$
2. $1 \leq N \leq 10^5$

3. $1 \leq a[i] \leq 10^{18}$

Example:

Input:
2
3
2 2 4
2
5 3
Output:
4
15
Explanation:
Test Case 1 : After adding 4 in the array,
LCM of all the elements will not change.

Increase the GCD

Difficulty: **Medium**Accuracy: **67.19%**Submissions: **439+**Points: **4**

Geek has an array **a[]** of **N** elements. Geek wants to find out minimum number of elements (possibly zero) such that after removing them GCD of all the elements of the array becomes greater than 1.
It is guaranteed that after removing minimum number of elements, **a[]** is non-empty.
Your task is to help Geek find the same.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^6$
3. $1 \leq a[i] \leq 10^5$
4. It is guaranteed that sum of N over all testcases does not exceed 10^6.

Example:

Input:

2
3
1 1 2
3
2 3 6

Output:

2
1

Explanation:

Test Case 1 : Geek has to remove first and second element, after which only 2 will be left in the array.

Test Case 2 : Geek will remove either first or second element.

Geometry Is Hard

Difficulty: **Easy**Accuracy: **69.77%**Submissions: **371+**Points: **2**

Geek struggles in doing every other geometry problem. Help him solve the following problem -

Given an equation of line **Ax + By = C**, find if there exists any integral solution to this equation. In other words, you have to find whether there exists any pair **{x, y}** where both **x** and **y** are integers such that they satisfy the equation of line.

If there exists any integral solution, print "**YES**" (without quotes), otherwise print "**NO**" (without quotes).

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains three space-separated integers **A**, **B**, and **C**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$
2. $1 \leq A, B, C \leq 10^9$

Example:

Input:

2

1 2 7

2 4 3

Output:

YES

NO

Explanation:

Test Case 1 : A possible valid pair of {x, y} is {1, 3} as 1 * 1 + 2 * 3 = 7.

Test Case 2 : Any pair of {x, y} will result in a even number. Hence, solution does not exist.

Find GCD of two numbers

Difficulty: **Easy**Accuracy: **59.53%**Submissions: **746+**Points: **2**

Given two positive integers A and B, find the GCD of A and B.

Example 1:

Input: A = 3, B = 6

Output: 3

Explanation: GCD of 3 and 6 is 3

Example 2:

Input: A = 1, B = 1

Output: 1

Explanation: GCD of 1 and 1 is 1

Your Task:

You don't need to read input or print anything. Complete the function **gcd()** which takes two positive integers as input parameters and returns an integer.

Expected Time Complexity: O(log(min(A, B)))

Expected Auxiliary Space: O(1)

Constraints:

$1 \le A, B \le 10^9$

Mathematical Principles

Yet Another Teams Dividing Problem

Difficulty: **Medium**Accuracy: **73.17%**Submissions: **406+**Points: **4**

There are **N** students in a class. Each student is given a work factor which is distinct number in the range **[1, N]**, in other words **a[]** is a permutation of integers from **1** to **N** where **a[i]** is the work factor of **i^{th}** student. As their project mentor, you want to divide **all** students in **X** teams where each team is a consecutive range of students. For example team formed by students at index {1, 2, 3} forms a valid team whereas team formed by students at index {1, 3, 7} is not a valid team. Team power is defined as maximum work factor of any student who is in that team. You want to maximise total team power i.e. sum of team powers of all teams. You have to find maximum total team power and number of ways in which you can divide teams to achieve that total team power, as this number can be very large find it modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **X**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print two integers - maximum total team power and number of ways in which you can divide teams to achieve that total team power, as this number can be very large print it modulo $10^9 + 7$.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq X \leq N \leq 10^5$
3. $1 \leq a[i] \leq N$

Example:

Input:

2
4 2
4 1 3 2
2 2
1 2

Output:

7 2
3 1

Explanation:

In the first test case, there are two ways of dividing teams in which maximum total team power of 7 can be achieved :
{4 1 | 3 2} and {4 | 1 3 2}.
In Second test case we have only one way to form two teams.
{ 1 | 2 }

Exam Seating

Difficulty: **Easy** Accuracy: **52.8%** Submissions: **161+** Points: **2**

Geek teaches a class of **N** students having roll numbers from **1** to **N**. Each student has a group number **a[i]**, if two students having roll number **i** and **j** have same group number *i.e.* if **a[i] = a[j]** then they are friends. Geek wants that no two friends must be seated in same room for giving exam.
As Geek is very busy in setting the question paper he gave you **Q** queries, each query consist of two integers **left** and **right**. He wants to know that if students having roll numbers from **left** to **right** can be seated in same room or not. You have to output "**YES**" (witjout quotes) if they can be seated in same room, otherwise output "**NO**" (without qoutes).

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two-space separated integers **N** and **Q**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.
4. Next **Q** lines contains two space-separated integers **left** and **right** denoting a query.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 1000$
2. $1 \leq N, Q \leq 10^3$
3. $1 \leq a[i] \leq 100$
4. $1 \leq left \leq right \leq 10^3$
4. It is guaranteed that sum of Q over all test cases does not exceed 10^5.

Example:

Input:

2

5 2

1 2 1 2 1

1 2

2 4

3 1

1 2 100

1 3

Output:

YES

NO

YES

Explanation:

Test Case 1 : Students with roll number 1 and 2 can be seated in same room but students with roll number 2,3 and 4 cannot be seated in same room because students with roll number 2 and 4 are friends.

Test Case 2 : All students can be seated in same room as none of them are friends.

Hard Question

Difficulty: **Easy**Accuracy: **44.57%**Submissions: **130+**Points: **2**

Geek teaches maths in an elementary school. He wants to give a hard divisibility question to his students, so he does not want to use integers which are divisible by 2, 3, 4 or 5 because otherwise question will become too easy to solve. He also wants to use integers in range [1, N] only as until now he has taught that many integers only. You have to find number of integers he can use.

Input:

1. The first line of the input contains a single integer **T** denoting the number of

test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the answer

Constraints:
1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^9$

Example:

Input:
2
7
5
Output:
2
1
Explanation:
Test Case 1: 1 and 7 are the only such numbers in the range [1, 7] which are not divisible by 2, 3, 4, and 5.
Test Case 2: 1 is the only integer in the range [1,5] which is not divisible by 2,3,4 and 5.

Beautiful Images

Difficulty: **Medium**Accuracy: **76.92%**Submissions: **347+**Points: **4**

A colored image is represented using **N * N** pixels where each pixel has one of the three colors - Red, Blue or Green.

Geek finds an image beautiful only if all three colors are present in atleast one of the pixels of the image.
Given **N**, find the number of beautiful images.
As answer can be very large, find it modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^5$
2. $2 \leq N \leq 10^5$

Example:

Input:
2
2
3
Output:
36
18150
Explanation:
Test Case 1 : By drawing all images, it can be easily seen that 36 images of 2*2 pixels are beautiful.

Work Ethic

Difficulty: **Easy**Accuracy: **87.97%**Submissions: **298+**Points: **2**

Geek wants to organise a competition in which **N** students participated. The work ethic of **i^{th}** student is given by **a[i]**, where **a[i] = i**. Geeks wants to make teams of two students. Work ethic of each team is sum of work ethics of students in the team. Geek wants to know how many minimum number of students he can randomly select from these **N** students such that he can make atleast one team with work ethic exactly **N + 1**.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $2 \leq N \leq 10^5$

Example:

Input:

2

2

4

Output:

2

3

Explanation:

Test Case 1 : a[] = {1, 2}, we have to
choose both the students such that we can
make atleast one team with work ethic = 3.

Test Case 2 : a[] = {1, 2, 3, 4}, we
can randomly choose 3 students with work ethics
either {1, 2, 3} or {1, 2, 4} or {1, 3, 4} or
{2, 3, 4} such that we can make atleast one team
with work ethic = 5.

Here, 2 cannot be the answer because if we
randomly select two students with work ethics
= {3, 4}, we cannot make atleast one team
with work ethic = 5.

Non-co-prime Numbers

Difficulty: **Medium**Accuracy: **82.24%**Submissions: **349+**Points: **4**

Given an array **a[]** of size **N**, find out the elements at two distinct indices such that their GCD is **not** 1, In other words, find out the two distinct array indices such that the elements at them are **not** co-prime and the number of array elements between them is minimum.

You have to output number of elements between the found elements. If two such indices are not found, print **-1**.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^4$
3. $2 \leq a[i] \leq 100$

Example:

Input:

2
3
2 3 4
2
2 5

Output:

1
-1

Explanation:

Test Case 1 : Two non-co-prime elements are first and third element and number of elements between both of them is 1.

Test Case 2 : Two Elements are not found such that they are not co-prime.

Christmas Swapping

Difficulty: **Medium** Accuracy: **83.7%** Submissions: **307+** Points: **4**

There are **N** people in a christmas office party. Each person is numbered from **1** to **N**. Each person brought a gift which he/she will give to another person. Gifts are also numbered from **1** to **N**. A person only gives gift to another person if

that person is his/her friend. You are given **M** pairs **{m_i, m_j}** denoting that people numbered **m_i** and **m_j** are friends.
There are arbitrary number of rounds in which people exchange gifts. Obviously no one wants to end up with a gift which they brought. Find number of different permutaions of gifts ending up with people numbered from **1** to **N** and since the number can be large, find it **modulo $10^9 + 7$.**

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers integers **N** and **M**.
3. The next **M** lines contains two space-separated integers integers **m_i** and **m_j**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10$
2. $1 \leq N \leq 10^3$
3. $1 \leq M \leq 10^3$
4. $1 \leq m_i, m_j \leq N$

Example:

Input:
2
2 1
1 2
3 1
2 3
Output:
1
0

Explanation:
Test Case 1 : There is only one way : person 1 will end up with gift person 2 brought and vice-versa.
Test Case 2 : There are zero ways as person 1 has no friend which means he can't have any gift.

DS Assignment

Difficulty: **Easy**Accuracy: **62.65%**Submissions: **83+**Points: **2**

Geek has to submit an assignment in Data Structure class in which he has to construct an array of size **N** of positive integers (greater than 0) such that the sum of none of its subarray is divisible by **K**. Since Geek is weak in DS, you have to help him construct the array.
Moreover, Geek also wants to impress the professor, so he wants the prefix array of the array to be lexicographically the smallest possible.

Input:
1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **K**.

Output: For each test case, print the array on a new line. If no such array exists, you have to print "-1" (without quotes).

Constraints:
1. $1 \le T \le 10^3$
2. $1 \le N, K \le 2*10^5$
2. The sum of N over all test cases will not exceed 10^6.

Example:

Input:

2
1 2
2 1

Output:

1
-1

Explanation:

Test Case 1 : There is only one subarray
and 1 is not divisible by 2.

Balls and Boxes

Difficulty: **Hard**Accuracy: **96.8%**Submissions: **252+**Points: **8**

Geek has **N** boxes numbered 1 to N. Each box contains a ball with some number written on it. The numbers written on the balls are a permutation of 1 to N. Geek started checking boxes randomly. Till now he has checked **K** boxes and to his surprise, none of the boxes, checked so far, contain the balls having the same number as the number written on their box.
Geek wants you to find the number of ways the balls can be put in the boxes such that none of the **K** boxes checked so far have the same number of balls inside them. Print the answer modulo 10^9+7.

Input Format:

- The first line contains an integer **T** - the number of test cases, then the test cases follow:
- The first line and the only line that contains two integers ***N and K.***

Output Format:

Print the answer by a newline character.
Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \leq T \leq 10^3$

$1 \leq N \leq 10^3$

$1 \leq S \leq \min(10,N)$

Example:

Input:

2

3 1

3 3

Output:

4

2

Explanation:

Test Case 1: Regardless of the box chosen,
out of the six possible combinations, there
will be 4 ways in which the chosen box does
not contain the ball having the same number.

Test Case 2: Possible assignments are:

{2, 3, 1}, {3, 1, 2}.

Find Three

Difficulty: **Medium**Accuracy: **53.16%**Submissions: **94+**Points: **4**

Given an array **Arr[ ]** of **N** integers. Check whether there exist three indexes **(1 ≤ i < j < k ≤ N)** such that **Arr[i] & Arr[j] ≠ 0, Arr[i] & Arr[k] ≠ 0,** and **Arr[j] & Arr[k] ≠ 0**.
Where **'&'** represents bitwise AND operator.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains the integer N.
- The next line contains N space-separated integers.

Output Format:

For each test case, print the answer (**"YES"** if indexes exit otherwise **"NO"**) to the problem followed by a new line character.

Note: The generated output is white space sensitive, do not add any extra spaces or unnecessary newline characters.

Constraints:

$1 \le T \le 1000$

$1 \le N \le 500$

$1 \le arr[i] \le 10^9$

*It is guaranteed that the sum of **N** over all test cases does not exceed **500***

Example:

Sample Input 1:

```
4
5
3 4 11 1 5
6
12 3 6 8 3 9
7
1 2 1 2 1 2 1
5
4 2 1 8 32
```

Sample Output:

```
YES
YES
YES
NO
```

Explanation:

For Test Case #1:-

N = 5

Arr [ ] = {3, 1, 11, 5, 4}

One possible set of three indexes are {1, 3, 5}

Number Theoretic Functions

Geek and Function ||

Difficulty: **Easy**Accuracy: **77.84%**Submissions: **361+**Points: **2**

Given an integer N, the task is to find f(N).

1. f(x) = Summation of g(y) for all the divisors of x, except x.
2. g(y) = Count of the number of integers from 1 to *y*, which are coprime to *y.*

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the answer

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^5$

Example:

Input:

3

6

16

1

Output:

4

8

0

Explanation:

Test Case 1: f(6) = g(1) + g(2) + g(3) = 1 + 1 + 2 = 4

Test Case 2: f(16) = g(1) + g(2) + g(4) + g(8) = 1 + 1 + 2 + 4 = 8

Yet Another Game

Difficulty: **Easy** Accuracy: **58.94%** Submissions: **95+** Points: **2**

Geek and his friend are playing yet another game in which both of them have two integers **A** and **B**. The game progresses turn by turn and Geek plays first.

In each turn, player chooses a positive integer not greater than the number had by opposite player (**A** in case of Geek and **B** in case of Geek's friend) which is co-prime to the number had by opposite player.

A player cannot choose the same number again.

The player which is unable to make a turn looses the game.

You have to find if Geek can win the game given **A** and **B** or not. Print "**Yes**" if Geek can win, otherwise "**No**" (without quotes).

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **A** and **B**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq A, B \leq 10^6$

Example:

Input:

2

1 3

3 4

Output:

Yes

No

Explanation:

Test Case 1 : Geek will choose 1, then his friend will choose 1, then Geek will choose 2, after which his friend will be unable to make a turn.

Test Case 2 : Geek will choose 1, then his friend will choose 1, then Geek will choose 3, then his friend will choose 2, after which Geek will be unable to make a turn.

Function of Function

Difficulty: **Medium**Accuracy: **92.9%**Submissions: **301+**Points: **4**

Geek had been given a function **$y(x) = x^2 - x$** for homework and that day they were taught Euler's Totient Function in class.

Euler's Totient Function **$\Phi(n)$** for an input **n** is count of numbers in {1, 2, 3, … , n} that are **relatively prime** to n. Given the value of **x**, help him find out the value of **$\Phi(y(x))$** .

Input:

1. The first line of the input contains a single integer **T** denoting the number of

test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **x**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^5$
2. $1 \leq x \leq 10^5$

Example:

Input:
2
3
100000
Output:
2
2592000000
Explanation:
Test Case 1 : y(3) = 6 and there are only
two numbers- 1 and 5 that are less than 6
and relatively prime to it.

Two and Five

Difficulty: **Easy**Accuracy: **61.03%**Submissions: **95+**Points: **2**

Euler's Totient function Φ(n) for an input **n** is the count of numbers in {1, 2, 3, …, n} that are relatively prime to n *i.e.* the numbers whose GCD (Greatest Common Divisor) with n is 1.
Geek's lucky numbers are **2**, **5,** and **x**. Help him as he wants to find out the **number of zeroes** in the number **y** which is equal to **$\Phi(2^x) * \Phi(5^x)$**.

Input:
1. The first line of the input contains a single integer T denoting the number of

test cases. The description of **T** test cases follows.
2. The first line of each test case contains an integer **x.**

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^3$
2. $0 \leq x \leq 10^5$

Example:

Input:
2
1
2
Output:
0
1
Explanation:
Test Case 1: $\Phi(2^1) * \Phi(5^1) = 1 * 4 = 4$ and
it has zero 0s in it.

Sum of XOR of all pairs

Difficulty: **Medium**Accuracy: **68.96%**Submissions: **423+**Points: **4**

Given an array **A** of **N** integers, find the sum of xor of all pairs of numbers in the array.

Input :

The first line consists of **T**, the number of testcases. Each test case is as follows:-

- The first line consists of **N**, the size of the array.
- The second line consists of **N** space-separated integers.

Output :

For each test case output in a single line the **sum** of XOR of all pairs.

Constraints :

$1 \leq \mathbf{T} \leq 1000$
$2 \leq \mathbf{N} \leq 10^5$
$1 \leq \mathbf{A[i]} \leq 10^5$

The sum of **N** over all testcases doesn't exceed $\mathbf{10^6}$

Example :

Input :
2
3
7 3 5
4
5 9 7 6
Output :
12
47
Example :
Testcase 1:- All possible pairs and their Xor
Value: (3 ^ 5 = 6) + (7 ^ 3 = 4)
+ (7 ^ 5 = 2) = 6 + 4 + 2 = 12

Binomial Coefficients

Choosing Items

Difficulty: **Easy** Accuracy: **70.01%** Submissions: **350+** Points: **2**

Geek has two bags of items. First bag contains **N** distinct items and second bag contains **M** distinct items. He wants to choose a total of **X** items from both the bags but he has a restriction - he will choose only even number of items from

first bag but he can choose any number of items from second bag.
Find number of ways he can choose items.
As the answer can be very large, print it modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains three space-separated integers **N**, **M** and **X**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 1000$
2. $1 \leq N, M \leq 100$
3. $0 \leq X \leq (N + M)$

Example:

Input:
2
2 3 3
1 1 2
Output:
4
0
Explanation:
Test Case 1 : There are four possible ways to choose items:
1. All items from second bag.

2. All items from first bag and first item from second bag.
3. All items from first bag and second item from second bag.
4. All items from first bag and third item from second bag.

Test Case 2 : There is no possible way to choose two items such that even number of items are chosen from first bag.

Subset XOR

Difficulty: **Medium** Accuracy: **50.27%** Submissions: **181+** Points: **4**

Geek has an array **a[]** of **N** elements. The array has a special property - every element in the array, **a[i]** is of the form 2^M where **M** is a non-negative integer. Geek challenges you to find number of ways you can choose a subset of elements of the array such that by taking bitwise XOR of all the choosen elements we will get **X**.

As answer can be very large, output it modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two-space separated integers **N** and **X**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^3$

3. $1 \leq a[i], X \leq 10^9$

Example:

Input:
2
4 3
1 2 2 1
2 10
2 4
Output:
4
0
Explanation:
Test Case 1 : There are four possible ways to choose a subset such that taking XOR of elementsâ€‹ choosen will give 3:
1. First and second element.
2. First and third element.
3. Fourth and second element.
4. Fourth and third element.
Test Case 2 : There is no way to choose a subset such that taking XOR of elements choosen will give 10.

Geek and Chocolates II

Difficulty: **Easy**Accuracy: **61.85%**Submissions: **152+**Points: **2**

As you know, Geek loves chocolates. So, he went to a shop and bought **K** chocolates. Geek wants to hide all his chocolates so that he does not have to share them with his friends. Now, Geek is wondering, in how many ways he can put **K** chocolates in **N** pockets which are on his jacket. A pocket can contain all the **K** chocolates or can be empty.

Note: All the chocolates are identical but All pockets are distinct.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two integers **N** and **K.**

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq N, K \leq 9$

Example:

Input:

2

2 1

3 2

Output:

2

6

Explanation:

Test Case 1: Geek can put his only chocolate in either first pocket or second pocket.
Hence, there are two ways to put chocolates.

Geek and Function

Difficulty: **Easy**Accuracy: **88.42%**Submissions: **259+**Points: **2**

Geek loves to solve maths problems but he got struck at a problem. As you are his friend, he came to you for help. Given an integer **N**, the task is to find summation of **f(i)** for all **3 ≤ i ≤ N**, where, $\mathbf{f(x) = x*{}^{N}C_{x} - (x-1)*{}^{N}C_{x-1} + (x-2)*{}^{N}C_{x-2} - (x-3)*{}^{N}C_{x-3}}$. As answer can be very large, you have to output it modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the answer.

Constraints:
1. $1 \le T \le 10^5$
2. $1 \le N \le 10^5$

Example:

Input:
2
5
7
Output:
15
70

Geek and Contest

Difficulty: **Easy**Accuracy: **74.35%**Submissions: **117+**Points: **2**

Geek wants to participate in a team contest. He has **N** friends. One wierd thing about Geek is that he is obsessed with prime numbers so he wants to choose only

2, 3 or 5 friends and form team with them to participate in the contest. Find number of ways he can form a team.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the answer

Constraints:
1. 1 ≤ T ≤ 500
2. 5 ≤ N ≤ 504

Example:

Input:
2
7
5
Output:
77
21

Geek and Combination

Difficulty: **Easy**Accuracy: **90.54%**Submissions: **255+**Points: **2**

Geek opted for a tough course at Geekland State University. Now he has to submit his homework which consists of a very hard problem. Now he asks you to help him solve the problem which is given two integers **N** and **K**, the task is to

find **A** from the given expression :

$\mathbf{K * {}^{N}C_{K} = N * A}$

Note: As the answer can be very large, output the answer modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **K**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$
2. $1 \leq K \leq N \leq 10^5$

Example:

Input:

2

4 2

100000 1000

Output:

3

796808364

Anagrams

Difficulty: **Medium**Accuracy: **69.82%**Submissions: **126+**Points: **4**

Geek's friend's name is **S**. Geek has a string **T** with him. He wants to give his friend a birthday gift - an anagram of his name, he wants to find out a number of

ways he can choose some subsequence of elements of **T** such that the resulting subsequence is an anagram of **S**.
Return answer modulo 1e9 + 7.

Input:
1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a string denoting **S**.
3. The second line of each test case contains a string denoting **T**.

Output: For each test case, print the answer.

Constraints:
1. 1 ≤ T ≤ 1000
2. 1 ≤ |S|, |T| ≤ 1000

Example:

Input:
2
a
aaab
abcd
abccc
Output:
3
0
Explanation:
Test Case 1 : There are three ways to choose

the subsequence - either choose the first element, second element or third element.
Test Case 2 : There is no way we can choose a subsequence as there is no "d" in T.

Count the Arrays

Difficulty: **Medium**Accuracy: **70.4%**Submissions: **135+**Points: **4**

Geek had a strictly increasing array **a[]** of **N** elements of distinct integers in the range **[1, M]**. He lost his array and only remembers one element **X** which was at position **y** (1-based indexing).
You have to find the number of arrays that can be a possibility for Geek's lost array.
It is possible that Geek makes a mistake in remembering element **X**.
As the answer can be very large, print it modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **M**.
3. The second line of each test case contains two space-separated integers **X** and **y**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^3$
2. $1 \leq N \leq M \leq 10^5$
3. $1 \leq y \leq N$

4. $1 \leq X \leq M$

Example:

Input:
2
3 5
2 1
3 3
3 2
Output:
3
0
Explanation:
Test Case 1 : There are three possible arrays which can be Geek's lost array : {{2, 3, 4}, {2, 3, 5}, {2, 4, 5}}
Test Case 2 : As 3 is at position 2 and array is sorted so there is no element left to be at position 3. Hence, no array exists such that it can be Geek's lost array.

Catalan Number

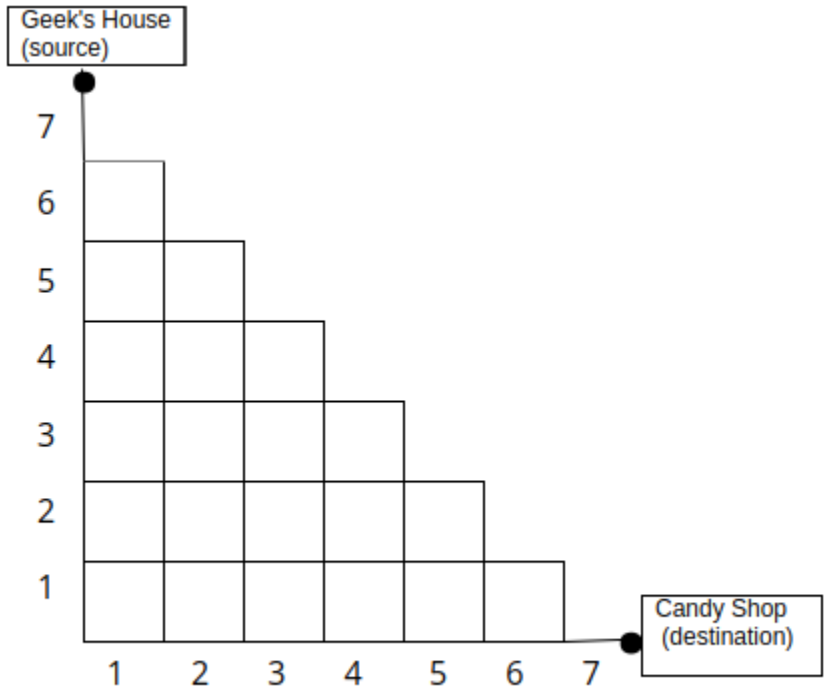
Triangular Colony

Difficulty: **Easy**Accuracy: **71.25%**Submissions: **306+**Points: **2**

Geek lives in colony whose height and width is **N** dekameter. The shape of the colony is a right angled triangle. There are horizontal and vertical roads running from left to right and top to bottom respectively at a distance of 1 dekameter each. Geek's house is located on the top end of the colony. Geek wants to go to the

candy shop to buy some candies. The candy shop is located on the opposite side of colony.

The following figure shows the map of a colony with height = 7 and width = 7 :



To reach the candy shop Geek can move only down or right. You have to find number of different paths Geek can take to reach the candy shop.
As the answer can be very large, find it modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^5$

Example:

Input:

3

1

2

3

Output:

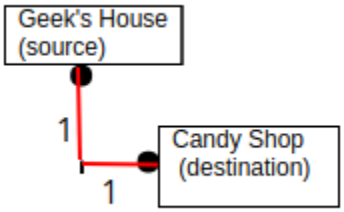
1

2

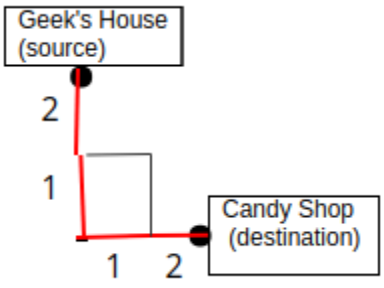
5

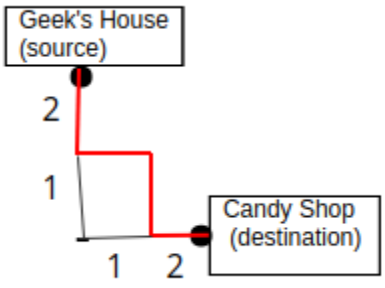
Explanation:

Test Case 1 : Geek can take only following path:



Test Case 2 : Geek can take following two paths:





Counting Binary Trees

Difficulty: **Medium**Accuracy: **85.86%**Submissions: **282+**Points: **4**

Geek has **N** numbers where **i^{th}** number is **a[i]**. Geek wants to make a binary tree of **N** nodes using all the numbers he has.

You have to find out how many distinct binary trees he can make.

As answer can be very large, find it modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^5$
3. $1 \leq a[i] \leq 10^4$
4. Sum of N over all test case does not exceed 10^6.

Example:

Input:

2
2
1 1
2
1 2

Output:

2

4

Explanation:

Test Case 1 : Geek can make following two Binary Trees:

```
  1       1
   \     /
    1   1
```

Test Case 2 : Geek can make following four Binary Trees:

```
 1     1   2   2
  \   /   /    \
   2 2   1      1
```

Geek and Bracket Sequence

Difficulty: **Medium**Accuracy: **64.43%**Submissions: **343+**Points: **4**

Geek has given you **N** opening brackets and **N** closed brackets, your task is to count the number of ways you can form balanced bracket sequences using all the given brackets.

Note: The answer can be very large. So, output answer modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N.**

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$
2. $1 \leq N \leq 10^4$

Example:

Input:
2
3
7
Output:
5
429
Explanation:
Test case 1: (((())), ()(()), ()()(),
(())() and (()()) are the balanced bracket
sequences.

Dyck Words

Difficulty: **Easy**Accuracy: **67.55%**Submissions: **37+**Points: **2**

Geek has **N 'X'** and **N 'Y'** characters. He wants to make a word using all his characters.
You have to find the probability that out of all the distinct words he can make, how many can be a dyck word.
A **Dyck word** is a string consisting of N X's and N Y's such that no prefix of the string has more Y's than X's.

NOTE: The answer can be represented as P / Q, you have to print P * Q^{-1} modulo $10^9 + 7$ where Q^{-1} is the modular multiplicative inverse of Q modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^9$

Example:

Input:

2

1

29381

Output:

500000004

305322989

Explanation:

Test Case 1 : Geek can make two words - "XY"
and "YX" out of which "XY" is a dyck word.

Christmas Decorations

Difficulty: **Easy**Accuracy: **93.59%**Submissions: **235+**Points: **2**

Geek had **2 * N** silver balls and **N** golden threads for Christmas decorations. He wanted to decorate his giant circular table so he placed the balls uniformly all around the edge of the table. All the balls had hooks and he wanted to attach a thread between any two pairs of balls such that the threads were stretched out on the table and the balls now dangled from the edge of the table and he had to make sure that none of the **N** threads intersected with each other otherwise the table will not look beautiful. Help Geek find out how many ways he could do that. As the answer can be very large, find it modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^3$
2. $1 \leq N \leq 10^3$

Example:

Input:
2
1
5
Output:
1
42
Explanation:
Test Case 1 : As there are only two balls,
a thread can be attached in only one way.

Playing with Polygons

Difficulty: **Medium**Accuracy: **89.77%**Submissions: **265+**Points: **4**

Geek was gifted a regular polygon with **N** vertices labeled from **1** to **N** in clockwise order. He was taught in school that the **triangulation** of a polygon is a set of triangles such that each vertex of each triangle is a vertex of the initial polygon and there is no pair of triangles such that they have any area in common and the total area of all triangles is equal to the area of the given polygon.
The **weight** of a triangulation is the sum of the weights of the triangles it consists of, where the weight of a triangle is denoted as the product of the labels of its

vertices.
Geek wanted to find the minimum weight among all the triangulations of the polygon by going through all of the possible ones. Let him know how many possibilities he has to go through and also the **weight** of
the **minimum triangulation.**
As numbers can be very large, print them modulo $10^9 + 7$.

Input:
1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print two space-separated integers denoting the **number of possibilities** and the **weight of the minimum triangulation** respectively.

Constraints:
1. $1 \leq T \leq 10^3$
2. $3 \leq N \leq 10^3$

Example:

Input:
2
3
4
Output:
1 6
2 18
Explanation:
Test Case 2 : A four sided polygon can
be triangulated in two ways by its 2
diagonals. The minimum weight triangulation

would be (1 * 2 * 3) + (1 * 3 * 4).

Modular Arithmetic

Yet Another Teams Dividing Problem

Difficulty: **Medium**Accuracy: **73.17%**Submissions: **406+**Points: **4**

There are **N** students in a class. Each student is given a work factor which is distinct number in the range **[1, N]**, in other words **a[]** is a permutation of integers from **1** to **N** where **a[i]** is the work factor of **i^{th}** student. As their project mentor, you want to divide **all** students in **X** teams where each team is a consecutive range of students. For example team formed by students at index {1, 2, 3} forms a valid team whereas team formed by students at index {1, 3, 7} is not a valid team. Team power is defined as maximum work factor of any student who is in that team. You want to maximise total team power i.e. sum of team powers of all teams. You have to find maximum total team power and number of ways in which you can divide teams to achieve that total team power, as this number can be very large find it modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **X**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print two integers - maximum total team power and number of ways in which you can divide teams to achieve that total team power, as this number can be very large print it modulo $10^9 + 7$.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq X \leq N \leq 10^5$
3. $1 \leq a[i] \leq N$

Example:

Input:

2

4 2

4 1 3 2

2 2

1 2

Output:

7 2

3 1

Explanation:

In the first test case, there are two ways of dividing teams in which maximum total team power of 7 can be achieved :

{4 1 | 3 2} and {4 | 1 3 2}.

In Second test case we have only one way to form two teams.

{ 1 | 2 }

Generous Donations

Difficulty: **Easy** Accuracy: **65.68%** Submissions: **67+** Points: **2**

Geek is the richest person on Geekland. He has so much money that sometimes it is impossible to represent the amount he has in a integer, so he uses a string to store the amount of money he has.

Geek is a very wise man and wishes to solve the problem of poverty on his planet

where infinite people live. He wants to donate every person $10^9 + 7$ geekian rupees till its possible, in other words, he will keep on donating $10^9 + 7$ geekian rupees until the amount of money he has become less than $10^9 + 7$. You have to find what amount of money will be left with him in the end.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains string **S** denoting amount of money Geek has.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10$
2. $1 \leq |S| \leq 5*10^5$

Example:

Input:
3
7263
100000000700
128173290008387367232311783 6
Output:
7263
0
540913798
Explanation:

Test Case 1 : Geek cannot donate anything so he will be left with the same amount he has initially.
Test Case 2 : Geek will donate to 100 persons after which he will be left with nothing.

Modulo

Difficulty: **Easy**Accuracy: **77.64%**Submissions: **309+**Points: **2**

Given three numbers A, B, and C. The task is to find $A^2 + B^2 - C^2$. Output can be very large, so, output answer modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains three space-separated integers **A, B,** and **C.**

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^5$
2. $0 \leq A, B, C \leq 10^9$

Example :

Input:

2
1 2 0
0 0 0

Output:

5
0

Large Factorials

Difficulty: **Easy**Accuracy: **40.54%**Submissions: **143+**Points: **2**

Geek wants to find out factorial of a number **N**. **N** can be large but he only knows how to calculate factorial of small numbers so he needs your help.
As answer can be very large, print it modulo **M**.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **M**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^9$
3. $1 \leq M \leq 10^5$

Example:

Input:

2
3 5
233 99997

Output:

1
29241

Explanation:

Test Case 1 : 3! = 6 and 6 % 5 = 1.

Interesting Ratio

Difficulty: **Medium**Accuracy: **55.69%**Submissions: **88+**Points: **4**

Geek has **R** items and **N** bags. He wants to find out the interesting ratio which is defined as the number of ways to put **R** items into **N** bags assuming all items and bags are distinct divided by the number of ways to put **R** items into **N** bags assuming all items are identical and all bags are distinct.

The answer can be represented in the form of **P/Q** where **P** and **Q** are co-prime. You have to output **P*Q^{-1} modulo $10^9 + 7$** where **Q^{-1}** is the modular multiplicative inverse of **Q modulo $10^9 + 7$**.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two integers **R** and **N.**

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$
2. $1 \leq R, N \leq 10^5$

Example:

Input:

2

1 2

2 2

Output:

1

333333337

Explanation:

Test Case 1: Number of ways to put 1 item into 2 bags assuming all items and bags to be distinct is 2 and Number of ways to put 1 item into 2 bags assuming all items to be identical and all bags to be distinct is 2.
P = 2 and Q = 2, so $P*Q^{-1} = 1$.

Test Case 2: Number of ways to put 2 items into 2 bags assuming all items and bags to be distinct is 4 and Number of ways to put 2 items into 2 bags assuming all items to be identical and all bags to be distinct is 3.
P = 4 and Q = 3, so $P*Q^{-1} = 333333337$.

A Power Problem

Difficulty: **Easy** Accuracy: **80.93%** Submissions: **299+** Points: **2**

Geek wants to solve a power problem. He has two integers **A** and **P** where **P** is always a prime number. He wants to find out A^P Modulo **P**.
A and **P** can be very large so they are represented using strings. Geek only knows how to handle small numbers, so he wants your help.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. Each test case contains 2 space-separated strings denoting **A** and **P**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$
2. $1 \leq A < P \leq 10^{200}$

Example:

Input:
2
2 3
1 5
Output:
2
1
Explanation:
Test Case 1 : $2^3 = 8$ and 8 % 3 = 2.
Test Case 2 : $1^5 = 1$ and 1 % 5 = 1.

Old Calculator

Difficulty: **Medium**Accuracy: **93.06%**Submissions: **260+**Points: **4**

Geek's friend is a scientist and for his current experiment, he needs the value of an integer **A** raised to the power of an integer **X** which could possibly be a very large number so he needs the answer modulo 10^9+7. The only problem is that he has an old calculator that only recognizes and supports operations on integers less than or equal to 10^9+7. Help him find out the answer nonetheless by finding out what number **R** he can raise **A** to, such that he can use that calculator for this calculation and still get the same answer as to his original question.

Input:
1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **A** and **X**.

Output: For each test case, print two space-separated integers, the number **R** followed by the answer to the calculation.

Constraints:
1. $1 \leq T \leq 10^3$

2. $1 \leq A \leq 10^9$

3. $1 \leq X \leq 10^{18}$

Example:

Input:

2

2 2

2436 10338868786567468

Output:

2 4

724534260 211252070

Explanation:

Test Case 2 : Raising 2436 to the power 724534260 would give the same answer 211252070 and all these numbers would be supported by the calculator.

Matrix Exponentiation

Geek's Coaching Classes

Difficulty: **Easy** Accuracy: **93.89%** Submissions: **246+** Points: **2**

Geek runs a coaching institution to prepare students for GEE - an entrance examination to take admission in one of the renowned engineering institutions in Geekland - GITs.

Recently **N** new students took admission in his coaching institution. He teaches students in **X** batches. He will assign a batch to each new student, so he wants to find how many possible ways are there to do that. As you are his friend help him find the answer.

As answer can be very large, you have to output it modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers **N** and **X**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^4$
2. $1 \leq N, X \leq 10^9$

Example:

Input:

2

2 2

2376 58

Output:

4

343136975

Explanation:

Test Case 1 : There are four possible ways of assigning two students in two batches :

1. Both students in first batch.
2. Both students in second batch.
3. First student in first batch and second student in second batch.
4. First student in second batch and second student in first batch.

What is the Password?

Difficulty: **Medium**Accuracy: **87.38%**Submissions: **286+**Points: **4**

Geek wants to play a game on his brother's mobile phone which is secured by a password. His brother is not telling him the password to open it, instead he told him that the password is **N**th Fibonacci Number modulo $10^9 + 7$.
Geek is not that good at maths, so he asked for your help to find the password.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^3$
2. $1 \leq N \leq 10^{18}$

Example:

Input:
3
4
5
1000000000000000000
Output:
3
5
209783453
Explanation: First 5 fibonacci numbers starting from N = 1 are :
$F_1 = 1$

$F_2 = 1$
$F_3 = 2$
$F_4 = 3$
$F_5 = 5$

Modulo ||

Difficulty: **Easy**Accuracy: **82.82%**Submissions: **274+**Points: **2**

Given three numbers **A, B,** and **C.** The task is to find $\mathbf{A^B + B^C - C^A}$. Output can be very large, so, output answer modulo $\mathbf{10^9 + 7}$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains three space-separated integers **A, B,** and **C.**

Output: For each test case, print the answer.

Constraints:

1. $1 \le T \le 10^3$
2. $1 \le A, B, C \le 10^9$

Example :

Input:

2
3 2 2
1 1 1

Output:

5

1

Modulo ||

Difficulty: **Easy**Accuracy: **82.82%**Submissions: **274+**Points: **2**

Given three numbers **A, B,** and **C.** The task is to find $A^B + B^C - C^A$. Output can be very large, so, output answer modulo $10^9 + 7$.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains three space-separated integers **A, B,** and **C.**

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^3$
2. $1 \leq A, B, C \leq 10^9$

Example :

Input:

2
3 2 2
1 1 1

Output:

5
1

Geek's Favourite Numbers

Difficulty: **Easy**Accuracy: **56.38%**Submissions: **55+**Points: **2**

Geek's favourite numbers are **2** and **3**. You want Geek to help you solve a problem in which the number **N** is used but Geek will not help you until **N** is given to him in the form of $2^X + 3^Y$ where **X** and **Y** are non-negative integers.

You have to find **X** and **Y**. If there are multiple possible pairs of **X** and **Y** find the pair in which **X** is maximum.

It is guaranteed that there exists at least one pair of **X** and **Y** such that **N** can be represented as $2^X + 3^Y$.

Input:

1. The first line of the input contains a single integer T denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print **X** and **Y** separated by a space in a new line.

Constraints:

1. $1 \le T \le 1000$
2. $1 \le N \le 10^{16}$

Example:

Input:

2

5

17179875745

Output:

2 0

34 8

Explanation:
Test Case 1 : There are two possible pairs of {X, Y} : {2, 0} and {1, 1} as $5 = (2^2 + 3^0)$ and $5 = (2^1 + 3^1)$. Since in {2, 0} value of X is maximum. Hence, {2, 0} is the answer.

Combinatorial Game Theory

Is this Nim?

Difficulty: **Medium**Accuracy: **58.33%**Submissions: **96+**Points: **4**

GeekA and *GeekB* are two programming geeks.

One day they decide to play a game.

In this game, they have exactly two piles of stones. One pile has **N** stones and other has **1** stone initially. Both of them play alternatively, starting with *GeekA*.

In each turn, the current player **must** perform the following steps :

- choose the pile with the less number of stones and throw away all stones from that pile.
- from the other pile, pick some stones such that at least one stone will remain in that pile and then transfer the picked stones to the emptied pile.
- if both piles have equal number of stones, then the player can choose any pile to empty.

The player who is unable to make a move, loses the game. Your task is to find who will win the game if both players play optimally.

Input Format :

First line contains integer T ,number of test cases.

Next T lines have one integer , denoting N for each test case.

Output Format :

Output should be name of the winner (either "*G*eekA" or "*G*eekB", without quotes).

Example 1:

Input:
2
2
3
Output:
GeekA
GeekB

Explanation:
In the first test case,
For the first move, *GeekA* willempty the pile having 1 stone and take 1 stone from other pileand keep it there. Now *GeekB* cannot make any move.

In the second test case,
it can be shown that , no matterwhat move *GeekA* does, *GeekB* will always win.

Constraints:

- $1 <= T <= 10^5$
- $1 <= N <= 10^{18}$

Chocolate Game

Difficulty: **Medium**Accuracy: **75.67%**Submissions: **321+**Points: **4**

GeekA and *GeekB* are two programming geeks and they have a crush on *Geekina*.

Today, they brought **N** chocolates for Geekina, and now Geekina wants them to play a game.

She kept all chocolates on a single pile. And now Geekina randomly chose **K** numbers which were not greater than **N**. Geekina told them, they can remove chocolates from the pile turn by turn but the number of chocolates that the player can remove must be one of those **K** distinct numbers! Whoever will take the last chocolate will win the game. It's guaranteed that one of those **K** numbers will be '**1**'.

GeekA and GeekB are now excited! They finally got the chance to impress Geekina! Now both GeekA and GeekB were programming nerds! So they both were going to play optimally. GeekA is supposed to move first.

Can you tell who will win this game and be able to impress Geekina ?

Remember, both of them play optimally.

Input Format :

- First line contains integer T, number of test cases.
- The first line of each test case contains N, denoting the number of chocolates and K , denoting the number of distinct chocolates which can be removed from the pile.
- The second line of each test case contain K distinct integers which represent the number of chocolates which can be removed from the pile in 1 move.

Output Format :

For each test case , print on a new line ,the name of the winner (either "GeekA" or "GeekB", without quotes).

Example 1:

Input:
2
5 3
1 4 3
22 3
1 8 3

Output:
GeekA
GeekB

Explanation:
For testcase 1 :
In first move, GeekA will remove 3 chocolates from pile.
In the next move, only 2 chocolates remains, so GeekB will be forcedto remove 1 chocolate. In the nextmove, GeekA will remove 1 chocolate
and 0 chocolates remain. So, GeekA wins!

For testcase 2:
It can be shown that GeekB will
always win.

Constraints:

- $1 <= T <= 10^3$
- $1 <= N <= 10^5$
- $1 <= K <= 10$
- $1 <=$ Distinct numbers $<= 10^3$
- Sum of N over all testcases doesn't exceed 10^6.

Chocolate Game

Difficulty: **Medium**Accuracy: **75.67%**Submissions: **321+**Points: **4**

GeekA and *GeekB* are two programming geeks and they have a crush on *Geekina*.

Today, they brought **N** chocolates for Geekina, and now Geekina wants them to play a game.

She kept all chocolates on a single pile. And now Geekina randomly chose **K** numbers which were not greater than **N**. Geekina told them, they can remove chocolates from the pile turn by turn but the number of chocolates that the player can remove must be one of those **K** distinct numbers! Whoever will take the last chocolate will win the game. It's guaranteed that one of those **K** numbers will be '**1**'.

GeekA and GeekB are now excited! They finally got the chance to impress Geekina! Now both GeekA and GeekB were programming nerds! So they both were going to play optimally. GeekA is supposed to move first.

Can you tell who will win this game and be able to impress Geekina ?

Remember, both of them play optimally.

Input Format :

- First line contains integer T, number of test cases.
- The first line of each test case contains N, denoting the number of chocolates and K , denoting the number of distinct chocolates which can be removed from the pile.
- The second line of each test case contain K distinct integers which represent the number of chocolates which can be removed from the pile in 1 move.

Output Format :

For each test case , print on a new line ,the name of the winner (either "GeekA" or "GeekB", without quotes).

Example 1:

Input:
2
5 3
1 4 3
22 3
1 8 3

Output:
GeekA
GeekB

Explanation:
For testcase 1 :
In first move, GeekA will remove 3 chocolates from pile.
In the next move, only 2 chocolates remains, so GeekB will be forcedto remove 1 chocolate. In the nextmove, GeekA will remove 1 chocolate
and 0 chocolates remain. So, GeekA wins!

For testcase 2:
It can be shown that GeekB will
always win.

Constraints:

- $1 <= T <= 10^3$

- $1 <= N <= 10^5$
- $1 <= K <= 10$
- $1 <= \text{Distinct numbers} <= 10^3$
- Sum of N over all testcases doesn't exceed 10^6.

Break the box II

Difficulty: **Hard**Accuracy: **76.11%**Submissions: **293+**Points: **8**

Alice and Bob are playing a game. There are **N** boxes arranged in a row in front of them. Each box has some number written on it. The game goes as follows

- Alice makes the first move
- In one move, player can break **exact one** box
- The box, that is being break in current move is not available for further moves.
- The game ends when there are exact **K** boxes remains.

If sum of numbers on remaining **K** boxes is **Odd**, **Alice wins** else **Bob wins**.

Your task is to find **winner** of the game if both players play optimally.

Input Format :

First line contains integer *t*, number of test cases.

Each test case contains 2 lines as follows -

- First line contains integers N and K.
- Next line contains N space separated integers, values written on boxes.

Output Format :

Print *t* lines, where *i-th* line represent output for *i-th* input.

print winner of the game.

Example :

Input:
2
3 1
2 2 1
2 2
53 20

Output:
Bob
Alice

Explanation:
Test 1: K=1 so remaining two boxes will be break. If Alice break the box withvalue 2, then Bob will break the boxwith value 1 and vice versa. In both the cases, remaining sum is2 which is even. Bob wins.

Test 2: N=K, so no move available. Initial sum is odd, Alice wins.

Constraints:

- $1 <= t <= 10^3$
- $1 <= K <= N <= 10^6$
- $1 <=$ numbers written on boxes $<= 10^9$
- sum of N over all test cases doesn't exceed 10^6

Kill the monster

Difficulty: **Medium**Accuracy: **72.93%**Submissions: **325+**Points: **4**

There is a monster in the Geekland whose power is **P**. Geek and Geekina are 2 people living in Geekland. Now, it's their task to kill the monster. The monster will be killed if all his power is lost (becomes **0**). Geek and Geekina will fire some bullets on him of the following type :

- Bullet of type-1 will decrease the power of monster by **1** (P = P-1)
- Bullet of type-2 will decrease the power of monster by **2** (P = P-2)
- Bullet of type-3 will decrease the power of monster by **k** (P = P-k)

When power of the monster becomes **0**, he is dead. There are infinite number of bullets of all three types. Both Geek and Geekina fire bullets alternatively, **starting with Geek**. In one move, shooter can choose **exact one** bullet of any type (type-1, type-2 or type-3) and fires it. After this bullet, power of monster decreases as mentioned above.

The player who is unable to fire a bullet, **loses**.

Note: Both of them won't use a bullet if it has more power than required

- If monster's power is 0, No more bullets will be fired.
- If monster's remaining power is 1, bullet of type-2 and type-3 won't be used.
- If monster's remaining power is less than k, bullet of type-3 won't be used.

Input Format :

First line contains integer *t*, number of test cases.

Next *t* lines contains one input per line as follows -

- Given two space separated integers *P* and *k*, power of monster and killing power of type-3 bullet respectively.

Output Format :

Print *t* lines, where *i-th* line represent output for *i-th* input.

Output should be name of the **winner** (either "Geek" or "Geekina", without quotes).

Example:

Input:
3
0 3
1 5
6 4

Output:
Geekina
Geek
Geekina

Explanation:
test 1: Monster is already dead (power is 0)
Geek won't fire any bullet, so
Geekina wins.

test 2: Geek will fire type-1 bullet and
monster will be dead. Geek wins.

test 3: Geekina will win. All possible
sequence of firing are:
{ Geek type-1, Geekina type-2,
Geek type-1, Geekina type-2, 0 }

{ Geek type-1, Geekina type-2,
Geek type-2, Geekina type-1, 0 }

{ Geek type-2, Geekina type-3, 0 }

{ Geek type-3, Geekina type-2, 0 }

Constraints:

- 1 <= t <= 1000
- 0 <= P <= 10^7
- 3 <= k <= 10^7

Definitely Not Nim!

Difficulty: **Hard**Accuracy: **95.58%**Submissions: **221+**Points: **8**

GeekA and GeekB are playing a game similar to Nim!

In this version, however, there is an infinitely long row, having infinite cells. Geekina, who is the crush of GeekA and GeekB has randomly placed some stones at some cells (only 1 stone at a place). Now GeekA and GeekB are playing the game to win the heart of Geekina!

The rules of the game are :

- The players take turns alternatively, starting with GeekA.
- You can take any stone from initial **cell i** and move it to cell j, such that **j>0** and **j** and all cells from j+1 to i-1 do not have any stone yet.
- The player who cannot make a valid move in his turn will lose.

You are the best friend of GeekA and you want him to win.

GeekB will play optimally and you will help GeekA play optimally. Determine if he can win and if he can win, then suggest him the move to win.

The winning move must be in the form X Y which means, move stone from position X to Y. If there are multiple winning moves, then choose the one with the minimum X and if there are multiple possible winning moves for the chosen X then choose the one with the minimum Y.

If GeekA cannot win no matter what, then print -1.

Input Format :

- First line contains integer T, number of test cases.
- First line of each test case will have an integer N, denoting the number of stones.
- The second line of each test case will have N space separated distinct integers which denote the initial positions of the stones. These integers are sorted in ascending order and are strictly increasing.

Output Format :

- Output should either be "-1" if GeekA cannot win otherwise X Y which denotes the first winning move from X to Y.

Example 1:

Input:
1
2
2 3
Output:
-1

Explanation:
In first turn , GeekA
has only 1 valid move.
Move from 2 to 1.
In next turn, GeekB will
move from 3 to 2.
In next turn, GeekA has no
valid move. GeekB wins.

Example 2:

Input:
1

3
1 3 5
Output:
5 4

Explanation:
The only winning move
for GeekA is to move
coin from 5 to 4.

Constraints:

-
  - 1 <= T <= 200
- 1 <= N <= 10^4
- 1 <= positions of coins <= 10^9

ProblemSubmissions

Impossible Nim

Difficulty: **Medium** Accuracy: **91.43%** Submissions: **247+** Points: **4**

You are given N rows. Each row has variable column size. The i'th row has C[i] column length. That means , each row has C[i] piles. The smallest pile has P[i] stones. The largest pile has P[i]+C[i]-1 stones. And all piles have distinct stones.

GeekA and GeekB started to play a turn-based game. The player take turns and make their move. In each move, the current player can choose any pile from any row and remove any amount of stones from it. After some turns, if some player cannot make a move then he loses.

Find out which player will win the game if GeekA starts the game.

Input Format :

- The first line will contain T,the number of test cases.
- The first line of each test case will contain N, the number of rows.
- The second line of each test case will contain N space separated integerswhere i'th integer denotes the column length of i'th row , i.e, C[i].
- The third line of each test case will contain N space separated integerswhere i'th integer denotes the smallest pile size of i'th row , i.e, P[i].

Output Format :

- For each test case , print on a new line ,the name of the winner (either "GeekA" or "GeekB", without quotes).

Example 1:

Input:

```
1
2
1 2
2 3
```

Output :

GeekA

Explanation :

There are 2 rows.
1st row has 1 columns and
that column has 2 sized pile.

2nd row has 2 columns, and
the smallest pile size is 3
and the largest pile size is 4.

It can be shown that
GeekA will always win.

Constraints :

- 1 <= T <=10
- 1 <= N <= 10^5
- 1 <= P[i],C[i] <=10^{12}
- The sum of N over all test cases doesn't exceed 10^5.

Time Limit : 1 sec.

Cursed Treasure!

Difficulty: **Medium**Accuracy: **91.36%**Submissions: **243+**Points: **4**

Once upon a time, there was a rich man in GeekLand. The name of the man was GeekyGeek! Before he died, he wanted to give his fortune to one of his two servants, GeekA and GeekB !!

He had a big garden whose shape was like a grid having N rows and M columns. So, he decided to divide all his fortune into N*M treasure boxes and bury one treasure box in each of the cells of the garden.

But there's a catch! One of the treasure boxes had the Cursed Pearl Of Doom! Whoever was going to take that treasure box, was going to have his life span reduced by 1 year!

GeekA and GeekB both knew which treasure box it was and even the location of the cell it was in! The location is R'th row and C'th column.

So, obviously, both of them wanted to not take that box. But one of them had to take it otherwise GeekyGeek wouldn't give any treasure to both!

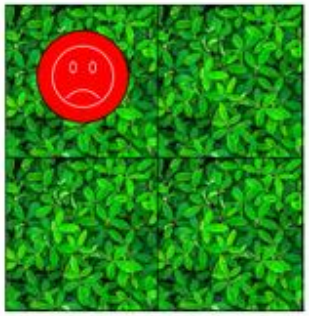
And they had to collect the treasure while playing a special game. The game was a turn-based game and GeekA was going to get the first turn. The rules of each turn were:

- The current player has to choose a part of the grid (i.e. garden) and collect all treasures from it.
- Out of the remaining part of the garden, the current player will choose an inner horizontal or vertical grid line and split the garden into two halves of non-zero area. The player can then choose to take any halve he wants. Either the top half or the bottom half in the case of a horizontal split, Or the left half or right half in the case of a vertical split.
- If only 1 cell is remaining in the garden, the current player has to take it!

See the following example case to understand the game better.



Turn 1 : GeekA splits at the 3rd horizontal grid-line and takes bottom part.



Turn 2 : GeekB splits at the 2nd vertical grid-line and takes right part.



Turn 3 : GeekA splits at the 2nd horizontal grid-line and takes bottom part.



Turn 4 : GeekB must take the only remaining cell.

Determine who will **not** take the Cursed Pearl Of Doom, if both of them play optimally.

Input Format :

The first line contains integer T, the number of test cases.

Next T lines have four integers, denoting N, M, R, and C for each test case.

Output Format :

The output should be the name of the servant who will not take the Cursed Pearl Of Doom (either "GeekA" or "GeekB", without quotes).

Example 1:

Input:
1
3 2 1 1
Output:
GeekA

Explanation:

The explanation for
this case is shown in
problem statement.

Constraints:

1 <= T <= 1000

1 <= N,M <= 10^9

1 <= R <= N

1 <= C <= M

Collect candies

Difficulty: **Medium**Accuracy: **88.61%**Submissions: **247+**Points: **4**

John and Harry are two programmers at Geekland. There are **N** baskets arranged in a row in front of them. All of the baskets have some candies. They are going to collect candies in the following way :

- Each of them takes turns (pick candies) alternatively.
- **John** goes **first**.
- In a move, candies can be taken from any **one** basket only.
- If currently the basket has **C** candies, then after taking some candies from it, the remaining candies must be in the range **[0, √C]** (√C is the floor value of the square root of C).
- The player who is unable to take candy, starts crying.

Harry is our friend and we can't see him crying. So we are going to add one more basket in the sequence such that Harry **must** win (and John cries).

Find the **minimum** number of candies that we are required to put in the newly added basket such that Harry always wins. (If no such basket is required, then the minimum number of candies in the new basket will be **0**).

Input Format :

The first line contains an integer **T** denoting the number of test cases

Each test case contains 2 lines as follows -

- The first line contains integer N, the number of baskets.
- The next line contains N space-separated integers, the number of candies in baskets.

Output Format :

Print *t* lines, where *i-th* line represents output for *i-th* input.

print the minimum number of candies to ensure Harry's win.

Example :

Input:
2
3
2 1 1
2
1 1

Output:
2
0

Explanation:
Test 1:
If No candies added :
- John takes 2 candies from basket1
- Harry can take 1 candy only

- John will take last 1 candy and Harry
  loses.

If 1 candy added :
- John takes 1 candy from basket1.
- Now 4 baskets have 1 candy each, that
  means exactly 4 turns. Which will ended
  up to John. Harry loses.

If 2 candy added :
- sequence will be 2 1 1 2 .
- The number of candies John takes, Harry
  will repeat and Harry will win for sure.

Test 2: Both of them take 1-1 candy ended
up to Harry. No additional candy
is required.

Constraints:

- $1 <= t <= 10^3$
- $1 <= N <= 10^5$
- 1 <= candies in each basket <= 10^9
- sum of N over test cases don't exceed 10^5

Balloon game

Difficulty: **Medium**Accuracy: **92.07%**Submissions: **241+**Points: **4**

geekA and *geekB* are two programming geeks. But they are getting bored with their daily routine and going to play a game.

In this game, they have some colorful balloons (filled with water) arranged in a row. Both of them play alternatively, starting with *geekA*.

In each turn, the player can move **one** of the following ways :

- shoot any one balloon
- shoot **exactly two** balloons of the **same** color
- shoot **all** balloons of the **same** color

After any shot, target balloon(s) get burst and can't be shot again.

The player who is unable to make a move loses the game. Your task is to find who will win the game if both players play optimally.

Input Format :

The first line contains integer t, number of test cases.

Next *t* lines contain one input per line as follows -

- you are given a string of capital Latin letters, where each letter represents the color of a balloon.

The same letter represents balloons with the same color.

Output Format :

Print *t* lines, where *i-th* line represents output for *i-th* input.

The output should be the name of the winner (either "geekA" or "geekB", without quotes).

Example 1:

Input:
GEEKS

Output:
geekA

Explanation:
In first move, *geekA* will shoot
one balloon of color E.

Now remaining 4 balloons have different
colors, so exact 4 more moves will be
there (ended to *geekA*).

geekB will be unable to make a move,
and lose the game.

Example 2:

Input:
CODERS

Output:
geekB

Explanation:
All 6 colors are different, so exact 6
moves will be there.
Starting with *geekA*, moves will be ended
to *geekB*.
geekA is unable to make a move and lose.

Constraints:

- 1 <= t <= 1000
- 1 <= length of string <= 10^5

- The sum of length of string over all test cases does not exceed 10^5

Geometric Algorithms

Angry Bird

Difficulty: **Medium**Accuracy: **88.55%**Submissions: **253+**Points: **4**

Geek lives in a 2 dimensional XY plane at co-ordinate (0,0). There are **N** crows living in that plane in their nests. The coordinates of the nests are given to you.
Geek wishes to break those nests because those crows annoy Geek every morning by making loud noises.

Geek has a sling-shot and a special ball which he's going to fling.

The special features of the ball are :

- It's trajectory isn't affected by gravity.
- It can change it's tragectory multiple times in mid-air but only upto a certain extent .
- More specifically, it can rotate it's velocity vector **clockwise**. However, it **cannot** rotate it's trajectory more after it has reached -90 degrees, with respect to the X-axis (aka. vertically downwards).

Geek is going to fling the ball at some angle and the ball can be programmed to **change trajectory at any time.** If the ball hits a nest, it breaks it and moves on without changing speed or direction. Determine, if Geek can break all nests in a single fling using the special ball.

Note : There can be some nests at (0,0). There can be multiple nests at a single position.

Input Format :

- The first line will contain T,the number of test cases.
- The first line of each test case will contain N, the number of nests.

- The second line of each test case will contain N space separated integers where i'th integer denotes the X coordinate of the i'th nests ,i.e, *X[i]*.
- The third line of each test case will contain *N* space separated integers where i'th integer denotes the Y coordinate of the i'th nests ,i.e, *Y[i].*

Output Format :

- For each test case , print on a new line ,the "YES" if all nests can be broken and "NO" if all nests cannot be broken.

Example 1:

Input:

2
3
1 2 4
1 2 0
3
1 2 4
1 3 0

Output :

YES
NO

Explanation :

For first test case,The locations for nests are (1,1),(2,2) & (4,0).Geek's ball will initially be at an angle of 45 degrees and after hitting (2,2), it willreduce angle and hit (4,0).
For second test case,there is no way to hit both(1,1) & (2,3) in same shot.If you start with (1,1), (2,3) is at higher slope andyou can't rotate counter-clockwise.If you start with (2,3), you cannotcome back to (1,1) because you cannotrotate beyond -90 degree slope wrt X-axis.

Constraints :

- $1 <= T <= 10^5$
- $1 <= N <= 10^5$
- $0 <= X[i],Y[i] <= 10^9$
- The sum of N over all test cases doesn't exceed 10^5.

Treasure Extractor

Difficulty: **Medium**Accuracy: **92.0%**Submissions: **237+**Points: **4**

Space pirate Geeky has landed on Geekopia planet. He's searching for treasures on the planet. He had found an old map which marked the 2-dimensional coordinates **(X,Y)** of **N** treasures on the planet surface. And Geeky wants all of those treasures.

For that, he has a gadget named Extractinator with him. He can point the gadget towards the planet's surface and draw a polygon on the gadget screen. The gadget will then extract all the treasures located inside that polygon! But the gadget will use **P** amount of power to extract from the surface. Here, **P** is equal to the perimeter of the polygon drawn (rounded down to nearest integer).

You are given the locations of the **N** treasures, you should determine the minimum power needed by the gadget to extract all treasures.

Input Format :

- The first line will contain T,the number of test cases.
- The first line of each test case will contain N, the number of treasures.
- Then N lines will follow, each line having 2 space separated integers denoting the coordinates of the i'th treasure.

Output Format :

- For each test case , print on a new line the minimum power.

Example 1:

Input:
1
4
1 0
0 1
1 1
0 0

Output :
4

Explanation :

The treasures are located
on the corner of a square
of side length 1. So, the
minimum perimeter is 4 here.

Constraints :

- $1 <= T <= 10^5$
- $3 <= N <= 2*10^5$
- $-10^9 <= X[i],Y[i] <= 10^9$
- The sum of N over all test cases doesn't exceed $2*10^5$.
- There will be at-least 3 points which are not colinear.

Time Limit : 1 sec.

Game of Functions!

Difficulty: **Hard**Accuracy: **89.09%**Submissions: **248+**Points: **8**

Geekina has bought some mathematical *functions* . The functions are in the from $f(x) = ax + b$.

Her friend Geek is a mathematics nerd. So, Geekina wants to test his skills. She will play a game with Geek. The game will consist of **N** moves.

Each move can be of two types :

- Geekina will write one function on paper.
  Input is in the form : 1 a b
- Geekina will give Geek an integer **x** and Geek will have to return the minimum value of **f(x)** among all functions currently written on the paper.
  Input is in the form : 2 x

For each move of type 2, you are supposed to give Geek the correct answer.

Note : First move will always be of type 1 & there will be at least one move of type 2.

Input Format :

- The first line will contain T,the number of test cases.
- The first line of each test case will contain N, the number of moves.
- Then N lines follow, each line may contain move of type 1 , in the form 1 a b or it may be a move of type 2, in the form 2 x.

Output Format :

- For each move of type 2, print the correct minimum value on a new line.

Example 1:

Input:
1
4
1 1 0
2 5
1 -1 0
2 5
Output :

5
-5

Explanation :

For first move of type 2,
At x=5, the minimum
value is 5, which is
given by the only function
written on paper.

For the second move of type 2
At x=5, the minimum
value is -5, which is
given by the 2nd function.

Constraints :

- 1 <= T <=10
- 2 <= N <= 10^5
- -10^5 <= a, b <=10^5
- 0 <= x <=10^5
- The sum of N over all test cases doesn't exceed 10^6.

Time Limit : 5 sec.

Hit the Buzzer

Difficulty: **Easy**Accuracy: **51.85%**Submissions: **27+**Points: **2**

A race is going to be held in Geekland very soon. It's a team race and you are leader of your team. So you are making some strategy to improve your chances of winning the race.

Every team has **N** players. Consider the ground (where the race is organized) as XY plane. All the players will get some positions (**not necessarily distinct**) on Y

axis. There are **N** buzzers placed on some positions (**not necessarily distinct**) on X axis. The race goes as follows:

- A member of a team starts running towards any buzzer and when he/she reaches , hit the buzzer.
- When *i-th* player hit its buzzer, *(i+1)-th* player starts running.
- Each player hits exactly one buzzer and each buzzer being hit by exactly one player.
- Total time taken by this team is **sum** of time taken by all N players.

If a player is at point (a, b) and runs towards buzzer at point (c, d) then time taken by this player is $(a-c)^2 + (b-d)^2$.

Your task is to make some strategy so that total time taken by your team is **minimum** possible.

Note: No player or Buzzer can be at origin (0, 0).

Input Format:

First line contains integer ***t***, number of test cases.

Each test case is given as follows -

- First line contains integer *N*, number of players and buzzers.
- Each of next *2*N* lines contains 2 integers *x, y* where *(x, y)* is position of a player or a buzzer. If given point is on *Y* axis then it's position of a player and If on *X* axis then position of a buzzer.

Note: It is guaranteed that given point is either on *X* axis or on *Y* axis (and Not origin).

Output Format:

Print *t* lines, where *i-th* line represent output for *i-th* test.

- Each line contains an integer, minimum possible time for your team.

Example 1:

Input:
1
2
0 1
1 0
0 -1
-2 0

Output:
7

Explanation:
Two Players are at (0, 1) and (0, -1).
Two Buzzers are at (1, 0) and (-2, 0).
Player at (0, 1) will run to Buzzer at
(1, 0) taking a time of 2 unit.
Player at (0, -1) will run to Buzzer at
(-2, 0) taking a time of 5 unit.
Total time taken is 7 unit, which is minimum
possible.

Example 2:

Input:
1
4
1 0
3 0
-5 0

6 0
0 3
0 1
0 2
0 4

Output:
101

Explanation:
4 Players are at (0, 1), (0, 2), (0, 3),
(0, 4).
4 Buzzers are at (1, 0), (3, 0), (6, 0),
(-5, 0).
Player at (0, 1) will run to Buzzer at
(1, 0) taking a time of 2 unit.
Player at (0, 2) will run to Buzzer at
(3, 0) taking a time of 13 unit.
Player at (0, 3) will run to Buzzer at
(-5, 0) taking a time of 34 unit.
Player at (0, 4) will run to Buzzer at
(6, 0) taking a time of 52 unit.
Total time taken is 101 unit, which is
minimum possible.

Constraints:

- $1 <= t <= 100$
- $1 <= N <= 10^5$
- $-10^6 <= x, y <= 10^6$
- sum of N over all test cases doesn't exceed 10^5

Is it closest pair problem?

Difficulty: **Hard**Accuracy: **92.95%**Submissions: **239+**Points: **8**

Geekina has asked Geek to solve a puzzle.

The puzzle contains **N** 2-dimensional points in the form of **(X,Y)** and you have to pick 2 points from that such that the cost minimizes.

Here, **cost** of picking 2 points (X_1,Y_1) and (X_2,Y_2) is equal to $\mathbf{(X_1+X_2)^2 + (Y_1+Y_2)^2}$.

But there's a catch, after you pick the points, **you can reflect the position vectors for any point across the coordinate axes any number of times.**

In other words, for example, you can convert (X_1,Y_1) to $(X_1,-Y_1)$ and (X_2,Y_2) to $(-X_2,Y_2)$.

Now you have to find and print out the minimum possible cost if you pick and rotate two points optimally.

Input Format :

The first line will contain *T*, the number of testcases.

- The first line of each testcase will contain *N*, the number of coordinates.
- Then *N* lines will follow, each line having 2 space separated integers denoting the coordinates of the i'th treasure.

Output Format :

- Print on a new line the minimum possible cost if you choose points optimally.

Example 1:

Input:

```
1
3
1 1
-1 1
```

5 5

Output :
0

Explanation :
We can choose point 1 and 2,
then reflext point 2 across
x-axis to make it (-1,-1) to
get minimum cost 0.

Constraints :

- $1 <= T <= 10^5$
- $3 <= N <= 10^5$
- $-10^9 <= X[i],Y[i] <= 10^9$
- Sum of N over all test cases do not exceed 10^5

Time Limit : 5 sec.

Yet another array problem.

Difficulty: **Hard**Accuracy: **92.73%**Submissions: **235+**Points: **8**

Geekina has an array **A** of length **N**. She want's to find the minimum beauty subarray of the array. The **length** of the subarray must be atleast **2**.

To compute the beauty of an subarray from **i** to **j** , such that j>i , we do the following steps :

- **Beauty = $(j-i)^2$ + $(A[i+1] + A[i+2] + A[i+3] + ... + A[j])^2$**

Geek wants to help Geekina, so he asked for your help to compute it. You have to tell him the minimum possible beauty.

Input Format :

- The first line will contain T,the number of test cases.
- The first line of each test case will contain N, the number of elements in the array.
- The next line has N integers, where i'th integer denotes the i'th element in the array.

Output Format :

- For each testcase, print the minimum beauty value on a new line.

Example 1:

Input:

1

4

1 2 -2 1

Output :

2

Explanation :

The least beautiful subarray is from index 2 to index 3 i.e. {-2 , 1}.

Sum of elements from index (2+1) = 1 (we consider sum of elements from $(i+1)^{th}$ element)

Beauty = $(3-2)^2 + (1)^2 = 2$

Constraints :

- $1 <= T <= 10^5$
- $2 <= N <= 10^5$
- $-10^4 <= A[i] <= 10^4$
- The sum of N over all test cases doesn't exceed 10^5.

Time Limit : 1 sec.

Find the point

Difficulty: **Medium**Accuracy: **82.85%**Submissions: **35+**Points: **4**

In a city of Geekland, there are **N** streets which are infinitely long and straight. Let's consider this city as XY plane.

Your task is to check if there exist any pair of streets for which following conditions are fulfilled:

- Streets are not parallel.
- Let (a, b) is a point where both streets meet.
- L < a < R for given range [L, R]

If there exist any pair of streets under above conditions, print **YES** else print **NO**.

You are given slope and intersept of streets (consider streets as straight line on XY plane), i.e given values of m and C for all N streets where *i-th* street will be of the form of y = mx + C .

Note: Given streets are pairwise distinct.

Input Format:

First line contains integer ***t***, number of test cases.

Each test case consist of *N+1* lines as follows -

- First line contains integer *N*, *L, R* as number of streets and range of required point respectively.
- Each of next *N* lines contains 2 integers *m, C* as slope and intercept of *i-th* street.

Output Format:

Print *t* lines, where *i-th* line represent output for *i-th* test.

- Each line contains "YES" or "NO".

Example:

Input:
2
2 1 3
1 0
0 3
2 -10 10
-1000000 1000000
1000000 -1000000

Output:
NO
YES

Explanation:
Test1 : Given range [L, R] is [1, 3] while given two lines y=x and y=3 intersects at point (3, 3). Point should be strictly inside range but it is on the edge, answer is NO.

Test2: Given two lines intersect at point (1, 0) and given range is [-10, 10]. x coordinate of intersection point is strictly in the range, answer is YES.

Constraints:

- 1 <= t <= 100
- 1 <= N <= 10^5
- -10^6 <= L, R, m, C <= 10^6
- sum of N over all test cases doesn't exceed 10^5

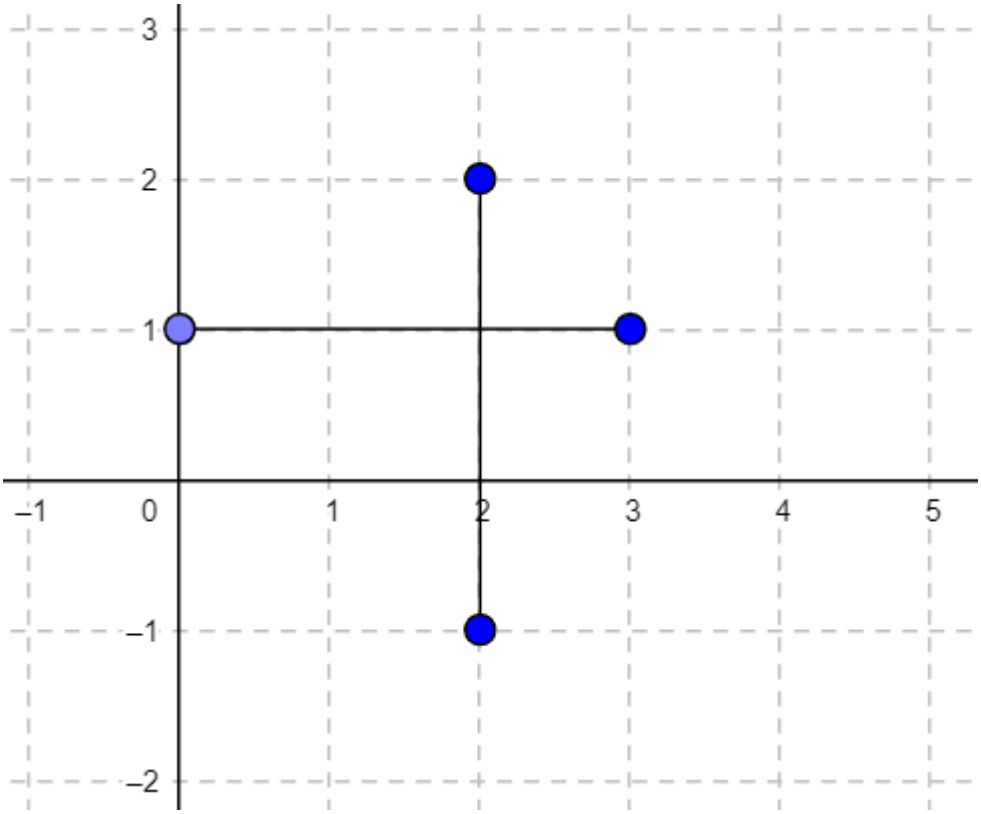
Count Intersections

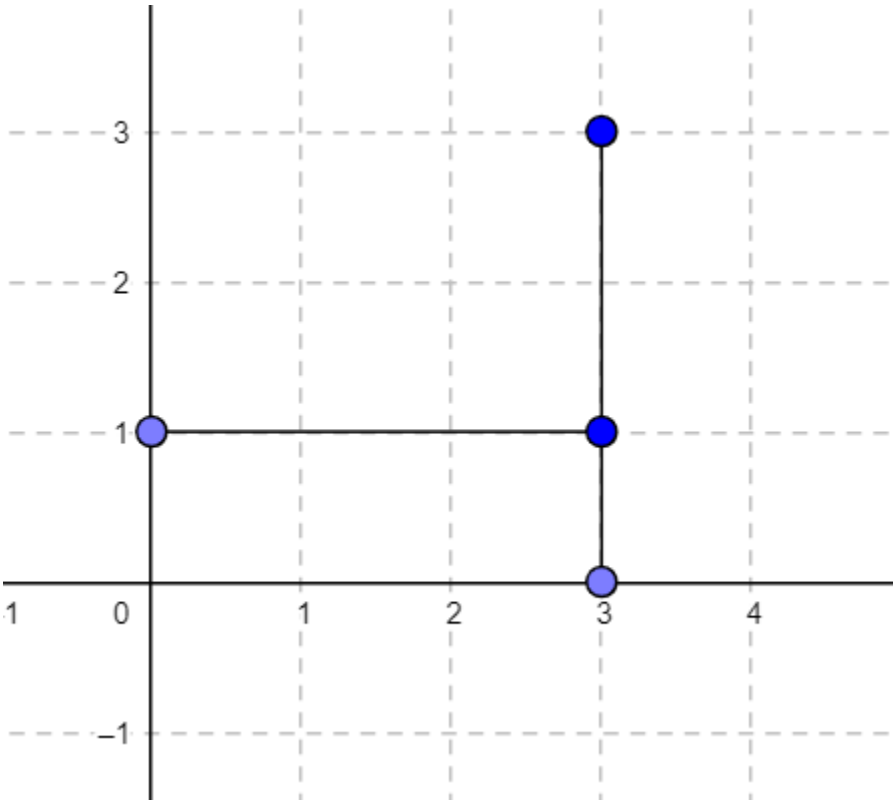
Difficulty: **Medium** Accuracy: **80.01%** Submissions: **40+** Points: **4**

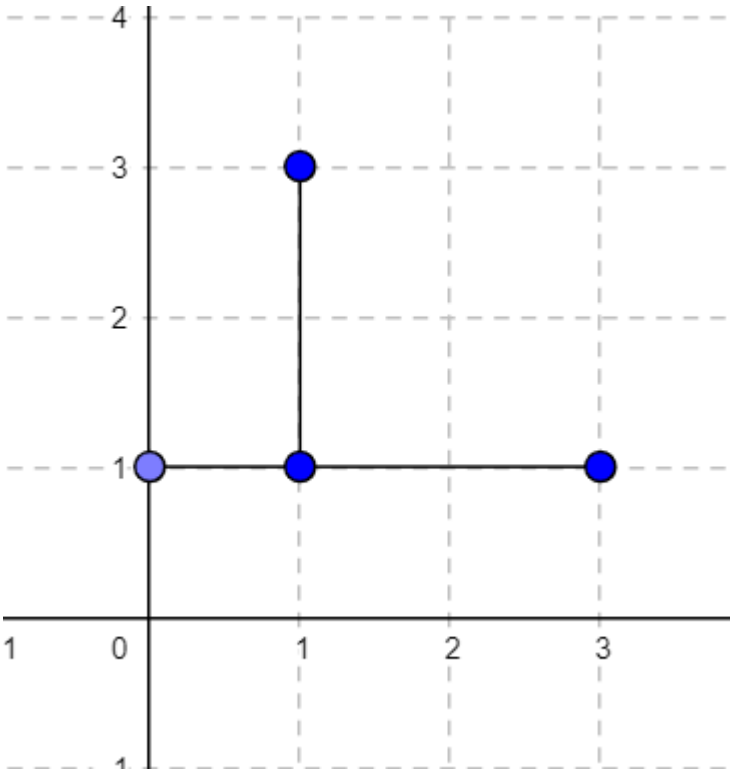
Given **N** lines in XY plane. Each line is either parallel to X axis or Y axis. Your task is to count number of Special intersections which is as follows :

- Suppose two lines are **A** and **B** which are not parallel to each other.
- Intersection point of A and B is **(p, q)** on XY plane.
- Point (p, q) **must** divide **at least** one line (among A and B) into two segments, then only it is considered as Special intersection.

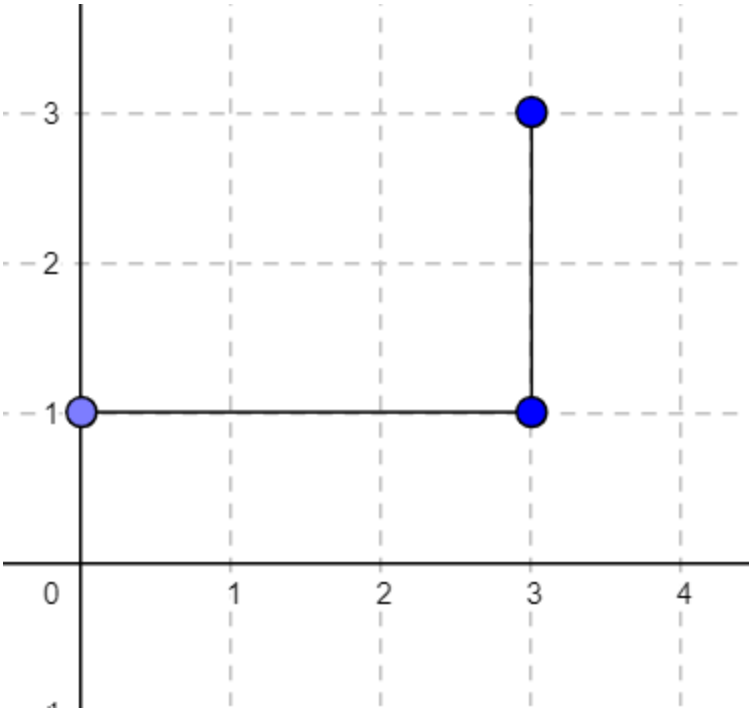
These are Special intersections :-


3
2
1
–1
–2
–1
0
1
2
3
4
5


3
2
1
–1
1
0
1
2
3
4



And, this is not :-



Note:

1. No two horizontal lines meet each other.
2. No two vertical lines meet each other.
3. Given lines have non-zero length.

Input Format:

First line contains integer ***t***, number of test cases.

Each test case consist of *N+1* lines as follows -

- First line contains integer *N*, number of lines
- Each of next *N* lines contains 4 integers x_1, y_1, x_2, y_2 where (x_1, y_1) and (x_2, y_2) are end points of *i-th* straight line.

Output Format:

Print *t* lines, where *i-th* line represent output for *i-th* test.

- Each line contains N integers, where *j-th* integer denote number of Special intersections for *j-th* straight line in *i-th* test.

Example:

Input:

1

4

2 1 2 6

1 3 5 3

4 1 4 4

1 5 5 5

Output:

2 2 1 1

Explanation:

$Line_1$ intersects with $Line_2$ and $Line_4$
$Line_2$ intersects with $Line_1$ and $Line_3$
$Line_3$ intersects with $Line_2$ only.
$Line_4$ intersects with $Line_1$ only.

Constraints:

- $1 <= t <= 100$
- $1 <= N <= 10^4$
- $1 <= x_1, y_1, x_2, y_2 <= 10^5$
- sum of N over all test cases doesn't exceed 10^4

Remove the tapes!

Difficulty: **Hard** Accuracy: **99.0%** Submissions: **219+** Points: **8**

Geek has finally finished building the Infinite wall. The infinite wall is an infinitely large 2D plane. It's his greatest creation!

But Geek's 3-year-old daughter is a mischievous girl. She took a cello-tape roll and put strips of cello tape at random positions. Of course, some strips may overlap on some other strips.

Now Geek has to remove those cello tapes from the wall. But he is very tired after working on the wall and cannot remove a cello-tape strip if some other strip is over it. **So, he'll have to remove the topmost strips first.**

You have to tell him which strips are initially on the topmost position (such that no other strip is taped over them) so that he can remove them.

You are given the location of **N** strips in the form **(X_1, Y_1)** and **(X_2, Y_2)** which denotes the location of endpoints of the strip. **The stips were pasted in the order they are given in input. So, if two strips intersect, then the strip that is given later in the input is on top of the other strip.** Also, **you can neglect the width of the strips** and consider them as line rays.

For example :

Fig 1 (Black strip below red and orange)

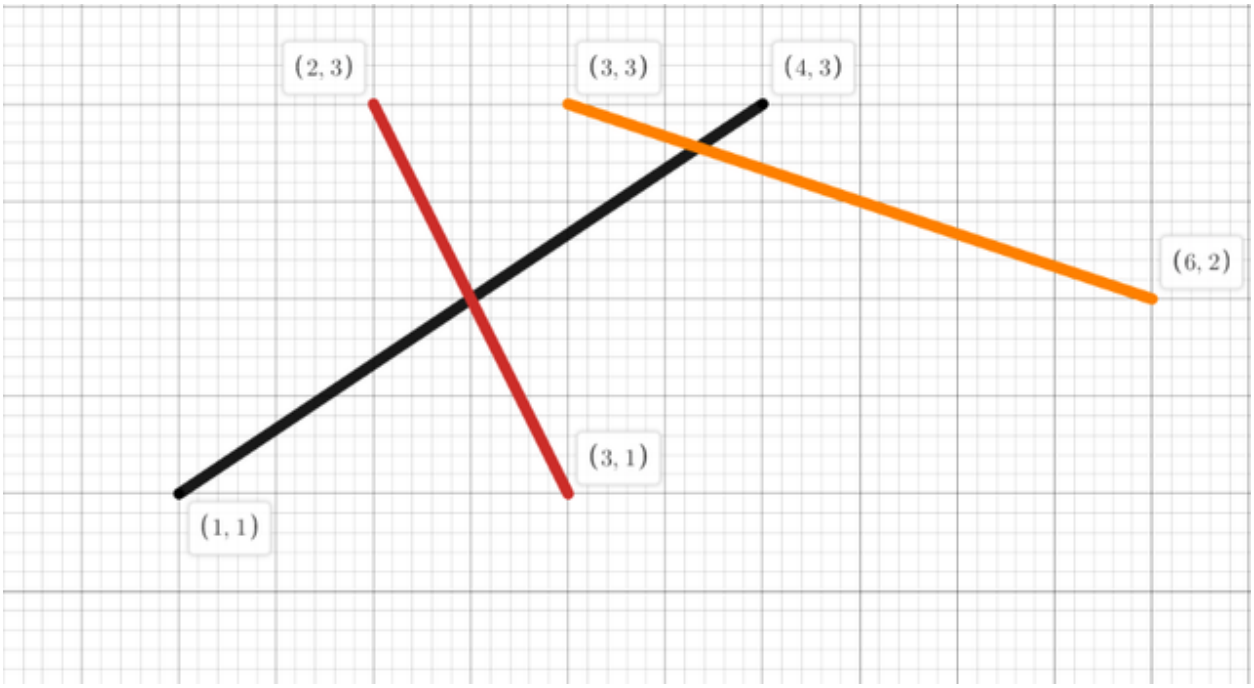
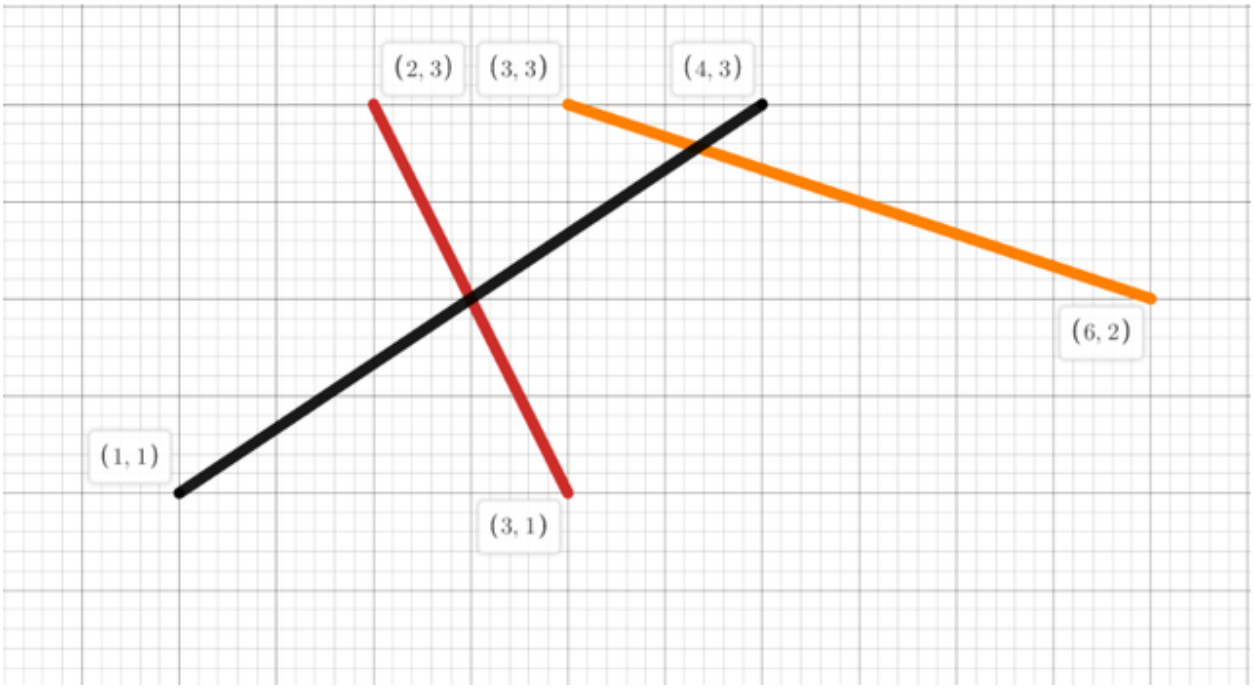


Fig 2 (Black strip above red and orange)



Print out the total number of strips that are on top and also the indices of the strip (**1-based**) according to their order in input.

Input Format :

- The first line will contain T, the number of test cases.
- The first line of each test case will contain N, the number of strips.
- Then N lines follow and the i'th line contains 4 integers which denote X_1, Y_1, X_2, and Y_2 respectively for the i'th index tape.

Output Format :

- For each test case, print on a new line M, the number of tapes that can be removed initially, and then on the next line print M space-separated integers sorted non-decreasingly denoting the indices of the tapes.

Example 1:

Input:

```
2
3
6 2 3 3
3 1 2 3
1 1 4 3
3
1 1 4 3
6 2 3 3
3 1 2 3
```

Output :

```
1
1
2
2 3
```

Constraints :

- $1 <= T <= 10^3$
- $1 <= N <= 10^3$
- $-10^4 <= X_1, Y_1, X_2, Y_2 <= 10^4$
- The sum of N^2 over all test cases doesn't exceed 10^6.

Time Limit: 1 sec.

Garden Area

Difficulty: **Easy**Accuracy: **61.11%**Submissions: **18+**Points: **2**

Mr. Geek is a very famous person in Geekland. He owns **N** gardens, which are represented as Rectangles on XY the plane. None of the gardens intersect or touch another garden.

You are given **4N** points (pairwise distinct) on the XY plane, which are corners of **N** gardens. These points are given in arbitrary order, your task is to identify these gardens and find the **Total area**.

Input Format:

The first line contains integer ***t***, number of test cases.

Each test case consists of *4*N+1* lines as follows -

- The first line contains integer *N*, the number of gardens
- Each of the next *4*N* lines contains 2 integers *x, y* where (x, y) is any one corner of any garden.

Output Format:

Print *t* lines, where *i-th* line represents output for *i-th* test.

- Each line contains an integer, the Total area of the gardens.
- If you can't find such N gardens, print **-1**.

Example 1:

Input:

1
3
5 2
0 15
15 7
10 15
5 7
10 10
7 10
20 10
20 15
7 15
15 2
0 10

Output:

135

Explanation:

$Garden_1$ with corners (0,10), (0,15), (7,10),
(7,15) has area 35 units.
$Garden_2$ with corners (10,10), (10,15),
(20,10), (20,15) has area 50 units.
$Garden_3$ with corners (5,2), (5,7), (15,2),
(15,7) has area 50 units.
Total area 35+50+50 = 135 units.

Example 2:

Input:

1
1

1 2
2 2
2 3
1 4

Output:
-1

Explanation:
No rectangular shape garden exist for given
corners.

Constraints:

- 1 <= t <= 1000
- 1 <= N <= 1000
- 0 <= x, y <= 10^6
- sum of N over test cases don't exceed 10^3

Shoot All

Difficulty: **Medium**Accuracy: **70.0%**Submissions: **20+**Points: **4**

There are **N** water balloons at some points in XY plane. Positions of all balloons are given as cartesian points (x_i, y_i). You have a cylindrical device (1 unit long) which shoots arrows from both sides.

Suppose this device is placed at point (a, b) horizontally then arrows will be shooted from a - 0.5 towards left and a + 0.5 towards right (because device is 1 unit long). Device can be placed at **any angle** initially (horizontally, vertically or any angle you want) .

Your task is to burst all N balloons using **minimum** number of moves, where one move is as follows :

- Rotate the device on any angle (only rotation is allowed, point (a, b) **must** be same) .
- Shoot any number of arrows from its ends.
- In current move, balloons which are colinear with this device can be shooted.
- One arrow will burst exactly one balloon.

Find **minimum** number of moves to burst all N balloons.

Note:

- It is not necessary to rotate device in each move.
- Point (a,b) (where devide is placed) doesn't have balloon.
- Two or more balloons may be placed at same position.

Input Format:

First line contains integer ***t***, number of test cases.

Each test case consist of *N+1* lines as follows -

- First line contains integer *N*, *a, b* as Number of balloons and Position of shooting device respectively.
- Each of next *N* lines contains 2 integers *x, y* where *(x, y)* is position of *i-th* balloon.

Output Format:

Print *t* lines, where *i-th* line represent output for *i-th* test.

- Each line contains an integer, minimum number of moves required.

Example 1:

Input:
1
4 0 0

1 1
2 2
2 0
-1 -1

Output:
2

Explanation:
Place device at (0, 0) at any angle.
In first move, rotate it horizontally
and burst balloon at (2, 0).
In second move rotate device 45 degree
anticlockwise and rest 3 balloons will
become colinear. Shoot all of them.

Example 2:

Input:
1
2 1 2
1 1
1 0

Output:
1

Explanation:
Place device at (1, 2) at any angle.
In first move, rotate it vertically and
both balloons will become colinear.
Shoot them.

Constraints:

- $1 <= t <= 100$
- $1 <= N <= 10^3$
- $-10^4 <= a, b, x, y <= 10^4$
- sum of N over all test cases doesn't exceed 10^3

Cheating In Exams

Difficulty: **Medium**Accuracy: **84.38%**Submissions: **32+**Points: **4**

In Geekland, Exam preparation are going on in full swing. There is an infinite size exam hall (consider as XY plane). To prevent cheating in exams, goverment of Geekland installed **N** cameras in exam hall. On the day of exam, **M** students will be giving exam. A student will be under the surveillance of a camera If and ony if Student is on right side of camera and its distance from camera in X direction is not less than distance in Y direction.

Suppose camera is mounted at point (a, b) and student is sitting at point (c, d) then $|b - d| <= a - c$ must hold for this student to be under the surveillance of this particular camera.

Initially, each student is under surveillance of **at least** one camera. But it came to notice that some cameras are unnecessarily installed i.e If these cameras are removed then also each student will be under surveillance of at least one camera.

Your task is to find **maximum** number of cameras that can be removed, so that goverment can save some money.

Note: Positions of all cameras and students are pairwise distinct.

Input Format:

First line contains integer ***t***, number of test cases.

Each test case is given as follows -

- First line contains two integers *N* and *M*, number of cameras and number of students respectively.
- Next *N* lines contains two integers *x* and *y*, where *(x,y)* will be position of *i-th* camera.
- Next *M* lines contains two integers *x* and *y*, where *(x,y)* will be position of *i-th* student.

Output Format:

Print *t* lines, where *i-th* line represent output for *i-th* test.

- Each line contains an integer, maximum number of cameras that can be removed.

Example:

Input:
1
3 3
-1 -1
-3 -1
-100 -150
0 0
-2 0
-94 -145

Output:
1

Explanation:
$Camera_1$ is unnecessarily installed, because
$student_1$ and $student_2$ will be under
surveillance of $camera_2$ and $student_3$ under

$camera_3$.

Constraints:

- 1 <= t <= 100
- 1 <= N, M <= $2*10^5$
- -10^8 <= x, y <= 10^8
- sum of N and sum of M over all test cases doesn't exceed 10^6

Miscellaneous

LIS and LDS

Difficulty: **Medium**Accuracy: **51.24%**Submissions: **41+**Points: **4**

Given an integer **N**, find a permutation of length **N** such that the sum of the length of the longest increasing subsequence (LIS) and length of the longest decreasing subsequence (LDS) of the permutation is minimum.

The sequence of **N** integers is called a permutation if it contains all integers from **1** to **N exactly** once.

You have to print the sum of the length of the longest increasing subsequence (LIS) and length of the longest decreasing subsequence (LDS) of the permutation.

For example given permutation of length 5 - {1, 4, 5, 2, 3}, possible longest increasing subsequence (LIS) is {1, 4, 5} and possible longest decreasing subsequence (LDS) is {5, 2} and sum of the length of the longest increasing subsequence (LIS) and length of the longest decreasing subsequence (LDS) of the permutation is 5.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^5$

Example:

Input:

3

4

5

20

Output:

4

5

9

Explanation:

Test Case 1 : One possible answer is {3, 4, 1, 2},
in which possible LIS is {3, 4} and possible
LDS is {4, 1} and sum of length of LIS and LDS
is 2 + 2 = 4.

Test Case 2 : Explained in the problem statement.

Chess Tournament

Difficulty: **Medium**Accuracy: **94.11%**Submissions: **238+**Points: **4**

There were **N** children in the class and a chess tournament had to be organized. Each child was to be assigned a team and had to play a match with **each of his or her teammates.** There were **M** teams and the teacher had to assign a team to each child in such a way that each team had atleast 1 child and the **total number of**

matches that took place (sum of matches over all teams) were **minimum**. Help her in this task and print this minimum possible number of matches that could take place.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space seperated integers **N** and **M.**

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^5$
2. $1 \leq M \leq N \leq 10^9$

Example:

Input:
2
5 2
7 1
Output:
4
21
Explanation:
Test Case 1 : When out of 5 , 3 children
are put in 1 team and 2 in the other then
total matches will be 3 + 1 = 4 which is
the minimum possible.

Tower Challenge
Difficulty: **Medium**Accuracy: **94.72%**Submissions: **229+**Points: **4**

Geek's mother challenged him and his sister for a game. Geek and his sister both had to make a tower of height **exactly N** each, by stacking up blocks each of height **1**. The conditions were that they had to request for blocks **exactly X** times and each time they could ask for any **non zero** number of blocks but mother would give Geek **twice** the number of blocks he asked for and would give his sister **one less than twice** the number of blocks she asked for. It was compulsory for them to use **all** the blocks they had got. If **atleast** one of them could possibly build the tower then they would win otherwise their mother would win. Print "**yes**" if they can win or "**no**" if they cannot (without quotes) .

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space seperated integers **N** and **X.**

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^5$
2. $1 \leq N , X \leq 10^9$

Example:

Input:
2
5 3
3 5
Output:
yes

no

Explanation:

Test Case 1 : Geek's sister can make it
by requesting for 1 , 1 and 2 blocks.

Rock, Paper and Scissors

Difficulty: **Easy**Accuracy: **36.36%**Submissions: **22+**Points: **2**

Geek and his friend are playing Rock, Paper and Scissors.
It is known that they played **N** rounds in which Geek played **A1** times rock, **B1** time paper and **C1** times scissors and his friend played **A2** times rock, **B2** time paper and **C2** times scissors. In each round, his friend plays first. Given **N, A1, B1, C1, A2, B2** and **C2,** find out if Geek can win the game if he plays optimally. To win he should win atleast **ceil(N / 2)** rounds. A draw is not considered a win.
Note that in rock-paper-scissors - rock beats scissors, paper beats rock and scissors beat paper.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains three space-separated integers **A1**, **B1** and **C1**.
4. The third line of each test case contains three space-separated integers **A2**, **B2** and **C2**.

Output: For each test case, print the "**Yes**" (without quotes) if Geek can win or "**No**" (without quotes) otherwise.

Constraints:

1. $1 \leq T \leq 10^5$
2. $1 \leq N, A1, B1, C1, A2, B2, C2 \leq 10^9$
3. $A1 + B1 + C1 = A2 + B2 + C2 = N$

Example:

Input:

2
3
1 1 1
1 1 1
10
8 1 1
6 3 1

Output:

Yes
No

Explanation:

Test Case 1 : When his friend will play paper, Geek will play scissors then when his friend will play rock, Geek will play paper and when his friend will play scissors, Geek will play rock. Geek can win 3 rounds, hence he can win the game.

Permutation Sorting

Difficulty: **Medium**Accuracy: **93.37%**Submissions: **227+**Points: **4**

Given a permutation **a[]** of size **N**, you have to sort the permutation using only following operation multiple number of times (possibly zero):

Choose any index **i**, remove the element **a[i]** and place it either in the front of the

array or at the back.
The sequence of **N** integers is called a permutation if it contains all integers from **1** to **N exactly** once.
For example - given a permutation of size 3 - {2, 1, 3}, we can choose index 2 and remove a[2] = 1 and place it in the beginning, then the permutation will become {1, 2, 3}.
You have to find the minimum number of operations required to sort the permutation.

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 100$
2. $1 \leq N \leq 10^6$
3. $1 \leq a[i] \leq N$
4. It is guaranteed that sum of N over all testcases does not exceed 10^6.

Example:

Input:
2
4

3 1 2 4

3

1 2 3

Output:

2

0

Explanation:

Test Case 1 : In the first operation, we will remove 2 and place it in the beginning, then array will become {2, 3, 1, 4}. Then we will remove 1 and place it in the beginning, then array will become {1, 2, 3, 4}.

Test Case 2 : The permutation is already sorted.

Team Representatives

Difficulty: **Medium**Accuracy: **69.23%**Submissions: **13+**Points: **4**

There were **N** team representatives standing in a straight line and i th representative had a[i] members in his team (including himself). They had to be grouped without changing their positions to form exactly **P** new teams where number of members in each was sum of team members of the representatives in that team. For example (1 ,4 , 5) are the number of team members of N = 3 representatives , then if P is 2 then they could be grouped as ((1, 4) , (5)) i.e 2 teams with 5 members each or as ((1) , (4 ,5)) i.e 2 teams with 1 and 9 members . A grouping would be valid if each of the new P teams had **odd** number of members. Print " **Yes** " if it was possible to form P valid new teams and " **No** " otherwise (without quotes).

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space separated integers **N** and **P**.

3. The second line of each test case contains **N** space separated integers which represent **a[ ]**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 100$
2. $1 \leq P \leq N \leq 10^5$
3. $1 \leq a[i] \leq 10^9$

Example:

Input:

2
4 3
1 2 3 4
3 2
1 4 5

Output:

No
Yes

Explanation:

Test Case 1 : After trying all possible groupings we can see that atleast 1 of the new teams would have even number of members.

Make it 120!

Difficulty: **Medium**Accuracy: **99.44%**Submissions: **209+**Points: **4**

Geek has an array **a[]** of **N** elements, where **a[i] = i** (1 based indexing). Geek wants to obtain number **120** by performing following operation **N - 1** times:
Choose two arbitrary elements of the array **a[i]** and **a[j] (i != j)**, remove both of

them from the array and insert either **a[i] + a[j]** or **a[i] - a[j]** or **a[i] *a[j]** in the array. After **N - 1** operations, only one element will remain in the array.
You have to find if Geek can obtain **120** at the end or not. Print "**Yes**" if it is possible or "**No**" otherwise (without quotes).

Input:
1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N**.

Output: For each test case, print the answer.

Constraints:
1. $1 \leq T \leq 10^5$
2. $1 \leq N \leq 10^5$

Example:

Input:
2
1
5
Output:
No
Yes
Explanation:
Test Case 1 : There is already only one element in the array i.e. 1.
Test Case 2 : The array is {1, 2, 3, 4, 5}.

During each operation Geek can choose two
elements multiply them and insert them back
in the array as 120 = 1 * 2 * 3 * 4 * 5.

Partners!

Difficulty: **Medium**Accuracy: **75.69%**Submissions: **37+**Points: **4**

There were **2*N** children in the class but there were only **N** computers in their lab so they had to form groups of two for that class. Now each student had a favourite number and wanted that either their partner 's number's parity should be **same** as their number' s parity or their numbers should differ by **atmost 1** otherwise they would be angry and not compatible. You will be given all the 2*N favourite numbers of the children in the class in order of their roll numbers and you have to print " **Yes**" if it was possible to make N pairs without making even a single child angry and " **No** " otherwise (without quotes).

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains one integer **N .**
3. The second line of each test case contains **2*N** space separated integers.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 50$
2. $1 \leq N \leq 10^5$
3. $1 \leq$ fav no [i] $\leq 10^5$

Example:

Input:
2
1

1 4
2
1 2 3 4

Output:

No
Yes

Explanation:

Test Case 1 : 1 and 4 neither have same parities nor differ by 1 so they will not be compatible.

Restricted Swaps

Difficulty: **Hard**Accuracy: **86.46%**Submissions: **251+**Points: **8**

Geek wants to sort an array **a[]** of size **N**. Almost all sorting algorithms include swapping of elements one way or another, but Geek is restricted to swap only **M** pair of elements where a pair is given by **{m_i, m_j}** which means we can swap **a[m_i]** and **a[m_j]**.

As his friend, Geek asked you to find out whether he can sort the array or not. Print "**Yes**" if it is possible or "**No**" otherwise (without quotes).

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains two space-separated integers integers **N** and **M**.
3. The second line of each test case contains **N** space-separated integers which represents **a[]**.
4. The next **M** lines contains two space-separated integers integers **m_i** and **m_j**.

Output: For each test case, print the answer.

Constraints:

1. $1 \le T \le 10$
2. $1 \le N \le 10^3$
3. $1 \le M \le 10^4$
4. $1 \le a[i] \le 10^9$
5. $1 \le m_i, m_j \le N$

Example:

Input:

2
2 1
1 10
1 2
3 1
8 5 1
2 3

Output:

Yes
No

Explanation:

Test Case 1 : The array is already sorted.
Test Case 2 : We can swap only 2nd and 3rd element which is not enough to sort the array.

Max Teams

Difficulty: **Medium**Accuracy: **52.0%**Submissions: **25+**Points: **4**

There were **N** children in the class with roll numbers from **1 to N**. Now the teacher wanted to divide them into groups following this rule , if the **absolute difference** between the roll numbers of any two children was **greater than 1** and a **divisor of N** then they had to be in the **same** team. Now she also wanted to **maximize** the total number of teams . Help her find out the maximum number of teams that could be formed **without** violating the rule.

Input:

1. The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases follows.
2. The first line of each test case contains a single integer **N**.

Output: For each test case, print the answer.

Constraints:

1. $1 \leq T \leq 10^2$
2. $1 \leq N \leq 10^{10}$

Example:

Input:

2
3
4

Output:

3
2

Explanation:

Test Case 1 : The difference between roll numbers 1 and 2 and the difference between 2 and 3 is 1 so they all can be in different teams and difference between 1 and 3 is 2

which is not a divisor of 3 so the rule is
again not violated.

Recursion

Number of Characters in Fibonacci String

Difficulty: **Medium**Accuracy: **57.59%**Submissions: **603+**Points: **4**

Given a number **n**, you have to calculate number of each character in **n'th** fibonacci string.

Fibonacci Strings are defined as follows: $F_0 = a$, $F_1 = b$, $F_i = F_{i-2}\ F_{i-1}$, **for i > 1.**

For Example, the first five Fibonacci strings are: **a**, **b**, **ab**, **bab**, **abbab**.

Input:

The first line of input contains one integer **n.**

Output:

You have to print two space-separated integers, the first one is the count of character 'a', and the second one is the count of character 'b'.

Example 1:

Input:
n = 5
Output:
2 3
Explaination:

Count of character 'a' = 2

Count of character 'b' = 3

Example 2:

Input:
n = 6
Output:
3 5

Expected Time Complexity: O(n)
Expected Auxiliary Space: O(n)

Constraints:
1 <= **n** <= 50

GCD of Factorials

Difficulty: **Easy**Accuracy: **72.66%**Submissions: **468+**Points: **2**

You are given 2 numbers **N** and **M.** You need to find the GCD of their factorial.

Note: We suggest you solve this problem using recursion.

Input

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains two integers **N, M**

Output

- Print **T** integers, the i-th should be the answer for the i-th test case.

Constraints

- 1 <= T <= 100000

- 1 <= N, M <= 18

Example 1

Input

```
2
3 4
2 3
```

Output

```
6
2
```

Explanation

```
For first testcase of example 1,
Factorial of 3 = 6,
Factorial of 4 = 24.
So, GCD(6, 24) = 6
```

Count Digits of Number

Difficulty: **Easy**Accuracy: **86.32%**Submissions: **392+**Points: **2**

You are given one number **N** and **Q** queries.

Each query is a number **X.** After adding **X** in **N**, you need to find the count of digits in **N.**

Note: We suggest you solve this problem using recursion.

Input

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains two integers **N, Q.**
- Next, **Q** lines contain one integer **X.**

Output

- Print corresponding answer for each queries.

Constraints

- 1 <= T, Q <= 100
- 1 <= N <= 100
- 1 <= X <= 100

Example 1

Input

```
1
1 3
6
51
45
```

Output

```
1
2
3
```

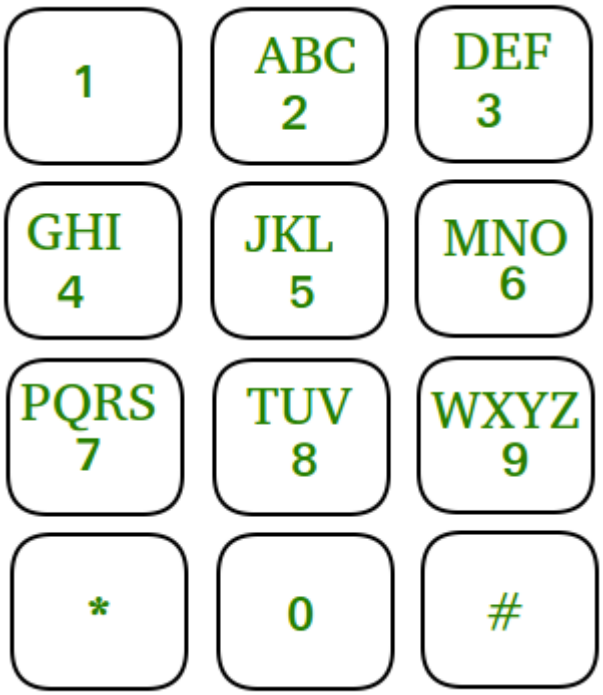
Explanation

```
For first testcase of example 1,
After 1st query, N = 1 + 6 = 7
After 2nd query, N = 7 + 51 = 58
After 3rd query, N = 58 + 45 = 103
```

All Possible Words From Mobile Keypad

Difficulty: **Hard**Accuracy: **78.3%**Submissions: **377+**Points: **8**

You are given a keypad, shown in the below diagram, and an array **A[]** of length **N**, containing a list of keys that you have to click.



You need to print all possible words by clicking on these keys in sorted order.

If you single-click on a key, any character written on it will be printed.

If you double-click on a key, it will print the corresponding number written on it.

Note: Two continuous same keys in the array can be considered as a double click.

Input:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains one integer **N.**
- The next line contains **N** integer **A[i].**

Output:

- For each test case print all possible words by clicking on these keys in sorted order.

Constraints:

- 1 <= T <= 10

- 1 <= N <= 10
- 2 <= A[i] <= 9

The sum of **N** over all test cases doesn't exceed 10.

Example 1:

Input:

```
1
3
2 2 3
```

Output:

```
2d
2e
2f
aad
aae
aaf
abd
abe
abf
acd
ace
acf
bad
bae
baf
bbd
bbe
bbf
bcd
bce
```

```
bcf
cad
cae
caf
cbd
cbe
cbf
ccd
cce
ccf
```

Explanation:

For first example all possible words
which can be formed are (Sorted order):
2d 2e 2f aad aae aaf abd abe abf acd
ace acf bad bae baf bbd bbe bbf bcd bce
bcf cad cae caf cbd cbe cbf ccd cce ccf

If 2 is pressed once then it can be print: **a, b, c,**
If 2 is pressed twice then it can be print: **2, aa, ab, ac ...**

Example 2:

Input:

```
2
1
9
2
3 3
```

Output:

```
w
x
```

```
y
z
3
dd
de
df
ed
ee
ef
fd
fe
ff
```

Valid Expression

Difficulty: **Medium**Accuracy: **49.64%**Submissions: **139+**Points: **4**

You have been given a mathematical expression and one of its possible solutions.

We can say the given expression is valid if we get the given solution from the expression by putting parentheses at different places.

You need to print "Yes" if the given expression is a valid expression, otherwise print "No".

Note: Given expression contains only digit and operators (' * ', ' + ' and ' - ')

Input:

- The first line contains one integer **t** (number of test cases).
- The first line of each test case contains one expression and one possible solution.

Output:

- Print t integers, the i-th should be the answer for the i-th test case.

Constraints:

- 1 <= t <= 100
- 1 <= length of expression <= 20

Example:

Sample Input

```
2
1*2+3*4 20
1+2*3 5
```

Sample Output

```
Yes
No
```

Explanation

```
Here all possible solutions of first testcase are :
(1*(2+(3*4))) =  14
(1*((2+3)*4)) =  20
((1*2)+(3*4)) =  14
((1*(2+3))*4) =  20 ( Same as given solution )
((1*2)+3)*4)  =  20 ( Same as given solution )
So, here answer of the first test case will be Yes.
```

Magnificent String

Difficulty: **Medium**Accuracy: **90.74%**Submissions: **312+**Points: **4**

Geek loves playing with strings, especially with strings consisting of lowercase Latin letters. One day, while playing with a string **s[1....n]** where **$n = 2^k$** for some k >= 0, Geek noticed an interesting property of certain strings called ch-magnificent strings. He discovered that a string s[1....n] is called **ch-magnificent** if at least one of the following three conditions is satisfied:

- The length of s is 1, and it consists of the character ch (i.e. s[1] = ch).

- The length of s is greater than 1, the first half of the string consists of only the character ch and the second half of the string is a (ch+1) - magnificent string.
- The length of s is greater than 1, the second half of the string consists of only the character ch and the first half of the string is a (ch+1) - magnificent string.

For example, "aabc" is 'a'-magnificent, "ffgheeee" is 'e'-magnificent.
Geek was so fascinated by this property that he decided to solve a problem related to it. He was given a string s[1....n] consisting of lowercase Latin letters and was tasked to find the **minimum number of moves** required to obtain an **'b'-magnificent** (i.e. ch-magnificent string ch = 'b') string from s.
In one move, he could choose one index i from 1 to n and **replace s_i with any lowercase Latin letter** (any character from 'a' to 'z').
Can you help Geek solve this problem ?

Note : It is guaranteed that the answer always exists.

Input Format:

The first line contains an integer **T** - the number of test cases. Then the test cases follows:

The first line contain an integer **n**, denoting length of string s. It is guaranteed that $n=2^k$ for some $k >= 0$.

The second line contains the string **s** consisting of n lowercase Latin letters.

Output Format:
For each test case, print minimum number of moves followed by newline character.

Constraints:
$1 \leq T \leq 1000$

$1 \leq n \leq 131072$

The sum of n over all test cases does not exceed $2 * 10^5$.

Example:

Input:

2

8

ccedbbbb

8

bbaaddcc

Output:

0

3

Explanation:

For test case 1, string s is already 'b'-magnificent string.

For test case 2, It is proven that minimum 3 moves are required.

In first move, replace s[3] with 'b', s = "bbbaddcc".

In second move, replace s[4] with 'b', s = "bbbbddcc".

In third move, replace s[5] with 'e', s = "bbbbedcc".

Binary Businessman

Difficulty: **Hard** Accuracy: **77.13%** Submissions: **341+** Points: **8**

Kaleen Bhaiya's son, Munna, is interested in learning mathematics to become a better businessman. To help him, Kaleen Bhaiya enrolls him in a prestigious coaching institute. However, before Munna can begin his classes, he is given a test by the institute to assess his eligibility.

The test consists of a list with a single element **num**. Munna must perform certain operations on this list. In each operation, he must remove any element x, such that **x > 1**, from the list and insert at the same position floor(x/2), x % 2,

floor(x/2) sequentially. He must continue with these operations until all the elements in the list are either **0** or **1**.

The institute wants to know the **total number of 1s in the range L to R** (1-indexed). Munna wants to become a better businessman, but unfortunately, he cannot solve this problem. Can you help Munna pass the test?

Input Format:

The first line contains an integer **T** - the number of test cases. Then the test cases follows:

The first and only line contains three integers **num**, **L** and **R**, denoting initial element and range.

Output Format:
For each test case, print number of 1s in given range followed by newline character.

Constraints:
$1 \leq T \leq 1000$
$0 \leq num \leq 2^{50}$
$1 \leq L \leq R$
$0 \leq R - L \leq 10^5$
It is guaranteed that R is not greater than the length of the final list.
Also it is guaranteed that sum of (R - L) over all test cases does not exceed 10^5.

Example:

Input:
2
7 2 6
10 3 7
Output:

5

4

Explanation:

For test case 1, [7] --> [3, 1, 3] --> [1, 1, 1, 1, 3] --> [1, 1, 1, 1, 1, 1, 1]

Number of 1's from position 2 to 6 is 5.

For test case 2, [10] --> [1, 0, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1]

Number of 1's from position 3 to 7 is 4.

String Equivalency Challenge

Difficulty: **Hard**Accuracy: **89.94%**Submissions: **317+**Points: **8**

Geek was attending a computer science class and learned about a new concept of string equivalency. Two strings, x and y, will be considered equivalent if:

1. They are exactly the same.
2. If we split string x into two equal halves, x1 and x2, and also split string y into two equal halves, y1 and y2, then one of the following conditions should be true:
    - x1 is equivalent to y1, and x2 is equivalent to y2.
    - x1 is equivalent to y2, and x2 is equivalent to y1.

As a part of their homework, the teacher asked each student to take two strings and determine if they are equivalent or not.

Geek, being a diligent student, has already solved the homework problem. Now, it's your turn to give it a try!

Input Format:

The first line contains an integer **T** - the number of test cases. Then the test cases follows:

The first two lines of the input contain two strings of same length given by the teacher.

Output Format:

For each test case, Print "YES" (without the quotes), if these two strings are equivalent, and "NO" (without the quotes) otherwise. followed by newline character.

Constraints:

$1 \leq T \leq 100$

$1 \leq |x| = |y| \leq 2 * 10^3$

The sum of |x| over all test cases does not exceed $2 * 10^3$.

Example:

Input:

2

aaba

abaa

aabb

abab

Output:

YES

NO

Explanation:

For test case 1, You should split the first string into strings "aa" and "ba", the second one into strings "ab" and "aa". "aa" is equivalent to "aa". "ab" is equivalent to "ba" as "ab" = "a" + "b", "ba" = "b" + "a".

Backtracking

Partitioning string into consecutively increasing integers

Difficulty: **Medium** Accuracy: **87.91%** Submissions: **306+** Points: **4**

Geek is given a string. Can you help him in finding out whether the given string can be partitioned into **two or more non empty substrings** in such a way that the integer values of the substrings are in the **strictly increasing order** and

the **difference** between the integer values of every two **adjacent substrings** is equal to 1.

For example, the string s = "00079080" can be partitioned into ["00079", "080"] with integer values [79,80]. The values are in increasing order and adjacent values differ by 1, so this way is valid.
Another example, the string s = "005" can be partitioned into ["0", "05"], ["00", "5"], or ["0", "0", "5"]. However all the ways are invalid because they have integer values [0,5], [0,5], and [0,0,5] respectively, all of which are not in increasing order.

Input:

The first line contains a **single integer t**, the number of test cases.

Each subsequent line contains a **string s**.

Output:

For each test case, print either **"true"** or **"false"**.

Constraints:

- 1 <= T <= 10^5
- 1 <= s.length() <= 10

Example:

Input:

```
2

000079080

005
```

Output:

true

false

Rescuing Geeks Kingdom

Difficulty: **Medium**Accuracy: **91.36%**Submissions: **303+**Points: **4**

The king of Geeks kingdom is in massive debt. To save his kingdom, he has to pay the debt's money as soon as possible. Therefore, he asked his minister to collect diamonds from a mine located near the kingdom to reduce the debt to some extent. Can you help the minister by finding the **maximum amount of diamond** that he can collect from the mine?

A mine is in the form of an **n*m matrix**; each cell in this mine has an integer representing the amount of diamond in that cell, **0** if it is empty.

The minister has to follow the following **rules** while collecting diamonds from the mine.

- He has to collect **all** the diamonds from a particular cell in which he is standing.
- He can walk one step to the **left, right, up, or down** from a given position.
- He can never visit a cell with **0** amount of diamond.
- He **can't** visit the **same** cell more than **once**.
- He can start and stop collecting diamonds from **any** position in the grid that has some amount of diamond.

Input:

The first line contains a single integer t, the number of test cases.

The first line of each test case contains two integers, n and m.

The next n lines of each test case contain m integers, each representing the amount of diamond in a particular cell.

Output:

Print the maximum amount of diamond that the minister can collect from the mine.

Constraints:

- 1 <= t <= 10
- 1 <= n, m <= 15
- 0 <= grid[i][j] <= 1000
- There are atmost **25** cells containing diamonds.

Example 1:

Input

```
3 3

0 8 0

7 8 13

0 12 0
```

Output

```
33
```

Explanation

```
Path to get the maximum amount of
diamonds, 12 -> 8 -> 13
```

Example 2:

Input

```
5 3

9 0 20
```

```
0 3 0

3 4 5

2 0 6

1 0 7
```

Output

```
28
```

Explanation

```
Path to get the maximum amount of
diamonds, 1 -> 2 -> 3 -> 4 ->5 -> 6 -> 7
```

Beautiful String

Difficulty: **Hard**Accuracy: **82.85%**Submissions: **321+**Points: **8**

Geek is provided with a string consisting of **lowercase English letters** and parentheses **'('** and **')'**. A string is called **beautiful** if it has a corresponding opening bracket for every closing bracket and vice versa.
Can you help Geek in making the string beautiful
by **removing** the **minimum** number of parenthesis? If there are **multiple** answers, you have to print all of them in **sorted** order.

Input:

The first line contains a **single integer t**, the number of test cases.

Each subsequent line contains a **string s**.

Output:

For each test case, print the beautiful strings in sorted order(each in separate line).

Constraints:

- 1 <= T <= 100
- 1 <= s.length() <= 20
- s consists of lowercase English letters and parentheses '(' and ')'.

Example:

Input:

```
3
()()()
(a)()()
)(
```

Output:

```
(()()
()()
(a()()
(a)()
```

Explanation:

```
In the last example we have to remove both the
paranthesis to make the string beautiful.
```

School Competition

Difficulty: **Hard**Accuracy: **91.3%**Submissions: **288+**Points: **8**

Geeks School is organizing a competition to solve the **crossword puzzle**. However, Chinmoy, who is taking part in this competition, found it challenging to solve. So he decided to cheat, and instead of solving the puzzle himself, he decided to write

a program to solve the puzzle quickly.
Can you help him making him win by writing a program on his behalf?
Following are the instructions for the puzzle:

- A **10*10** crossword grid is provided, along with a **set of words** that need to be **filled** into the grid.
- Cells in the grid are marked either with **'+'** or **'-'**.
- Cells marked with a **'-'** needs to be filled with the word list.
- A word present in the word list needs to be fitted **horizontally** or **vertically** in the crossword grid.

Input:

Each of the first 10 lines represents grid[i], each of which has 10 characters, grid[i][j]

The following line contains n represents the length of the word list.

The following n lines contain the word present in the word list.

Output:

Print the 10*10 grid after fitting all the words present in the word list.

Constraints:

- 1 <= n <= 10
- grid[i][j] = {+,-}
- 'A' <= words[i][j] <= 'Z'

Note: You can assume that only one output is possible for the given set of testcases and it will always exist.

Example 1:

Input:

```
+----+++++
```

```
++++-+++++
++++-+++++
++++------
++++-+++-+
++++-+++-+
++++-+++-+
++++-+++-+
++++-+++++
++++++++++
4
TREE
ELEPHANTS
PICKLE
LEMON
```

Output:

```
+TREE+++++
++++L+++++
++++E+++++
++++PICKLE
++++H+++E+
++++A+++M+
++++N+++O+
++++T+++N+
++++S+++++
++++++++++
```

Example 2:

Input:

```
+-+++++++
+-------++
+-++-+++++
```

```
+-------++
+-++-++++-
+-++-++++-
+-++------
+++++++++-
++++++++++
++++++++++
6
ANDAMAN
MANIPUR
ICELAND
ALLEPY
YANGON
PUNE
```

Output:

```
+M++++++++
+ANDAMAN++
+N++L+++++
+ICELAND++
+P++E++++P
+U++P++++U
+R++YANGON
+++++++++E
++++++++++
++++++++++
```

Burger Shop

Difficulty: **Medium**Accuracy: **82.74%**Submissions: **324+**Points: **4**

Geek opens a burger shop where he is offering **n** burgers along with **m** toppings. A customer comes to the shop and plans to spend around **x** amount of money.

The customer should order **exactly one burger** and may order **zero or more toppings**. The customer can order **each topping at most twice.**

Given the lists of **prices** of available **burgers** and **toppings**, what is the price closest to **x** out of all the possible orders? Here, a price is said to be **closer to x** when the **difference from x is smaller**.

Input:

The first line contains an integer **T** representing the number of test cases.

The first line of each test case has two integers, **n** and **m**, representing the length of the burger and toppings list.

The second line of each test case contains **n integers** representing the burger's cost.
The third line of each test case contains **m integers** representing the toppings cost.
The fourth line of each test case contains an integer **x** representing the approximate amount of money the customer would spend.

Output:

For each test case you have to return the **price closest to x** out of all the possible orders. If there are multiple answers then return the lower one.

Constraints:

- 1 <= T <= 100
- 1 <= n, m <= 10
- 1 <= burgers[i], toppings[i] <= 10^4
- 1 <= x <= 10^4

Note: The customer is allowed to make an order that costs more than x.

Example:

Input:

```
3
2 3
8 9
5 4 50
24
2 2
1 7
3 4
7
2 2
3 10
2 5
9
```

Output:

```
23
7
8
```

Explanation:

1st test case:

Consider the following combination (all 0-indexed):

Choose burger 1: cost 9

Take 1 of topping 0: cost 2 x 5 = 10

Take 2 of topping 1: cost 1 x 4 = 4

Take 0 of topping 2: cost 0 x 100 = 0

Total: 9 + 10 + 4 + 0 = 23. Exact 24 can't be made.

2nd test case:

Don't take any topping. Take only burger

which cost 7.

3rd test case:

It is possible to get costs 8 and 10.

Return 8 as it is the lower cost.

Kabaddi Match

Difficulty: **Easy**Accuracy: **83.03%**Submissions: **325+**Points: **2**

Geek is a coach of Kabaddi game in his village school. He wants to organize a Kabaddi match between the group of **n** students. For the match, he wants to divide the students into two groups of equal size The **ith** student has a strength **A[i]**. Geek wants a fun and exciting game; for this, the **absolute difference** between the group's strengths should be **minimum** as possible. A group's strength is the **sum of the strengths** of the students in the group.
Can you help Geek in finding the **minimum absolute difference** between the two groups?

Input:

The first line of input contains a single integer **'T'**, representing the number of test cases.

Then the **'T'** test cases follow.

The first line of each test case contains a positive integer **'n'**, which represents the number of students.

The next line contains **'n'** single space-separated integers representing the strength of the students.

Output:

For each test case, print the minimum absolute difference between the group's strength.

The output of each test case is printed in a separate line.

Constraints:

- 1 <= T <= 10
- 2 <= n <= 20
- 1 <= Arr[i] <= 10^5

Note: If N is odd, then the size of one group should be (N-1)/2, and the size of another group should be (N+1)/2.

Example:

Input:

```
3
2
3 5
3
2 6 3
4
4 2 3 1
```

Output:

```
2
1
0
```

Explanation:

```
In the 1st test case, the first group contains students
with strength {5} and the second group contains students
with strength {3}, the absolute difference between
the group's strength is (5)-(3)=2.

In the 2nd test case, the first group contains students
with strength {6} and the second group contains students
with strength {2,3}, the absolute difference between
```

the group's strength is (6)-(2+3)=1.

In the 3rd test case, the first group contains students with strength {1,4} and the second group contains students with strength {2,3}, the absolute difference between the group's strength is (5)-(5)=0.

Good Strings

Difficulty: **Hard** Accuracy: **86.62%** Submissions: **292+** Points: **8**

A string str is called a **good string** if there **does not** exist three difference indices i, j, k such that i, j, k are consecutive three terms of an arithmetic progression and **str[i] = str[j] = str[k]**.

You are given a string **s** and two integers **K**, **A**. You have to **find the number of good strings t with alphabet size A** (i.e. if A = 1, then string can only have a's and if A = 2, string can only contain 'a' or 'b'. e.g. if A = 2, t can be aa, ab, bb etc., for A = 3, string can only contain 'a', 'b' and 'c', i.e. it can be aab bac, bbb, cbb etc.) of length |s| such that hamming distance between strings s and t is **less than or equal to K**. Hamming distance between two strings of same size is defined as the number of indices where their corresponding characters differ. Formally, for two strings s, t of length N, hamming distance will be number of indices $1 \leq i \leq N$ such that s[i] != t[i].

Input Format:

The first line contains an integer **T** - the number of test cases. Then the test cases follows:

The first line will contain two integers **A, K**.

The second line will contain the string **s** consisting of lowercase characters only.

Output Format:

For each test case, print number of good strings followed by newline character.

Constraints:

$1 \leq T \leq 50$

$1 \leq A \leq 3$

$0 \leq K \leq |s|$

$1 \leq |s| \leq 50$

Example:

Input:

2

1 0

a

2 2

aba

Output:

1

5

Explanation:

For test case 1, only string of length 1 and alphabet size 1 will be 'a'. Hamming distance of "a" with "a" is zero which is ? 1. So answer is 1.

For test case 2, there can be total 8 strings. 6 of them are good. aaa and bbb are not good, remaining all are good. From the 6 good strings, bab has a hamming distance of 3, remaining other 5 have hamming distance <= 2. Hence answer is 5.

Power Sum

Difficulty: **Easy**Accuracy: **93.77%**Submissions: **292+**Points: **2**

Alice is a math enthusiast who is fascinated with the properties of natural numbers. One day, she decided to explore a problem related to **unique sums of nth**

powers of natural numbers. She picked an integer x and an exponent n, and wanted to find out how many unique ways there are to express x as the sum of nth powers of distinct natural numbers.

For example, suppose Alice chose x = 13 and n = 2. She realized that there is only one way to express 13 as the sum of distinct squares of natural numbers: 13 = 2^2 + 3^2. She wanted to know if there are other values of x and n for which there are multiple ways to express x as the sum of distinct nth powers of natural numbers.

Can you help Alice solve this problem by writing a program that computes the number of unique ways to express a given x as the sum of distinct nth powers of natural numbers?

Input Format:
The first line contains an integer **T** - the number of test cases. Then the test cases follows:
The first line contains two positive integers **X** and **N**.

Output Format:
For each test case, print the number of unique ways followed by newline character.

Constraints:
1 ≤ T ≤ 10
1 <= X <= 1000
2 <= N <= 10

Example 1:

Input:
2
10 2
100 2

Output:

1

3

Explanation:

For test case 1,

X = 10 and N = 2, we need to find the number of ways that 10 can be represented as the sum of squares of unique numbers.

$10 = 1^2 + 3^2$

This is the only way in which can be expressed as the sum of unique squares.

For test case 2,

$100 = 10^2 = (6^2 + 8^2) = (1^2 + 3^2 + 4^2 + 5^2 + 7^2)$

Divide and Conquer – CP

Binary Search

Minimum median in K moves

Difficulty: **Medium**Accuracy: **82.2%**Submissions: **349+**Points: **4**

You are given an array **A[]** of size **N.** **N** is an odd number. If **A** is sorted in non-decreasing order, then middle element of **A** is known as **median** of **A.** For example if **A[]={1, 2, 3, 4, 3}** after sorting **A[] = {1, 2, 3, 3, 4}.** Here, **3** is the median of **A[]** as it is the middle element when we sort **A.**

You have **K** moves, in one move you can change any element of **A** to any other **positive number**.

Print the minimum median you can achieve in atmost **K** moves.

Input:

1. First line of input contains two integers **N** and **K**, size of **A[]** and number of moves respectively.

2. Next line contains **N** space separated integers, **i^{th}** of which is value of **A[i]**

Output:

Output must contain only one integer, minimum median possible by using atmost **K** moves.

Constraints:

$1 <= K, N <= 10^5$

$1 <= A[i] <= 10^9$

Example:

Input:

```
7 4

3 2 1 4 5 3 2
```

Output:

```
1
```

Explanation:

Sorted **A[]** is {1, 2, 2, 3, 3, 4, 5} with median **3.** We can set **A[2], A[3]** and **A[4]** in sorted **A[]** to **1 . A[]** becomes {1, 1, 1, 1, 4, 5}. So median is **1** in **3** moves.

Maximum value of the function

Difficulty: **Medium**Accuracy: **90.55%**Submissions: **309+**Points: **4**

Find **maximum value** function **F(x)** defined below, can achieve for some **x** such that **- Π/2 <= x < = Π/2.**

$F(x) = a \cos(x) / (bx^2 + x + 1)$

Here **a** and **b** are constants, given in input.

NOTE:

Print the answer rounded to exactly 4 decimal places. Take value of **Π = 3.14159**

Input

- First line of input contains **T** number of testcases, **T** lines follow.
- Each testcase contains two integers **a** and **b**

Output

- For every testcase print on a new line maximum value **F(x)** can achieve rounded to **exactly 4 decimal places**

Constraints

- **1<=T<=10000**
- **1<=a,b<=100**

Example 1

Input

```
2

1 3
```

3 4

Output

1.0778

3.1777

Explanation

For **a=1** and **b=3 , F(x) = cos(x)/ (3x² + x + 1)** which achieves maximum value of **1.07780** at **-0.144.** Therefore answer for the first testcase rounded to exactly **4 decimal places** is **1.0778.**

For **a=3** and **b=4 , F(x) = 3cos(x)/ (4x² + x + 1)** which achieves maximum value of **3.177659** at **-0.112.** Therefore answer for the second testcase rounded to exactly **4 decimal places** is **3.1777.**

Kth element in Grid

Difficulty: **Medium**Accuracy: **68.5%**Submissions: **394+**Points: **4**

Given two arrays **A** and **B** of size **N** each. We create a **2D** grid **G** of dimensions **NxN,**

such that, **G[i][j] = A[i]*B[j].**

Find K_{th} smallest element in **G.** In other words, If we sort all the elements of **G** in non-decreasing order, find the K_{th} element in that order.

Input

- First line contains an integer **T** , **T** testcases follow.
- First line of each test case contains two integers **N** and **K**
- Second line contains **N** space separated integers denoting elements of **A**
- Third line contains **N** space separated integers denoting elements of **B**

Output

- Print an integer in a single line denoting K_{th} smallest element in **G**

Constraints

- $1 <= N <= 3*10^4$
- $1 <= K <= N^2$
- $1 <= A_i, B_i <= 10^9$

Example 1

Sample Input

```
1

3 4

1 2 3

3 4 2
```

Sample Output

```
4
```

Explanation

A[]={1, 2, 3} and **B[]={3, 4, 2}.** Values in **G** will be **{2, 3, 4, 4, 6, 8, 6, 9, 12} .**
K=4, so **4th** smallest element is **4.**

NOTE : Answer may **not fit** in a **32-bit integer**

Lower Bound

Difficulty: **Easy**Accuracy: **85.51%**Submissions: **333+**Points: **2**

Given an array **A** and **K** queries.

Each query is a number **X.** Print value of the minimum number in **A** that is greater than or equal to **X.**

Input

- First line contains an integer **T**, number of testcases
- First line of each testcase contains two integers **N, K** size of **A** and number of queries
- Second line of each testcase contains **N** space separated integers denoting elements of **A**
- In next **K** lines, each lines contains an integer **X**

Output

- For each number **X** in a query, print **minimum number** in **A** that is greater than or equal to **X**
- If there is no such number then print **-1**

Constraints

- **1<=T<=20**
- **1<=N,K<=10^5**
- **1<=A[i], X <= 10^9**

Example 1

Input

```
1

5 4

1 3 7 1 2

2

5

4

9
```

Output

```
2

7

7

-1
```

Explanation

For first query **2** is the smallest number greater or equal to **2**, in second and third query **7** is the smallest number that is greater or equal to **5** and **4.**

In 4th query answer will be **-1** as there is no number in **A** which is greater or equal to **9**

Upperbound

Difficulty: **Easy**Accuracy: **79.61%**Submissions: **329+**Points: **2**

Given an array **A** and **K** queries.

Each query is a number **X.** Print value of the minimum number in **A** that is **not** less than or equal to **X.**

Input

- First line contains an integer **T**, number of testcases
- First line of each testcase contains two integers **N, K** size of **A** and number of queries
- Second line of each testcase contains **N** space separated integers denoting elements of **A**
- In next **K** lines, each lines contains an integer **X**

Output

- For each number **X** in a query, print **minimum number** in **A** that is **not** less than or equal to **X**
- If there is no such number then print **-1**

Constraints

- **1<=T<=20**
- **1<=N,K<=10^5**
- **1<=A[i], X <= 10^9**

Example 1

Input

```
1
```

```
5 4

1 3 7 1 2

1

5

7

2
```

Output

```
2

7

-1

3
```

Explanation

For first query **2** is the smallest number in **A** that is not less than or equal to **1**, in second query **7** is the smallest number that is not less than or equal to **5.**

In 3rd query, there is no number in **A** which is greater than **7** , so answer is **-1**

In 4th query, **3** is the smallest number in **A** which is not less than or equal to **2**

Equal Ranges

Difficulty: **Easy**Accuracy: **84.27%**Submissions: **318+**Points: **2**

Given an array **A** and **K** queries.

Each query consists of two numbers **L** and **R .** Print count of numbers in **A** which are greater than or equal to **L** and less than or equal to **R.** More formally count distinct **i** such that **L<=A[i]<=R**

Input

- First line contains an integer **T**, number of testcases
- First line of each testcase contains two integers **N, K** size of **A** and number of queries
- Second line of each testcase contains **N** space separated integers denoting elements of **A**
- In next **K** lines, each lines contains two integers **L** and **R**

Output

- For each pair **L** and **R** in a query, print **count of A[i]** such that **L<=A[i]<=R**

Constraints

- **1<=T<=20**
- **1<=N,K<=10^5**
- **1<=A[i], L, R <= 10^9**

Example 1

Input

```
1
5 2
1 3 7 1 2
1 4
9 30
```

Output

```
4
0
```

Explanation

For first query answer is **4** as there are **4** numbers **1, 1, 2** and **3** between **1** and **4**

In second query answer is **0** as no number in **A** is in the range **[9,30]**

Equal Ranges II

Difficulty: **Hard**Accuracy: **56.7%**Submissions: **439+**Points: **8**

Given an array **A** consisting of **N** integers. You have to answer **Q** queries. Each query consists four space separated integers **L , R , X, Y**.
For a query **L R X Y**, count **A[i]** such that **L<=i<=R** and **X<=A[i]<=Y.**

Input

- First line of input contains one integer **T** denoting number of independent test cases
- First line of each test case contains two space separated integers, **N** and **Q** , size of **A** and number of queries respectively.
- Second line of each test case contains **N** space separated integers denoting elements of **A**
- Each of the next **Q** lines contain four space separated integers **L, R, X** and **Y**, denoting the query.

Output

- For each query print the answer in a new line

Constraints

- **1<=T<=5**
- **1<=N,Q<=10^5**
- **1<=A[i]<=10^9**

- **1<=L<=R<=N**
- **1<=X<=Y<=10^9**

Sample Input

```
1
6 3
5 3 2 2 1 2
1 6 2 4
2 4 3 3
3 5 1 100
```

Sample Output

```
4
1
3
```

Explanation

First query is **1 6 2 4** i.e. find number of integers in range [2,4] which lie between index 1 and 6 inclusive. There are 4 such integers viz. 3,2,2,2
Second query is **2 4 3 3** i.e. find number of integers in range [3,3] which lie between index 2 and 4 inclusive. There is only one such integer A[2] = 3.

Frequency range query

Difficulty: **Medium**Accuracy: **85.16%**Submissions: **316+**Points: **4**

Given an array **A** and **K** queries.

Each query consists of three numbers **L, R** and **X.** Print frequency of **X** in range **L** and **R.** More formally count number of **i** in range **[L, R]** such that **A[i]** equals **X.**

Input

- First line contains an integer **T**, number of testcases
- First line of each testcase contains two integers **N, K** size of **A** and number of queries
- Second line of each testcase contains **N** space separated integers denoting elements of **A**
- In next **K** lines, each lines contains three integers **L, R** and **X**

Output

- For each query **L, R** and **X** in a query, print **frequency** of **X** in range **[L, R]** in **A**

Constraints

- **1<=T<=5**
- **1<=N,K<=10^5**
- **1<=L<=R <=N**
- **1<=A[i], X <= 10^5**

Example 1

Input

```
1
5 3
1 3 7 1 2
1 2 3
3 5 9
1 5 7
```

Output

```
1
0
1
```

Explanation

First query asks for occurence of **3** in range **A[1:2]** , as **A[2]** is **3** answer is **1**

In **2nd** query answer there is no **9** in **A[3:5]** therefore answer is **0**

For **3rd** query there is **1** occurence of **7** in **A[1:5]** so answer is **1**

Find the critical point

Difficulty: **Hard**Accuracy: **78.02%**Submissions: **333+**Points: **8**

Critical point of a function **F(x)** is such a value of **x** where **derivative of F(x)** is zero. We have **derivative** of a **bi-quadratic polynomial function** of which we want to find the critical point.

Interestingly, this **function** has **only one critical point.**

If you don't know anything about calculus, here is the problem statement for you

Given a **cubic polynomial $ax^3+ bx^2+cx+d$** which is guaranteed to have only **one root**. Find the root of the polynomial. More formally find value of **x** such that **$ax^3+ bx^2+cx+d = 0$.**

It is guaranteed that there exists exactly one such **x** for the given polynomial.

Input

- First line contains an integer **T**, number of testcases
- First line of each testcase contains four integers **a, b, c** and **d**

Output

- For each testcase print in a new line an integer **Y** such that **$aY^3+bY^2+cY+d = 0$**
- It is guaranteed that there is only one solution and it is an integer

Constraints

- **$1<=T<=10^4$**
- **$-10^5 <=a, b, c, d<=10^5$**

Example 1

Input

```
2
1 0 0 -1
2 -2 -3 -2
```

Output

```
1
2
```

Explanation

First polynomial is x^3-x and it is zero only when $x=1$, thus answer for first testcase is **1**

Second polynomial is $2x^3-2x^2-3x-2$ and it is zero only when $x=2$, therefore answer for second testcase is **2**

Allocate Minimum Number of Pages

Difficulty: **Medium** Accuracy: **69.17%** Submissions: **386+** Points: **4**

You are given **N** number of books. Every **ith** book has **Ai** number of pages. The books are arranged in ascending order.

You have to allocate contiguous books to **M** number of students. There can be many ways or permutations to do so. In each permutation, one of the M students will be allocated the maximum number of pages. Out of all these permutations, the task is to find that particular permutation in which the maximum number of pages allocated to a student is the minimum of those in all the other permutations and print this minimum value.

Each book will be allocated to exactly one student. Each student has to be allocated at least one book.

Note: Print **-1** if a valid assignment is not possible, and allotment should be in contiguous order (see the explanation for better understanding).

Input Format:

The first line of the input contains a single integer **T** ($1 \leq T \leq 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains a single integer ***N.***
- The following line contains ***N*** space separated integers denoting the array elements.
- The following line contain a single integer ***M*.**

Output Format:

For each test case print the required answer as described in the problem statement.

Constraints:

$1 <= N, T <= 10^5$

$1 <= A [i] <= 10^6$

$1 <= M <= 10^5$

Sum of N over all test cases will not exceed 2 x 10^5

Example 1:

Input:

2

4

12 34 67 90

2

3

15 17 20

2

Output:
113
32

Explanation:
Test Case 1 -
Allocation can be done in following ways:
{12} and {34, 67, 90} -> Maximum Pages = 191
{12, 34} and {67, 90} -> Maximum Pages = 157
{12, 34, 67} and {90} -> Maximum Pages =113.
Therefore, the minimum of these cases is 113, which is selected
as the output.

Test Case 2 -
Allocation is done as {15,17} and {20}

Who's Above Me

Difficulty: **Medium**Accuracy: **83.04%**Submissions: **312+**Points: **4**

Given an array **A** of consisting of **N** integers .Find largest possible **A[j]<=2*A[i]** such that i!=j for every 1<=i<=n . Find maximum possble **A[j]** for each **A[i]** , if there doesn't exist any **A[j]** , the answer is **-1** for that given **A[i]**. Print an array of integers you have calculated for the array **A**.

Input Format:

The first line of the input contains a single integer T denoting the number of test cases.

The first line of every test case consists of integer **N** denoting the number of elements in **A**.

The second line of every test case consists of **N** space separated integers which are the elememts of array **A**.

Output Format:

Print **N** space separated integers. The answer of each testcase must be printed on newline.

Constraints:-

$1<=T<=10^5$

$1<=N<=10^5$

$1<=A[i]<=10^6$

The sum of N over all test cases doesn't exceed 10^6

Example:
Input:
2
8
95 73 36 83 79 74 27 5
3
54 56 96

Output:
83 95 27 95 95 95 36 -1
96 96 56

Explaination:
1st test case: -
The largest possible number for 95 is 190. And the largest
value less than equal to 190 available is 83.
For 5 the largest possible number is 10. But there is isn't any

number other than 5 available so answer is -1.

Aggresive Cows

Difficulty: **Medium**Accuracy: **50.47%**Submissions: **541+**Points: **4**

Farmer John has built a new long barn, with N stalls. The stalls are located along a straight line at positions x1,...,xN .

His C cows don't like this barn layout and become aggressive towards each other once put into a stall. To prevent the cows from hurting each other, FJ wants to assign the cows to the stalls, such that the minimum distance between any two of them is as large as possible. What is the largest minimum distance?

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integer ***N.***
- For Next N lines, each line contain $\mathbf{A_1,A_2,...,A_n}$**.**

Output Format:

For each test case, print the answer to the problem followed by a newline character.

Constraints:

$2 <= \mathbf{N} <= 10^5$

$1 <= \mathbf{C_i} <= 10^5$

$0 <= X_i <= 1,000,000,000$

Example:

Input:

1

```
5 3
1 2 8 4 9

Output:
3
```

Farthest number

Difficulty: **Medium**Accuracy: **94.74%**Submissions: **292+**Points: **4**

Given an array **Arr[]** of size **N**. For every element in the array, the task is to find the index of the farthest element in the array to the right which is smaller than the current element. If no such number exists then print **-1**.

Note: 0 based indexing.

Input Format:

The first line consists of a single integer **T** denoting the number of test cases. The description of each test case is as follows -

- The first line consist of a single integer N denoting the size of the array.
- Second line consists of N space separated integers, where the ith element corresponds to Arr[i].

Output Format:

For each test case print N space separated integers, where the ith integer corresponds to the answer of Arr[i].

Constraints:

$1 \le T \le 10^4$

$1 \le N \le 10^5$

$0 \le Arr[i] \le 10^9$

Example:

```
Input:
2
```

5
3 1 5 2 4
5
1 2 3 4 0

Output:
3 -1 4 -1 -1
4 4 4 4 -1

Explanation:
Test case 1 - Arr[3] is the farthest smallest element to the right of Arr[0]. Arr[4] is the farthest smallest element to the right of Arr[2]. And for the rest of the elements, there is no smaller element to their right.
Test case 2 - Arr[4] is the farthest samlles element for first 4 elements, and there exists no smaller element in the right of Arr[4], hence Ans[4] = -1.

Changed Subsegment

Difficulty: **Medium**Accuracy: **94.55%**Submissions: **258+**Points: **4**

A cat is situated on an infinite 2D plane, initially in the cell **(0, 0)**. It can go in the following four directions:

1. U move from **(x, y)** to **(x, y + 1)**
2. D move from **(x, y)** to **(x, y – 1)**
3. L move from **(x, y)** to **(x – 1, y)**
4. R move from **(x, y)** to **(x + 1, y)**

You have got a sequence of **N** operations. You have to modify this sequence so after performing it the cat will **end** up in **(x, y)**.

You have to change the sequence so the length of changed subsegment is **minimum** possible. The length can be calculated as follows: **maxID – minID + 1**, where

maxID is the maximum index of a changed operation, and **minID** is the minimum index of a changed operation.

For example, if RRRRRRR changes to RLRRLRL, then the operations with indices 2, 5 and 7 are changed, so the length of changed subsegment is 7 − 2 + 1 =6.

Another example: if DDDD changes to DDRD, then the length of changed subsegment is 1.

If there are no changes, then the length of changed subsegment is 0. Changing an operation means replacing it with some operation (possibly the same); You can't insert new operations into the sequence or remove them.

Calculate the **minimum length of subsegment** that needs to be changed so that the cat will go from **(0, 0)** to **(x, y)**, or tell if it's impossible.

Input format:

The first line should contain **T**, denoting the number of test cases.

The first line of each test case should contain an integer **n**, denoting the number of operations.

The second line of each test case should contain a string of **n** characters, denoting the sequence of operations.

The third line of each test case should contain two integers **x, y**, denoting the coordinates of the cell where the cat should end it's path.

Output format:

Print an integer, denoting the **minimum possible length of subsegment** that can be changed so the resulting sequence of operations moves the cat from (0, 0) to (x, y). If this change is impossible, print **–1**.

The output of each test case must be printed in a separate line.

Example:

Input:

```
3
5
RURUU
-2 3
4
RULR
1 1
3
UUU
100 100
```

Output:

```
3
0
-1
```

Explanation:

1st test case: Sequence can be changed to LULUU.
So the length of the changed subsegment is
3 − 1 + 1 = 3.
2nd test case: Sequence already leads the cat
to (x, y), so the length of the changed
subsegment is 0.
3rd test case: Cat can't end his path
in the cell (x, y).

Constraints:

- $1 <= T <= 10^3$

- $1 <= N <= 2 * 10^5$
- $-10^9 <= x, y <= 10^9$
- Each character in the sequence of operations is either U, D, L or R.

Rearrange Array

Difficulty: **Medium**Accuracy: **89.63%**Submissions: **280+**Points: **4**

Given two arrays **A** and **B**

You are allowed to swap adjacent elements of **A** in one move, find the minimum number of moves required to convert **A** to **B**.

It is guaranteed that **A** can always be converted into **B** in finite moves.

Input

- First line contains an integer **T**, denoting number of testcases
- First line of each testcase is an integer **N** size of **A** and **B**
- Second line of each testcase contains **N** space separated integers denoting elements of **A**
- Third line of each testcase contains **N** space separated integers describing **B**

Output

- Print a single integer denoting minimum number of adjacent swaps to convert **A** into **B**

Constraints

- **$1<=T<=10$**
- **$1<=N<=10^5$**
- **$1<=A[i],B[i]<=10^9$**

Example

Input

```
2
4
1 2 3 2
3 2 1 2
3
1 2 3
2 3 1
```

Output

```
3
2
```

Explanation

In first test, we can do following swaps, (numbers to be swapped are in bold at every step)

[1, **2**, **3**, 2] -> [**1**, **3**, 2, 2] -> [3, **1**, **2**, 2] -> [3, 2, 1, 2] . Therefore it took only **3** steps to convert **A** into **B.** It can be shown that number of minimum steps is **3**

In second case, we can proceed by swapping 2 and 1 and then 1 and 3 i.e. [**1**, **2**, 3] -> [2, **1**, **3**] -> [2, 3, 1]. Thus, conversion can be accomplished in **2** swaps. It is obvious in this testcase that lesser answer than **2** is not possible.

Cut or Add

Difficulty: **Medium**Accuracy: **86.72%**Submissions: **293+**Points: **4**

You have **N** magic sticks. You can either increase or decrease the length of a magic stick but everytime you increase or decrease one unit of $\mathbf{i_{th}}$ it will cost you **cost[i]** units.

Length of $\mathbf{i^{th}}$ stick is **L[i]** and cost of adding or cutting one unit from $\mathbf{i_{th}}$ stick is **cost[i].**

You want to make length of all sticks equal. Calculate the minimum cost of making all sticks of equal length.

Input

- First line contains **T**, number of testcases to follow
- First line of each testcase contains **N** number of sticks
- Second line of each testcase contains **N** space separated integers, $\mathbf{i_{th}}$ of which is **L[i]**
- Third line of each testcase contains **N** space separated integers $\mathbf{i_{th}}$ of which is **cost[i]**

Output

- For each testcase print in a new line an integer **X** representing minimum cost to make length of all sticks equal

Constraints

- **1<=T<=10**
- **1<=L[i]<=10^5**
- **1<=cost[i]<=10^5**
- **1<=N<=10^5**

Example

Input

```
2
4
2 3 5 6
1 2 3 4
5
50 12 10 2 8
40 33 0 10 7
```

Output

11
1648

Explanation

In first testcase we can achieve a cost of **11** by making all sticks equal to **5.**

11 = |5-2|*1+|5-3|*2+|5-5|*3+|5-5|*4

In second testcase minimum cost that can be achieved is **1648** by making every stick equal to **12**

Polynomial Multiplication

Difficulty: **Hard**Accuracy: **92.31%**Submissions: **273+**Points: **8**

Given two polynomials **A** and **B** both of degree **N.** Find the product of **A** and **B**

Representation:

A polynomial **P** of degree **N** is represented as a sequence of **N+1** integers **P[0], P[1], ..., P[N]**, where **P[i]** is coefficient of x^i

As value of coefficients of product polynomial can be large, output your answer modulo **998244353**

NOTE : It is guaranteed that **N+1** is a power of **2**, i.e. **N+1 =** 2^a for **a** in **[1,17]**

Input

- First line contains an integer **T**, denoting number of testcases two follow
- First line of each testcase contains an integer **N,** denoting degree of both the polynomials
- Second line of each testcase contains **N+1** space separated integers representing polynomial **B**

- Third line of each testcase contains **N+1** space separated integers representing polynomial **B**

Output

- For each testcase print in a new line **2*N-2** integers coefficients of the polynomial which is product of polynomials **A** and **B.**

Constraints

- **1<=T<=5**
- **1<=N<=2^{17} (N+1 = 2^a** for **a** in **[1,17])**
- **0<=A[i],B[i]<=10^4**

Sample Input

```
2
3
1 2 0 2
3 1 1 0
3
9 10 5 8
7 5 7 5
```

Sample Output

```
3 7 3 8 2 2 0

63 115 148 196 125 81 40
```

Explanation

In first test, we have **A = 1 + 2x + 0x² + 2x³** and **B = 3 + x + x² + 0x³** and product of **A** and **B** is

A.B = 3 + 7x + 3x² + 8x³ + 2x⁴ + 2x⁵ + 0x⁶

Optimal Meeting Point

Difficulty: **Hard**Accuracy: **93.63%**Submissions: **266+**Points: **8**

You are given the positions and preparation times of n people living on a coordinate line. Each person wants to meet at a common position x_0.

The time taken by the i-th person to reach the meeting place is determined by their distance from the meeting place ($|x_i - x_0|$) and their individual preparation time (t_i). The total time for an individual is calculated as $t_i + |x_i - x_0|$.

Your task is to find the optimal meeting position x_0 that minimizes the total time for all n people to gather at the meeting place.

Input Format:

The first line contains an integer T denoting the number of test cases. Each test case contains an integer n representing the number of people. The next line contains n space separated integers representing the positions of the people and next line contains n space separated integers denoting the time people need to get dressed.

Output Format:

For each test case, print the optimal meeting point. The relative error must not exceed 10^{-6}.

Constraints:

$1<=T<=10^3$

$1<=n<=10^5$

$0<=x_i<=10^8$

$0<=t_i<=10^8$

Example:

Input:

3
1
0
3
2
3 1
0 0
3
1 2 3
4 1 2

Output:

0.0
2.0
1.0

Explanation:

In the 1st test case there is one person, so it is efficient to choose his or her position for the meeting place. Then he or she will get to it in 3 minutes, that he or she need to get dressed.

In the 2nd test case there are 22people who don't need time to get dressed. Each of them needs one minute to get to position 2.

In the 3rd test case, total time taken for all to gather will be {4, 5, 6} for positions {1, 2, 3} respectively. Position 1 requires minimum time to reach.

Stacks and Queues – CP

Stack

Sum of XOR of all pairs

Difficulty: **Medium**Accuracy: **68.96%**Submissions: **423+**Points: **4**

Given an array **A** of **N** integers, find the sum of xor of all pairs of numbers in the array.

Input :

The first line consists of **T**, the number of testcases. Each test case is as follows:-

- The first line consists of **N**, the size of the array.
- The second line consists of **N** space-separated integers.

Output :

For each test case output in a single line the **sum** of XOR of all pairs.

Constraints :

$1 \leq \mathbf{T} \leq 1000$
$2 \leq \mathbf{N} \leq 10^5$
$1 \leq \mathbf{A[i]} \leq 10^5$

The sum of **N** over all testcases doesn't exceed $\mathbf{10^6}$

Example :

Input :
2
3

7 3 5
4
5 9 7 6
Output :
12
47
Example :
Testcase 1:- All possible pairs and their Xor
Value: (3 ^ 5 = 6) + (7 ^ 3 = 4)
+ (7 ^ 5 = 2) = 6 + 4 + 2 = 12

Geek and The Followers

Difficulty: **Medium**Accuracy: **89.95%**Submissions: **278+**Points: **4**

There are N geeks are sitting in a row. You have been given the scores of all the geeks.

Each geek follows one geek who sits on the next and has a higher score.

You need to find the followers of each geek.

Note: You need to print the index of followers in 1-index format.

Input:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains one number **N** (number of geeks).
- The second line of each test case contains scores of all geeks.

Output:

- For each test case print **N** lines. **i-th** line contains the number of followers of **i-th** geeks and the index of all followers.

Constraints:

- 1 <= **T** <= 100

- $1 <= \mathbf{N} <= 10^5$
- $0 <= \mathbf{Score[i]} <= 10^5$

Example:

Input:

```
2
4
7 8 1 4
7
5 2 6 3 1 0 4
```

Output:

```
0
1 1
0
1 3

0
0
2 1 2
0
0
0
3 4 5 6
```

Explanation

```
In first testcase,
first geek follow second geek,
third geek follow fourth geek,
So, 1st and 3rd geek has 0 followers,
2nd and 4th geek has 1 follower.
```

Valid Parenthesis

Difficulty: **Easy**Accuracy: **86.3%**Submissions: **322+**Points: **2**

Given an expression string **X**. Examine whether the pairs and the orders of “{“,”}”,”(“,”)”,”[“,”]” are correct in exp.

For example, you have to print "Valid" for exp = “[()]{}{[()()]()}” and 'Invalid' for exp = “[()]”.

Input:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains one expression **X**.

Output:

- Print **T** lines, the i-th should be the answer for the i-th test case.

Constraints:

- 1 <= **T** <= 100
- 0 <= | **X** | <= 32000

Example:

Input:

```
2
[()]{}{[()()]()}
[()]
```

Output:

```
Valid
Invalid
```

Palindromic Folder Path

Difficulty: **Medium**Accuracy: **89.84%**Submissions: **293+**Points: **4**

Given the path of a folder. Examine whether the path is palindromic or not. A path is called palindromic, if we reverse the path we get the same path.

For example, you have to print "**Yes**" for path "**/root/geek/root**" and "**No**" for path "**/home/geek/stack**".

Input:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains one **Path**.

Output:

- Print **T** lines, the i-th should be the answer for the i-th test case.

Constraints:

- 1 <= **T** <= 10^3
- 0 <= **| Path |** <= 10^6

Example:

Input:

```
2
/root/geek/root
/home/geek/stack
```

Output:

```
Yes
No
```

Battle of Monsters

Difficulty: **Medium**Accuracy: **87.96%**Submissions: **283+**Points: **4**

You have been given the power of N monsters, and everyone is moving on giving directions.

If two monsters face each other, then they attack each other and the monster with more power will kill the other monster. If both have the same power then both will die.

After the battle is over you need to find out the number of living monsters and the power of each monster.

Note The movement speed of all monsters are same.

Input:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains one Integer number of monsters.
- The next line contains **N** integers (power of each monster)
- Next line containing **N** characters (**'R'** for moving right and **'L'** for moving left).

Output:

- Print **T** lines, the i-th should be the answer for the i-th test case.
- The first line of i-th test case contains one integer **K** (number of living monsters after the battle end)
- The second line of i-th test case contains the power of **K** monsters.

Constraints:

- $1 <= \mathbf{T} <= 10^2$
- $1 <= \mathbf{N} <= 10^5$

Example:

Input:

```
3
5
5 6 2 3 4
R L R R L
4
5 2 2 5
R R L L
3
```

```
5 1 8
R L L
```

Output:

```
2
6 4
0
1
8
```

Explanation:

In first test case 2-nd monstart will kill 1-st monsters and 5-th will kill 3-rd and 4-th monsters.

Queue

Geeks and Problem Ladder

Difficulty: **Easy**Accuracy: **83.06%**Submissions: **271+**Points: **2**

You have been given **M** problems of given difficulty on a ladder and **N** Geek with problem-solving strength on a circular row.

One geek can solve only one problem with less or equal difficult problems than their strength.

On a problem ladder, to get access of the i_{th} problem on the ladder, you must solve the $(i-1)_{th}$ problem.

First, the geek in front of the row will try the problem at the top of the ladder.

If he solved it, the problem and geek will be removed from the list.

Otherwise, the next geek will try this problem and the current geek will join at the end of the row.

You need to find the number of solved problems.

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains two integers **N** and **M.**
- The next line contains **N** integers (strength of each geek in range 1 to 10^5)
- Next line containing **M** integers (difficulty of each problem in range 1 to 10^5).

Output format:

- Print **T** lines, the i_{th} should be the answer for the i_{th} test case.

Constraints:

- 1 <= **T** <= 10^2
- 1 <= **N, M** <= 10^5

Example Input:

```
2
3 3
5 6 3
6 3 2
5 6
10 5 6 8 4
9 10 5 9 2 5
```

Example Output:

```
3
1
```

Explanation:

In the first test case,

1st step: front geek with strength 5 is unable to solve the first problem of difficulty 6

after 1st step, row : [6 3 5], ladder : [6 3 2]

2nd step: front geek will solve the top problem

after 2nd step, row : [3 5 6], ladder : [3 2]

3rd step: front geek will solve the top problem

after 3rd step, row : [5 6 3], ladder : [2]

4th step: front geek will solve the top problem

after 4th step, row : [6 3 5], ladder : [ ]

Garden Visit

Difficulty: **Medium**Accuracy: **81.96%**Submissions: **305+**Points: **4**

You are given **N** Garden, connected circularly (first garden is next to $\mathbf{N_{th}}$ garden) and $\mathbf{i_{th}}$ garden contains **arr[i]** plants.

On $\mathbf{i_{th}}$ day you will visit continuous **K** nurseries, starting from $\mathbf{i_{th}}$ garden.

For each day you need to print an index of the last visited Garden with the maximum number of plants on that day.

Note: You need to print the index of the garden in one index format for N days.

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains two integers **N** and **K.**
- The next line contains **N** integers.

Output format:

- Print **T** lines, the i_{th} should be the answer for the i_{th} test case.

Constraints:

- $1 <= \mathbf{T} <= 10^2$
- $1 <= \mathbf{N, K} <= 10^5$

- $1 <= \mathbf{arr[i]} <= 10^5$

Example Input:

```
2
5 3
1 2 4 4 6
6 3
3 3 3 3 3 3
```

Example Output:

```
3 4 5 5 5
3 4 5 6 1 2
```

Explanation:

For the first test case,

On 1_{st} day, you will visit gardens with plant : [1 2 4], max = 4, So ans = 3

On 2_{nd} day, you will visit gardens with plant : [2 4 4], max = 4, So ans = 4

On 3_{rd} day, you will visit gardens with plant: [4 4 6], max = 6, So ans = 5

On 4_{th} day, you will visit gardens with plant : [4 6 1], max = 6, So ans = 5

On 5_{th} day, you will visit gardens with plant : [6 1 2], max = 6, So ans = 5

Design Queue

Difficulty: **Easy**Accuracy: **79.91%**Submissions: **314+**Points: **2**

Given Q number of queries, you need to perform operations on queue(**FIFO**). Operations to be performed are listed below:

Query Type	**:**	**Operation**
1	:	EnQueue (insert from rear)
2	:	DeQueue (remove from front)
3	:	Front(To print front)

Input :

First line of input contains number of testcases **T**. For each testcase, first line contains **Q** (Number of Queries). Next **Q** lines contains the queries of any of the three type (**1, 2, 3**) with element to be inserted (during the EnQueue operation).

Output :

For each testcase, if the operation is to **EnQueue(1)** then insert the elments, if the operations is to **DeQueue(2)** then remove the element and if it is **Front(3)**, then print the element at the front of Queue(If Queue is empty, print "**-1**", without quotes).

Constraints :

$1 <= T <= 100$

$1 <= q_i <= 10^6$

$1<=Q<=200$

Example :

Input :

1
6
3
1 5
1 6
1 7
2
3

Output :

-1
6

Explanation :

First query is to print Front, but since Queue is empty, so we print -1. Next three

queries are to EnQueue 5, 6, and 7, so we insert them in Queue. Next query is DeQueue, so we remove 5 from Queue. Final query is to print the Front, so 6 is there at the Front and thus printed.

Priority Queue

Geek and Movies

Difficulty: **Easy**Accuracy: **84.72%**Submissions: **313+**Points: **2**

Theatres open after an extended shutdown due to coronavirus pandemic. As Geek is very fond of movies, he decided to watch a movie in the theatre along with his family. After reaching the theatre, he saw that many people have lined up for the tickets as the theatres opened after a huge gap. He knows that there are **N** rows in the theatre, each having a different seat capacity. **The price of the ticket is equal to the number of vacant seats in a particular row.** For instance, if there are X vacant seats in a row, the ticket price (if the person will sit in this row) is X rupees. Now, each person standing in the line will get a ticket one by one.
You are provided with the seating capacities of different rows. Would you please help the theatre's staff distribute the tickets in such a way so that they can get the maximum amount of money by selling them?

Input:

The first line contains **T**, representing the number of test cases.
The first line of each test case contains two integers **N** and **M**, representing the number of rows and the number of people standing in a line, respectively.
The second line of each test case
contains **N** integers **X[1],X[2],X[3]....X[N]** where **X[i]** represents the number of empty seats initially in the ith row.

Output:

Print the maximum amount of money obtained by selling the tickets. The output of each test case should be printed in a separate line.

Constraints:

$1 <= T <= 10$

$1 <= N <= 10^5$

$1 <= M <= 10^5$

$1 <= X[i] <= 10^5$

Sum of **X[i]** for all **1 <= i <= N** will always be greater than **M**.

Example:

Input:

```
3
3 4
2 5 6
5 8
2 6 10 12 11
10 20
2 2 4 4 3 2 3 4 1 1
```

Output:

```
20
82
47
```

Explanation:

```
In the first test case the number of rows is 3
and the number of people in the line is 4.
Since the maximum number of empty seats is 6
in the 3rd row. Therefore the first person will
```

buy the ticket for the 3rd row. Now the number of empty seats(also the maximum) in the 2nd and 3rd row are equal which is 5. Therefore the second person will buy the ticket either from the 2nd row or 3rd row. Suppose that the second person buy the ticket from the 2nd row. Now the third person will buy the ticket from the 3rd row as it has the maximum number of empty seats till now which is 5. Now again the fourth person can buy the ticket either from the 2nd or 3rd row as they have the maximum number of empty seats. Therefore the total amount of money earned after selling the tickets: 6 + 5 + 5 + 4 = 20.

Diwali Sale

Difficulty: **Medium**Accuracy: **78.04%**Submissions: **332+**Points: **4**

Geeks classes are organizing a Diwali sale in which they are giving 75% off on their online courses. Seeing this great opportunity, Chinmoy decides to purchase as many courses as he can. Each course is reprsented by a pair **[duration_i, expiryDay_i]** indicating that the ith course should be taken continuously for **duration_i** days and must be finished before or on **expiryDay_i**. Suppose that Chinmoy purchases **N** courses and starts learning from **1st day** and can't learn the **two or more courses at once**, find out the **maximum** number of courses he can learn.

Input:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains **N**, representing the number of courses.

The following **N** lines of each test case consist of a pair **[duration_i,expiryDay_i]**, representing the **ith** course.

Output:

Print the maximum number of courses that Chinmoy can learn. The output of each test case must be printed in a separate line.

Constraints:

1 <= T <= 100

1 <= N <= 10000

1 <= duration_i, expiryDay_i <= 10000

Example:

Input:

```
3
4
2 3
3 17
11 13
10 3
2
5 3
6 2
1
9 10
```

Output:

```
3
0
1
```

Explanation:

1st test case: Chinmoy should start with 1st course (considering 1 based indexing). Then from 3rd day he should start learning 3rd course. From 14th day he should start learning 2nd course. He won't be able to learn the 4th course.

Geek and his trip

Difficulty: **Medium** Accuracy: **93.04%** Submissions: **275+** Points: **4**

Geek is going on a trip with his friends by car. They want to reach their destination location as soon as possible. Therefore they will take the minimum number of stops while traveling. They will only stop to fill up the fuel in the car at different petrol pumps. The petrol pumps are represented by an array where **petrolPump[i]=[location_i,fuel_i]**, indicates that the ith petrol pump is location_i distance from the initial location and has fuel_i liters of fuel. You can assume that the car fuel tank has an **infinite** amount of capacity, and it consumes **1 liter for traveling 1 km.** Also the car can only move towards the destination location. It can't go backward.
You are provided with the initial fuel in the car, the destination location and the petrol pumps array(elements are in the **increasing** order in accordance with the location_i). Can you help Geek and his friends in finding the minimum number of stops?

Note: If the car reaches the petrol pump with 0 fuel, then the car can fill up with the fuel at the petrol pump. Also, if the car reaches the destination with 0 fuel, it will still be considered to have arrived.

Input:

The first line contains **T,** representing the number of test cases.

The first line of each test case contains **N**, representing the size of **petrolPump** array.

The following **N** lines contains the elements of the petrolPump array.

The next line of each test case contains **destination,** representing the distance of the destination location from the initial position.

The next line of each test case contains **initial_fuel,** representing the initial fuel present in the car.

Output:

Print the **minimum** number of stops to reach the destination location. Print **-1**, if you are **unable** to reach the final destination. The output of each test case must be printed in a separate line.

Constraints:

$1 <= T <= 100$

$1 <= N, destination, initial_fuel <= 10^5$

$1 <= fuel_i <= 10^5$

$1 <= location_i <= destination$

Example:

Input:

```
2
3
5 20
10 30
30 20
50
10
1
```

```
20 100
30
1
```

Output:

```
2
-1
```

Explanation:

```
1st test case: It contains 3 petrol pumps. The
destination location is 50km away from the
initial location. The initial fuel in the car is
10 liter. The car needs to stop at petrol pumps
located at 10km and 30km to reach the destination at
minimal stops.
2nd test case: It contains 1 petrol pump which is
located at 20 km. The destination location is 30km
away from the initial location. The initial fuel in the
car is 1 liter. It is not possible to reach the car at
the destination location.
```

Card Game

Difficulty: **Medium** Accuracy: **95.96%** Submissions: **271+** Points: **4**

Alexa and Siri are playing a game where they have to choose one card, alternatively with Alexa starting first. On choosing a card, the player will get points equal to the value written on the card, and then that card will be **removed** from the deck.

You are given **two** lists each of length **N**, **alexaValues** and **siriValues**.

Each **alexaValues[i]** and **siriValues[i]** represents the value written by Alexa and Siri on the **ith** card.

The winner of the game is the one who receives the maximum points at the end of the game. If both receive equal points, then the game will be considered

a **draw**. Both the players will play **optimally**, and both **know** the other's player's values.
Your task is to determine the **winner** of the game.

Input:
The first line contains **T**, representing the number of test cases.
The first line of each test case contains **N**, representing the length of lists alexaValues and siriValues.
The second line of each test case contains **N** integers; **ith** integer represents the **alexaValues[i]**.
The third line of each test case contains **N** integers; **ith** integer represents the **siriValues[i]**.

Output:
You have to print:
"Alexa" if Alexa wins
"Siri" if Siri wins
"Draw" if the game results in a draw.
The output of each test case should be printed in a **separate line**.

Constraints:

- $1 <= T <= 100$
- $1 <= N <= 10^5$
- $1 <= alexaValues[i], siriValues[i] <= 10^5$

Example:

Input:

```
3
2
2 5
7 2
2
```

```
3 8
9 4
3
2 5 6
4 7 5
```

Output:

```
Draw
Siri
Alexa
```

Explanation:

1st test case: If Alexa chooses 1st card(considering 1-based indexing), then he will get 2 points, and then Siri will only be left with 2nd card, and she will also get 2 points. This case results in a draw. Considering another case where Alexa chooses 2nd card and then Siri chooses 1st card. This will also result in a draw as here also both will get 2 points.
2nd test case: If Alexa chooses 1st card(considering 1-based indexing) and then Siri chooses 2nd card. In this case, Alexa will get 2 points, and Siri will get 4 points. In this case, Siri wins. Considering another case when Alexa chooses 2nd card and then Siri chooses 1st card. Here Alexa will get 8 points, and Siri will get 9 points. In this case, also Siri wins.

Card Game II

Difficulty: **Medium**Accuracy: **79.69%**Submissions: **345+**Points: **4**

Alexa and Siri are **alternatively** playing a game, with Alexa beginning **first**.

At first, there are **N cards** in a deck. On every player's turn, that player takes action comprising of **eliminating** any **non-zero square number** of cards from a deck.

The player will lose the game if they aren't able to make the next move.

Your task is to determine the winner of the game, assuming that both the players will play **optimally**.

Input:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains **N**, which **initially** represents the number of cards in the deck.

Output:

You have to print:

"Alexa" if Alexa wins

"Siri" if Siri wins

Constraints:

- 1 <= T <= 100
- 1 <= N <= 10^5

Example:

Input:

```
3
4
5
1
```

Output:

Alexa
Siri
Alexa

Explanation:

1st test case: Since N=4 is already a perfect square.
Therefore Alexa can win by removing the 4 cards from
the deck.
2nd test case: Siri can win if she plays optimally.
Consider a case when Alexa starts the game by removing
only 1 card. Then the remaining cards will be 4, and
since it is a perfect square, Siri can remove all the
cards and win the game. Consider another case when
Alexa starts the game by eliminating 4 cards. Then
the remaining cards will be 1, and Siri can win by
removing this card from the deck as, after that,
Alexa doesn't have any moves left.
3rd test case: Alexa can remove 1 card winning the
game because Siri doesn't have any moves left.

Geeks and Primes

Difficulty: **Hard**Accuracy: **91.93%**Submissions: **286+**Points: **8**

Geeks recently read about prime numbers in his mathematics class. Now he is solving a problem related to them in which he is given a **sorted** list **arr** in **strictly increasing** order containing only **prime** numbers. For every **p<q** in the list, the **valid** fraction is p/q. He has to find the **Kth smallest valid** fraction. Can you help Geeks in solving this task?

Input:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains **N**, representing the size of the list.

The following line of each test case contains **N integers**, representing the sorted list containing only prime numbers in strictly increasing order.

The following line contains **K**.

Output:

For each test case, first print the **numerator** and then
the **fraction's denominator** with a **space** between them.

Contraints:

- 1 <= T <= 100
- 2 <= N <= 1000
- 1 <= arr[i] <= 1000000(only prime numbers in this range)
- 1 <= K <= (N*(N-1))/2

Example:

Input:

```
2
2
5 13
1
5
2 3 5 7 11
4
```

Output:

```
5 13
2 5
```

Explanation:

```
2nd test case: The valid fractions in sorted order are:
2/11, 3/11, 2/7, 2/5, 3/7, 5/11...
```

The fourth fraction is 2/5.

Trees

Inorder Traversal

Difficulty: **Basic**Accuracy: **67.15%**Submissions: **156K+**Points: **1**

Given a Binary Tree, find the In-Order Traversal of it.

Example 1:

Input:

```
    1
  / \
 3   2
```

Output: 3 1 2

Example 2:

Input:

```
     10
   /    \
  20     30
 /  \   /
40   60 50
```

Output: 40 20 60 10 50 30

Your Task:

You don't need to read input or print anything. Your task is to complete the function **inOrder()** that takes root node of the tree as input and returns a list containing the In-Order Traversal of the given Binary Tree.

Expected Time Complexity: O(N).

Expected Auxiliary Space: O(N).

Constraints:

1 <= Number of nodes <= 10^5

0 <= Data of a node <= 10^5

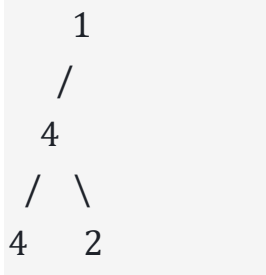
Preorder Traversal

Difficulty: **Basic**Accuracy: **62.73%**Submissions: **161K+**Points: **1**

Given a binary tree, find its preorder traversal.

Example 1:

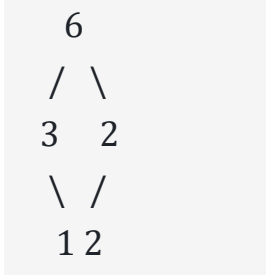
Input:



Output: 1 4 4 2

Example 2:

Input:



Output: 6 3 1 2 2

Your Task:

You just have to complete the function **preorder()** which takes the root node of the tree as input and returns an array containing the preorder traversal of the tree.

Expected Time Complexity: O(N).

Expected Auxiliary Space: O(N).

Constraints:

1 <= Number of nodes <= 10^4

0 <= Data of a node <= 10^5

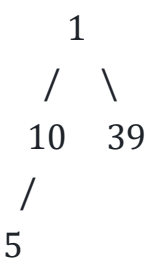
Postorder Traversal

Difficulty: **Basic**Accuracy: **74.96%**Submissions: **115K+**Points: **1**

Given a binary tree, find the Postorder Traversal of it.

For Example, the postorder traversal of the following tree is:

5 10 39 1



Example 1:

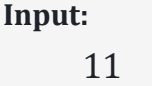
Input:



Output: 11 13 10 8 19

Example 2:

Input:



```
    /
   15
  /
 7
```

Output: 7 15 11

Your Task:

You don't need to read input or print anything. Your task is to complete the function **postOrder()** that takes **root node** as input and returns an array containing the postorder traversal of the given Binary Tree.

Expected Time Complexity: O(N).

Expected Auxiliary Space: O(N).

Constraints:

1 <= Number of nodes <= 10^5

0 <= Data of a node <= 10^6

Level order traversal

Difficulty: **Easy**Accuracy: **70.31%**Submissions: **206K+**Points: **2**

Given a **root** of a binary tree with **n** nodes, find its level order traversal.
Level order traversal of a tree is [breadth-first traversal](#) for the tree.

Example 1:

Input:

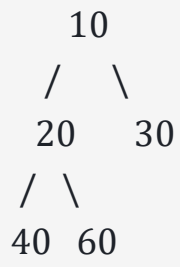
```
  1
 /  \
3    2
```

Output:

1 3 2

Example 2:

Input:



Output:

10 20 30 40 60

Your Task:

You don't have to take any input. Complete the function **levelOrder()** that takes the root node as input parameter and returns a list of integers containing the level order traversal of the given Binary Tree.

Expected Time Complexity: O(n)

Expected Auxiliary Space: O(n)

Constraints:

$1 \leq n \leq 10^5$

$0 \leq$ Data of a node $\leq 10^9$

Minimum between two nodes

Difficulty: **Hard**Accuracy: **84.11%**Submissions: **290+**Points: **8**

There is an undirected connected tree with **N** vertices. Given Q queries of type **(u, v).** Calculate the node with minimum value in the simple path from **u** to **v** in the tree.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains integer **N**, the size of the tree.
- Each of the next **N - 1** lines contains two integers **u** and **v**, indicating an edge between vertices **u** and **v**.
- The next line contains integer **Q**, the number of queries.
- Each of the next **Q** lines contains two integers **u** and **v.**

Output Format:

For each test case, print Q lines, answer to each query followed by a newline character.

Note: The generated output is white space sensitive, do not add any extra spaces or unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 10^4$

$2 \leq \mathbf{N, Q} \leq 2 * 10^5$

$1 \leq \mathbf{u, v\ (u \neq v)} \leq N$

*It is guaranteed that the sum of **N** and **Q** over all test cases does not exceed **$5 * 10^5$*** and the given edges form a connected tree.

Example:

Sample Input:

```
1
5
1 2
1 3
3 4
4 5
```

2
2 4
5 5

Sample Output:
1
5

Explanation:
Testcase 1:
Simple path between 2 and 4 is
2 -> 1 -> 3 -> 4
the minimum value node in the simple path is 1

Lowest Common Ancestor in O(log N)

Difficulty: **Medium** Accuracy: **81.09%** Submissions: **307+** Points: **4**

Given an array 'arr' that represents a tree, where arr[i] = x denotes that there exists a directed edge from arr[i] to i. Where arr[0] will always be -1 as 0 is the root node of the tree.
Answer Q queries, where each query is of type LCA(u, v) i.e. finding the Lowest Common Ancestor of the nodes u and v.

Input Format:
The first line of the input contains a single integer T, denoting the number of test cases. The description of T test cases is as follows:

- The first line of each test case contains two space-separated integers N and q.
- Then a single line consisting of n integer represents the edges present in the tree.
- The following q lines contain two space-separated integers u and v.

Output Format:

For each test case, print Q lines each consisting of a single integer, where the ith line is the answer of the ith query.

Constraints:

$0 < T < 100$

$0 < n < 10^5$

$0 < q < 10^5$

Example:

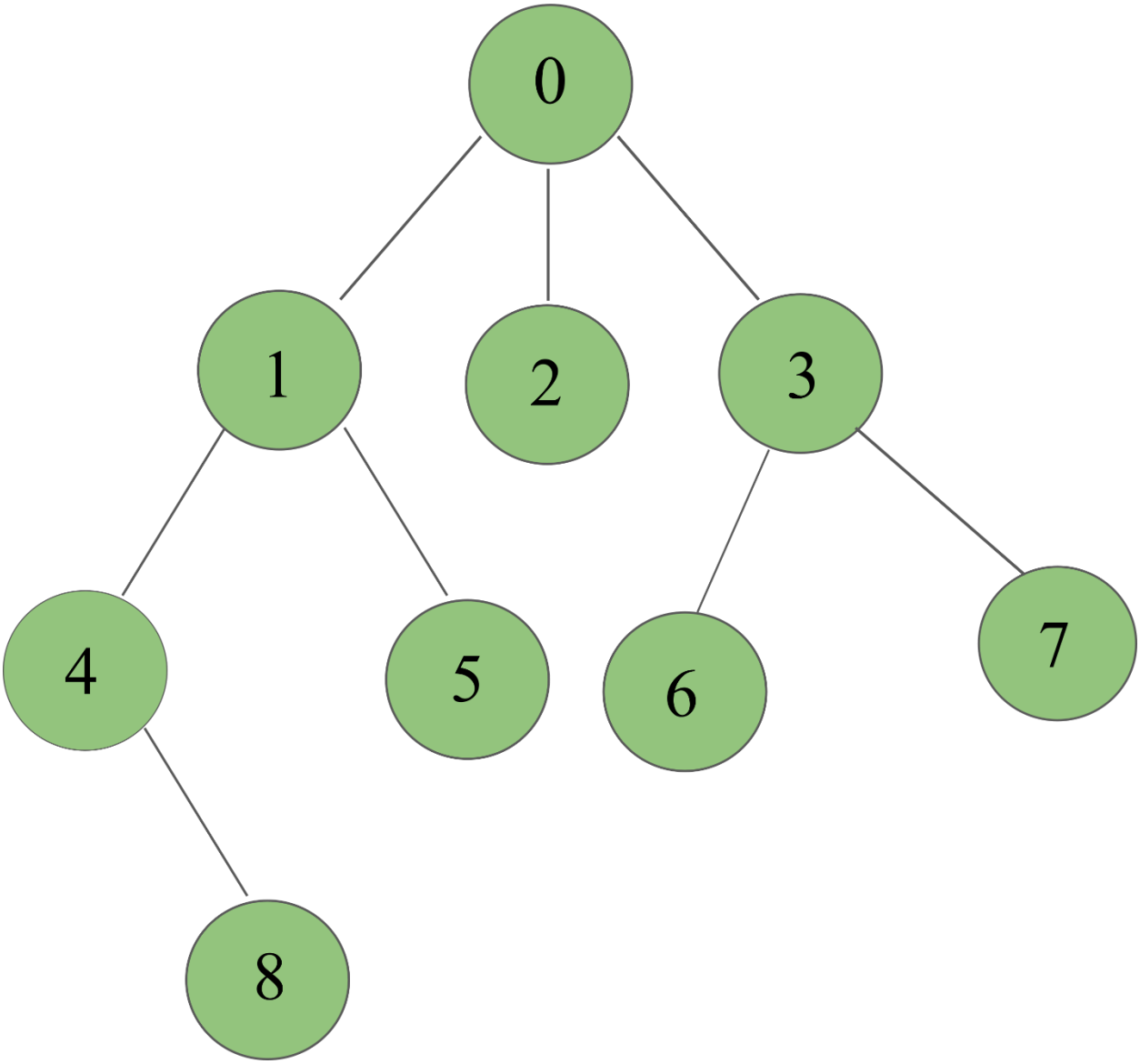
1
9 4
-1 0 0 0 1 1 3 3 4
4 5
5 6
4 7
8 1

Output:

1
0
0
1

Explanation:

The answer to the queries can be easily understood from the below-given image.



Minimum Cost

Difficulty: **Medium**Accuracy: **46.43%**Submissions: **28+**Points: **4**

There are **N** cities in Treeland numbered from **0** to **N - 1**. **M** parcels are present on **source[i]** and are to be delivered at **destination[i]**.

The structure of Treeland is given by array **p[ ]** and **w[ ]** of size **N** where **p[i]** is the direct ancestor of **i^{th}** city and **w[i]** is the cost of road connecting **p[i]** and **i. p[0] = -1** and **w[0] = 0** as **0th** city is the capital.

You can swap the cost of any two roads as many times as you want. So, your task is to minimize the total cost of parceling all the items.

Input Format:

The first line of the input contains a single integer **T**, denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers ***N*** and ***M*.**
- The second line contains **N** space-separated integers $p_0, p_1, ..., p_{N-1}$.
- The third line contains **N** space-separated integers $w_0, w_1, ..., w_{N-1}$.
- The third line contains **M** space-separated integers $source_0, source_1, ..., source_{N-1}$.
- The third line contains **M** space-separated integers $destination_0, destination_1, ..., destination_{N-1}$.

Note: p[0] = -1 and w[0] = 0.

Output Format:

For each test case, print the minimum possible cost followed by a newline character.

Note: The generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \le T \le 10^4$

$1 \le N, M \le 10^5$

$0 \le p[i], source[i], destination[i] < N$

$1 \le w[i] \le 10^5$

p[0] = -1 and w[0] = 0

The sum of **N** and **M** over all the test cases will not exceed $2 * 10^5$.

Example:

Sample Input:

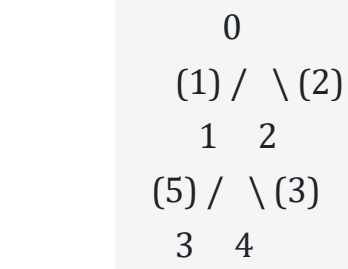
2
5 2
-1 0 0 1 1
0 1 2 3 4
1 2
0 4
7 1
-1 0 0 1 1 2 2
0 1 1 1 1 1 1
3
5

Sample Output:

7
4

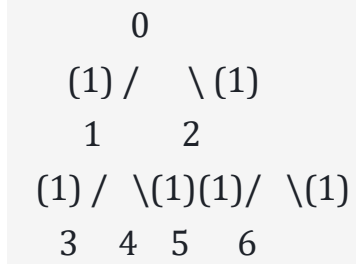
Explanation:

Testcase 1:
Treeland looks like:



One of the optimal weight distribution
is shown above for source = 1 and
destination = 0 cost = 1 for source = 2
and destination = 4 cost = 2 + 1 + 3 = 6
hence, total cost = 7

Testcase 2:
Treeland looks like:



One of the optimal weight distribution
is shown above for source = 3 and
destination = 5 cost = 1 + 1 + 1 + 1 = 4
hence, total cost = 4

Subtree Updates

Difficulty: **Hard**Accuracy: **88.46%**Submissions: **243+**Points: **8**

You are given a connected tree of **N** nodes numbered from **0** to **N-1** rooted at the **0th** node, where **p[i]** is the parent of **the ith** node and **p[0] = -1** as the 0th node is the root.
In the given tree each node has one of the two colors - black or white, if **A[i] = 1** then **the ith** node has **black** color else if **A[i] = 0** then **the ith** node has **white** color.
Each node also has a value, initially, the value of all nodes is 0.
You have to perform **Q** queries on the tree where the **ith** query is either of the two type:

1. **1 u x col**: you have to add value = **x** to every node of color = **col** which is in the subtree of node **u (node inclusive).** ($0 \le u < N$, $1 \le x \le 10^9$ and $0 \le col \le 1$)
2. **2 u**: find the value of the node **u**. ($0 \le u < N$)

Input Format:
The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- First-line contains two space-separated integers **N** and **Q**.
- Second-line contains **N** space-separated integers **p[ ]**.
- The third line contains **N** space-separated integers **A[ ]**.
- Next, **Q** lines denote a query and contain four space-separated integers if the query is of type 1 or two space-separated integers if the query is of type 2.

Output Format:

For each test case,

- print the answer to each query of type 2 in the order they appear in the input separated by white spaces

followed by a newline character.

Note: It is guaranteed that the last query is of type 2. The generated output is white space sensitive, do not add any extra spaces or unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 10^4$

$1 \leq N, Q \leq 10^5$

$0 \leq p[i] < N$

$p[0] = -1$

$0 \leq A[i] \leq 1$

*It is guaranteed that the sum of **N** and **Q** over all test cases does not exceed **$5 * 10^5$*** and the given edges form a connected tree.

Example:

Sample Input:

```
1
3 4
-1 0 0
```

```
1 0 1
1 0 5 1
2 0
2 1
2 2
```

Sample Output:

```
5 0 5
```

Explanation:

Testcase 1:
Structure of the tree is:

```
  0
 / \
1   2
```

In the first query, we add value = 5 to every node in the subtree of 0th node (0th node inclusive) which is of color black. In the subsequent queries, we print values of each node.

Greedy

Connecting the dots

Difficulty: **Medium** Accuracy: **86.82%** Submissions: **296+** Points: **4**

There are **N** cities on a plane. Every city is represented by two integers **X** and **Y**, denoting location of city on the plane. You want to construct some 2-way roads between the cities such that it is possible to go from every city to any other city. Cost of laying road of length **L** is **L** rupees.

Find the minimum cost of creating a road network that connects all cities. Report the cost exactly upto **two** decimal places.

Input

- First line of each testcase contains **T** , number of testcases
- First line of each testcase contains **N** number of cities
- Next **N** line of each testcase contains two integers **X** and **Y** denoting location of the cities
- No two cities share the same location

Output

- For each testcase, print in a new line cost of connecting all the cities.
- Report the cost rounded exactly upto two decimal places

Constraints

- **1<=T<=5**
- **1<=N<=1000**
- **1<=X,Y<=1000**

Sample Input

```
1
3
1 2
5 4
3 2
```

Sample Output

```
4.83
```

Explanation

We can build a road connecting city (1,2) to (3,2) with cost **2** and one road connecting (3,2) to (5,4) with cost **2.83** thus total cost would be **4.83**

0-1 MST

Difficulty: **Hard**Accuracy: **91.12%**Submissions: **259+**Points: **8**

A **spanning tree** of a graph is a **subgraph** that is a tree and **includes all vertices of the graph**.

Let's define **weight of a spanning tree** as sum of weights of all edges in that tree. **Minimum spanning tree** is a spanning tree with minimum weight

Given a graph **G**, for every edge find if it belongs to some **minimum spanning tree**

Note : There can be more than one **MST**s possible for a graph.

Input

- First line contains **T**, number of testcases
- First line of each testcase contains two integers **N, M** number of vertices in the graph and number of edges respectively
- Each one of next **M** lines contains three integers **u, v** and **w**. Representing an edge between vertex **u** and **v** with weight **w.** There are no self-loops.

Output

- For each testcase print in **M** lines, i_{th} line must contain **1** if i_{th} edge belongs to some **MST** otherwise it should contain **0**

Constraints

- **1<=T<=10**
- **1<=N<=10^5**
- **N-1<=M<=10^5**
- **1<=u,v<=N (u is not equal to v)**
- **1<=w<=10^9**

Sample Input

```
2
4 4
```

```
1 2 3
1 3 2
3 4 5
2 4 3
3 3
1 2 1
2 3 1
3 1 1
```

Sample Output

```
1
1
0
1
1
1
1
```

Explanation

In first test, there is only one **minimum spanning tree** 3----1----2---4 , with weight 8, therefore only 1st, 2nd and 4th edges belong to some MST and answer for them is **1** and 3rd edge doesn't belong to any **MST**, so answer is **0** for 3rd edge

In second test one MST is 1---2----3 with weight **2** so first and second edge belong to a MST, now another possible MST is 2----3---1 with weight **2** so third edge also belongs to a MST, therefore answer for all 3 edges is **1**

Pizza delivery

Difficulty: **Medium**Accuracy: **84.43%**Submissions: **283+**Points: **4**

Raghav has started a pizza delivery franchise, it has **K** outlets which look over **N** delivery locations (delivery locations also include the location of the outlet, obviously). He wants to serve his customers as fast as possible because it saves money on fuel and earns his company a good reputation.

Given location of **K** outlets and map of the city. For each one of the **N** location in the city print the **minimum time it takes to reach an outlet** or **minimum time it takes to be reached from an outlet.**

City can be represented as a graph of **N** nodes and **M** two-way edges, each edge has a positive weight **W** which denotes the time to travel between endpoints of that edge when we use the given edge.

Input

- First line contains an integer **T** denoting number of testcases, **T** testcases follow
- First line of each test contains three integers **N**, **M** and **K**
- Second line contains **K** integers in range **[1,N]** denoting the nodes which are pizza outlets.
- Each of the next **M** lines contain three integers **u, v** and **w** denoting there is a two way path between location **u** and **v** which costs you time **w**. **u** is not equal to **v** and it is possible to reach from every location in city to any other location.

Output

- For each test print **N** space separated integers in a single line, i_{th} of which is a minimum time to reach i_{th} node from nearest outlet

Constraints

- **1<=T<=5**
- **1<=N, M<=10^5**
- **1<=u,v,K<=N**

- **$1<=w<=10^9$**

Sample Input

```
1
5 5 2
2 4
1 2 2
1 3 2
3 4 1
2 4 3
2 5 2
```

Sample Output

```
2 0 1 0 2
```

Here we have two outlets at location **2** and **4,** so minimum time to reach location **2** and **4** is **0**, **1** and **5** can be reached from **2** in **2** units of time, thus answer for them is **2** and nearest outlet to **3** is **4** with distance of **1**

Infinite paths

Difficulty: **Medium**Accuracy: **94.44%**Submissions: **251+**Points: **4**

Given a directed weighted graph **G** and a source node **src** in the graph, for every node **x** in **G** , find the cost of minimum path from **src** to **x**.

Input

- First line of each test contains **T** , number of testcases. **T** tests follow.
- First line of each test contains three integers **N** (number of nodes), **M** (number of edges) and **src** (source node)
- Next **M** lines contain description of **M** edges in form of three integers **u, v** and **w** , indicating that there is a directed edge from **u** to **v** with cost **w**

Output

- For every testcase print **N** lines.
- If **i_{th}** node is reachable from **src** print the minimum cost to reach **i** in the **i_{th}** line.
- If **i_{th}** node is not reachable print "UNREACHABLE" without quotes in **i_{th}** line.
- If **i_{th}** node is reachable but there is no minimum cost to reach there, print "-inf" without quotes in **i_{th}** line.

Constraints

- **1<=T<=10**
- **1<=N<=3000**
- **1<=M<=6000**
- **1<=u,v<=N**
- **-10^6<=w<=10^6**

Sample Input

```
1
6 6 1
1 2 2
2 5 -2
5 3 1
3 2 -3
4 2 2
1 6 3
```

Sample Output

```
0
-inf
-inf
UNREACHABLE
-inf
3
```

Explanation

In the graph given in sample input, **1** is **src** and cost of reaching **1** is 0 (this may not be the case always), we can reach **6** by following a direct edge from 1 to 6 with weight **3**, thus cost of reaching **6** is **3**.
4 is unreachable from **1**
2, 3 and **5** are reachable but there is **no minimum weight** that can be assigned to the path between **1** and them therefore we print **-inf** for **2, 3** and **5**

Torrent Freak

Difficulty: **Hard**Accuracy: **93.23%**Submissions: **266+**Points: **8**

Naman is a torrent freak, he downloads a lot of files on bittorrent network. In a torrent network, users can be defined as nodes. Each node has two functions **leeching** and **seeding**.

Node which is sharing **(seeding)** the file is called **seeder** and the node downloading **(leeching)** the file is called **leecher** . A node can act as both **seeder** and **leecher** at the same time.

For the sake of simplicity in this problem, Naman is the **only leecher** on the network .

There are **N** seeders Naman is connected to, he has to download **M** files, size of i_{th} file is **F[i]**.
Every seeder has a **limit on the data** it can send to someone, i_{th} seeder can seed atmost **D[i]** megabytes of data.

All files are not available with every seeder (check input section for details on which files are available with which seeder), you have to find out the **maximum data Naman can use if he downloads files optimally.**

Input

- First line contains **T** number of testcases, **T** testcases follow

- First line of each testcase contains **N** and **M** number of seeders and number of files respectively
- Second line of each testcase contains **N** space separated integers $\mathbf{i_{th}}$ of which is **D[i]** the maximum amount of data $\mathbf{i_{th}}$ seeder can send in megabytes
- Third line of each testcase contains **M** space separated integers $\mathbf{i_{th}}$ of which is **F[i]** , size of $\mathbf{i_{th}}$ file in megabytes
- Now **N** lines follow, $\mathbf{i_{th}}$ of which represents files available with $\mathbf{i_{th}}$ seeder, each line starts with an integer **X**, number of files the seeder has then **X** integers in the range **[1,M]** follow in the same line representing the index of those **X** files which the current seeder has. (Refer sample input for further explanation)

Output

- For every testcase print only one integer in a new line denoting maximum data that Naman can use.

Constraints

- **1<=T<=10**
- **1<=N,M<=500**
- **1<=D[i],F[i]<=1000**
- **1<=X<=M**

Sample Input

```
1
3 2
3 1 1
4 5
1 1
2 1 2
1 2
```

Sample Output

```
5
```

Explanation

In first and the only testcase
First line contains 3 and 2 (i.e N is 3 and M is 2) so, there are 3 seeders and 2 files.
Second line contains 3 integers denoting limits on the data that can be sent by every seeder - First seeder can share only 3MB, second and third seeder can share atmost 1 MB of data.

Third line represents size of the files, as there are only two files this line contains two integers, size of first file is 4MB and second file's size is 5MB
Now in next three lines show what all files are present with every seeder. First of them is "1 1" denoting that first seeder has 1 file and index of that file is one (we are considering 1-based indices here) which means first seeder has the first file.

Thus second seeder has both the files and third seeder has only the second file.

Answer for this test is **5** because, Naman can download first file completely by downloading 3MB of the first file from the first seeder (who is capped exactly at 3MB) and download remaining 1MB of the first file from the second seeder as he also has the first file and is capped at 1MB.

Now second seeder can't send anymore data and the only file that remains to be downloaded is second file which is owned by second and third seeder. So Naman downloads 1MB of the second file from third seeder . Therefore total data Naman was able to use is 5MB.

Adding One

Difficulty: **Easy**Accuracy: **91.1%**Submissions: **281+**Points: **2**

Given a non-negative integer represented as an array of digits. Your task is to add 1 to the number (increment the number by 1). The digits are stored such that the most significant digit is at the starting index of the array.

Example:

If the array has [4, 5, 6], the resultant array should be [4, 5, 7] as 456 + 1 = **457**.

Input:

The first line of input contains an integer **T** denoting the number of test cases. The first line of each test case contains a single integer N denoting the size of the array. The second line contains **N** space-separated integers $\mathbf{A_1}$, $\mathbf{A_2}$, ..., $\mathbf{A_N}$ denoting the elements of the array.

Output:

Print the resultant array.

Constraints:

$1 \le T \le 300$

$1 \le N \le 10^7$

$0 \le A[i] \le 9$

Example:

Input:

2

4

5 6 7 8

3

9 9 9

Output:

5 6 7 9

1 0 0 0

Factorial or Not

Difficulty: **Easy**Accuracy: **94.18%**Submissions: **277+**Points: **2**

For a given number **N**, find whether it is a factorial number or not. A **Factorial number** is a number equal to other numbers' factorial value.

Input Format:

- The first line contains an integer **T** - the number of test cases, then the test cases follow:
- The first and only line contains an integer ***N.***

Output Format:

Print 1 if N is a factorial number, or 0 otherwise, followed by a newline character.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

1 <= N,T <= 100000

Example:

Input:

2

6

5

Output:

1

0

Explanation:

Test Case 1: 6 is factorial of 3

Test Case 2: no number's factorial is 5.

DP – Basics

Subset Sum

Difficulty: **Medium**Accuracy: **79.37%**Submissions: **383+**Points: **4**

Given an array **A** of size **N** of non-negative integers, and a value **sum**, determine if there is a subset of the given set with sum equal to given **sum**.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integer ***N.***
- The Second line denote N number $a_1, a_2, \ldots a_n$,
- The third line of each test case contains an integer **sum**

Output Format :

Print 1 if there exists a subset with given sum and 0 otherwise.

Constraints:

1 <= **T** <= 10

1 <= **N** <= 100

1<= **A[i]** <= 100

1<= **sum** <= 10^5

Example

Input:

2

6

3 34 4 12 5 2

9

6

3 34 4 12 5 2

30

Output :
1
0

Explanation:
For testcase 1 :- Here there exists a subset with
sum = 9, 4+3+2 = 9.
For testcase 2 :- There is no subset with sum 30.

Longest Common Subsequence

Difficulty: **Easy**Accuracy: **90.22%**Submissions: **325+**Points: **2**

Given two sequences, find the length of longest subsequence present in both of them. Both the strings are of uppercase.

Input Format:

The first line of the input contains a single integer T, denoting the number of test cases. The description of T test cases is as follows:

- The first line consists of two numbers (say n and m) separated by space denoting the lengths of sequences.
- The second line consists of a squence of length n.
- Thrid line consist of another squence which is of length m.

Output Format:

For each test case print the length of longest common squence in a separate line.

Constraints:

1 <= T <= 100
1<=size(str1),size(str2)<=10^3

Example:

Input:
2
6 6
ABCDGH
AEDFHR
3 2
ABC
AC

Output:
3
2

Explanation:
Test Case 1 - LCS for input Sequences ABCDGH and AEDFHR is ADH of length 3.
Test Case 2 - LCS of "ABC" and "AC" is "AC" of length 2.

Longest Increasing Subsequence with no more than K distinctness

Difficulty: **Medium**Accuracy: **68.57%**Submissions: **337+**Points: **4**

Given an array a[] of size N, find the length of longest subsequence such that the following conditions are met -

- The elements in the subsequence are strictly increasing.
- The difference between adjacent elements of array should not exceed a given integer 'K'.

Input Format:
The first line of input consists of a single integer T, denoting the number of test cases. The description of each test case is as follows -

- The first line of each test case contains two space-separated integers N and K.
- Following line consist N space-separated integers, denoting the array elements.

Output Format:

For each test case print the size of the longest increasing subsequence present in the corresponding array.

Constraints:

$1 < T < 100$

$1 \leq N \leq 10^5$

$1 \leq A[i] \leq 10^5$

$0 \leq k \leq 10^3$

Sum of N over all test cases will not exceed $5 * 10^5$

Example:

Input:

2
16 5
1 8 4 12 2 10 6 14 1 9 5 13 3 11 7 15
6 2
5 8 3 7 9 1

Output:

6
3

Explanation:

Test Case 1 - Longest increasing subsequence is 1 2 6 9 13 15, which has length **6** and none of adjacent elements have difference greater than 4.

Test Case 2 - Longest increasing subsequence is 5 7 9, with length **3** and none of adjacent elements have difference greater than 2.

Longest Increasing Subsequence

Difficulty: **Medium**Accuracy: **80.04%**Submissions: **361+**Points: **4**

Given an array of integers, find the **length** of the **longest (strictly) increasing subsequence** from the given array.

Input Format:

The first line of input consists of a single integer T, denoting the number of test cases. The description of each test case is as follows -

- The first line of each test case contains a single integer N, denoting the size of the array.
- Following line consist N space-separated integers, denoting the array elements.

Output Format:

For each test case print the size of the longest increasing subsequence present in the corresponding array.

Constraints:

$1 \le T \le 100$

$1 \le N \le 10^5$

$0 \le A[i] \le 10^6$

Sum of N over all test cases will not exceed $5 * 10^5$

Example:

Input:

2

16

0 8 4 12 2 10 6 14 1 9 5 13 3 11 7 15

6

5 8 3 7 9 1

Output:

6

3

Explanation:

Test Case 1 - Longest increasing subsequence is 0 2 6 9 13 15, which has length **6**

Test Case 2 - Longest increasing subsequence is 5 7 9, with length **3**

Edit Distance

Difficulty: **Medium**Accuracy: **85.95%**Submissions: **321+**Points: **4**

Given two strings **s** and **t.** Return the minimum number of operations required to convert **s** to **t.**

The possible operations are permitted:

1. Insert a character at any position of the string.
2. Remove any character from the string.
3. Replace any character from the string with any other character.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- A line containing two strings **s** and **t** respectively separated by a space

Output Format :

Print the minimum number of operations required to convert **s** to **t.**

Constraints:

1 <= **T** <= 50

1 ≤ Length of both strings ≤ 100

Both the strings are in lowercase.

Example

Input:
2
geek gesek
gfg gfg

Output :
1
0

Explanation:
For testcase 1 :- One operation is required
inserting 's' between two 'e's of str1.
For testcase 2 :- Both strings are same.

Geek and Jumping Game

Difficulty: **Medium**Accuracy: **62.5%**Submissions: **408+**Points: **4**

Geek is playing a jumping game in which he is provided with an integer array **arr(0-indexed)** of size **N** and an integer **K**. He initially has to start from index **0** and reach the index **n-1**. He can jump **at most K** steps ahead in a single move without going outside the array boundaries that means if he is standing at index **i**, he can jump to any index in the range **[i+1, min(n-1,i+k)] inclusive**. The game's **score** is the **sum** of all **arr[i]** for each index i which the Geek visited in the array.

You need to calculate the **maximum** score that the Geek can achieve in this game.

Input:

The first line contains **T**, denoting the number of test cases.

The first line of each test case contains **N**, representing the length of integer array **arr**.

The second line of each test case contains **N** integers representing the elements of integer array **arr**.

The third line of each test case contains **K**.

Output:

You have to print the **maximum** score that the Geek can achieve.

The output of each test case must be printed in a **separate** line.

Constraints:

- 1 <= T <= 100
- 1 <= N, K <= 10^5
- -10^4 <= arr[i] <= 10^4

Example:

Input:

```
2
5
1 2 3 4 5
2
6
2 -2 4 5 -5 6
3
```

Output:

```
15
17
```

Explanation:

1st test case: Geek should visit every index to achieve the maximum score. Maximum score: 1+2+3+4+5=15
2nd test case: Geek should visit the cells with values 2, 4, 5, and 6, which results in a maximum score of 17.

Card Game II

Difficulty: **Medium**Accuracy: **79.69%**Submissions: **345+**Points: **4**

Alexa and Siri are **alternatively** playing a game, with Alexa beginning **first**.

At first, there are **N cards** in a deck. On every player's turn, that player takes action comprising of **eliminating** any **non-zero square number** of cards from a deck.

The player will lose the game if they aren't able to make the next move.

Your task is to determine the winner of the game, assuming that both the players will play **optimally**.

Input:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains **N**, which **initially** represents the number of cards in the deck.

Output:

You have to print:

"Alexa" if Alexa wins

"Siri" if Siri wins

Constraints:

- $1 <= T <= 100$
- $1 <= N <= 10^5$

Example:

Input:

```
3
4
5
1
```

Output:

```
Alexa
Siri
Alexa
```

Explanation:

1st test case: Since N=4 is already a perfect square. Therefore Alexa can win by removing the 4 cards from the deck.
2nd test case: Siri can win if she plays optimally. Consider a case when Alexa starts the game by removing only 1 card. Then the remaining cards will be 4, and since it is a perfect square, Siri can remove all the cards and win the game. Consider another case when Alexa starts the game by eliminating 4 cards. Then the remaining cards will be 1, and Siri can win by removing this card from the deck as, after that, Alexa doesn't have any moves left.
3rd test case: Alexa can remove 1 card winning the game because Siri doesn't have any moves left.

Jumping on Stone 1

Difficulty: **Medium**Accuracy: **88.49%**Submissions: **312+**Points: **4**

There are N stones, numbered $1, 2, .., N$. For each x ($1 \le x \le N$), the height of Stone x is A_x.

There is a frog who is initially on Stone **1**. He will repeat the following action some number of times to reach Stone N:

- If the frog is currently on Stone x, jump to Stone $x + 1$ with cost $|h_x - h_{x+1}|$ or Stone $x + 2$ with cost $|h_x - h_{x+2}|$.

Find the minimum possible total cost required to reach Stone **N**.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integer, **N**.
- The second line has **N** space-separated positive integers $A_1, A_2, ..., A_N$.

Output Format:

Print the minimum cost for each test case, followed by a newline character.

Note: The generated output is white space sensitive, do not add extra spaces or unnecessary newline characters.

Constraints:

$1 \le T \le 10^4$

$1 \le N \le 10^5$

$1 \le A_i \le 10^4$

*The sum of **N** over all test cases does not exceed $2 * 10^5$.*

Example:

Sample Input:

```
1
6
```

30 10 60 10 60 50

Sample Output:
40

Explanation:
Testcase 1:
If we follow the path
1 → 3 → 5 → 6, the total
cost would be
30−60+60−60+60−50 = 40

Jumping on Stone 2

Difficulty: **Medium**Accuracy: **59.82%**Submissions: **423+**Points: **4**

There are N stones, numbered $1, 2, .., N$. For each x ($1 \le x \le N$), the height of Stone x is A_x.
There is a frog who is initially on Stone **1**. He will repeat the following action some number of times to reach Stone N:

- If the frog is currently on Stone **x**, jump to Stone **x + 1, x + 2, x + 3...., x + K** with cost $|h_x - h_y|$ where y is the stone to land on.

Find the minimum possible total cost required to reach Stone **N**.

Input Format:
The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers, **N** and **K**.
- The second line has **N** space-separated positive integers $A_1, A_2, ..., A_N$.

Output Format:

Print the minimum cost for each test case, followed by a newline character.
Note: The generated output is white space sensitive, do not add extra spaces or unnecessary newline characters.

Constraints:
$1 \leq \mathbf{T} \leq 1000$
$1 \leq \mathbf{N} \leq 10^4$
$1 \leq \mathbf{K} \leq 100$
$1 \leq \boldsymbol{A_i} \leq 10^4$
The sum of ***N*** *over all test cases does not exceed* $\boldsymbol{6 * 10^4}$.

Example:

Sample Input:
1
5 3
10 30 40 50 20
Sample Output:
30

Explanation:
Testcase 1:
If we follow the path $1 \rightarrow 2 \rightarrow 5$,
the total cost incurred would be
$10-30+30-20=30$.

0 - 1 Knapsack Problem
Difficulty: **Medium**Accuracy: **86.08%**Submissions: **309+**Points: **4**
You are given weights and values of **N** items, and put these items in a knapsack of capacity **W** to get the maximum total value in the knapsack. Note that we have only **one quantity of each item**.
In other words, given two integer arrays **val[0..N-1]** and **wt[0..N-1]** which represent

values and weights associated with **N** items respectively. Also given an integer W which represents knapsack capacity, find out the maximum value subset of **val[]** such that the sum of the weights of this subset is smaller than or equal to **W.** You cannot break an item, **either pick the complete item or don't pick it (0-1 property)**.

Input:

The first line consists of an integer T i.e the number of test cases.

- The first line of each test case consists of two integers **n** and **w**.
- The second line contains n integers for the **val[]** array.
- The third line contains n integers for the **wt[]** array.

Constraints:

1<=T<=1000
1 ≤ N ≤ 1000
1 ≤ W ≤ 1000
1 ≤ wt[i] ≤ 1000
1 ≤ v[i] ≤ 1000

Example 1:
Input:
N = 3
W = 4
values[] = {1,2,3}
weight[] = {4,5,1}
Output: 3

Example 2:
Input:
N = 3

W = 3
values[] = {1,2,3}
weight[] = {4,5,6}
Output: 0

Coin Change

Difficulty: **Medium**Accuracy: **73.04%**Submissions: **349+**Points: **4**

Given an integer array of **coins[ ] of size N** representing different denominations of currency and an integer **sum**, find the number of ways you can make **sum** by using different combinations from coins[ ].
Note: Assume that you have an infinite supply of each type of coin.

Input:

- The first line consists of an integer T i.e the number of test cases.
- The first line of each test case consists of two integers sum and N.
- The next line consists of N-spaced integers.

Output:

For each test case print a single integer denoting the number of ways to make change for a given sum of money.

Constraints:

1<=T<=10
1<=sum, N<=10

Example 1:

Input:
sum = 4 ,
N = 3
coins[] = {1,2,3}
Output: 4

Explanation: Four Possible ways are:
{1,1,1,1},{1,1,2},{2,2},{1,3}.

Example 2:

Input:
Sum = 10 ,
N = 4
coins[] ={2,5,3,6}
Output: 5
Explanation: Five Possible ways are:
{2,2,2,2,2}, {2,2,3,3}, {2,2,6}, {2,3,5}
and {5,5}.

Minimum Cost Path

Difficulty: **Hard**Accuracy: **57.4%**Submissions: **399+**Points: **8**

Given a square grid of size N, each cell of which contains an integer cost that represents a cost to traverse through that cell, we need to find a path from the top left cell to the bottom right cell by which the total cost incurred is minimum. From the cell (i,j) we can go (i,j-1), (i, j+1), (i-1, j), (i+1, j).

Note: It is assumed that negative cost cycles do not exist in the input matrix.

Input:

- The first line consists of an integer T i.e the number of test cases.
- The first line of each test case consists of an integer N, the size of the square grid.
- Each of the next N lines consists of N-spaced integers.

Output:

For each test case print a single integer denoting the minimum cost as described.

Constraints:

$1 \le n \le 500$

$1 \le$ cost of cells ≤ 1000

Example 1:

Input:

grid =
{{9,4,9,9},{6,7,6,4},
{8,3,3,7},{7,4,9,10}}

Output:

43

Explanation:

The grid is-
9 4 9 9
6 7 6 4
8 3 3 7
7 4 9 10
The minimum cost is-
9 + 4 + 7 + 3 + 3 + 7 + 10 = 43.

Example 2:

Input:

grid = {{4,4},{3,7}}

Output:

14

Explanation:

The grid is-
4 4
3 7
The minimum cost is- 4 + 3 + 7 = 14.

Longest Palindromic Substring

Difficulty: **Medium**Accuracy: **81.33%**Submissions: **326+**Points: **4**

Given a String, find the longest palindromic subsequence.

Input:

The first line consists of an integer T i.e the number of test cases.
The first line of each test case contains a string S.

Output:

For each test case print a single integer denoting the length of the longest palindromic subsequence.

Constraints:

1<=T<=20
$1 \leq |S| \leq 10^4$

Example 1:

Input:
S = "bbabcbcab"
Output: 7
Explanation: Subsequence "babcbab" is the
longest subsequence which is also a palindrome.

Example 2:
Input:
S = "abcd"
Output: 1
Explanation: "a", "b", "c" and "d" are

palindromic and all have a length 1.

Longest Common Substring

Difficulty: **Medium**Accuracy: **79.87%**Submissions: **333+**Points: **4**

Given two strings. The task is to find the length of the longest common substring.

Input:

The first line consists of an integer T i.e the number of test cases.

- The first line of each test case consists of an integer N and M, the size of input stings s1 and s2.
- The second line of each test case contains sting s1.
- The third line of each test case contains sting s2.

Output:

For each test case print a single integer denoting the length of the longest common substring

Constraints:

1<=T<=100
1<=n, m<=1000

Example 1:
Input: S1 = "ABCDGH", S2 = "ACDGHR", n = 6, m = 6
Output: 4
Explanation: The longest common substring
is "CDGH" which has a length 4.

Example 2:
Input: S1 = "ABC", S2 "ACB", n = 3, m = 3
Output: 1

Explanation: The longest common substrings
are "A", "B", and "C" all having length 1.

2D-DP

Minimum Cost in Colorful Grid

Difficulty: **Easy**Accuracy: **97.43%**Submissions: **269+**Points: **2**

You are given a colorful grid of size n * n filled with 'R', 'G', 'B'.

Here 'R', 'G', 'B' mean red, green, blue respectively.

From each cell, you can either move only to the right or down.

If you move to a different color cell, the cost will be 1.

You are currently in the top-left cell, find the minimum cost to reach the bottom-right cell of the grid.

Input:

- The first line contains one integer **t** (number of test cases).
- The first line of each test case contains two space-separated integers **r** (number of rows) and **c** (number of columns).
- Next **r** lines of each test case contain strings of length **c,** consisting of the characters 'R', 'G', 'B'.

Output:

- Print t integers, the i-th should be the answer for the i-th test case.

Constraints:

- 1 <= t <= 100
- 1 <= r, c <= 100

Example:

Sample Input

```
3
3 3
RGB
GBR
RRB
4 3
BRR
GBG
BBB
RGG
2 3
GGG
BBG
```

Sample Output

```
3
3
0
```

Explanation

Here all possible paths of the first test case are :

1. RGRRB (Cost of this path = 3)
2. RGBRB (Cost of this path = 4)
3. RGBRB (Cost of this path = 4)
4. RGBRB (Cost of this path = 4)
5. RGBRB (Cost of this path = 4)
6. RGBRB (Cost of this path = 4)

So, here answer of the first test case will be 3.

Multi Level Game

Difficulty: **Easy** Accuracy: **92.05%** Submissions: **279+** Points: **2**

Geek is playing a multi-level game in which there are **N** levels. In each level**(1<= i <= N)**, he will get some points by playing any of the following three games.

- Shooting Game: Get x_i points.
- Racing Game: Get y_i points.
- Puzzle Game: Get z_i points.

As Geek gets bored quickly, he can't play two same games in **two** or **more consecutive** levels.

You have to find the **maximum** points which Geek can gain by playing this multi-level game.

Input Format:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains **N**, representing the number of levels in the multi-level game.

The following N lines of each test case contain three integers x_i, y_i, **and** z_i representing the points gain in the **ith** level in shooting, racing, and puzzle games.

Output Format:

Print the **maximum** points which Geek can gain by playing this multi-level game.

The output of each test case must be printed in a separate line.

Constraints:

- $1 <= T <= 10^3$
- $1 <= N <= 10^5$
- $1 <= x_i, y_i, z_i <= 10^4$

Example:

Input:

```
2
3
10 20 70
30 40 90
50 60 110
1
200 300 400
```

Output:

```
220
400
```

Explanation:

```
1st test case: Geek will gain 220 points by playing
puzzle game in 1st level(70 points), a racing game
in 2nd level(40 points), and a shooting game in 3rd
level(110 points).
```

Tom and Jerry

Difficulty: **Easy**Accuracy: **71.33%**Submissions: **34K+**Points: **2**

Tom and Jerry being bored in this pandemic, decides to play a game. Given an **integer N**. On each player's turn, that player makes a move by **subtracting a divisor of current N** (which is less than N) from current N, thus **forming a new N for the next turn**. The player who does not have any divisor left to subtract loses the game.

The game begins with Tom playing the first move. Both Tom and Jerry play optimally. The task is to determine who wins the game. Return **1** if Tom wins, else return **0**.

Example 1:

Input:
N = 2
Output:
1
Explanation:
Tom subtracts 1 from N to make N = 1.
Now, Jerry isn't left with any possible
turn so Tom wins the game, and therefore
the Output is 1.

Example 2:

Input:
N = 4
Output:
1
Explanation:
1st turn: Tom subtract 1 from N as 1 is
a divisor of 4 and less than 4.

2nd turn: N=3, Jerry has to subtract 1 as 1
is the only divisor of 3 which is less than 3.

3rd turn: N=2, Tom subtract 1 as 1 is the
only divisor of 2 which is less than 2.

4th turn: N=1, Jerry can't subtract any value.
So, Tom wins.

Your Task:
You don't need to read input or print anything. Your task is to complete the

function numsGame() which takes an Integer N as input and returns 1 if Tom wins else returns 0.

Expected Time Complexity: $O(1)$

Expected Auxiliary Space: $O(1)$

Constraints:

$1 < N \leq 10^8$

Number of Ways to partition the string

Difficulty: **Hard**Accuracy: **93.18%**Submissions: **262+**Points: **8**

Geek is given a string. Can you help him in finding out the number of ways the given string can be partitioned into **one or more non empty substrings** in such a way that the integer values of the substrings are in the **non decreasing order** and **no integer** value must have **leading zero**.

Input:

The first line contains a **single integer t**, the number of test cases.

Each subsequent line contains a **string s**.

Output:

For each test case, print the number of ways to partition the string. Since the answer can be very large, print it modulo **1e9+7**.

The output of each test case must be printed in a separate line.

Constraints:

- $1 <= T <= 10^2$
- $1 <= s.length() <= 10^3$

Example:

Input:

```
3
789
26
045
```

Output:

```
3
2
0
```

Explanation:

1st test case: The given string can be partitioned into 3 ways- [**789**, **7 | 8 | 9**, **7 | 89**].
2st test case: The given string can be partitioned into 2 ways- [**26**, **2 | 6**].
3st test case: The given string can't be partitioned. We can't partition it like this - [**045**, **04 | 5**] as the integer value of the substrings are starting with zero.

Coloring Walls

Difficulty: **Medium** Accuracy: **86.62%** Submissions: **112+** Points: **4**

There is a row of **n** walls in GeeksLand. Few walls are colored, and few are uncolored. The walls are colored with colors ranging from **1 to c.**

The **beauty** of the group of walls is defined as the minimum number of adjacent groups; we can partition all the given n walls such that each group contains wall of the same color. For example, if the walls are colored as [2, 2, 3, 3, 2, 1, 1], then the beauty of the group of walls is 4 as we can divide the walls into 4 groups [{2, 2}, {3, 3}, {2}, {1, 1}].

You are provided with array **walls**, an **n*c** matrix **cost**, and an integer **k** where

- **walls[i]** represent the color of the **ith** wall and 0 if it is uncolored.
- **cost[i][j]** represents the cost to color the **ith** wall with color **j+1**.

Your task is to find the **minimum** cost for painting the uncolored walls such that the beauty of the group of walls is exactly **k.**

Note: The colored walls should not be colored again, and the uncolored walls can be colored only with colors ranging from **1 to c.**

Input Format:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains three integers **n,c, and k,** representing the number of walls, number of colors, and the final expected value of the beauty of the group of walls.

The second line of each test case contains **n** integers walls[0], walls[1].... walls[n-1], representing the initial colors of the walls.

Then the following lines of each test case contain the **n*c cost** matrix.

Output Format:

Print the minimum cost to color all the given n walls. If there is no way to color the given walls with beauty **k**, then print **-1**.

The output of each test case must be printed in a separate line.

Constraints:

- 1 <= T <= 10
- 1 <= n <= 100
- 1 <= c <= 20
- 1 <= k <= 100

- 0 <= walls[i] <= c
- 1 <= cost[i][j] <= 10^4

Example:

Input:

```
3
3 2 2
0 0 2
1 3
2 4
3 5
3 2 1
2 1 2
4 3
9 6
10 15
3 2 3
2 1 2
4 3
9 6
10 15
```

Output:

```
3
-1
0
```

Explanation:

1st test case: Color the 0th and 1st wall with color
1 minimizes the cost (1 + 2 = 3) of coloring the
walls,and also the resulting beauty of a group of walls
will be 2 as the final coloring will be [1, 1, 2] which

matches the expected beauty of a group of walls(k = 2).
2nd test case: Here, all the walls are colored initially, and the current beauty of the group of walls is 2, but the expected beauty of the group of walls is 3(k = 3), so there is no valid coloring, and the answer is - 1.
3rd test case: Here, all the walls are colored, and the beauty of the group of walls match k, so no paint is used, and the answer is 0.

Minimum sum

Difficulty: **Medium**Accuracy: **17.14%**Submissions: **96K+**Points: **4**

Given an array Arr of size N such that each element is from the range 0 to 9. Find the minimum possible sum of two numbers formed using the elements of the array. All digits in the given array must be used to form the two numbers.

Example 1:

Input:
N = 6
Arr[] = {6, 8, 4, 5, 2, 3}
Output: 604
Explanation: The minimum sum is formed by numbers 358 and 246.

Example 2:

Input:
N = 5
Arr[] = {5, 3, 0, 7, 4}

Output: 82
Explanation: The minimum sum is
formed by numbers 35 and 047.

Your Task:
You don't need to read input or print anything. Your task is to complete the function **solve()** which takes **arr[]** and **n** as input parameters and returns the minimum possible sum. As the number can be large, return string presentation of the number without leading zeroes.

Expected Time Complexity: O(N*logN)
Expected Auxiliary Space: O(N)

Constraints:
$1 \leq N \leq 10^7$
$0 \leq Arr_i \leq 9$

Another Knapsack Problem

Difficulty: **Medium**Accuracy: **83.45%**Submissions: **284+**Points: **4**

Geek is given **N** books, and each book is associated with some weight and some id number which ranges from **1 to M**. Now he has to put **M** books, one of each id, in a knapsack whose maximum weight-bearing capacity is **K**. Can you help him in finding out the minimum vacant space left in the knapsack after putting M books, one of each id?

Input Format:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains three integers **N, M, and K**, representing the number of books, range of id number, and maximum knapsack capacity, respectively.

The second line of each test case contains **N** integers $\mathbf{w_1, w_2 .. w_n}$, representing the weights associated with each book.

The third line of each test case contains **N** integers $\mathbf{id_1, id_2, id_n}$, representing the ids associated with each book.

Output Format:

Print the minimum unused capacity of the knapsack. If there is no way to fill the knapsack, print **-1**.

The output of each test case must be printed in a separate line.

Constraints:

- 1 <= T <= 100
- 1 <= M <= N <= 100
- 1 <= K <= 10000
- 1 <= w[i] <= 100
- 1 <= id[i] <= M

Example:

Input:

```
2
5 5 11
2 2 2 2 2
1 2 3 4 5
9 3 21
3 3 3 5 5 5 7 7 7
1 2 3 1 2 3 1 2 3
```

Output:

```
1
0
```

Explanation:

1st test case: Since we have 5 books of each id,
we have to select them all. Therefore the total
weight will be 10 (2+2+2+2+2). The minimum unused
capacity = 11-10 = 1.
2nd test case: Choose books of weight 7 of each id.
The resultant weight will become 21 (7+7+7). The
minimum unused capacity = 21-21 = 0.

Breaking Stones

Difficulty: **Medium**Accuracy: **96.38%**Submissions: **261+**Points: **4**

You are provided with **N** stones indexed from **0 to N-1**. A number is written on each stone. The numbers on the stones are represented by a **stones** array; **stones[i]** denote the number written on the **ith** stone.

You have to break all the stones. On breaking **ith** stone, you will get **stones[left]*stones[i]*stones[right]** amount of money.
Here **left** and **right** are **adjacent** indices of **i**. After the burst, the **left** and **right** will become **adjacent** to each other.

You have to determine the **maximum** amount of money you can get by **optimally** breaking all the stones.

Note:

You can imagine **stones[-1] = stones[n] = 1**. They are not real; therefore, you can't break them.

Input Format:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains **N**, representing the number of stones.

The second line of each test case contains **N** integers **stones[0], stones[1] ..... stones[n-1]**, representing the number written on the stones.

Output Format:

Print the **maximum** amount of money gained by optimally breaking all the stones .

The output of each test case must be printed in a separate line.

Constraints:

- 1 <= T <= 100
- 1 <= N <= 100
- 0 <= stones[i] <= 100

Example:

Input:

```
2
4
4 2 1 3
2
2 10
```

Output:

```
46
30
```

Explanation:

1st test case:

Following should be the order of breaking of stones:
Initial Array: [4, 2, 1, 3]
Step 1: [4, 2, 3] Amount Gained: 6
Step 2: [4, 3] Amount Gained: 24
Step 3: [4] Amount Gained: 12
Step 4: [] Amount Gained: 4
Total Amount: 6 + 24 + 12 + 4 = 46

2nd test case:

Following should be the order of breaking of stones:
Initial Array: [2, 10]
Step 1: [10] Amount Gained: 20
Step 2: [] Amount Gained: 10
Total Amount: 20 + 10 = 30

Combinatorics

Spare Boxes

Difficulty: **Hard**Accuracy: **91.94%**Submissions: **274+**Points: **8**

Geek found an infinite stock of spare boxes with integer dimensions. He wants to arrange them in a straight line of dimension Ax2 where 1 ≤ **A** ≤ **N**. He wants to find out the sum of all possible ways of putting the boxes.
More formally, Geek wants to find out the sum of the number of ways to arrange the boxes in a grid with dimension Ax2 where 1 ≤ **A** ≤ **N.**
Since the answer may be large, print it modulo 10^9+7.

Input Format:

- The first line contains an integer ***T*** denoting the number of test cases
- Then each test case contains a single integer **N** - the maximum length of the grid.

Output Format:

For each test case print the answer in a new line.
Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:
$1 \leq T \leq 100$
$1 \leq N \leq 100,000$

Example:

Sample Input:
3
1
2
200

Sample Output:
2
10
629325950

Explanation:
In the first test case, there are 2
ways:- one by using a box of dimension
2x1 and another by using two boxes of
dimension 1x1.
In the second test case, there are 8
ways to arrange boxex in a grid of dimension
2x2, so total number of ways is 10(sum of
i equal to 1 and 2)

Coin Change

Difficulty: **Medium**Accuracy: **91.66%**Submissions: **288+**Points: **4**

Given an integer array **coins** of size **N** representing distinct denominations of currency and an integer **sum**, find the number of ways you can make **sum** by using different combinations from coins.
Since the answer can be very large, return the ans 10^9+7.
Note: Assume that you have an infinite supply of each type of coin.

Input :

The first line of input consists of **T**, the number of testcases. Each testcase is as follows :-

- The first line of each testcase consists of two space separated integers **sum** and **N** respectively.
- Second line consists of **N** space separated integers.

Output :

For each testcase print the number of ways to make change for given sum of money.

Constraints :

1 <= **T** <= 10

1 <= **sum, N** <= 10^3

Example :

Input :
2
4 3
1 2 3
10 4

2 5 3 6

Output :
4
5

Explaination :
Testcase 1 : Four Possible ways are:
{1,1,1,1},{1,1,2},{2,2},{1,3}.
Testcase 2 : Five Possible ways are:
{2,2,2,2,2}, {2,2,3,3}, {2,2,6}, {2,3,5}
and {5,5}.

Diplomatic Grouping

Difficulty: **Hard**Accuracy: **88.86%**Submissions: **279+**Points: **8**

N diplomats have arrived in Geek's country. Since the matter is political, diplomats can have healthy or bad relations with each other. The Prime Minister wants to test the problem-solving ability of Geek. SO he asks him to assign rooms to all of them. He will then score his assignment in the following manner:

- If diplomats **i** and **j (i!=j)** are in the same room, then A_{ij} will be added to Geek's score.
- If there is only 1 diplomat in a room then 0 will be added to Geek's score for that room.

Find the **maximum** score that Geek can get.

Input Format:

- The first line contains an integer ***T*** denoting the number of test cases
- The first line of each test case contains an integer **N** - the number of diplomats.
- Next **N** lines contain **N** space-separated integers.

Output Format:

For each test case print the answer in a new line.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 10$

$1 \leq \mathbf{N} \leq 15$

$0 \leq \mathbf{A_{ij}} \leq 10^9$

Example:

Sample Input:

```
2
4
0 10 -7 4
10 0 -1 9
-7 -1 0 7
4 9 7 0
2
0 -1
-1 0
```

Sample Output:

```
23
0
```

Explanation:

In the first test case, we can group
them in the following manner:- {1,2,4},{3}

So, Geek's score will be:
$(A_{12}+A_{14}+A_{24}) + (A_{33}) = 23$
In the second test case, We'll keep
each one of them in saperate rooms.
So, Geek's Score will be:
$(A_{11})+(A_{22}) = 0$

Friends and Party

Difficulty: **Hard**Accuracy: **93.46%**Submissions: **274+**Points: **8**

Geek fand Geekina has invited their friends for a party. The main event is the Best Pair dance competition. Both Geek and Geekina have invited **N** friends and they want them to pair in such a way that each pair has a member invited by Geek and a member invited by Geekina. Unfortunately, not everyone is comfortable in pairing with anyone. More formally,

- If $\mathbf{A_{ij}} = 1$, then **i-th** of Geek's friend is comfortable in pairing with **j-th** of Geekina's friend, otherwise they are not comfortable in pairing.

Find the number of ways in which their friends can pair such that every member of a pair is comfortable in with the other. Since the answer can be large, print it modulo $\mathbf{10^9+7}$.

Input Format:

- The first line contains an integer ***T*** denoting the number of test cases
- The first line of each test case contains two integers **N** - the number of friends invited by Geek and Geekina.
- Each of the next **N** lines contains **N** space-separated integers.

Output Format:

For each test case print the answer in a new line.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 20$

$1 \leq \mathbf{N} \leq 18$

$0 \leq \mathbf{A_i} \leq 1$

Example:

Sample Input:

```
2
3
0 1 0
1 1 1
0 1 1
2
1 1
1 1
```

Sample Output:

```
1
2
```

Explanation:

In the first test case, there is only 1 ways:- {1,2},(2,1},(3,3},

In the second test case, there are two ways:

{1,2} {2,1}

{1,1} {2,2}

Shipping Problem

Difficulty: **Hard**Accuracy: **73.39%**Submissions: **308+**Points: **8**

Geek fhas to ship **K** boxes. He has called **N** trucks for shipping. The ***i*-th** truck can carry at most A_i boxes. Geeks wonders how many ways are there such that the **K** boxes are shipped all at once.
Since the answer can be large, print it modulo 10^9+7.

Input Format:

- The first line contains an integer ***T*** denoting the number of test cases
- The first line of each test case contains two integers **N,K** respectively.
- The second line of each test case consist **N** space-separated integers - $A_1,A_2,...,A_N$ - the capacity of each truck

Output Format:

For each test case print the answer in a new line.
Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \le T \le 100$
$1 \le N \le 100$
$1 \le K \le 100,000$
$1 \le A_i \le K$

Example:

Sample Input:

```
2
3 2
1 2 1
5 10
```

1 2 3 2 1

Sample Output:

4

0

Explanation:

In the first test case, there are 4 ways:- {0,2,0},(1,0,1},(1,1,0},{0,1,1}

In the second test case, even if all N trucks carry boxes at their maximum capacity, they won't be able to transport all boxes at once.

Yet another coloring problem

Difficulty: **Medium**Accuracy: **94.01%**Submissions: **268+**Points: **4**

There is a tree with N vertices, numbered 1 .... N. For each i (1<= i <= N-1) , i-th edge connect vertex x_i and y_i.

You have to paint each vertex in white or black. Here, it is not allowed to paint two adjacent vertices both in black.

Find the number of ways in which vertices can be painted modulo (10^9+ 7).

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- First-line contains two space-separated integers **N**.
- Following N-1 contains 2 space separated integers x_i and y_i denoting an edge between nodes x_i and y_i.

Output Format:

For each test case,

- print the number of ways to paint modulo (10^9+ 7).

followed by a newline character.

Note: The generated output is white space sensitive, do not add any extra spaces or unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 10^4$

$1 \leq N \leq 10^5$

$1 \leq x_i, y_i \leq N$

It is guaranteed that the sum of ***N*** *over all test cases does not exceed* ***5 * 10^5*** and the given edges form a connected tree.

Example:

Sample Input:

```
1
3
1 2
1 3
```

Sample Output:

```
5
```

Explanation:

Testcase 1:

Structure of the tree is:

```
  1
 / \
```

2 3
If we assume tree to be like
2-1-3.Then, we can paint in
W W W
W B W
B W W
W W B
B W B
ways where W is white and B is
black.There are 5 ways in total.

Maximum Element

Difficulty: **Medium**Accuracy: **88.18%**Submissions: **288+**Points: **4**

Geek was teaching his class about Expected value. He gave the class an array of size **N**. He defined the beauty of any non-empty subset of the array as the maximum value present in the subset. He then asked the class to find the expected value of the beauty of a subset.

You, being the class topper, told Geek that to find the expected value of the beauty of a subset you just need to divide the sum of the beauty of all non-empty subsets by the total number of non-empty subsets.

Geek then asked you to find the sum of beauty of all non-empty subsets of the given array **A.** Since the answer can be huge, print it modulo 10^9+7.

Input Format:

- The first line contains an integer **T** - the number of test cases. Then the test cases follows:
- The first line contains two integers ***N.***
- The second line contains **N** space-separated positive integers $A_1,A_2,...,A_N$.

Output Format:

Print the answer followed by a newline character.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \leq N,T \leq 10^5$

$1 \leq A_i \leq 10^9$

The sum of N over all test cases does not exceed $2*10^5$.

Example:

Input:

2

2

6 10

1

7

Output:

26

7

Explanation:

Test Case 1: All possible non-empty subsets are {6},{10},{6,10} and their sum of beauty are 6+10+10 = 26.

Test Case 2: There is only one possible subset.

More Problem Solving

Array and Counting

Difficulty: **Hard**Accuracy: **82.67%**Submissions: **305+**Points: **8**

You are given an array of integers $A_1, A_2, ..., A_N$. Your task is to find the number of indices **x** *(1 ≤ x ≤ N)* such that there exists some index **y** *(1 ≤ y ≤ N)* and $A_x \,\&\, A_y = 0$.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integer ***N.***
- The second line contains **N** space-separated positive integers $A_1, A_2, \ldots, A_N$.

Output Format:

For each test case, print the answer to the problem followed by a newline character.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 10^4$

$1 \leq \mathbf{N} \leq 10^5$

$1 \leq \mathbf{A_i} \leq N$

The Sum of **N** over all the test cases is $\leq 2 * 10^5$

Example:

Sample Input:

```
1
3
1 2 3
```

Sample Output:

```
2
```

Explanation:

(1 & 2) = 0

Club Activity

Difficulty: **Medium**Accuracy: **96.41%**Submissions: **265+**Points: **4**

The club is open for N days. For each i day, Rohit will choose one of the following activities and do it on the *ii*-th day:

- A: Play Cricket and Gain a_i points of happiness.
- B: Play Tennis and Gain b_i points of happiness.
- C: Play Football Gain c_i points of happiness.

As Rohit gets bored easily, he cannot do the same activities for two or more consecutive days.

Find the maximum possible total points of happiness that Rohit gains.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integer ***N.***
- For Next N lines, each line contain $\mathbf{A_i, B_i , C_i.}$

Output Format:

For each test case, print the answer to the problem followed by a newline character.

Constraints:

$1 <= \mathbf{T} <= 10^4$

$1 <= \mathbf{N} <= 10^5$

$1 <= \mathbf{A_i} <= N, 1<=\mathbf{B_i} <=N, 1<= \mathbf{C_i} <=N$

Example:

Input:

```
3
10 40 70
20 50 80
30 60 90

Output:
210

Explanation:
If Taro does activities in the order
C, B, C, he will gain 70 + 50 + 90 = 210
points of happiness.
```

Jerry Pick

Difficulty: **Medium**Accuracy: **95.87%**Submissions: **244+**Points: **4**

There are *N* Cakes, numbered 1,2...N Initially, for each **i(1<=i<=N)**, Cake i has a_i **(1 <=a_i <=3)** pieces of Jerry on it.

Alex will perform the following operation repeatedly until all the pieces of sushi are eaten:

- Roll a die that shows the numbers **1,2 ... N** with equal probabilities, and let **i** be the outcome. If there are some pieces of Jerry on Cake i, eat one of them; if there is none, do nothing.

Find the expected number of times the operation is performed before all the pieces of Jerry are eaten.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integer ***N.***
- The Second denote N number $a_1, a_2, ...a_n$,

Output Format:

Print the expected number of times the operation is performed before all the pieces of Jerry are eaten. The output is considered correct when the relative difference is not greater than 10^{-9}.

Constraints:

$1 <= \mathbf{T} <= 1500$

$1 <= \mathbf{N} <= 300$

$1 <= \mathbf{A_i} <= 3$

Example:

Input:
1
3
1 1 1

Output:
5.500000000

Explanation:

The expected number of operations before the first piece of sushi eaten, is *1*. After that, the expected number of operations before the second sushi is eaten, is *1.51.5*. After that, the expected number of operations before the third sushi is eaten, is *3*. Thus, the expected total number of operations is *1 + 1.5 + 3 = 5.51+1.5+3=*5.500000000.

Stone game

Difficulty: **Medium**Accuracy: **88.75%**Submissions: **266+**Points: **4**

Given an array ***A*** consisting of ***N*** positive integers. Geek and Geekina will play the following game against each other. Initially, they have a pile consisting of ***K*** stones. The two players perform the following operation alternately, starting from Geek:

- Choose an element ***x*** in ***A***, and remove exactly ***x*** stones from the pile.

A player loses when he becomes unable to play. Assuming that both players play optimally, determine the winner.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers, **N** and **K**.
- The second line contains **N** space-separated positive integers $A_1, A_2, ..., A_N$.

Output Format:

Print the winner, i.e., "Geek" or "Geekina" without the quotes if both play optimally for each test case, followed by a newline character.

Note: The generated output is white space sensitive, do not add extra spaces or unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 100$

$1 \leq \mathbf{N} \leq 100$

$1 \leq \mathbf{K} \leq 10^5$

$1 \leq A_1 < A_2 < ..., < A_N. \leq K$

It is guaranteed that the sum of ***N*** *over all test cases does not exceed* ***100*** and t*he sum of* ***K*** *over all test cases does not exceed* 10^5.

Example:

Sample Input:

1

2 7

2 3

Sample Output:

Geek

Explanation:

Testcase 1:

Geek should remove two stones. Then, whatever Geekina does in her operation, Geek wins, as follows:

- If Geekina removes two stones, Geek can remove three stones to make Geekina unable to make a move.
- If Geekina removes three stones, Geek can remove two stones to make Geekina unable to make a move.

Grid With Blockage

Difficulty: **Medium** Accuracy: **94.43%** Submissions: **267+** Points: **4**

Given an N*M grid with obstacles i.e. if gird[i][j] = -1 it means there is an obstacle at cell (i, j) and it is not possible to walk through it.

Geek is standing at the top-left cell and wants to count the total number of distinct ways to reach the bottom-right cell if in one move he can either move to the right cell or down cell.

Input Format:

The first line of input consists of a single integer T, denoting the total number of test cases. The description of each test is as follows -

- The first line consists of two space-separated integers N and M denoting the number of rows and columns in the grid respectively.
- The following N lines consist of M integers each where the jth value in the ith row denotes the value of cell (i, j).

Output Format:

For each test case, print a single integer mod $10^9 + 7$ representing the number of distinct ways to reach the bottom-right cell.

Constraints:

$0 < T < 100$

$0 < N, M <= 1000$

grid[i][j] = {0, 1}

Example:

Input:

```
2
3 3
0 0 0
0 -1 0
0 0 0
2 2
0 -1
-1 0
```

Output:

```
2
0
```

Explanation:

Test Case 1 - The only possible two ways to reach cell (2, 2) from (0, 0) are -

1. (0, 0) -> (0, 1) -> (0, 2) -> (1, 2) -> (2, 2)
2. (0, 0) -> (1, 0) -> (2, 0) -> (2, 1) -> (2, 2)

Test Case 2 - There is no possible way to reach cell (1, 1) starting from cell (0, 0).

Knapsack 2

Difficulty: **Medium**Accuracy: **13.79%**Submissions: **87+**Points: **4**

There are ***N*** items, numbered ***1, 2,..., N***. For each ***i*** **(*1 ≤ i ≤ N*)**, Item ***i*** has a weight of w_i and a value of v_i.

Geek has decided to choose some of the ***N*** items and carry them home in a knapsack. The capacity of the knapsack is ***W***, which means that the sum of the weights of items taken must be at most ***W***.

Find the maximum possible sum of the values of items that Geek takes home.

Input Format:

- The first line contains an integer **T** - the number of test cases, then the test cases follows:
- The first contains two integers ***N,W.***
- Next, **N** line contains **N** pairs of integers- w_i v_i for **(*1 ≤ i ≤ N*).**

Output Format:

Print the answer followed by a newline character.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \le T \le 10^3$

$1 \le N \le 100$

$1 \le \mathbf{W} \le 10^9$

$1 \le \mathbf{w_i} \le \mathbf{W}$

$1 \le \mathbf{v_i} \le 10^3$

The sum of v_i over all test cases does not exceed 10^6.

Example:

Sample Input:

```
2
3 28
7 10
28 7
25 7
4 52
23 10
37 1
5 9
8 3
```

Sample Output:

```
10
22
```

Explanation:

For 1st test case, we can choose 1st
item.
For 2nd test case, we can choose 1st,
3rd and 4th item.

Trucks and Boxes

Difficulty: **Medium**Accuracy: **84.49%**Submissions: **298+**Points: **4**

Geek has to ship a few boxes from one city to another. There are **N** boxes in total. The weight of the ***i*-th** box is w_i and the amount geeks is paid for shipping the ***i*-th** box is c_i. Geek has a truck with a maximum weight carring capacity of **W** kgs. Find the maximum amount geek can earn on his first trip.

Constraints:

$1 \leq N \leq 500$

$1 \leq W \leq 10^5$

$1 \leq w_i \leq W$

$1 \leq c_i \leq 10^9$

The sum of N over all test cases does not exceed 500.

The sum of W over all test cases does not exceed 10^5.

Input Format:

- The first line contains an integer **T** - the number of test cases. Then the test cases follows:
- The first line contains two integers ***N,W.***
- The second line contains **N** space-separated positive integers $w_1, w_2, \ldots, w_N$.
- The third line contains **N** space-separated positive integers $c_1, c_2, \ldots, c_N$.

Output Format:

Print the answer followed by a newline character.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Example:

Input:

2

3 28
7 12 28
7 1 7
4 37
6 10 20 36
7 3 10 2

Output:

8
20

Explanation:

Test case 1:Geek can carry the 1st
and 2nd box with total weight 19 and
total earning 8.
Test case 2:Geek can carry the 1st,
2nd and 3rd box with total weight
36 and total earning 20.

Advanced DP

Range DP

Special String operation

Difficulty: **Hard**Accuracy: **91.94%**Submissions: **211+**Points: **8**

Geek was making **Special String**. For this he started with empty string then he did any one of the two following operations repeatedly:

- Add any lowercase letters at the end of string twice.
- If string is not empty, add any lowercase letters in between two adjacent elements in string twice.

Now, From the final **Special String,** Geek deletes some characters and gives to you.

Determine the minimum possible number of character deleted by geek.

Input Format:

The first line of the input contains a single integer **T** (1 ≤ T ≤ 10), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integers ***N* .**
- The next line contains a string.

Note: The sum of **N** over all the test cases will not exceed 500.

Output Format:

For each test case, print the minimum possible number of character deleted, followed by a newline character.

Example:

Sample Input:

1

4

aabc

Sample Output:

2

Explanation:

Testcase 1:

Special string would have been

aabccb

Maximum Profit

Difficulty: **Medium**Accuracy: **51.35%**Submissions: **37+**Points: **4**

You are given an array **A[ ]** of **N** distinct integers. You can choose two adjacent elements **A[i]** and **A[i+1]** and replace them with one element with a value equal to the greatest

common divisor of **A[i]** and **A[i+1]**, profit added in that case will be equal to **A[i] + A[i+1]**. You can do this operation until the size of the array becomes 1.

Print the **Maximum Profit** which we can get.

Input Format:
The first line of the input contains a single integer **T** ($1 \leq T \leq 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integers ***N*** **.**
- The following lines contains array of integers.

Output Format:
For each test case, print the **Maximum Profit** which we can get.

Example:

Sample Input:
T = 1
N = 3 A[] = {1, 2, 4}
Output: 9
Explanation: The optimal way is to merge 2nd and
3rd element so the new array will be {1,2} and
profit at this point will be 6. Again merging
1st and 2nd element, new array will be {1}, profit
at this point will be 6 + 3 = 9. Now we can't do
anymore moves.

Palindromic Partitioning

Difficulty: **Hard**Accuracy: **93.11%**Submissions: **217+**Points: **8**

Given a string **str**, a partitioning of the string is a palindrome partitioning if every sub-string of the partition is a palindrome. Determine the fewest cuts needed for palindrome partitioning of the given string.

Input Format:
The first line of the input contains a single integer **T** ($1 \le T \le 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains a string **Str .**

Output Format:
For each test case, print the minimum number of partitions required.

Example:

Sample Input:
T = 1
str = "ababbbabbababa"
Output: 3
Explaination: After 3 partitioning substrings
are "a", "babbbab", "b", "ababa".

Matrix Chain Multiplication

Difficulty: **Hard**Accuracy: **96.57%**Submissions: **211+**Points: **8**

Given a sequence of matrices, find the most efficient way to multiply these matrices together. The efficient way is the one that involves the least number of multiplications.

The dimensions of the matrices are given in an array **arr[]** of size **N** (such that N = number of matrices + 1) where the **i^{th}** matrix has the dimensions **(arr[i-1] x arr[i])**.

Input Format:
The first line of the input contains a single integer **T** ($1 \le T \le 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains an integers ***N*** **.**
- The following lines contains array of integers.

Output Format:

For each test case, print the minimum number of multiplication operations needed to be performed.

Example:

Sample Input:

T = 1

N = 5

arr = {40, 20, 30, 10, 30}

Output: 26000

Explaination: There are 4 matrices of dimension 40x20, 20x30, 30x10, 10x30. Say the matrices are named as A, B, C, D. Out of all possible combinations, the most efficient way is (A*(B*C))*D.

The number of operations are -

20*30*10 + 40*20*10 + 40*10*30 = 26000.

Bitmask DP

Find Optimally 2

Difficulty: **Hard**Accuracy: **91.19%**Submissions: **193+**Points: **8**

This is the hard version of this problem. In this version, we have larger constraints.

You are given **N** strings $S_0, S_1, .. S_{N-1}$. You have to answer **Q** independent queries, each query is described below:

- You are given a string **Query[i]** having all characters distinct. The task is to find the minimum index **x** such that at least one permutation of **Query[i]** is the **subsequence** of S_x. If there is no such index possible then print **-1**.

Note: Consider 0-based indexing and every string consists of the first **20** lowercase Latin characters.

*A **subsequence** is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements. The list of all subsequences for the word "**apple**" would be "a", "ap", "al", "ae", "app", "apl", "ape", "ale", "appl", "appe", "aple", "apple", "p", "pp", "pl", "pe", "ppl", "ppe", "ple", "pple", "l", "le", "e", "" (empty string).*

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- First-line consists of integers N and Q.
- Next, N line contains the string S_i
- Next, Q line contains the string $Query_i$

Output Format:

For each test case, For each query, print the answer followed by a newline character.

Note: The generated output is white space sensitive, do not add any extra spaces or unnecessary newline characters.

Constraints:

$1 \le T \le 10$

$1 \le N \le 10^6$

$1 \le Q \le 10^5$

$1 \le$ sum of the length of all strings $S[i] \le 3 * 10^6$

*It is guaranteed that the sum of **N** over all test cases does not exceed **$2 * 10^5$***

Example:

Sample Input:

1

3 1

bba
aabc
def
ad

Sample Output:
-1

TSP

Difficulty: **Medium**Accuracy: **32.16%**Submissions: **408+**Points: **4**

Given a matrix cost of size n where cost[i][j] denotes the cost of moving from city i to city j. Your task is to complete a tour from city 0 (0-based index) to all other cities such that you visit each city at most once and then at the end come back to city 0 at min cost.

Input Format:
The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains the integer N.
- Next, the N line contains N space-separated integers.

Output Format:
For each test case, print the answer to the problem followed by a new line character.
Note: The generated output is white space sensitive, do not add any extra spaces or unnecessary newline characters.

Constraints:
$1 \leq \mathbf{T} \leq 10$

$1 \le \mathbf{N} \le 20$

$1 \le \mathbf{cost[i][j]} \le 1000$

It is guaranteed that the sum of ***N*** *over all test cases does not exceed* ***20***

Example:

Sample Input:

1
2
0 111
112 0

Sample Output:

223

Explanation:

We can visit 0->1->0 and
cost = 111 + 112.

Subset Game

Difficulty: **Hard** Accuracy: **86.1%** Submissions: **216+** Points: **8**

Elena is the topper of the class. Once her teacher asked her a problem. He gave Elena an array of integers of length n. He calls a subset of array arr good if its product can be represented as a product of one or more distinct prime numbers. He asked her to find the number of different good subsets in arr modulo 109 + 7.
As a good friend of Elena help her to solve the problem.

Input format:

The first line contains an integer T - the number of test cases. Then the test cases follows:|
The first line of the input contains one integers N, denoting number of

elements in the array arr.the matrix
The next line contains n array elements, arr[i] (for 0<=i<n)

Output format:
For each test case, print the answer folllowed by newline character.

Constraints:
1 <= N <= 10^5
1< = arr[i] <= 30

Example :

Input:

2
4
1 2 3 4
3
2 2 3

Output:
6
5

Explanation:

1st - The good subsets are:

- [1,2]: product is 2, which is the product of distinct

prime 2.

- [1,2,3]: product is 6, which is the product of

distinct primes 2 and 3.

- [1,3]: product is 3, which is the product of distinct

prime 3.

- [2]: product is 2, which is the product of distinct

prime 2.

- [2,3]: product is 6, which is the product of distinct

primes 2 and 3.

- [3]: product is 3, which is the product of distinct

prime 3.

2nd - The good subsets are : {2}, {2}, {2, 3}, {2, 3}, {3}

Subsets with XOR value

Difficulty: **Medium**Accuracy: **47.16%**Submissions: **53+**Points: **4**

Given an array arr of N integers and an integer K, find the number of subsets of arr having XOR of elements as K.

Input format: The first line contains an integer T - the number of test cases. Then the test cases follows: The first line of the input contais two integers - length of the array - N & the given integer - K.

The next line contains n array elements, arr[i] (for all 0<=i<arr[l])

Output format:
For each test case, print the answer folllowed by newline character.

Constraints:
1<=T<=100
1<=N<=20
1<=K<=100
0<=arr[i]<=100

Example:
Input:
2
4 6
6 9 4 2
5 4
1 2 3 4 5

Output:
2
4

Explanation:
1st - The subsets are
{4,2} and {6}
2nd - The subsets are {1, 5},
{4}, {1, 2, 3, 4},
and {2, 3, 5}

Probability DP

Faulty Delivery

Difficulty: **Medium**Accuracy: **82.76%**Submissions: **29+**Points: **4**

Azamon delivery services, serves **N** orders by **N** users everyday, it is known that order of every user is **different** from order of every other user.

There can be some errors and order of some user can be delivered to other user.

Find the **probability** that **every user receives order of someone else.**

Assume that it is **equally likely** for an order to be received by any user.

Input

- First line of input contains an integer **T** number of testcases,
  then **T** testcases follow
- First line of each testcase contains an integer **N** number of orders to be served

Output

- For every **N** print the probability that every user receives order of someone else
- If probability is **P/Q** where **P** and **Q** are integers print the result as $P*Q^{-1} \bmod 10^9+7$

Constraints

- $1<=T<=10^3$
- $1<=N<=10^5$

Sample Input

```
2
3
10
```

Sample Output

```
333333336
906629471
```

Explanation

In first case there are **3** orders which can be distributed in **6** ways, there are only two scenarios possible such that everyone receives order from some other user [2,1,3] and [3,1,2] i_{th} element in these lists means the order i_{th} user received
So probability is **2/6** and $2*6^{-1} \bmod (10^9+7) = 333333336$
Therefore answer for first test is **333333336**

Prince's Shooting Game

Difficulty: **Hard**Accuracy: **79.26%**Submissions: **212+**Points: **8**

On a warm and sunny day, Prince Alexander decided to visit the shooting gallery in Central Park. He was not skilled at shooting and sought assistance in winning the main prize: a large pink plush panda. Luckily, Prince Alexander had you by his side to help him.

The shooting gallery was set on an infinite vertical plane, with a Cartesian coordinate system. Targets would appear at various points on this plane. Each target was defined by its coordinates (x_i, y_i), the time of its appearance (t_i), and a number (p_i) representing the probability of Prince Alexander hitting that target if he aimed at it.

Targets would appear and disappear instantly, meaning Prince Alexander could only hit a target if his gun sight was aimed at (x_i, y_i) precisely at the time t_i. The gun sight moved at a constant speed of 1 unit per time interval. Prince Alexander possessed complete knowledge about all the targets in advance, thanks to his royal status. He aimed to play optimally, maximizing the expected value of the number of targets hit. He could choose to aim at any target starting from time 0.

Input Format:

First line contains an integer T denoting the number of test cases. Each test case consists of first line as an integer n denoting the amount of targets in the shooting gallery. Next n line contains 4 space separated values - x, y, t and p.

Output Format:

For each test case, print the maximized expected value.

Constraints:

$1<=n<=1000$

$-1000<=x_i, y_i<=1000$

$0<=t_i<=10^9$

$0<=p_i<=1$

Example:

Input:

2

1

0 0 0 0.5

2

0 0 0 0.6

5 0 5 0.7

Output:

0.5

1.3

Digit DP

Prime Sum Numbers - I

Difficulty: **Hard**Accuracy: **79.14%**Submissions: **211+**Points: **8**

Count number of integers in range **[L,R]** whose sum of digits is a prime number.

A prime number is a number which has only two factors **1** and the number

itself. **1** is neither prime nor composite.

Input

- First line contains **T** ,number of testcases, **T** testcases follow
- First line of each testcase contains two space separated integers **L** and **R**

Output

- For each testcase print a single integer in a newline ,count of integers in **[L,R]** with prime sum of digits.
- As answer can be large print it modulo **1000000007**

Constraints

- **1<=T<=10**
- **1<=L<=R<10^{200}**

Sample Input

```
2
2 20
98 9999
```

Sample Output

```
9
2974
```

Explanation

In range **[2,20]** there are **9** integers sum of digits of which is a prime number **2, 3, 5, 7, 11, 12, 14, 16** and **20** respectively.

In range **[98, 9999]** there are **2974** integers

Prime Sum Numbers - II

Difficulty: **Hard**Accuracy: **87.26%**Submissions: **188+**Points: **8**

Find K_{th} integer in range **[L,R]** whose sum of digits is a prime number.
A prime number is a number which has only two factors **1** and the number itself. **1** is neither prime nor composite.

Input

- First line contains **T** ,number of testcases, **T** testcases follow
- First line of each testcase contains three space separated integers **L, R** and **K**

Output

- For each testcase print a single integer in a newline ,K_{th} integer in **[L,R]** with prime sum of digits.
- If there are less than **K** integers with prime sum of digits in **[L,R]** print **-1**

Constraints

- **1<=T<=10**
- **1<=L<=R<10^{18}**

Sample Input

```
3
2 20 3
98 9999 2
2 2 3
```

Sample Output

```
5
101
-1
```

Explanation

For first testcase , in range **[2,20]** there are **9** integers sum of digits of which is a prime number **2, 3, 5, 7, 11, 12, 14, 16** and **20** respectively, so the **3rd** integer among

them is **5**, therefore answer for this test is **5**
In second testcase , range **[98, 9999]** has **2974** integers, first is **98** itself with sum **17** and second is **101** with sum **2**, so answer is **2** for the second case
In third case, there is no answer as range **[2,2]** contains only **1** integer and **K** is **3** therefore we print **-1**

Magical Number

Difficulty: **Hard**Accuracy: **95.96%**Submissions: **175+**Points: **8**

The problem requires determining the count of "d-magic" numbers within a specified range [a, b] that are also divisible by m. A number is considered d-magic if the digit d appears exclusively at even positions in its decimal representation and does not occur elsewhere.

For example, consider the scenario where d = 7. In this case, the numbers 1727374, 17, and 1 would be classified as 7-magic since the digit 7 is only present at even positions in their decimal representations and is absent elsewhere. On the other hand, numbers like 77, 7, 123, 34, and 71 would not qualify as 7-magic numbers. Additionally, a single-digit number containing d, such as 7, would be categorized as 0-magic, while numbers like 123, 34, and 71 would be 2-magic, 4-magic, and 1-magic, respectively.

The objective is to compute the number of d-magic numbers within the range [a, b] that are divisible by m. Since the answer might be quite large, it is required to find the remainder when dividing the result by 10^9 + 7, effectively calculating the value modulo 10^9 + 7.

Constraints:

1<=m<=2000

0<=d<=9

a, b are positive integers

Example:

Input:

3

2 6

10 99

2 0

1

9

19 7

1000

9999

Output:

8

4

6

Explanation:

The numbers from answer of the first example are 16, 26, 36, 46, 56, 76, 86 and 96.

The numbers from the answer of the second example are 2, 4, 6 and 8.

The numbers from the answer of the third example are 1767, 2717, 5757, 6707, 8797 and 9747.

Graphs – CP

Walkthrough & BFS

Delivery In city

Difficulty: **Easy**Accuracy: **88.17%**Submissions: **330+**Points: **2**

You have been given **N** cities and **M** pairs of cities are connected in a one-way path.

Your friend Ram has a supermarket in city 1. He needs your help to find out if he can offer delivery services in all the cities.

Help Ram and print **"Yes"** if your friend can offer delivery service in all cities otherwise print **"No".**

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains three integers **N,** M.
- Next **M** lines contain two integer **X & Y. (**path from **X** to **Y**).

Output format:

- Print **T** lines, the i_{th} should be the answer for the i_{th} test case.

Constraints:

- 1 <= **T** <= 10^2

- $1 <= N, M <= 10^4$

Example Input:

```
2
5 4
1 2
1 3
3 4
3 5
5 3
1 2
1 3
4 5
```

Example Output:

```
Yes
No
```

Explanation:

For the first test case, If we connect all paths, we will get the below graph,

So, Ram can reach all cities from city 1.

Covid Containment Zones

Difficulty: **Medium**Accuracy: **80.31%**Submissions: **183+**Points: **4**

You are given an **N*M** 2D grid contains '**0**' and '**1**'. Where '**1**' refers to land and '**0**' refers to water.

An island is constructed by linking neighboring lands horizontally or vertically and is surrounded by water. You can presume that the grid's four edges are all surrounded by water.

We call an island a containment zone if there are more than **K** land with covid positive patients.

Also, you are given the location of **Q** lands with covid positive patients (in one index format).

You need to find the number of Containment zones (Islands).

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains four integers **N, M, Q, K.**
- Next **N** lines contains **M** integers ('0' or '1').
- Next, **Q** lines contain index of lands with covid positive patients

Output format:

- Print **T** lines, the $\mathbf{i_{th}}$ should be the answer for the $\mathbf{i_{th}}$ test case.

Constraints:

- 1 <= **T** <= 10^2
- 1 <= **N, M, K** <= 10^4
- 1 <= **Q** <= **N*M**
- 1 <= **X** <= N
- 1 <= **Y** <= M

Example Input:

```
2
5 5 5 1
1 1 1 0 0
1 0 1 0 0
0 1 0 1 1
1 1 0 0 1
1 0 0 0 1
0 2
1 0
2 4
3 4
```

```
4 0
3 3 3 3
1 1 1
0 1 0
1 1 1
0 0
0 2
2 2
```

Example Output:

```
2
0
```

Explanation:

For first test case,
If we mark red all lands with covid positive patients,
we will get the following grid,

1 1 1 0 0
1 0 **1** 0 0
0 1 0 **1 1**
1 1 0 0 **1**
1 0 0 0 **1**

Here, only two islands has more than k land with covid positive patients.

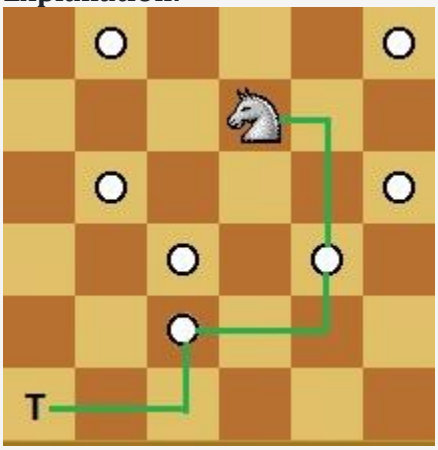
Knight Walk

Difficulty: **Medium**Accuracy: **37.61%**Submissions: **45K+**Points: **4**

Given a square chessboard, the initial position of Knight and position of a target. Find out the minimum steps a Knight will take to reach the target position.If it cannot reach the target position return -1.

Note:
The initial and the target position co-ordinates of Knight have been given accoring to 1-base indexing.

Example 1:

Input:
N=6
knightPos[ ] = {4, 5}
targetPos[ ] = {1, 1}
Output:
3
Explanation:


Knight takes 3 step to reach from
(4, 5) to (1, 1):
(4, 5) -> (5, 3) -> (3, 2) -> (1, 1).

Example 2:

Input:
N=8
knightPos[ ] = {7, 7}
targetPos[ ] = {1, 5}
Output:

4

Explanation:
Knight takes 4 steps to reach from
(7, 7) to (1, 5):
(4, 5) -> (6, 5) -> (5, 3) -> (7, 2) -> (1, 5).

Your Task:
You don't need to read input or print anything. Your task is to complete the function **minStepToReachTarget()** which takes the inital position of Knight (KnightPos), the target position of Knight (TargetPos) and the size of the chess board (N) as an input parameters and returns the minimum number of steps required by the knight to reach from its current position to the given target position.If it cannot reach the target position return -1.

Expected Time Complexity: $O(N^2)$.
Expected Auxiliary Space: $O(N^2)$.

Constraints:
1 <= N <= 1000
1 <= Knight_pos(X, Y), Targer_pos(X, Y) <= N

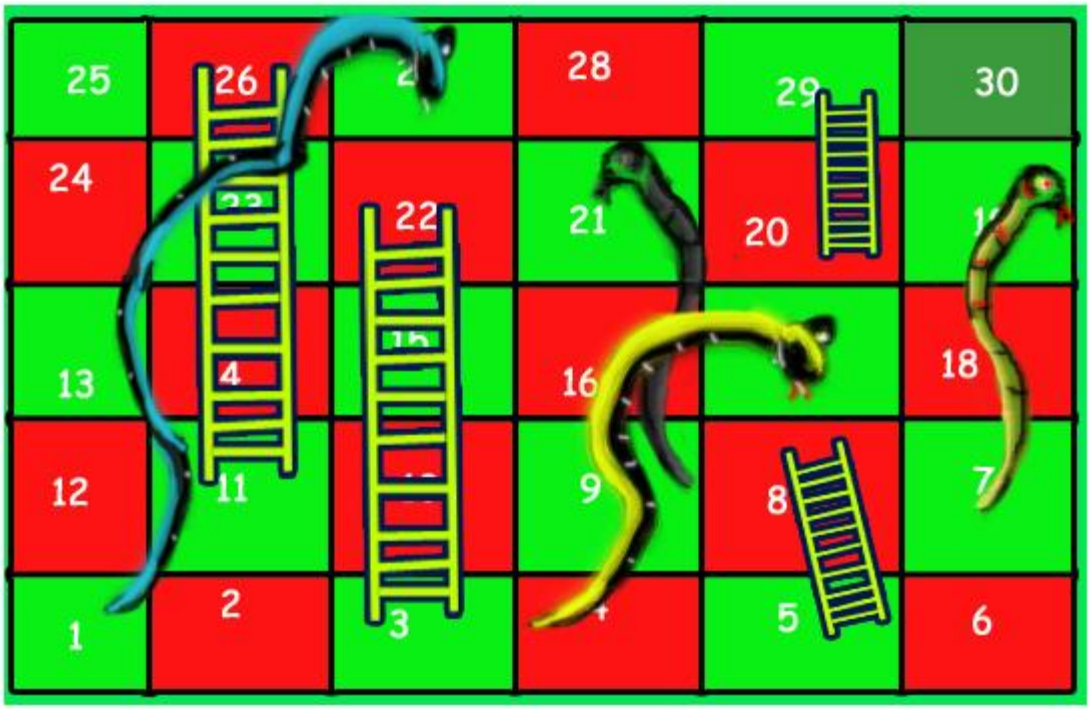
Snake and Ladder Problem

Difficulty: **Medium**Accuracy: **34.65%**Submissions: **53K+**Points: **4**

Given a **5x6** snakes and ladders board, find the minimum number of dice throws required to reach the destination or last cell (**30th** cell) from the source (1st cell).

You are given an integer **N** denoting the total number of snakes and ladders and an array **arr[]** of **2*N** size where **2*i** and **(2*i + 1)th** values denote the starting and ending point respectively of **ith** snake or ladder. The board looks like the following.

Note: Assume that you have complete control over the 6 sided dice. No ladder starts from 1st cell.



Example 1:

Input:
N = 8
arr[] = {3, 22, 5, 8, 11, 26, 20, 29,
17, 4, 19, 7, 27, 1, 21, 9}
Output: 3
Explanation:
The given board is the board shown in the figure. For the above board output will be 3.
a) For 1st throw get a 2.

b) For 2nd throw get a 6.
c) For 3rd throw get a 2.

Your Task:
You do not need to read input or print anything. Your task is to complete the function **minThrow()** which takes **N** and **arr** as input parameters and returns the minimum number of throws required to reach the end of the game.

Expected Time Complexity: $O(N)$
Expected Auxiliary Space: $O(N)$

Constraints:
$1 \leq N \leq 10$
$1 \leq arr[i] \leq 30$

Level of Nodes

Difficulty: **Medium**Accuracy: **50.95%**Submissions: **57K+**Points: **4**

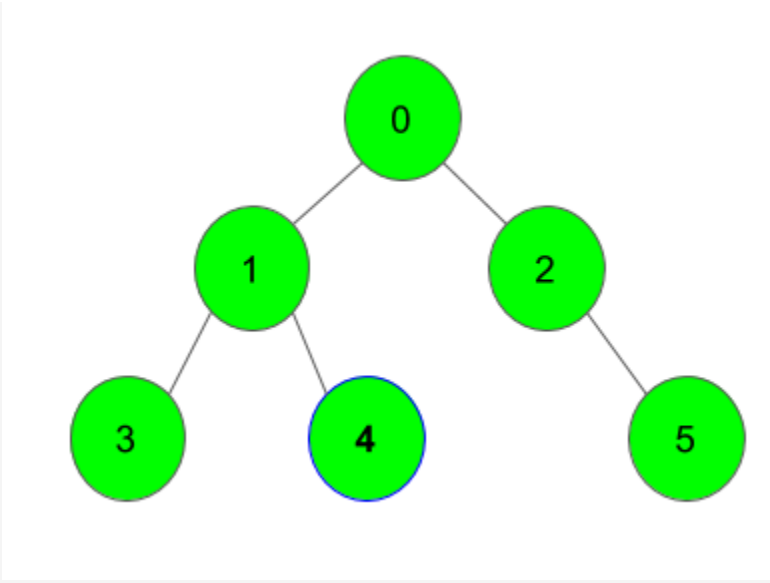
Given an integer **x** and an undirected acyclic graph with **v nodes**, labeled from **0** to **v-1**, and **e** edges, return the **level** of node labeled as **x.**

Level is the **minimum** number of edges you must travel from the node 0 to some target.

If there doesn't exists such a node that is labeled as **x, return -1**.

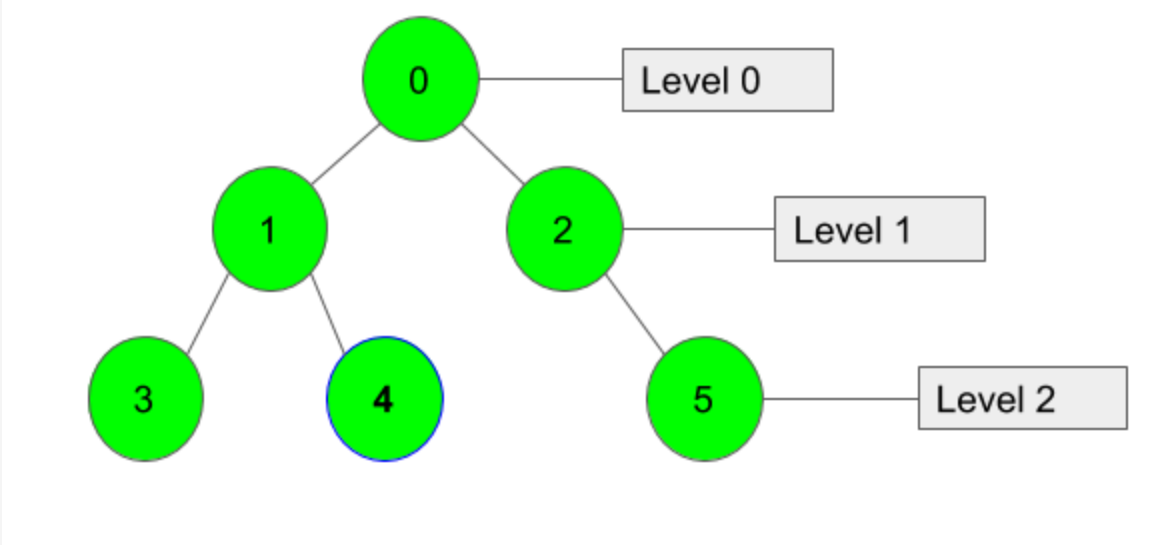
Examples :

Input: x = 4



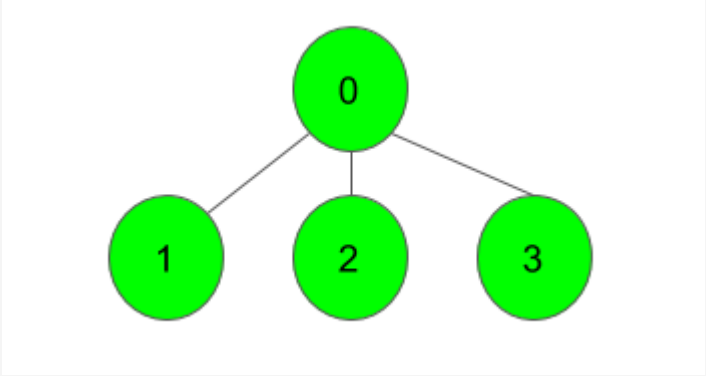
Output: 2

Explanation:



We can clearly see that Node 4 lies at Level 2.

Input: x = 1



Output: 1

Explanation: Node 1 lies at level 1, immediate after Node 0.

Expected Time Complexity: O(v)

Expected Auxiliary Space: O(v)

Constraints:

$2 \leq v \leq 10^4$

$1 \leq e \leq 10^4$

$0 \leq adj_{i,j} < v$

$1 \leq x < v$

Note : Graph doesn't contain multiple edges and self loops.

DFS

Number of connected components

Difficulty: **Easy** Accuracy: **87.37%** Submissions: **301+** Points: **2**

You are given an undirected graph. You have to find the number of connected componenets in the given graph.

Input Format:

The first line of the input contains a single integer **T** ($1 \leq T \leq 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers ***N*** $(1 \le N \le 10^5)$ *and* ***M*** $(1 \le M \le 10^5)$.
- The following ***M*** lines contain 2 space separated integers **u** $(1 \le u \le N)$ and **v** $(1 \le v \le N)$ denoting
  denoting that there is an edge between node **u** and node **v**.

Note: Same edge will not be repeated in the input. No self-loops will be there in the input.The sum of **N** and **M** over all test cases doesn't exceed 10^5. The generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

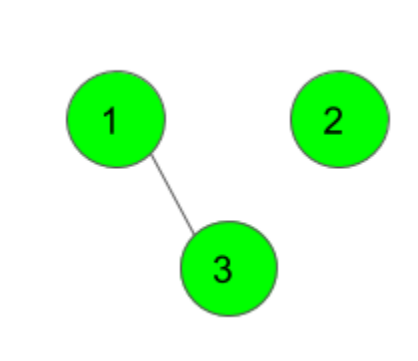
Output Format:

For each test case, for each test case , print the number of connected components followed by a newline character.

Example 1:

Sample Input:

```
1
3 1
1 3
```



Sample Output:

```
2
```

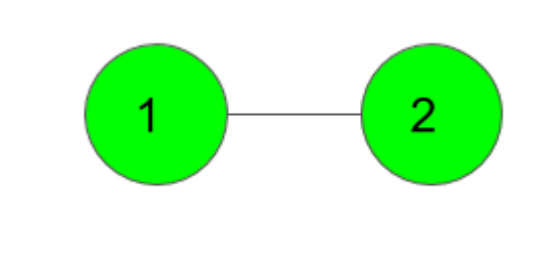
Explanation:

Testcase 1:
given graph has 3 nodes. One component
has node 1 and node 3.Another component
has node 2. Hence, we have 2 components
in total.

Example 2:

Sample Input:

1
2 1
1 2



Sample Output:

1

Explanation:

Testcase 1:
given graph has 2 nodes. All of them
are connected and hence, we got 1 connected
component.

Constraints:

$1 \leq \mathbf{T} \leq 10^4$
$1 \leq \mathbf{N,M} \leq 10^5$

Problem Suggestions

Difficulty: **Medium**Accuracy: **62.86%**Submissions: **35+**Points: **4**

You have been given a tree of **N** problems at Geeksforgeeks. Every problem has a difficulty level **D[i]**, You can solve this problem if it is less difficult than your strength **S**, otherwise, you can skip almost **K** problems.

For each problem, there are some suggestions for the next problem. You need to pick any of them as the next problem, which isn't visited before. If there isn't any suggestion for the next problem you can leave the problem tree and this will be called an exit point.

You have to find the number of exit points in the given problem tree.

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains three integers **N, S, K.**
- The next line contains **N** integers (difficulty of each problem in range 1 to 10^5)
- Next **N** lines contain one integer **Q**, number of suggestions of $\mathbf{i_{th}}$ problem, and one array of length **Q** containing its suggestions (index of problems in 1 index format).

Output format:

- Print **T** lines, the $\mathbf{i_{th}}$ should be the answer for the $\mathbf{i_{th}}$ test case.

Constraints:

- 1 <= **T** <= 10^2
- 1 <= **N, S, K, Q** <= 10^5

Example Input:

```
1
5 10 1
10 11 7 8 9
3 2 3 4
```

```
1 5
0
0
0
```

Example Output:

```
2
```

Explanation:

Delivery In city

Difficulty: **Easy**Accuracy: **88.17%**Submissions: **330+**Points: **2**

You have been given **N** cities and **M** pairs of cities are connected in a one-way path.

Your friend Ram has a supermarket in city 1. He needs your help to find out if he can offer delivery services in all the cities.

Help Ram and print "**Yes**" if your friend can offer delivery service in all cities otherwise print "**No**".

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains three integers **N,** M**.**
- Next **M** lines contain two integer **X & Y. (**path from **X** to **Y**).

Output format:

- Print **T** lines, the $\mathbf{i_{th}}$ should be the answer for the $\mathbf{i_{th}}$ test case.

Constraints:

- $1 <= \mathbf{T} <= 10^2$
- $1 <= \mathbf{N, M} <= 10^4$

Example Input:

```
2
5 4
1 2
1 3
3 4
3 5
5 3
1 2
1 3
4 5
```

Example Output:

```
Yes
No
```

Explanation:

For the first test case, If we connect all paths, we will get the below graph,

So, Ram can reach all cities from city 1.

Covid Containment Zones

Difficulty: **Medium**Accuracy: **80.31%**Submissions: **183+**Points: **4**

You are given an **N*M** 2D grid contains '**0**' and '**1**'. Where '**1**' refers to land and '**0**' refers to water.

An island is constructed by linking neighboring lands horizontally or vertically and is surrounded by water. You can presume that the grid's four edges are all surrounded by water.

We call an island a containment zone if there are more than **K** land with covid positive patients.

Also, you are given the location of **Q** lands with covid positive patients (in one index format).

You need to find the number of Containment zones (Islands).

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains four integers **N, M, Q, K.**
- Next **N** lines contains **M** integers ('0' or '1').
- Next, **Q** lines contain index of lands with covid positive patients

Output format:

- Print **T** lines, the i_{th} should be the answer for the i_{th} test case.

Constraints:

- 1 <= **T** <= 10^2
- 1 <= **N, M, K** <= 10^4
- 1 <= **Q** <= **N*M**
- 1 <= **X** <= N
- 1 <= **Y** <= M

Example Input:

```
2
5 5 5 1
1 1 1 0 0
1 0 1 0 0
0 1 0 1 1
1 1 0 0 1
1 0 0 0 1
0 2
1 0
2 4
3 4
4 0
3 3 3 3
```

```
1 1 1
0 1 0
1 1 1
0 0
0 2
2 2
```

Example Output:

```
2
0
```

Explanation:

For first test case,
If we mark red all lands with covid positive patients,
we will get the following grid,

1 1 1 0 0
1 0 **1** 0 0
0 1 0 **1 1**
1 1 0 0 **1**
1 0 0 0 **1**

Here, only two islands has more than k land with covid positive patients.

Deliverable Cities

Difficulty: **Medium**Accuracy: **78.39%**Submissions: **162+**Points: **4**

You have been given **N** cities and **M** pairs of cities are connected in a one-way path.

Your friend Ram has a supermarket in town 1. Ram wants to launch a delivery service of its own. We call a city deliverable if the delivery partner can return to

the supermarket from this city. He needs your help to find out all deliverable cities.

NOTE: All inputs are given in one index format.

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains three integers **N,** M**.**
- Next **M** lines contain two integer **X & Y. (**path from **X** to **Y**).

Output format:

- The first line of i-th test case output contains the number of deliverable cities (**Q**).
- The second line of i-th test case containing **Q** integers, index of all deliverable cities.

Constraints:

- 1 <= **T** <= 10^2
- 1 <= **N , M** <= 10^4
- 1 <= **X , Y** <= **N**

Example Input:

```
2
4 5
1 2
2 3
2 4
4 1
4 3
3 3
1 2
2 3
3 1
```

Example Output:

```
3
1 2 4
3
1 2 3
```

Explanation:

For the first test case, If we connect all paths, we will get that delivery partner can return to the supermarket from this city 1, 2, 4.

Chocolate Party

Difficulty: **Medium**Accuracy: **30.94%**Submissions: **84+**Points: **4**

Abhi has an array **arr[]** of **N** chocolate bags
contains **arr[i]** chocolates and **K** friends. He invited them to a party. Now he doesn't want any of his friends to come back from the party angry.
So he decided that he would share N chocolate bags among his friends so that each of his friends would get an equal number of chocolates.
Now print "**Yes**" if his all friends will be happy, otherwise print "**No**".

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains two integers **N, K.**
- The next line contains **N** integer **arr[i]**.

Output format:

- Print **T** lines, the $\mathbf{i_{th}}$ should be the answer for the $\mathbf{i_{th}}$ test case.

Constraints:

- $1 <= \mathbf{T} <= 10^4$
- $1 <= \mathbf{K} <= \mathbf{N} <= 20$

- $0 <= arr[i] <= 10^3$

Example Input:

```
2
5 3
1 2 2 3 4
4 3
1 2 3 4
```

Example Output:

```
Yes
No
```

Explanation:

For the first test case, It's possible to divide all chocolate bags among 3 friends : [{ 1, 3 }, { 2, 2 }, { 4 }]

Topological Sort

Problem Solving

Difficulty: **Medium** Accuracy: **86.67%** Submissions: **285+** Points: **4**

You're given **N** problems and an array arr of **M** pairings with arr[i] = { **X** , **Y** }, indicating that problem **Y** must be solved before problem **X**.

If you are able to solve all of the problems, print "**YES**" otherwise, print "**NO**".

Note: All inputs are given in 1-index format.

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains two integers **N, M.**
- The next **M** lines contain a pair { **X , Y** }.

Output format:

- Print **T** lines, the i_{th} should be the answer for the i_{th} test case.

Constraints:

- $1 <= T <= 10^2$
- $1 <= N, M <= 10^3$
- $1 <= X, Y <= N$

Example Input:

```
2
2 1
2 1
3 4
1 2
1 3
2 1
2 3
```

Example Output:

```
Yes
No
```

Explanation:

Geek and His Degree

Difficulty: **Medium**Accuracy: **30.65%**Submissions: **75+**Points: **4**

Geek needs to complete **N** courses to get his degree. Each course has specific prerequisites i.e. If course X has pre-requisite Y then it is mandatory for Geek to complete Y before enrolling for X.
In Geekland, they follow a unique semester system i.e. Geek can enroll in any number of courses such that Geek meets pre-requisites for that course.

Now you need to determine the minimum number of semesters Geek will require to get his degree. Also, print the print the order of courses. More details about this are in the output format section.

Input Format:
The first line of input contains a single integer i.e. T, the number of test cases. The description of each test case is as follows -

- The first of each test case consists of two space-separated integers i.e. N and M denoting the total number of courses (1 - N) and the number of prerequisites.
- The following M lines consist of two space-separated integers X and Y denoting that course Y has X as its pre-requisite.

Output Format:
For each test case -

- Print IMPOSSIBLE it is not possible for Geek to get his degree.
- Otherwise, on the first line print, the total number of semesters required is such that Geek can get his degree (say N). The following N lines should contain the courses (in non-decreasing order) in which Geek should enroll to get his degree.

Constraints:

$0 < T < 100$
$0 < N <= 10^5$
$0 < M <= 10^5$
$1 <= X, Y <= N$
Sum of N over all test cases will not exceed $2 * 10^5$

Example:
Input:
2
4 4

1 2
1 3
2 4
3 4
2 2
1 2
2 1

Output:

3
1
2 3
4
IMPOSSIBLE

Explanation:

Test Case 1 - Geek can pursue course 1 in his first semester then he can pursue courses 2 and 3 in the second semester. Then in the third semester, he can pursue course 4. It can be shown that there is no other possible way by which Geek can get his degree in less than 3 semesters.

Test Case 2 - Since there are only two courses and both of them are prerequisites of each other. Geek can't get his degree.

Sequence Uniqueness

Difficulty: **Hard** Accuracy: **94.05%** Submissions: **203+** Points: **8**

Michael, a brilliant computer scientist from Algoville, came across an intriguing problem. He was given a **sequence A** of length N, which was a permutation of integers from 1 to N. Additionally, he was provided with **M** pairs of indices (X_i, Y_i) with the property **$A[X_i] < A[Y_i]$.**

Michael wanted to determine if the sequence A could be uniquely determined. Can you help him solve this problem?

Input Format:

The first line contain an integer T - the number of test cases. Then the test cases follows:

The first line contains two integers **N** and **M** denoting length of sequence A and number of pairs.
Next line contains N space separated integers denoting **A[i]** i.e sequence A.

Output Format:

Print **"Yes"** without quotes if the sequence A can be uniquely determined, along with the values of A.
If A cannot be uniquely determined, print **"No"** without quotes.

Constraints:

$1 \leq T \leq 1000$
$2 \leq N \leq 1 * 10^5$
$1 \leq M \leq 1 * 10^5$
$1 \leq Xi, Yi \leq N$
There is an A consistent with the input.
All pairs are **distinct**.
The sum of N and M over all test cases in a test file does not exceed $2 * 10^5$.

Example:

Input:

```
2
3 2
2 3
3 1

3 2
3 1
```

3 2

Output:

Yes

3 1 2

No

Explanation:

For test case 1, We can uniquely determine that A=(3,1,2).

For test case 2, Two sequences (2,3,1) and (3,2,1) can be A.

Lowest Common Ancestor (LCA)

Weighted cycle queries

Difficulty: **Medium**Accuracy: **42.87%**Submissions: **49+**Points: **4**

Given a connected, undirected and weighted graph **G** with **N** nodes and **N-1** edges, with no self loops and multiple edges.

You have to answer **Q** independent queries of form **u, v, w**

For every query print the **length of the longest cycle** formed if we add an undirected edge with weight **w** between nodes **u** and **v**

NOTE : All queries are independent, no edges added in some query is carried forward to other query.

Input

- First line of input contains an integer **T** , number of testcases
- First line of each test contains one integer **N**
- Each of the next **N-1** lines contain three space separated integers **u, v** and **w** denoting there is an edge in the graph **G** between **u** and **v** with weight **w**

- Next line contains **Q** number of queries, then **Q** queries follow
- Each of the next **Q** lines contain 3 space separated integers **u,v** and **w** denoting parameters for the query

Output

- For every query print in a new line length of the longest cycle in the graph if given edge is added.

Constraints

- **1<=T<=10**
- **1<=N,Q<=10^5**
- **1<=u,v<=N**
- **1<=w<=10^9**

Sample Input

```
1
6
1 3 1
3 4 1
1 2 1
2 5 2
2 6 1
3
6 5 3
3 2 2
1 2 2
```

Sample Output

```
6
4
3
```

Explanation

In first query an edge of weight **3** is added between **6** and **5** it creates a cycle 2---

6---5-----2 with weight **6** so answer is **6** for the first query
In second query a cycle 1---3---2----1 is created with weight **4**, therefore answer for second query is **4**

LCA Pairs

Difficulty: **Medium**Accuracy: **74.29%**Submissions: **35+**Points: **4**

You are given a connected tree of **N** nodes numbered from **0** to **N-1** rooted at the **0th** node, where **p[i]** is the parent of **the ith** node and **p[0] = -1** as the 0th node is the root.

Calculate the numbers of pairs of nodes (a,b) $0 \leq a < N$, $a \leq b < N$ such that LCA(a,b) = x , for each x where $0 \leq x < N$.

The LCA or lowest common ancestor between two nodes a and b is defined as the lowest node in tree that has both a and b as descendants (where we allow a node to be a descendant of itself).

Input:

The first line contains the number of test cases. ($1<=T<=10^4$).

- The first line contains N, the number of vertices.
- The next line contains N space-seperated integers denoting the parents of each node.

Output:

Print N integers denoting the number of pairs of LCA as asked for each x from 0 to N-1.

Example 1:

Input:
2

3

-1 0 0

5

-1 0 0 1 1

Output:

4 1 1

8 4 1 1 1

Explanation:

Test Case 1:

The structure of tree is :

```
  0
 / \
1   2
```

For LCA = 0, we have 4 pairs (0,0), (0,1),
(0,2), (1,2)

For LCA = 1, we have only 1 pair (1,1)

For LCA = 2, we have only 1 pair (2,2)

Test Case 2:

The structure of tree is :

```
    0
   / \
  1   2
 / \
3   4
```

For LCA = 0, we have 8 pairs (0,0), (0,1),
(0,2),(0,3),(0,4),(1,2),(2,3),(2,4)

For LCA = 1 , we have 4 pairs (1,1), (1,3),
(1,4),(3,4)

FOR LCA = 2, we have only 1 pair (2,2),

FOR LCA = 3, we have only 1 pair (3,3)
For LCA = 4, we have only 1 pair (4,4)

Constraints:
$1 \leq N \leq 10^5$
$0 \leq p[i] < N$ where $0 < i < N$
p[0]=-1

Bridges and Articulation Point

Critical Roads

Difficulty: **Medium** Accuracy: **90.17%** Submissions: **243+** Points: **4**

Geekland is made up of **N** cities linked by **M** bidirectional highways. A city has no roads connecting it to itself, and each pair **{ X, Y }** of cities has just one route connecting them.

There may be more than one road connecting two cities.

A critical road is one that, if closed, would prohibit certain cities from communicating with one another.

You need to find the number of critical roads in Geekland.

Note: All inputs are given in 1-index format.

Input format:

- The first line contains one integer T (number of test cases).
- The first line of each test case contains two integers N, M.
- The next M lines contain a pair { X, Y }.

Output format:

- Print T lines, the ith should be the answer for the ith test case.

Constraints:

- $1 <= \mathbf{T} <= 10^2$
- $1 <= \mathbf{N}, \mathbf{M} <= 10^3$
- $1 <= \mathbf{X}, \mathbf{Y} <= \mathbf{N}$

Example Input:

```
2
5 5
1 2
1 3
1 4
2 3
3 5
3 3
1 2
1 3
2 3
```

Example Output:

```
2
0
```

Critical Cities

Difficulty: **Medium** Accuracy: **92.11%** Submissions: **231+** Points: **4**

Geekland is made up of **N** cities linked by **M** bidirectional highways. A city has no roads connecting it to itself, and each pair { **X**, **Y** } of cities has just one route connecting them.

There may be more than one road connecting two cities.

A critical city is one that, if locked down, would prohibit certain cities from communicating with one another.

You need to find all critical cities of Geekland in sorted order.

Note: All inputs are given in 1-index format.

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains two integers **N, M.**
- The next **M** lines contain a pair { **X, Y** }.

Output format:

- Print **T** lines, the $\mathbf{i_{th}}$ line should contain **K,** the number of critical cities and **K** integers, index of each critical city.

Constraints:

- 1 <= **T** <= 10^2
- 1 <= **N, M** <= 10^3
- 1 <= **X , Y** <= **N**

Example Input:

```
2
5 5
1 2
1 3
1 4
2 3
3 5
3 3
1 2
1 3
2 3
```

Example Output:

```
2 1 3
0
```

Euler path and circuit

Check whether Graph is Euler or not

Difficulty: **Easy**Accuracy: **83.04%**Submissions: **289+**Points: **2**

You are given an **undirected** graph. You have to find wheather the given graph is euler graph, semi - euler graph or none of them.

Eulerian Path is a path in graph that visits every edge exactly once. Eulerian Circuit is an Eulerian Path which starts and ends on the same vertex. A graph is called Eulerian if it has an Eulerian Cycle and called Semi-Eulerian if it has an Eulerian Path.

Input Format:

The first line contains **T**, denoting the number of test cases.

The first line of each test case contains 2 integers **V** and **E**, representing the vertices and edges of graph.

The following **E** lines of each test case contains two integers **u** and **v**, representing an edge between the nodes **u** and **v**.

Output Format:

You have to print **"2"(without quotes)** if the given graph is **euler** graph, print **"1"(without quotes)** if the given graph is **semi euler** and print **"0"(without quotes)** if the graph is neither euler nor semi euler.

Constraints:

- $1 <= T <= 10$
- $1 <= V <= 10^5$
- $0 <= E <= 10^5$
- $0 <= u,v < V$

Example 1:

Input:

```
2
3 3
0 1
1 2
0 2
3 2
0 1
0 2
```

Output:

```
2
1
```

Example 2:

Input:

```
1
6 5
0 1
0 2
1 3
2 3
4 5
```

Output:

```
0
```

Unlocking Mystery Box

Difficulty: **Medium**Accuracy: **78.79%**Submissions: **297+**Points: **4**

Geek is given a mystery box, and he is very curious to open it, but to open it he has to solve a puzzle. In a puzzle, he is given **N** words and he has to arrange these **N** words so that every word begins with the same letter as the previous word ends. For example, suppose he is provided with 3 words [**hbc**, **def**, **fgh].** Now he can arrange the words in this way **[def, fgh, hbc]** to satisfy the required condition.

Your task is to determine whether Geek can solve the puzzle or not.
Print **"Possible"(without quotes)** if Geek can solve the puzzle
and **"Impossible"(without quotes)** otherwise.

Input Format:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains **N**, representing the number of words.

The following **N** lines of each test case contains a string **s**, representing the word.

Output Format:

Print **"Possible"**(without qoutes) if Geek can solve the puzzle
and **"Impossible"**(without qoutes) otherwise.

Constraints:

- $1 <= T <= 10^3$
- $1 <= N <= 10^5$
- $1 <= s.length <= 10^2$

Example:

Input:

```
2
```

```
4
abc
ghi
cde
efg
2
geeksforgeeks
geeksforgeeks
```

Output:

```
Possible
Impossible
```

Explanation:

Test case 1:
Geek can arrange the words in this order:
[abc, cde, efg, ghi].

Bipartite Graph

Geek Classes
Difficulty: **Medium**Accuracy: **92.11%**Submissions: **254+**Points: **4**

In Geek school, there are **N** students and **M** friendships amongst them. Your task is to split the students into two classes, with no two students in each class being friends. You have complete control over the class sizes.

If it is possible to split all students into two classes, print "**Yes**" otherwise, print "**No**".

Note: All inputs are given in 1-index format.

Input format:

- The first line contains one integer **T** (number of test cases).

- The first line of each test case contains two integers **N, M.**
- The next **M** lines contain a pair { **X, Y** }.

Output format:

- Print **T** lines, the $\mathbf{i_{th}}$ should be the answer for the $\mathbf{i_{th}}$ test case.

Constraints:

- 1 <= **T** <= 10^2
- 1 <= **N, M** <= 10^4
- 1 <= **X , Y** <= **N**

Example Input:

```
2
5 3
1 2
1 4
4 5

4 4
2 3
1 3
1 2
2 4
```

Example Output:

```
Yes
No
```

Explanation:

For the first test case, one possible answer is {1, 5} and {2, 4}.

So, you need to print **Yes**.

The Bipartite Graph

Difficulty: **Medium** Accuracy: **94.03%** Submissions: **250+** Points: **4**

You are given N & E, denoting the number of nodes & bi-directional edges in a graph. Check whether the graph is bipartite or not.
If the graph is bipartite, print 1, otherwise print 0.

Input Format:
The first line contains an integer T - the number of test cases. Then the test cases follows:
The first line of the input contain two integers V & E, denoting number of vertices & edges.
The next E lines contains u & v, denoting, there is a bi-directional between u & v.

Output Format:
For each test case, Print "1" (without the quotes) if the graph is bipartite, otheriwsie "0" (without the quotes) Folllowed by newline character.

Constraints:
1<=T<=100
$1 \le V, E \le 10^5$

Example
Input:
```
2
3 2
0 1
1 2
4 4
0 2
0 3
2 3
1 3
```

Output:
```
1
0
```

Explanation:
1st - The given graph can be colored in two colors so, it is a bipartite graph.

2nd - The given graph cannot be colored in two colors such that color of adjacent vertices differs.

Find Maximum Bipartite Matching

Difficulty: **Medium**Accuracy: **53.85%**Submissions: **13+**Points: **4**

There are M job applicants and N jobs. Each applicant has a subset of jobs that he/she is interseted in. Each job opening can only accept one applicant and a job applicant can be appointed for only one job. Given a matrix G where G(i,j) denotes ith applicant is interested in jth job. If G(i,j) is 1, the applicant is interested, otherwise not.

Find an assignment of jobs to applicant in such that as many applicants as possible get jobs. Print the maximum number of applicants who can get the job.

Input Format:

The first line contains an integer T - the number of test cases. Then the test cases follows:|
The first line of the input contain two integers N & M, denoting rows & columns of the matrix G.
The next N lines contains M integers each.

Output Format:

For each test case, print the answer folllowed by newline character.

Constraints:

1<=T<=100
1 ≤ N, M ≤ 100

Example:

Input:

2
3 5

1 1 0 1 1
0 1 0 0 1
1 1 0 1 1
6 2
1 1
0 1
0 1
0 1
0 1
1 0

Output:

3
2

Explanation:

1st - There is one of the possible assignment-
First applicant gets the 1st job.
Second applicant gets the 2nd job.
Third applicant gets the 3rd job.

2nd - There is one of the possible assignment-
First applicant gets the 1st job.
Second applicant gets the 2nd job.

Disjoint Sets

Merging One's

Difficulty: **Medium**Accuracy: **40.73%**Submissions: **54+**Points: **4**

You are given a binary string **s** of size **N** consisting of only **0's** and **1's**. You are asked to perform **Q** queries which are of the following types:

Type 1: **"1"**(without quotes): Print the length of the longest substring consisting of all 1's.
Type 2: **"2 X"**(without quotes and X varies from 1 to N): Change the **Xth** position of string **s** with **'1'**. It can also be the case that the character at **Xth** position is already **'1'**.

Input Format:

First line consists of two integers **N** and **Q**, representing the length of string and the number of queries to be performed respectively.

Second line consists of the string **s**.

Next **Q** lines consists of queries of any one type(1 or 2).

Output Format:

For each query of type 1, print the length of the longest substring consisting of all 1's in a new line.

Constraints:

- 1 <= N, Q <= 10^5
- 1 <= X <= N

â€‹**Example:**

Input:

```
6 7
011000
1
2 4
1
2 5
1
```

```
2 6
1
```

Output:

```
2
3
4
5
```

Explanation:

```
Initially the string s = "011000". So the answer to the
1st query is 2.
In 3rd query the string s = "011100". So the answer to
the 3rd query is 3.
In 5th query the string s = "011110". So the answer to
the 5th query is 4.
In 7th query the string s = "011111". So the answer to
the 7th query is 5.
```

Amazing Strings

Difficulty: **Medium**Accuracy: **43.9%**Submissions: **41+**Points: **4**

Geeks recently read about **amazing** strings.

A string is called **amazing** if it fulfills the following 3 conditions:

- Length of the string should be N.
- Each character of the string must vary from **'a'** to **'z'**.
- For each pair [X,Y] in input, substring [X, Y] of this string must be a palindrome.

You have to find the number of **amazing** strings.

Input Format:

The first line of each test case contains **T**, representing the number of test cases.

The first line of each test case contains 2 integers **N** and **M**, representing the length of the string and the number of pairs [X, Y].

The following **M** lines of each test case contain two integers [X, Y].

Note: X and Y are 1 based indexing of string.

Output Format:

Print the number of **amazing** strings. Since the answer can be too large, print it modulus **1000000007**(1e9+7).

The output of each test case must be printed in a separate line.

Constraints:

- 1 <= T <= 10
- 1 <= N <= 1000
- 0 <= M <= 200000
- 1 <= X <= Y <= N

â€‹**Example:**

Input:

```
1
6 3
1 5
5 6
1 6
```

Output:

```
26
```

Removing Roads

Difficulty: **Medium**Accuracy: **72.72%**Submissions: **22+**Points: **4**

There are **n** cities in Geeksland connected by **m two-way** roads. You are asked to perform **k** operations in the sequence which are of the following types:

Type 1: "**remove a b**"(without quotes). Remove the road between **a** and **b**. It is always ensured that there is a direct road between the two cities.

Type 2: "**ask a b**"(without quotes). Check whether **a** and **b** cities are connected. In other words, check whether we can reach **a** from **b** and vice versa.

After all the operations are applied, all the roads will be removed.

You have to find the result of each query of type **ask**.

Input Format:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains **n** and **m**, respectively representing the number of cities and the number of roads connecting them.

The following **m** lines of each test case contain 2 integers, **a** and **b**, representing a 2-way road between two cities **a** and **b**.

The following line of each test case contains **k**, representing the number of operations to be performed.

The following **k** lines of each test case contain the operation of any of the types (1 or 2) described above.

Output Format:

For each of operation of type **ask** output "**YES**"(without quotes), if two cities are connected, and "**NO**"(without quotes) otherwise. The order of the answers should correspond to the order of operations of type **ask** in input.

Note: You can assume that no two cities are **directly** connected by multiple roads.

Constraints:

- 1 <= T <= 10
- 1 <= n <= 50000
- 0 <= m <= 100000
- m <= k <= 150000

â€‹**Example:**

Input:

```
1
3 3
1 2
2 3
3 1
6
ask 3 3
remove 1 2
ask 1 2
remove 1 3
remove 2 3
ask 3 1
```

Output:

```
YES
YES
NO
```

Assigning Id's

Difficulty: **Hard**Accuracy: **93.81%**Submissions: **241+**Points: **8**

You are given an **N*M** matrix **mat**. You have to assign **id's** to each of the element in the matrix by following the below 3 rules:â€‹

- **id** must be a **positive** integer.
- If two elements x and y are in the same row or column, then

1. If x > y, then id(x) > id(y)
2. If x == y, then id(x) == id(y)
3. If x < y, then id(x) < id(y)

- The **id** assigned should be as small as possible.

â€‹You can assume that the unique answer will exist under the given rules.

Input Format:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains two integers **N** and **M**, representing the size of the matrix.

The following **N** lines of each test case contains **M** integers, representing the elements of the matrix.

Output Format:

Print an **N*M** matrix representing the id of each element. The element in **ith** row and **jth** column should represent the id of element **mat[i][j]**.

The output of each test case must be printed in a separate line.

Constraints:

- 1 <= T <= 10
- 1 <= N, M <= 400
- -10^5 <= mat[i][j] <= 10^5

â€‹**Example:**

Input:

```
2
2 2
4 4
4 4
3 3
1 2 3
4 5 6
7 8 9
```

Output:

```
1 1
1 1
1 2 3
2 3 4
3 4 5
```

Friend Circle

Difficulty: **Easy**Accuracy: **41.01%**Submissions: **39+**Points: **2**

A quiz competition is organized between **N** students in a school. There are **M** pairs that represent the friendly relation between the students. You will be given two integers **X** and **Y**, in each friendship relation, indicating that **X** and **Y** are friends.

Now the jury has to decide the top **2** winners of the competition so that the winners should not belong to the same friend circle. Find the number of ways in which they can do this.

Note:

If **P** and **Q** are friends, **Q** and **R** are friends, then **P, Q, R** will have the same friend circle.

The same friend relation can appear multiple times in the input; however, **X** and **Y** will always be different.

Two combinations are considered different if either first or second ranker is different.

Input Format:

The first line contains **T**, representing the number of test cases.

The first line of each test case contains two space-separated integers **N** and **M**, representing the number of students and the number of friendly relations.

The following M lines of each test case contain two integers **X** and **Y**, representing that **X** and **Y** are friends.

Output Format:

For each test case, print the number of ways in which the jury can choose the top two winners.

The output of each test case must be printed in a separate line.

Constraints:

- 1 <= T <= 20000
- 2 <= N <= 200000
- 1 <= M <= 500000
- 1 <= X, Y <= N

The sum of N over all the test cases won't exceed 200000 and M won't exceed 500000.

Example:

Input:

```
2
2 1
1 2
6 3
1 2
3 4
5 6
```

Output:

```
0
```

Strongly connected components

Web Crawler

Difficulty: **Medium**Accuracy: **68.3%**Submissions: **41+**Points: **4**

Nikhil made a simple **web crawler** which opens a webpage **finds all the links on that webpage** and then opens those links and repeats the same process on the new page.
This crawler has a memory and **doesn't visit the same page twice** to avoid infinite loops.

For the scope of this task web can be defined as collection of **N** web pages.
A **cluster** is a collection of web pages such that if crawler starts from any web page in a cluster, it can visit all other web pages in that cluster.
Nikhil wants to know how many clusters exist on the web and size of each cluster.

Input

- First line of input contains an integer **T**, number of testcases, **T** tests follow
- First line of each test contains **N** number of webpages.
- Next **N** lines contain description of each webpage.
- Each webpage is described by the links to other webpages it has.
- Description of $\mathbf{i_{th}}$ webpage contains an integer **L** which is the number of links the web page has, followed by **L** integers $\mathbf{j_{th}}$ of which is the index (1-based) of the page the $\mathbf{j_{th}}$ link points to.
- A page doesn't point to itself.

Output

- For each test first print an integer **C** in a new line denoting number of clusters on the web

- In second line print **C** integers denoting size of the clusters in non-increasing order.

Constraints

- **1<=T<=10**
- **1<=N<=10^5**
- **0<=L[i]<=10^5**
- **Sum of L[i] for all pages doesn't exceed 2*10^5**

Sample Input

```
2
5
1 2
1 3
2 1 4
1 5
1 4
3
0
0
0
```

Sample Output

```
2
3 2
3
1 1 1
```

Explanation

In first test there are two clusters , largest cluster has **3** pages 1, 2 and 3 , second cluster has two pages 4 and 5

In second case there are 3 clusters as no page points to any other page therefore every page forms a cluster with size **1**

XOR the graph

Difficulty: **Medium**Accuracy: **93.82%**Submissions: **213+**Points: **4**

Given a graph **G** consisting of undirected edges, every edge has weight **1** or **0** .
You can do following operation on the graph arbitrary number of times

- Pick a node **X** on the graph
- Xor the weight of all the edges that **X** is part of with **1**

Find out if it is possible to make weight of all edges equal to **1**

Input

- First line of input contains an integer **T** number of testcases
- First line of each testcase contains two integers **N** and **M** number of nodes and number of edges respectively
- Next **M** lines contain 3 integers **u, v** and **w** denoting there is an edge between node **u** and **v** with weight **w**
- **u** is not equal to **v**

Output

- For every testcase print "Possible" in a new line without quotes if it is possible to make weight of all the edges equal to **1** otherwise print "Impossible" without quotes in a new line

Constraints

- **1<=T<=10**
- **1<=N<=1000**
- **1<=M<=(N*(N-1))/2**
- **1<=u,v<=N, u is not equal to v**
- **0<=w<=1**

Sample Input

```
2
3 3
1 2 1
```

```
2 3 0
1 3 1
4 4
1 2 1
2 3 1
3 4 1
4 2 1
```

Sample Output

```
Impossible
Possible
```

Explanation

In first case there is no way that weight of every edge can be made equal to 1, in second case all edges are already 1.

Graduate Geek

Difficulty: **Hard** Accuracy: **92.69%** Submissions: **245+** Points: **8**

Geek is planning to graduate from a university that offers **N** subjects and **M** courses. Each course consists of exactly **two subjects**.
She has to **take at least one subject in every course**.

Every subject has a unique id a natural number from **1** to **N**
This university has a special rule that exam for $\mathbf{i_{th}}$ subject is on $\mathbf{i_{th}}$ day when examinations start.

Geek wants some break in between exams so that she can prepare for next exam. So for every **i** in range **[1,N/2]** , she will take either $\mathbf{2*i - 1_{th}}$ or $\mathbf{2*i_{th}}$ subject so that she has some breaks between exams.

In other words she distributes exams in **N/2** sets viz. {1,2}, {3,4}, {5,6}, ... {N-1,N} or (*{N} if N is odd*), and take only one exam from each set.

Find out if it is possible to sign up for all **M** courses satisfying the above criterion.

Input

- First line of input contains an integer **T**, number of testcases, **T** testcases follow
- First line of each testcase contains two space separated integers **N** and **M**
- Next **M** lines describe **M** courses in form of two space separated integers **u** and **v**, which means that subject **u** and **v** are offered in this course. **u** is not equal to **v**

Output

- For each test print "POSSIBLE" in a new line without quotes if it is possible to sign up for all **M** courses satisfying the rules stated above otherwise print "IMPOSSIBLE" in a new line without quotes

Constraints

- **1<=T<=10**
- **1<=N<=10^5**
- **N-1<=M<=2*10^5**
- **1<=u,v<=N**

Sample Input

```
2
4 3
1 2
2 3
3 4
4 4
```

```
1 3
3 4
2 4
2 3
```

Sample Output

```
Possible
Possible
```

Explanation

In first case Geek can take subject **1** and **4** which covers all the **3** courses and the Geek's criterion stated above.
In second case, if she chooses **{1 , 3}** or **{1, 4}** all courses won't be covered, she can't choose **{3,4}** or **{2,1}** but choosing **{2,3}** covers all the courses.

Shortest Path Algorithms

Djikstra Algorithms

Difficulty: **Medium**Accuracy: **48.28%**Submissions: **29+**Points: **4**

You are given a weighted, undirected and connected graph of **V** vertices and **E** edges and each edge has a edge weight **W** . You are given the source vertex **S** and you have to find the shortest distance of all the vertex's from the source vertex **S**. The vertices are numbered from 0 to (**V**-1).You have to print a list of integers denoting shortest distance between **each node** and Source vertex **S**.

Input Format :

The first line of the input contains a single integer **T** denoting the number of test cases.

Each testcase is as follows :-

- The first line consists of **V**,**E**

- E lines follow each containing **u,v,W** (where **u,v** are vertices and **W** is edge weight)
- The last line consists of source vertex **S**

Output Format:

Print **V** numbers separated by a space each containing distance from source node **S** to the vertex numbered i (0 <= i <= **V**-1)

Constraints:

1 ≤ **T** ≤ 100
1 ≤ **V** ≤ 1000
1 ≤ **W** ≤ 1000
1 ≤ **E** ≤ [(V*(V - 1)) / 2]
0 ≤ **S** < V

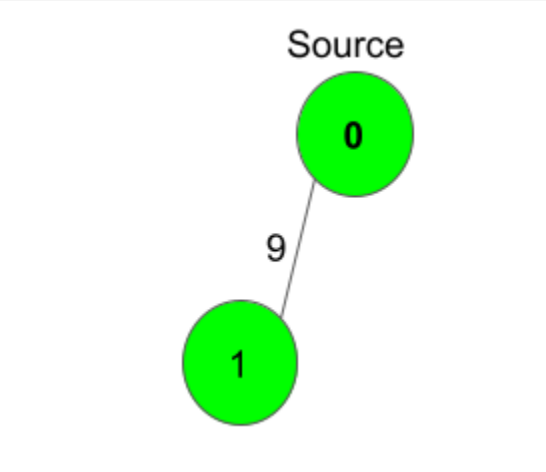
Example :

Input :
2
2 1
0 1 9
0
3 3
0 1 1
0 2 6
1 2 3
2

Output :
0 9
4 3 0

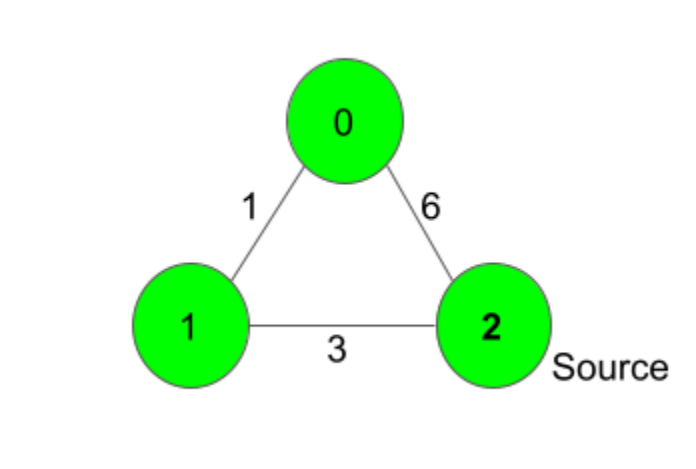
Explaination :

First test case :-



The source vertex is 0. Hence, the shortest distance of node 0 is 0 and the shortest distance from node 1 is 9.

Second test case :-



For nodes 2 to 0, we can follow the path-2-1-0. This has a distance of 1+3 = 4, whereas the path 2-0 has a distance of 6. So, the Shortest path from 2 to 0 is 4.
The shortest distance from 0 to 1 is

min(3,6+1) = 3.

Distance from the Source (Bellman-Ford Algorithm)

Difficulty: **Medium**Accuracy: **34.62%**Submissions: **26+**Points: **4**

Given a weighted, directed and connected graph of V vertices and E edges, Find the shortest distance of all the vertices from the source vertex S.

Note: If the Graph contains a negative cycle then print -1.

Input Format -

The first line consists of a single integer **T** determines the number of test cases. The description of the following T test cases is as follows:

- First line consists of two space-separated integers **V** and **E** denoting the number of vertices in the graph and the number of edges in the graph respectively.
- Next **E** line consists of three space-separated integers **u, v,** and **wt** denoting that there exists an edge from u to v having a weight of wt.
- Next line consists of a single integer denoting the source vertex.

Output Format:

For each test case print V integers, where the ith integer denotes the minimum cost to reach the ith vertex from the source. Also, if a negative weight cycle exists in the graph print only a single integer i.e. -1.
After every test case print a newline character.

Constraints:

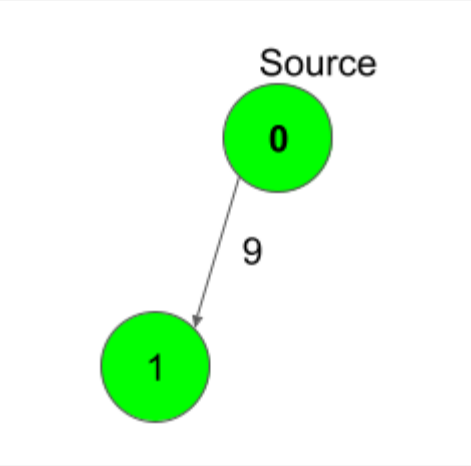
$1 \le T \le 100$
$1 \le V \le 500$
$1 \le E \le V*(V-1)$
$-1000 \le adj[i][j] \le 1000$
$0 \le S < V$

Example:

Input:
2
2 1
0 1 9
0
3 4
0 1 5
1 0 3
1 2 -1
2 0 1
2

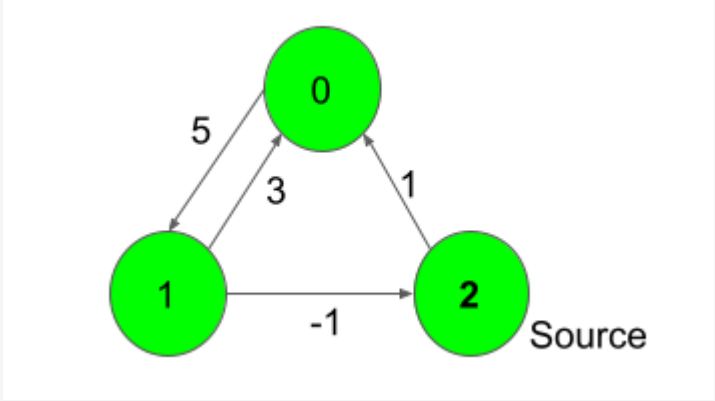
Output:
0 9
1 6 0

Explanation:
Test Case 1 -



Shortest distance of all nodes from source is printed.

Test Case 2 -



For nodes 2 to 0, we can follow the path- 2-0. This has a distance of 1. For nodes 2 to 1, we cam follow the path- 2-0-1, which has a distance of 1+5 = 6,

Floyd Warshall

Difficulty: **Medium**Accuracy: **47.06%**Submissions: **17+**Points: **4**

The problem is to find the shortest distances between every pair of vertices in a given **edge-weighted directed** graph. The graph is represented as an adjacency matrix of size **n*n. Matrix[i][j]** denotes the weight of the edge from **i to j.** If **Matrix[i][j]=-1,** it means there is no edge from **i to j.**

Input Format:

The first line of the input consists of a single integer T denoting the number of test cases. The description of each test cases is as follows -

- The first line consists of a single integer n denoting the number of vertices in graph.
- Following n lines consists of n integers separated by sapce. Where the jth number in i^{th} line denotes the weight of edge between vertex i and j.

Output Format:

For each test case print n lines, each consisting of n space separated integers. Where the jth number in i^{th} line denotes the minimum possible cost to reach vertex j from i or vice-versa.

Constraints:

1 <= T <= 30
1 <= n <= 100
-1 <= matrix[i][j] <= 1000

Example:

Input:
2
2
0 25
-1 0
3
0 1 43
1 0 6
-1 -1 0

Output:
0 25
-1 0
0 1 7
1 0 6
-1 -1 0

Explanation:
Test case 1 - The shortest distance between every pair is already given(if it exists).
Test case 2 - We can reach 3 from 1 as 1->2->3 and the cost will be 1+6=7 which is less than 43.

Shortest Path in Constrained Matrix

Difficulty: **Medium**Accuracy: **17.39%**Submissions: **23+**Points: **4**

Given an **n * m** matrix (say a[][]), where each cell of the matrix has exactly one door which leads to any other cell (maybe out of the matrix). The value in the cell a[i][j] means -

- **0:** The door from the cell a[i][j] leads to the cell a[i][j+1]
- **1:** The door from the cell a[i][j] leads to the cell a[i][j-1]
- **2:** The door from the cell a[i][j] leads to the cell a[i+1][j]
- **3:** The door from the cell a[i][j] leads to the cell a[i-1][j]

However, it costs 1 coin to construct a door that leads to one of the adjacent cells (Up, Down, Left, or Right).

Print another n * m matrix (say ans[][]), where ans[i][j] represents the minimum cost to reach the cell (i, j) from the cell (0, 0).

Input Format:

The first line of the input contains a single integer T denoting the number of test cases.

The first line of every test case consists of two space-separated integers i.e. '**n**' and '**m**'. Followed by 'n' lines consisting of '**m**' space-separated integers.

Output Format:

Print '**n**' lines, each consisting of '**m**' integers. Where the jth value of the ith line denotes the minimum cost to reach the cell (i, j) from the cell (0, 0).

Constraints:

0 < n, m < 1000
0 <= a[i][j] <= 3

Example:

Input:

2
4 4
0 0 0 0
1 1 1 1
2 2 2 2
3 3 3 3
---
2 2
1 0
3 2

Output:

0 0 0 0
1 1 1 1
2 2 2 2
2 2 2 2
---
0 1
1 2

Explanation:

First Test Case -

Starting from (0, 0) we can go anywhere in the 0th row,
hence the cost to reach any cell in the 0th row is 0.
To reach the first row, it is required to construct doors

to downward cells. Hence at least 1 coin is required to reach any cell of 1st row.
Similarly to reach the 2nd row we will again construct doors. Now to reach cells of the 3rd row, the door already exists and hence we can reach any cell of the third row with the cost of 2.

Second Test Case -

From (0, 0) the existing door leads to (0, -1) i.e. out of the matrix, hence to reach either of the cell (0, 1) or (1, 0) requires constructing a door. Similarly from cells (1, 0) or (0, 1), we can go outside of the matrix by using the existing doors. Therefore, we will again construct a door to reach (1, 1).

Note: "---------" is only for visualization purpose, the actual test cases do not contain of "-------".

Minimum Cost

Difficulty: **Hard** Accuracy: **100.0%** Submissions: **1+** Points: **8**

Given a directed graph (not necessarily connected and can have self-loops) with **N** vertices numbered from **1 to N** and **M** edges. Given an array **A[ ]** of size **N** where the **i^{th}** integer denotes the cost of building a directed edge from the vertex, **i** to vertex **1** (A_1 is 0).

You can add some directed edges (possibly zero) from vertex **x** to vertex **1** (1 < x <= N) with the cost of A_x. Please find the minimum total cost such that for every vertex **y** (1 < y <= N), there is a path from vertex **y** to vertex **1**.

Note: If you don't add any edge, the minimum cost will be zero.

Input Format:

The first line of the input contains a single integer **T** ($1 \le T \le 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers, ***N*** and ***M***.
- The following **M** lines contain two space-separated integers, ***u*** and ***v***, denoting a directed edge from ***u*** to ***v***.
- The last line contains **N** space-separated positive integers $A_1, A_2, \ldots, A_N$.

Note: The sum of **N** and **M** over all the test cases will not exceed 10^6.

Output Format:

For each test case, print the minimum cost such that for every vertex **y** (1 < y <= N), there is a path from vertex **y** to vertex **1** followed by a newline character.

Note: The generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \le T \le 10^4$

$1 \le N \le 2 * 10^5$

$1 \le M \le 2 * 10^5$

$1 \le u, v \le N$

$0 \le A[i] \le 10^9$

Example:

Sample Input:

```
2
4 3
2 1
3 4
```

3 1
0 2 3 4
3 2
3 1
2 1
0 1 0

Sample Output:

4
0

Explanation:

Testcase 1:
Their is already a path from 2 and
3 to 1 and no path from 4 to 1 so
we add and edge from 4 to 1 with
cost of 4.

Testcase 2:
There is allready a path from 2
and 3 to 1.

Spanning Tree

Minimum Spanning Tree

Difficulty: **Medium**Accuracy: **35.0%**Submissions: **20+**Points: **4**

Given a weighted, undirected, and connected graph of **V** vertices and **E** edges. The task is to find the sum of weights of the edges of the Minimum Spanning Tree.

Input Format:

The first line consists of a single integer T denoting the number of total test cases. The description of each test case is as follows -

- The first line contains two space-separated integers (say n and m) denoting the number of vertices and the total number of edges present in the graph.
- The following m lines consist of three space-separated integers (say u, v, and w) denoting that there exists a bi-directional edge of weight w between u and v.

Output Format:

For each test case print the sum of weights of edges of the Minimum Spanning Tree in a single line.

Constraints:

$2 \leq V \leq 1000$

$V-1 \leq E \leq (V*(V-1))/2$

$1 \leq w \leq 1000$

Graph is connected and doesn't contain self-loops & multiple edges.

Example 1:

Input:

```
2
3 3
0 1 5
1 2 3
0 2 1
2 1
0 1 5
```

Output:

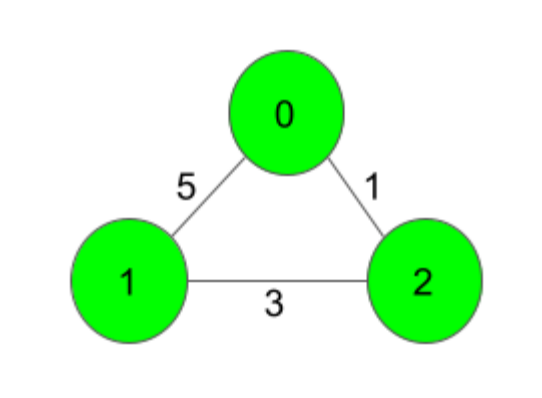
```
4
```

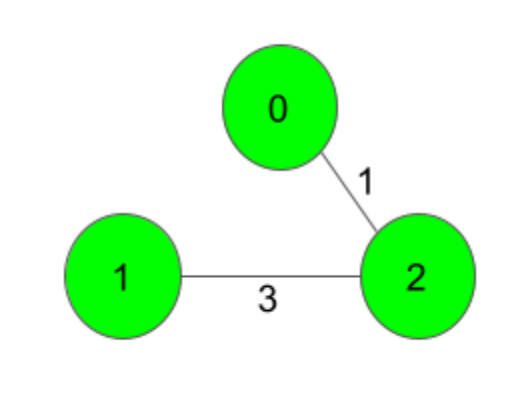
5

Explanation:

Test Case 1 -

The corresponding graph is -

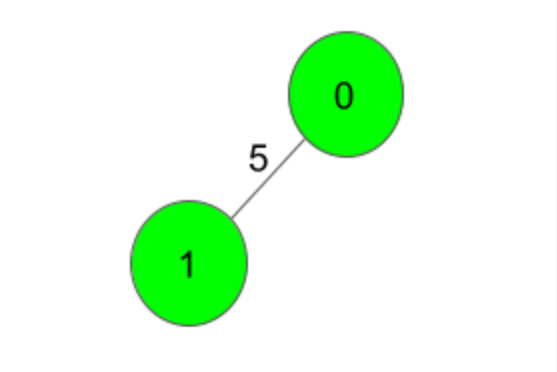




The resulting Spanning Tree of weight 4 is shown above.

Test Case 2 -

The corresponding graph is -



Here, only one Spanning Tree is possible which has a weight of 5.

Minimal Difference Tree

Difficulty: **Hard**Accuracy: **28.57%**Submissions: **7+**Points: **8**

Consider a weighted undirected graph with **N** nodes. Find a spanning tree in the graph so that the difference between the maximal and the minimal edges is minimal.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers, **N** and **M**.
- Next, **M** lines contain the description of edges, one per line. Edge **i** is described by three space-separated integers **u**, **v**, and **w** the ends of the edge and its weight, respectively.

Output Format: For each test case, if the spanning tree exists, then the first line of the output should contain "YES", and the second line should contain one integer the minimal difference between the maximal and the minimal edges of the spanning tree. Otherwise, the sole line of the output should have "NO".

Followed by a newline character for each test case.

Note: The generated output is white space sensitive, do not add extra spaces or unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 10^4$

$1 \leq \mathbf{N, M} \leq 1000$

$1 \leq \mathbf{u, v} \leq N, -10^9 \leq \mathbf{w} \leq 10^9$

*The sum of **N * M** over all test cases does not exceed **5 * 10^5,** and at least one query of the 2nd type is in each test case.*

Example:

Sample Input:

```
1
3 4
1 2 3
2 3 -1
1 3 2
1 1 -10
```

Sample Output:

```
YES
1
```

Explanation:

Testcase 1:

The optimal spanning tree

contains edge 1 2 with weight 3

and 1 3 with weight 2

Number Of Spanning Trees In A Graph

Difficulty: **Medium**Accuracy: **41.66%**Submissions: **12+**Points: **4**

Given a connected undirected graph of N vertices and M edges. The task is the find the total number of spanning trees possible in the graph.

Note: A spanning tree is a subset of Graph G, which has all the vertices covered with the minimum possible number of edges. Hence, a spanning tree does not have cycles and it cannot be disconnected. By this definition, we can draw a conclusion that every connected and undirected Graph G has at least one spanning tree.

Input format:

The first line contains an integer T - the number of test cases. Then the test cases follows:|
The first line of the input contain two integers N & M, denoting number of vertices & edges of the matrix G.
The next M lines contains u & v, denoting there is an edge between u & v.

Output format:

For each test case, print the answer folllowed by newline character.

Constraints:

1<=T<=10
1 <= N <= 10
N - 1 <= M <= N *(N - 1) / 2
0 <= graph[i][0], graph[i][1] <= N - 1

Example

Input:

1
6 5
0 3
0 1

1 2
1 5
3 4

Output:
1

Explanation: Only one spanning tree is possible,i.e. the graph itself. {{0, 1}, {0, 3}, {1, 2}, {1, 5}, {3, 4}}

String Algorithms

Substring Counting

Difficulty: **Hard**Accuracy: **90.91%**Submissions: **253+**Points: **8**

Geek has found a string **S** of length **N**. His friend asks him the following question **Q** times:

- Each query contains an integer **X,** find the number of ways to choose **X** substrings from **S** such that all of them are equal.

Since the answer can be large, print it modulo **10^9+7.**

Input Format:

- The first line contains an integer ***T*** denoting the number of test cases
- The first line of each test case contains two integers **N, Q** respectively - the length of the string and the number of queries.
- The second line of each test case consists of a string of length **N** containing only lowercase English letters.
- Next, **Q** lines contain an integer **X** - the value for the query.

Output Format:

For each query print the answer in a new line.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 200$

$1 \leq \mathbf{N} \leq 2000$

$1 \leq \mathbf{Q} \leq 10{,}000$

$1 \leq \mathbf{X_i} \leq 10^9$

Example:

Sample Input:

```
1
4 2
abab
2
1
```

Sample Output:

```
3
10
```

Explanation:

In the first test case, there are 3 ways:- {'a','a'},('b','b'},('ab','ab}

In the second test case, we can choose any substring.

Substring Matching

Difficulty: **Hard**Accuracy: **87.66%**Submissions: **252+**Points: **8**

Geek has found a string **S** of length **N**. Find the longest substring **T** of **S** which satisfies all of the following conditions.

- **T** is a prefix of **S**.
- **T** is a suffix of **S.**
- **T** is the substring of the string **S[2,3,...., N-1]** (Notice we're not considering 1st and last character of the string.

If no such substring exists then print "**-1**".

Input Format:

- The first line contains an integer ***T*** denoting the number of test cases
- The first line of each test case contains two integers **N** - the length of the string.
- The second line of each test case consists of a string of length **N** containing only lowercase English letters.

Output Format:

For each test case print the answer in a new line.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 1000$

$1 \leq \mathbf{N} \leq 100{,}000$

S contains only lowercase English characters.

Example:

Sample Input:

```
2
5
```

ababa
4
abab

Sample Output:
a
-1

Explanation:
In the first test case, the answer
is 'a'
In the second test case, no valid
substring exist.

Geek and Strings

Difficulty: **Hard**Accuracy: **96.48%**Submissions: **227+**Points: **8**

Geek was playing with strings. He decided to create a new convention of palindromes. He defined **K** different types of palindromes as follows:
1-drome is a string which reads the same as backward as forward.
If we have two K-drome string **S,T** which are equal. i.e. **S = T**, then we can obtain a (K+1)-drome in the following manner:

- Concatenate **S** and **T** i.e. **S+T.**
- Concatenate **S** and **T** with a character in between them i.e. **S+c+T,** where **c** is any lowercase English character.

Given a string **S** of length **N**, find a sequence of **N** integers such that **i-th** integer denote the number of substrings of **S** which are **i-drome**.

Input Format:

- The first line contains an integer ***T*** denoting the number of test cases

- The first line of each test case contains a single integer **N** - the length of the string
- The second line of each test case contains a string **S** of size **N** containing only lower-case English alphabets.

Output Format:

For each test case print **N** integers **i-th** integer denote the number of substrings of **S** which are **i-drome.** After every integer print a white-space character(after the last integer of the array too) and then a newline character.

Note: Generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Constraints:

$1 \leq \mathbf{T} \leq 50$

$1 \leq \mathbf{N} \leq 1000$

S contains only lower-case English alphabet

Example:

Sample Input:

```
2
4
abba
7
abacaba
```

Sample Output:

```
6 1 0 0
12 4 1 0 0 0 0
```

Explanation:

In the first test case, 1-dromes are:
a,b,b,a,bb,abba, then substring bb is
also a 2-drome.

Longest Invalid Search Keyword

Difficulty: **Hard**Accuracy: **90.46%**Submissions: **242+**Points: **8**

Geeksforgeeks has **N** articles. You don't want to read all the articles. So you are searching for some articles using the search bar.
To search for articles you need to type in some keywords and this will return all articles containing this keyword as a substring in title. A keyword is said to be **invalid** when it returns all **N** articles.

You are given title of **N** articles as string. You need to find the length of longest invalid search keyword.

Input format:

- The first line contains one integer **T** (number of test cases).
- The first line of each test case contains one integer **N.**
- The next **N** lines contain a string (title of i-th article).

Output format:

- Print **T** lines, the i_{th} should be the answer for the i_{th} test case.

Constraints:

- 1 <= **T** <= 10^2
- 1 <= **N** <= 10^3
- 1 <= **length of title <=** 10^3

Example Input:

```
2
2
abcd
ababcab
```

```
3
cdxy
cde
adbcde
```

Example Output:

```
3
2
```

Explanation:

For the first test case, longest common keyword is : "abc"

So, you need to print 3.

Treasure Hunt

Difficulty: **Medium**Accuracy: **86.47%**Submissions: **243+**Points: **4**

Alexa, Bob and their temporary buddies Mohan and Ram has finally found the ancient temple. However, its doors were firmly locked and even Bob had no luck opening them.

A little later they found a string ***s***, carved on a rock below the temple's gates. Alexa supposed that that's the password that opens the temple and read the string aloud. However, nothing happened. Then Alexa supposed that a password is some substring ***t*** of the string ***s***.

Ram supposed that the substring *t* is the beginning of the string *s*; Mohan supposed that the substring ***t*** should be the end of the string ***s***; and Bob supposed that ***t*** should be located somewhere inside the string ***s***, that is, ***t*** is neither its beginning, nor its end.

Alexa chose the substring ***t*** so as to please all his companions. Besides, from all acceptable variants Alexa chose the longest one. When Alexa read the substring ***t*** aloud, the temple doors opened.

You know the string ***s***. Find the substring ***t*** or determine that such substring does not exist and all that's been written above is just a nice legend.

Input Format:

The first line should contain **T**, denoting the number of test cases.

The second line contain string **s**.

Output Format:

For each test case output the string ***t***. If a suitable ***t*** string does not exist, then print "**Just a legend**" without the quotes.

Example:

Input:
2
fixprefixsuffix
abcdabc

Output:
fix
Just a legend

Explanation:
In the first test case the longest substring
which satisfy all the friends condition is "fix".
In the second test there is no substring which
satisfy all the friends condition so the output
is Just a legend.

Constraints:

- 1 <= T <= 1000
- 1 <= s.length <= 10^6

Sahil loves gfg

Difficulty: **Easy**Accuracy: **89.66%**Submissions: **261+**Points: **2**

Sahil is done with his internship and now it's time to attend Geeks Classes to enhance knowledge of Data Structures. However, to join this class, he has been given a task to complete. The task is to count the occurrences of "gfg" in the given string, and if he completes this task, he qualifies for Geeks Classes. But, Sahil is facing some problem in solving this task, so help him.

Input :

First line of input contains "T" testcases. Next "T" lines contains Strings with lowercase characters.

Output :

For each testcase, output the count the ccurrences of "gfg"(continuous) present in string.

Constraints :

1 <= T <= 10

1<= length_of_string <= 100

Example :

Input :

2

geefgfgksforgfgeeks

dsajslsdlsfgf

Output :

2

-1

Explanation :
#Test Case 1: In the first testcase, "gfg" occurs two times in the String. First ccurrence starts at index 4 and next starts at 12.
#Test Case 2: In second testcase, there is no occurences of "gfg" in given string. So, it returned -1.

Rolling Hash

Difficulty: **Hard**Accuracy: **72.94%**Submissions: **310+**Points: **8**

You are given a string **S** and a Pattern **P**. You need to find all matches of **hash** of **P** in string **S**. Also, print the **index (0 based)** at which the pattern's hash is found. If **no** match is found, **print -1**.

Note: All the matches should have same length as pattern P.

*The hash of pattern **P** is calculated by **summing** the values of **characters** as they appear in the **alphabets table**. For reference, a is 1, b is 2, ...z is 26. Now, using the mentioned values, hash of ab is 1+2=3.*

Input:

The first line of input contains T denoting the number of testcases. T testcases follow. Each testcase contains two lines of input. The first line contains the string **S**. The second line contains the pattern **P**.

Output:

For each testcase, in a new line, print the matches and index separated by a space. All the matches should be printed in their own separate lines.

Constraints:
$1 <= T <= 100$
$1 <= |S|, |P| <= 10^5$

Examples:

Input:
1
bacdaabaa
aab
Output:
aab 4
aba 5
baa 6

Explanation:
Testcase1: P is aab, and S is bacdaabaa
Now, the hash of P: aab is 1+1+2=4
In the string S, the hash value of 4 is obtained by the following:
aab=1+1+2=4, at index 4
aba=1+2+1=4, at index 5
baa=2+1+1=4, at index 6

Range Query Problems

Sparse Table

Minimum Partition

Difficulty: **Hard**Accuracy: **20.98%**Submissions: **167+**Points: **8**

Given an array **A[ ]** of **N** integers, integer **K** and **Q** query where the **i^{th}** query is described as:

- **l r**: partition the subarray $A_l, A_{l+1}, A_{l+2}, ..., A_r$ into **P** number of subarrays such that each subarray has a sum at most **K**. The value of **P** should be minimum.

Your task is to calculate the value of **P** for each query.

Input Format:

The first line of the input contains a single integer **T** denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains three space-separated integers, **N, K,** and **Q**.
- The second line has **N** space-separated positive integers $A_1, A_2, \ldots, A_N$.
- Next, **Q** lines contain two space-separated integers denoting the ith query.

Output Format:

For each test case,

- print the answer to each query in the order they appear in the input separated by white spaces

followed by a newline character.

Note: The generated output is white space sensitive, do not add extra spaces or unnecessary newline characters.

Constraints:

$1 \leq T \leq 10^4$

$1 \leq N, Q \leq 10^5$

$1 \leq A[\,] \leq 10^9$

$\max(A[\,]) \leq K \leq 10^{14}$

$1 \leq l \leq r \leq N$

*The sum of **N** and **Q** over all test cases does not exceed $2 * 10^5$.*

Example:

Sample Input:

```
1
4 6 2
4 3 1 2
```

1 3

1 4

Sample Output:

2 2

Explanation:

Testcase 1:

1st Query:

l = 1, r = 3, A[l...r] = {4, 3, 1}

We can partition the subarray into two subarrays: {4} and {3, 1}.

2nd Query:

l = 1, r = 4, A[l...r] = {4, 3, 1, 2}

We can partition the subarray into two subarrays: {4} and {3, 1, 2}.

Range Minimum Query - O(1)

Difficulty: **Hard**Accuracy: **87.82%**Submissions: **280+**Points: **8**

Given an array immutable a[] of size **n**, you need to answer **Q** queries where each query is of type **query[i] = {L, R}** and you need to tell the minimum element present in the array in the range [L, R] in **O(1) time**.

Input Format:

The first line of the input consists of a single integer T, denoting the number of test cases. The description of each test case is as follows:

- The first line consists of two numbers n and Q denoting the size of the array and number of queries respectively.
- The second line consists of n space separated integers denoting the array elements.
- Follow Q lines consists of two integers each i.e. L and R for each query.

Output Format:

For each query print the minimum element is the corresponding range in a single line.

Constraints:

$0 < T < 100$

$0 < n <= 10^5$

$0 <= a[i] <= 10^9$

$0 < Q <= 10^5$

Sum of n over all Test Cases will not exceed $2 * 10^5$

$0 <= L <= R < n$

Example:

Input:

```
2
5 3
3 2 1 4 5
0 2
3 4
0 4
6 2
3 9 2 3 4 5
3 3
2 5
```

Output:

```
1
4
1
3
```

2

Explanation:
Test Case 1 - Minimum of the range [0, 2] is 1, while that of range [3, 4] is 4 and for the range [0, 4] i.e. the complete array minimum is 1.
Test Case 2 - Minimum for the range [3, 3] is 3, while that of range [2, 5] is 2.

Minimum Sum

Difficulty: **Medium**Accuracy: **82.43%**Submissions: **859+**Points: **4**

You are given an array **A[ ]** of **N** integers. You can change the sign of any element any number of times. Compute the minimum value of ***abs(sum of all numbers in the array)***.
Here **abs(x) = -x** if (x < 0) else **abs(x) = x.**

Input:
N = 5
A[] = {3, 1, 4, 5, 0}
Output:
1
Explanation:
Change the sign of 1 and 5
abs(summation of all elements of array)
= abs(3 + (-1) + 4 + (-5) + 0) = 1

Example 2:

Input:
N = 2
A[] = {1, -1}
Output:
0
Explanation:
No need to change the sign.

Your Task:
You don't need to read, input, or print anything. Your task is to complete the function **solve( ),** which takes **N** and **A[ ]** as input parameters and returns the minimum possible answer to the problem.

Constraints:
$1 \le N \le 100$
$-N \le A[i] \le N$

Magic Balls over Hills

Difficulty: **Medium**Accuracy: **65.71%**Submissions: **140+**Points: **4**

Geek faces an unusual challenge. In the land of Geekland, there are numerous hills and villages. The ruler of Geekland has assigned Geek with the task of transporting enchanted balls from one village to another. However, Geek needs to transport a large number of balls, and carrying them on foot would be incredibly time-consuming. To overcome this obstacle, Geek devises a clever plan - launching them.
Geekland is divided into **intervals**, each consisting of a village and a hill.
Geek can launch an ball accurately from Village A to Village B if there is **no** hill between them (**exclusive**) that is higher than the hill in Village A.

Input Format:

The first line contain an integer T - the number of test cases. Then the test cases follows:

The first line contains two integers **N** and **M**, representing the number of hills and the number of intervals, respectively.
Next line contains N space separated integers where **H[i]** denote the height of hill in i-th village.
Next M lines, each contains integers A and B (1 <= A < B <= N), which indicate the

village number **from** which Geek wants to launch the ball and the village number **to** which Geek wants to launch the ball.

Output Format:

For each test case, print a single integer representing the **count** of balls that Geek can successfully launch.

Constraints:

$1 \leq T \leq 1000$

$2 \leq N \leq 10^5$

$1 \leq M \leq 10^5$

The sum of N and M over all test cases in a test file does not exceed $2 * 10^5$.

Example:

Input:

2
7 3
2 3 5 4 2 1 6
3 5
2 5
4 6
5 2
5 2 3 6 3
1 4
3 5

Output:

2
1

Subarray Count

Difficulty: **Hard**Accuracy: **89.46%**Submissions: **266+**Points: **8**

Given an integer array **A** of size **N** and an integer value **K**, we need to find the number of sub-arrays of A such that the **difference** between the maximum value and minimum value in the sub-array is less than or equal to K.

Note : A subarray is the sequence of consecutive elements of the array.

Input Format:
The first line contain an integer T - the number of test cases. Then the test cases follows:

The first line contains two integers **N** and **K**.
Next line contains N space separated integers denoting **A[i]** i.e array A.

Output Format:
Print the number of sub-arrays satisfying above mentioned condition.

Constraints:
$1 \leq T \leq 1000$
$1 \leq N \leq 10^5$
$0 \leq K \leq 10^9$
$1 \leq Xi, Yi \leq N$
The sum of N over all test cases in a test file does not exceed $2 * 10^5$.

Example:

Input:
2
3 2
7 6 14
5 0
3 3 3 3 3
Output:
4
15

Explanation:

For test case 1, The sub-arrays which statisfy the conditon are
[7] difference = 0,
[6] difference = 0,
[14] difference = 0,
[7, 6] difference = 1.

Segment Trees

Range LCM Queries

Difficulty: **Medium**Accuracy: **70.77%**Submissions: **3K+**Points: **4**

You are given an array of size **N** and **Q** queries. Solve the queries to find LCM for the given range alongside the queries to update the array values.

Note: 0-based indexing is used.

Example 1:

Input:
N = 6, Q = 3
arr[] = {2,3,4,6,8,16}
Queries: getLCM(0,2)
updateValue(3,8)
getLCM(2,5)

Output:
12
16

Explanation: There are 3 queries:
Query 1 : lcm(2, 3, 4) = 12
Query 2 : 6 changes to 8
Query 3 : lcm(4, 8, 8, 16) = 16

Your Task:

Complete **updateValue()** and **getLCM()** function. The **segment tree array has been**

prebuilt for you. You only need to use it to complete the queries.
getLCM() returns the LCM of elements in range [L,R].
updateValue() processes the point update query.

Expected Time Complexity: O(Q*Log(N)*Log(N)).
Expected Auxiliary Space: O(1).

Constraints:
1 <= **N** <= 10^4
1 <= **Q** <= 10^5
0 <= **L, R, index** <= N-1
1 <= **arr[i], value** <= 10^4

Add and update

Difficulty: **Medium**Accuracy: **94.78%**Submissions: **233+**Points: **4**

Given an array **A** ($-10^9 \le A_i \le 10^9$) of size **N** ($1 \le N \le 10^5$) and **q** ($1 \le q \le 10^5$) queries. Each query is as described bellow

- **1 i x**: set the element with index **i** to **x** ($1 \le i \le N$, $-10^9 \le x \le 10^9$).
  2 l r : print the sum of element in the range **l** to **r** ($1 \le l \le r \le N$).

Input Format:
The first line of the input contains a single integer **T** ($1 \le T \le 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers ***N*** and ***q*****.**
- The second line contains **N** space-separated integers $A_1, A_2, \ldots, A_N$.
- The following ***q*** lines contain the queries as described in the problem statement
  either of type **1** or of type **2**.

Note: The sum of **N** and **q** over all the test cases will not exceed $2 * 10^5$.

Output Format:

For each test case, for each query of type 2 , print the sum of elements in the given range followed by a newline character.

Note: There is at least one query of the second type in each test case. The generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Example:

Sample Input:

```
1
3 2
-1 4 5
1 2 3
2 1 2
```

Sample Output:

```
2
```

Explanation:

Testcase 1:
given array A = {-1, 4, 5}
after first query A = {-1, 3, 5}
second query asks for sum of
elements in range [1, 2] after 1st
update. So, after 1st update arr
was {-1, 3, 5}. Answer is (-1 + 3) = 2.

Number of Maximum in a Range

Difficulty: **Hard**Accuracy: **94.74%**Submissions: **246+**Points: **8**

Given an array **A** ($1 \le A_i \le 10^9$) of size **N** ($1 \le N \le 10^5$) and **q** ($1 \le q \le 10^5$) queries. Each query is one of the two types

- **1 l x**: set the element with index **l** to **x** ($1 \le l \le N$, $1 \le x \le 10^9$).
- **2 l r**: calculate the number of elements equal to the maximum of elements with indices from **l** to **r** ($1 \le l \le r \le N$).

Input Format:

The first line of the input contains a single integer **T** ($1 \le T \le 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers ***N*** and ***q***.
- The second line contains **N** space-separated positive integers $A_1, A_2, \ldots, A_N$.
- The following ***q*** lines contain the queries as described in the problem statement.

Note: The sum of **N** and **q** over all the test cases will not exceed $2 * 10^5$.

Output Format:

For each test case, For each query of the second type, print the number of elements equal to the maximum in the range followed by a newline character.

Note: There is at least one query of the second type in each test case. The generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Example:

Sample Input:

```
2
2 2
1 10
1 1 10
```

2 1 2
3 2
8 8 1
2 2 3
1 1 100

Sample Output:

2
1

Explanation:

Testcase 1:
given array A = {1, 10}
first query is of type 1
update 10 on the first index
now array A = {10, 10}
second query is of type 2
maximum in the range (1, 2) is
10 and the count of the presence
of 10 in this range is 2.

Order of Heights

Difficulty: **Hard**Accuracy: **48.0%**Submissions: **25+**Points: **8**

There are *N* persons standing in the queue. They are numbered from 1 to *N*. *Hi* is the height of the person *i*. *Pi* is the number of persons who are taller than *i* and standing in front of *i*.

Given these two arrays *H* and *P*, your task is to find the actual order of the queue.

It is guaranteed that the height of each person will be unique.

Input:

- The first line of input contains *N* - the number of persons.
- The second line of input contains *N* space-separated numbers *H1, H2, ..., HN*.

- The third line of input contains *N* space-separated numbers *P1, P2, ..., PN*.
- It is guaranteed that the answer will always exist.
- The sum of N over all test cases dosent exceed 10^5

Output:

Print *N* space-separated integers *p1, p2, ..., pN* - representing the actual order of the queue, where *pi* is the person number standing at the *i*th position in the queue.

Constrains:

$1 \le N \le 10^5$

$1 \le H_i \le 10^8$

$0 \le P_i \le N$

Example

Input:

```
6
5 3 2 6 1 4
0 1 2 0 3 2
```

Output:

```
1 2 3 5 4 6
```

Min Queries

Difficulty: **Medium**Accuracy: **48.14%**Submissions: **27+**Points: **4**

You have given an array A of n integers. Your task is to answer q queries. Each query contains an integer x.

For each query, print the minimum index i such that Ai ≥ x, and then decrease the value at position i by x, i.e., Ai = Ai - x. If there doesn't exist any index i satisfying the condition, then simply print 0.

Input:

The first line contains T, the number of test cases. (1<=T<=100).

- The first input line has two integers n and q: the number of values and queries.
- The second line has n integers A_1, A_2, …, A_n: the array values.
- Finally, there are q integers describing the queries. x_1, x_2, …, x_q.
- The sum of n over all test cases dosent exceed 2*10^5.

Output:

Print the index *i* as mentioned in the problem statement. If such an index doesn't exist, then print 0.

Constraints:

$1 \le n, q \le 2 \times 105$
$0 \le Ai, xi \le 10^6$

Example

Input:

```
8 5
3 2 4 1 5 5 2 6
4 4 7 1 1
```

Output:

```
3 5 0 1 1
```

Merge Sort Tree

Count Inversions

Difficulty: **Medium**Accuracy: **67.54%**Submissions: **355+**Points: **4**

Given an array of integers. Find the Inversion Count in the array.

Inversion Count: For an array, inversion count indicates how far (or close) the array is from being sorted. If array is already sorted then the inversion count is 0. If an array is sorted in the reverse order then the inversion count is the maximum.
Formally, two elements a[i] and a[j] form an inversion if a[i] > a[j] and i < j.

Input Format:

The first line of the input consist a single integer T denoting the number of test cases. The description of each test case is as follows -

- The first line consists of a single integer N denoting the size of the array.
- Second line consists of N space separated numbers denoting the array elements.

Output Format:

For each test case print the number of inversions.

Constraints:

$1 \leq N \leq 5*10^5$

$1 \leq arr[i] \leq 10^{18}$

Example:

Input:

2

5

2 4 1 3 5

5
2 3 4 5 6

Output:
3
0

Explanation:
Test case 1 - The sequence 2, 4, 1, 3, 5 has three inversions (2, 1), (4, 1), (4, 3).
Test case 2 - As the sequence is already sorted so there is no inversion count.

Lazy Propagation

Range Update 1

Difficulty: **Hard** Accuracy: **81.64%** Submissions: **278+** Points: **8**

There is an array **A** of size **N** ($1 \le N \le 10^5$) initially filled with zeros and **q** ($1 \le q \le 10^5$) queries. Each query is one of the two types

- **1 l r v**: assign value **v** to all elements on the segment from **l** to **r** ($1 \le l \le r \le N$, $0 \le v < 2^{30}$).
- **2 l r**: find the maximum on the segment from **l** to **r** ($1 \le l \le r \le N$).

Input Format:
The first line of the input contains a single integer **T** ($1 \le T \le 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers ***N*** and ***q***.
- The following ***q*** lines contain the queries as described in the problem statement.

Note: The sum of **N** and **q** over all the test cases will not exceed $2 * 10^5$.

Output Format:

For each test case, For each query of the second type, print the answer followed by a newline character.

Note: There is at least one query of the second type in each test case. The generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Use fast I/O.

Example:

Sample Input:

```
1
5 4
1 1 3 3
2 2 2
1 2 4 4
2 2 3
```

Sample Output:

```
3
4
```

Explanation:

Testcase 1:
initially array A = {0, 0, 0, 0}
first query is of type 1
aray A = {3, 3, 3, 0}
second query is of type 2
so the answer will be 2

Range update 2

Difficulty: **Hard**Accuracy: **61.85%**Submissions: **333+**Points: **8**

There is an array **A** of size **N** ($1 \le N \le 10^5$) initially filled with zeros and **q** ($1 \le q \le 10^5$) queries. Each query is one of the two types

- **1 l r v**: apply the operation $A_i = A_i \mid v$ (bitwise OR) to all elements from l to r ($1 \le l \le r \le N$, $0 \le v < 2^{30}$).
- **2 l r**: find bitwise AND of elements in the range from **l** to **r** ($1 \le l \le r \le N$).

Input Format:

The first line of the input contains a single integer **T** ($1 \le T \le 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers ***N*** and ***q***.
- The following ***q*** lines contain the queries as described in the problem statement.

Note: The sum of **N** and **q** over all the test cases will not exceed $2 * 10^5$.

Output Format:

For each test case, For each query of the second type, print the answer followed by a newline character.

Note: There is at least one query of the second type in each test case. The generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Example:

Sample Input:

```
1
5 4
1 1 3 3
2 2 2
```

1 2 4 4

2 2 3

Sample Output:

3

7

Explanation:

Testcase 1:

initially array A = {0, 0, 0, 0}

first query is of type 1

array A = {3, 3, 3, 0}

second query is of type 2

so the answer will be 2

third query is of type 1

array A = {3, 6, 7, 4}

fourth query is of type 2

so the answer will be 7

Range Update 3

Difficulty: **Hard**Accuracy: **80.74%**Submissions: **270+**Points: **8**

There is an array **A** of size **N** ($1 \le N \le 10^5$) initially filled with zeros and **q** ($1 \le q \le 10^5$) queries. Each query is one of the two types

- **1 l r v**: assign value v to all elements on the segment from l to r ($1 \le l \le r \le N$, $0 \le v < 2^{30}$).
- **2 l r**: find the sum on the segment from **l** to **r** ($1 \le l \le r \le N$).

Input Format:

The first line of the input contains a single integer **T** ($1 \le T \le 10^4$), denoting the number of test cases. The description of **T** test cases is as follows:

- The first line of each test case contains two space-separated integers ***N*** and ***q***.

- The following ***q*** lines contain the queries as described in the problem statement.

Note: The sum of **N** and **q** over all the test cases will not exceed $2 * 10^5$.

Output Format:

For each test case, For each query of the second type, print the answer followed by a newline character.

Note: There is at least one query of the second type in each test case. The generated output is white space sensitive, do not add extra spaces on unnecessary newline characters.

Example:

Sample Input:

```
1
5 4
1 1 3 3
2 2 2
1 2 4 4
2 2 3
```

Sample Output:

```
3
8
```

Explanation:

Testcase 1:
initially array A = {0, 0, 0, 0}
first query is of type 1
array A = {3, 3, 3, 0}
second query is of type 2
so the answer will be 3
third query is of type 1

array A = {3, 4, 4, 4}
fourth query is of type 2
so the answer will be 8.

Special Graph

Difficulty: **Medium**Accuracy: **87.14%**Submissions: **226+**Points: **4**

You are given an array **A[ ]** of **N** integers. Consider a complete graph of N nodes numbered from **1** to **N**, where the weight of edge connecting nodes ***x*** and ***y*** *(x≠y)* is **| x * y - A[x] % A[y] - A[y] % A[x] |**.Your task is to find the cost of the minimum spanning tree of this graph.

The cost of the spanning tree is **the sum of the weights of all the edges in the tree**. There can be many spanning trees. Minimum spanning tree is the spanning tree where the cost is minimum among all the spanning trees.

Example 1:

Input:
N = 5
A[] = {1, 3, 5, 5, 1}
Output:
10
Explanation:
One of the minimum spanning tree will be

```
      1
    / | \
   2  5  4
  /
 3
```

Cost will be weight of edge 1 - 2 +
weight of edge 1 - 4 +
weight of edge 1 - 5 +
weight of edge 2 - 3

= 1 + 3 + 5 + 1 = 10

Example 2:

Input:
N = 3
A[] = {1, 1, 1}
Output: 5
Explanation:
One of the minimum spanning tree will be

```
    1
   / \
  3   2
```

Cost will beweight of edge 1 - 3 +
weight of edge 1 - 2
= 2 + 3 = 5

Your Task:
You don't need to read, input, or print anything. Your task is to complete the function **solve( ),** which takes **N** and **A[ ]** as input parameters and returns the answer to the problem.

Constraints:
$1 \leq N \leq 5 * 10^4$
$1 \leq A[i] \leq N$

Fenwick Trees

Count Inversions

Difficulty: **Medium**Accuracy: **67.54%**Submissions: **355+**Points: **4**

Given an array of integers. Find the Inversion Count in the array.

Inversion Count: For an array, inversion count indicates how far (or close) the array is from being sorted. If array is already sorted then the inversion count is 0. If an array is sorted in the reverse order then the inversion count is the maximum.
Formally, two elements a[i] and a[j] form an inversion if a[i] > a[j] and i < j.

Input Format:

The first line of the input consist a single integer T denoting the number of test cases. The description of each test case is as follows -

- The first line consists of a single integer N denoting the size of the array.
- Second line consists of N space separated numbers denoting the array elements.

Output Format:

For each test case print the number of inversions.

Constraints:
$1 \le N \le 5*10^5$
$1 \le arr[i] \le 10^{18}$

Example:

Input:
2
5
2 4 1 3 5
5
2 3 4 5 6

Output:
3

0

Explanation:
Test case 1 - The sequence 2, 4, 1, 3, 5 has three inversions (2, 1), (4, 1), (4, 3).
Test case 2 - As the sequence is already sorted so there is no inversion count.

Curious Robin Hood

Difficulty: **Medium**Accuracy: **93.07%**Submissions: **245+**Points: **4**

Robin Hood likes to loot rich people since he helps the poor people with this money. Instead of keeping all the money together he does another trick. He keeps **N** sacks where he keeps this money. The sacks are numbered from **0** to **N-1**. Now each time he can he can do one of the three following tasks:

1. Give all the money of the **i-th** sack to the poor, leaving the sack empty.
2. **Add** new amount (given in input) in the ith sack.
3. Find the **total** amount of money from **i-th** sack to **j-th** sack.

Since he is not a programmer, he seeks your help.

Input Format:

The first line contain an integer **T** - the number of test cases. Then the test cases follows:

The first line contains two integers **N** and **Q** denoting the number of sacks and number of queries respectively.
The Next line contains N space separated integers in the range [1, 10^9] denoting initial amount of money in the **i-th** sack.

Each of the next q lines contains a task in one of the following form:

- **1 i** - give all the money of the i-th sack.
- **2 i v** - add money **v** ($1 \le v \le 10^9$) to the i-th ($0 \le i < n$) sack.

- **3 i j** - find the total amount of money from the ith sack to the jth sack ($0 \le i \le j < n$).

Output Format:

For each test case, If the query **type** is **1**, then print the amount of money given to the poor. If the query **type** is **3**, print the total amount from i-th to j-th sack.

Constraints:

$1 \le T \le 1000$

$1 \le N, Q \le 10^5$

Each test case contains **atleast one** query of type 1 or type 3.

The sum of N and Q over all test cases does not exceed $2 * 10^5$.

Example:

Input:

```
1
5 6
3 2 1 4 5
1 4
2 3 4
3 0 3|
1 2
3 0 4
1 1
```

Output:

```
5 14 1 13 2
```

Hanoi Factory

Difficulty: **Hard** Accuracy: **88.91%** Submissions: **225+** Points: **8**

Of course you have heard the famous task about Hanoi Towers, but did you know that there is a special factory producing the rings for this wonderful game? Once upon a time, the ruler of the ancient Egypt ordered the workers of Hanoi Factory

to create as high tower as possible. They were not ready to serve such a strange order so they had to create this new tower using already produced rings.
There are **N** rings in factory's stock. The **i-th** ring has **inner radius a_i, outer radius b_i** and **height h_i.**
The goal is to select some subset of rings and arrange them such that the following conditions are satisfied:

- Outer radiuses form a non-increasing sequence, i.e. one can put the j-th ring on the i-th ring only if $b_j \leq b_i$.

- Rings should not fall one into the the other. That means one can place ring j on the ring i only if $b_j > a_i$.
- The total height of all rings used should be **maximum** possible.

Input Format:

The first line contain an integer **T** - the number of test cases. Then the test cases follows:

The first line contain an integer **N** denoting the number of rings in factory's stock. Next **N** lines each contains three integers a_i, b_i and h_i denoting inner radius, outer radius and the height of the i-th ring respectively..

Output Format:

For each test case, print the maximum height of the tower that can be obtained followed by newline character.

Constraints:

$1 \leq T \leq 1000$
$1 \leq N \leq 10^5$

$1 \le a_i, b_i, h_i \le 10^9$

The sum of N over all test cases does not exceed $2 * 10^5$.

Example:

Input:

2

3

1 5 1

2 6 2

3 7 3

4

1 2 1

1 3 3

4 6 2

5 7 1

Output:

6

4

Explanation:

For test case 1, the optimal solution is to take all the rings and put them on each other in order 3, 2, 1.

For test case 2, one can put the ring 3 on the ring 4 and get the tower of height 3, or put the ring 1 on the ring 2 and get the tower of height 4.

Ranged Queries Problems

Reading Books

Difficulty: **Medium**Accuracy: **79.61%**Submissions: **308+**Points: **4**

Geek has **N** books, every book has infinite number of pages. Books are numbered from **1** to **N.** Geek wants to monitor his progress, so he asks you **Q** queries of two types. :

- **1 L R :** Geek reads **1** page from books numbered **L, L+1, L+2, L+3, ……., R**
- **2 X :** Find out how many pages have been read in the book numbered **X.**

You have to print answers to each query of the second type. Initially Geek hasn't started reading any book so number of pages read in every book is equal to **0**.

Input

- First line of input contains an integer **T** number of testcases
- First line of each test case contains two space separated integers **N** and **Q**
- Each of the next **Q** lines contains three or two space separated integers depending on the query.
- If first integer of the query is **1** then it is followed by two intgers **L** and **R** denoting the range of the books Geek has read one page from
- If first integer of the query is **2** then it is followed by only one integer **X** asking the number of pages read in $\mathbf{X_{th}}$ book

Output

- For every query of type **2** print an integer in new line denoting number of pages read in the book **X.**

Constraints

- **1<=T<=10**
- **1<=N,Q<=10^5**
- **1<=L<=R<=N**
- **1<=X<=N**

Sample Input

```
2
```

```
3 3
1 2 3
2 3
2 1
4 4
1 1 4
1 4 4
2 1
2 4
```

Sample Output

```
1
0
1
2
```

Explanation

In first testcase, Geek reads one page from book **2** and **3** in first query after that in second query Geek asks for number of pages read in book **3** ,answer is **1.** Third query asks for number of pages read in book **1** answer is **0** as no page has been read in book **1.**

In second test case Geek reads one page from book **1, 2, 3** and **4** in first query **.** After that in second query **1 4 4** Geek reads one more page from the book **4** .
Third query asks for pages read in book **1** , answer is **1.**
Fourth query asks for pages read in book **4** , answer is **2** as book **4** was read in both first and second query.

City of Skyscrapers

Difficulty: **Medium**Accuracy: **84.56%**Submissions: **266+**Points: **4**

Geek is travelling in a city of skyscrapers. City can be represented as a 2D square grid of dimensions **N** X **N**. Every location in city can be described with 2D co-ordinates **(x, y)**, such that **1<=x<=N** and **1<=y<=N**.

Every moment a new building is erected in the city. Geek is curious about what will be the height of the tallest building he will see, if he starts travelling from some position **(u,v)** in the city.

Geek can move only in two directions **South** and **East.** If he is currently standing at **(x, y)** he can move towards **east** by going to location **(x+1, y)** or he can move **south** by going to location
(x, y+1)

You have to process **Q** queries. Queries can be of two types

- **1, X, Y, H** - a building of height **H** is erected at location **(X,Y)**
- **2, X, Y** - print the tallest building Geek can see if he starts travelling from **(X, Y)**

Input

- First line of input contains an integer **T,** number of testcases
- First line of each test case contains two space separated integers **N** and **Q** , size of grid and number of queries respectively
- Each of the next **Q** lines contain either **3** or **4** space separated integers depending upon the type of query. First integer of every query denotes the type of the query.

Output

- For every query of type **2** print an integer in a new line denoting height of tallest building, Geek can visit starting from the location given in query.

NOTE: Query of type **1** can repeat for a **(X, Y)** when it happens, old building on that location is demolished and a new building with height **H** is created in its place.

Constraints

- **1<=T<=10**
- **1<=N<=2000**
- **1<=Q<=10^5**
- **1<=H<=10^9**
- **1<=X,Y<=N**

Sample Input

```
1
4 6
2 1 1
1 4 3 4
2 1 1
1 2 3 5
2 1 1
2 3 3
```

Sample Output

```
0
4
5
4
```

Explanation

There is only one test case in the sample input with **N** equal to **4** and **6** queries.
Answer to first query is zero, because there is no building in the city at the time of its execution.
In second query a building of height **4** is created at **(4, 3)**
In third query Geek can visit the building created in the last query at the

location **(4, 3)** if he starts from **(1, 1).** So answer to this query is **4.**
In fourth query a new building of height **5** is created at location **(2, 3)**
In fifth query if Geek starts from **(1,1)** he can visit **(2, 3)** and then **(4,3)** the tallest building he will visit will have height **5** (the building at (2,3))
In sixth query Geek starts from **(3, 3)** as he can only move North and East he will be able to visit only one building **(4, 3)** so answer to this query is **4**

Indexed Multiset

Difficulty: **Hard**Accuracy: **81.51%**Submissions: **281+**Points: **8**

Implement a data structure called **indexed multiset** that supports following three operations.

- **INSERT X :** insert integer **X**
- **REMOVE X :** remove one instance of **X,** it is guaranteed that **X** exists in multiset
- **PRINT K :** print value of $\mathbf{K_{th}}$ smallest integer currently in the multiset.

X is the $\mathbf{K_{th}}$ smallest number in the multiset if we sort the contents of the multiset in non-decreasing order and **X** occupies $\mathbf{K_{th}}$ position.
Note that task is to implement a multiset so there can exist multiple copies of an integer in the structure.

Input

- First line of input contains **T** number of test cases.
- First line of each test case contains an integer **Q** number of queries
- Each of the next **Q** lines contain a string followed by an integer denoting a query

Output

- For every query of type **PRINT** print in a new line $\mathbf{K_{th}}$ smallest element in the set as defined above.
- If there are less than **K** elements in the multiset print **-1**

Constraints

- **1<=T<=10**
- **1<=Q<=10^5**
- **1<=K, X<=10^5**

Sample Input

```
1
9
INSERT 5
INSERT 2
PRINT 1
PRINT 3
INSERT 10
INSERT 10
REMOVE 2
PRINT 3
PRINT 2
```

Sample Output

```
2
-1
10
10
```

Explanation

There is only 1 test case in the sample input with 9 queries.
After first two insert queries multiset looks like {2, 5} when asked to **print 1** in 3rd query we print **2** as it is the first element in the multiset. When asked to **print 3** we print **-1** as multiset has only **2** elements.
After inserting **10** two times in next two queries set becomes {2, 5, 10, 10} now we process **remove 2** query and set becomes {5, 10, 10}

Therefore answer to second last query **print 3** is **10** and last query **print 2** is also **10.**

Bitset - Easy Version

Difficulty: **Medium**Accuracy: **74.47%**Submissions: **145+**Points: **4**

Geek invented a data structure called bitset. A bitset of length **N** has **N** bits arranged sequentially numbered from **1** to **N**. Initially all bits in the bitset are unset (set to zero).

He wants to implement two operations on this data structure which take two integers **L** and **R** as parameters.

Geek wants to implement following operations:

- **SET L R :** Set all bits in the range **L** to **R**
- **COUNT L R :** Count number of set bits in range **L** to **R**

Input

- First line of each input contains an integer **T** , number of test cases , **T** test cases follow
- First line of each test case contains two space separated integers **N** and **Q** , size of bitset and number of queries
- Each of the next **Q** lines contains one string denoting type of query followed by two space separated integers **L** and **R**

Output

- For each query of type **COUNT** print the result in a new line.

Constraints

- **1<=T<=10**
- **1<=N,Q<=10^5**
- **1<=L<=R<=N**

Sample Input

```
1
```

```
6 5
COUNT 2 3
SET 4 5
COUNT 1 5
SET 1 3
COUNT 2 4
```

Sample Output

```
0
2
3
```

Explanation

For first query in the first test case, bitset looks like [000000] and
therefore **COUNT 2 3** prints **0**
Second query is **SET 4 5** so bitset becomes [000110]
Third query **COUNT 1 5** so answer is **2** as two bits **4** and **5** were set in previous operation
Fourth query is **SET 1 3** bitset becomes [111110]
Fifth query is **COUNT 2 4** answer for it is **3** as bits 2,3 and 4 are set

Friendly Visit

Difficulty: **Hard**Accuracy: **93.95%**Submissions: **246+**Points: **8**

Geek's friend live in different cities. He wants to pay them a visit. There are **N** cities with **N-1** roads connecting them, no two cities are connected by more than **1** road.
Geek doesn't want to go to a friend's house empty-handed. There is a gift shop in every city. Gift shop in city **i** sells a gift for **C[i]** rupees.
Cost of the gifts in every shop keeps on changing throughout the day.

Geek wants to find the minimum cost of gift he can buy if he starts his trip from city **u** and goes to visit a friend in city **v** using the shortest path between two

cities.

Input

- First line of input contains an integer **T**, number of testcases
- First line of each test case contains two space separated integers **N** and **Q**, denoting number of cities and number of queries you have to answer.
- Next line contains **N** space separated integers, $\mathbf{i_{th}}$ of which denotes **C[i]** cost of a gift in $\mathbf{i_{th}}$ city.
- Next **N-1** lines contain description of the cities in form of two space separated integers **u** and **v** denoting there is a bi-directional road connecting **u** and **v**
- Next **Q** lines contain queries in the form of three space separated integers **type, i, j.**
- If **type** equals **1** then it means cost of gift at city **i** has changed to **j**
- If **type** equals **2** then print in a new line minimum cost of gift Geek can buy travelling from city **i** to city **j**

Output

- For every query where **type** equals **2** print in a new line minimum cost of gift Geek can buy travelling from city **i** to city **j**

Constraints

- **$1<=T<=5$**
- **$2<=N<=10^5$**
- **$1<=Q<=10^5$**
- **$0<=C[i]<=10^9$**
- **$1<=type<=2$**
- **$1<=u,v<=N$**
- **$1<=i,j<=N$ if type==2**
- **$1<=i<=N$ and $0<=j<=10^9$ if type==1**

Sample Input

```
1
3 4
1 2 3
1 2
2 3
2 1 3
1 1 2
2 1 3
2 3 2
```

Sample Output

```
1
2
2
```

Explanation

First query asks for minimum cost of gift on path from **1** to **3** the answer is **1** because Geek can buy a gift with cost **1** at city **1.**
Second query changes the cost of gift at city **1** from **1** to **2** .
Third query asks again for minimum cost of gift on path from **1** to **3** now the minimum cost is **2** because cost of gift at city **1** was changed in the last query.
Fourth query asks for minimum cost of gift on path from **3** to **2** , answer is **2** because cost at city **2** is **2** and **3** is **3** , clearly minimum cost is **2**

Greater and Smaller

Difficulty: **Medium**Accuracy: **83.32%**Submissions: **126+**Points: **4**

Given an array of integers **A** . We define following two functions on **A** :

F(i) - number of integers to the **left of A[i]** which are **greater than A[i]**
G(i) - number of integers to the **right of A[i]** which are **smaller than A[i]**

Find array **B** such that **B[i]=F(i)-G(i)**

Input

- First line of input contains an integer **T** , number of testcases, **T** test cases follow
- First line of each test contains integer **N**, size of **A**
- Second line of each test contains **N** space separated integers $\mathbf{i_{th}}$ of which denotes **A[i].**

Output

- For each test print **N** space separated integers in a new line $\mathbf{i_{th}}$ of which denotes **B[i].**

Constraints

- **1<=T<=10**
- **1<=N<=10^5**
- **1<=A[i]<=10^9**

Sample Input

```
2
3
1 2 3
4
2 3 1 3
```

Sample Output

```
0 0 0
-1 -1 2 0
```

Explanation

In first testcase, there is no smaller element to the right and no greater element to the left of any number so **F(i)** and **G(i)** are zero for every **i** therefore every element of **B** is zero

In second testcase

F(1) - G(1) = 0 - 1 = -1

F(2) - G(2) = 0 - 1 = -1

F(3) - G(3) = 2 - 0 = 2

F(4) - G(4) = 0 - 0 = 0

Frequency of elements

Difficulty: **Medium**Accuracy: **74.81%**Submissions: **139+**Points: **4**

You are given an array **arr** of **n** integers. You have to perform **q** queries. Each query consists of two space-separated integers **i** and **j**. For each query you have to determine the number of occurrences of the most frequent value among the integers in the range $\mathbf{arr_i}$ **.....** $\mathbf{arr_j}$.

Input Format:

The first line contains two space-separated integers **n** and **q**, representing the number of integers and queries respectively.

The second line consists of **n** elements, representing the number of elements in the array.

The following **q** lines consists of two integers **i** and **j**, representing the boundary indices of each query.

Output Format:

For each query, print the number of occurrences of the most frequent value within the given range.

The output of each query must be printed in a separate line.

Constraints:

- $1 <= n <= 10^5$

- $1 <= q <= 10^5$
- $1 <= arr[i] <= 10^5$
- $0 <= i <= j <= n-1$

Example:

Input:

```
10 5
1 2 3 3 2 1 1 1 2 3
0 5
1 3
6 9
5 8
2 9
```

Output:

```
2
2
2
3
3
```

Bitset - Hard Version

Difficulty: **Hard**Accuracy: **91.12%**Submissions: **249+**Points: **8**

NOTE : This is the hard version of the problem, only difference between hard and easy version is constraints on N which also implies constraints on L and R

Geek invented a data structure called bitset. A bitset of length **N** has **N** bits arranged sequentially numbered from **1** to **N**. Initially all bits in the bitset are unset (set to zero).
He wants to implement two operations on this data structure which take two

integers **L** and **R** as parameters.
Geek wants to implement following operations:

- **SET L R :** Set all bits in the range **L** to **R**
- **COUNT L R :** Count number of set bits in range **L** to **R**

Input

- First line of each input contains an integer **T** , number of test cases , **T** test cases follow
- First line of each test case contains two space separated integers **N** and **Q** , size of bitset and number of queries
- Each of the next **Q** lines contains one string denoting type of query followed by two space separated integers **L** and **R**

Output

- For each query of type **COUNT** print the result in a new line.

Constraints

- **1<=T<=10**
- **1<=Q<=10^5**
- **1<=N<=10^9**
- **1<=L<=R<=N**

Sample Input

```
1
6 5
COUNT 2 3
SET 4 5
COUNT 1 5
SET 1 3
COUNT 2 4
```

Sample Output

```
0
```

```
2
3
```

Explanation

For first query in the first test case, bitset looks like [000000] and
therefore **COUNT 2 3** prints **0**
Second query is **SET 4 5** so bitset becomes [000110]
Third query **COUNT 1 5** so answer is **2** as two bits **4** and **5** were set in previous operation
Fourth query is **SET 1 3** bitset becomes [111110]
Fifth query is **COUNT 2 4** answer for it is **3** as bits 2,3 and 4 are set

Online distinct queries

Difficulty: **Hard** Accuracy: **86.57%** Submissions: **261+** Points: **8**

Given an array **A** consisting of **N** integers. You have to answer **Q** queries. Each query consists two space separated integers **L** & **R** .
For a query **L R** , count number of distinct integers in this subarray **A[L], A[L+1], ..., A[R]**

NOTE : You have to answer queries online. To enforce this, **L** and **R** for **ith** query will be xored with the answer of previous query, i.e. if answer for first query is **X** , and input for second query is **L** and **R** then the real range in which you have to find answer is **(L xor X)** and **(R xor X)**.

Input

- First line of input contains one integer **T** denoting number of independent test cases
- First line of each test case contains two space separated integers, **N** and **Q** , size of **A** and number of queries respectively.
- Second line of each test case contains **N** space separated integers denoting elements of **A**

- Each of the next **Q** lines contain two space separated integers **L** and **R**, denoting the query.
- Except first query of each test case **L** and **R** of every query is written in input by xoring it with result of the preceding query.

Output

- For each query **L R** print in a new line number of distinct integers in the subarray **A[L], A[L+1],.....,A[R]. L** and **R** discussed here are decrypted.

Constraints

- **1<=T<=5**
- **1<=N,Q<=10^5**
- **1<=A<=10^9**
- **1<=L<=R<=N (when xored with result of previous query)**

Sample Input

```
1
6 3
5 3 2 2 1 2
1 6
6 0
1 7
```

Sample Output

```
4
2
2
```

Explanation

First query is unencrypted, so **L = 1** and **R = 6** there are **4** distinct integers in range **[1, 6].**
Now **L=6** and **R=0** in second query, response of previous query is **4**

L xor 4 = 6 xor 4 = 2, and
R xor 4 = 0 xor 4 = 4 thus range for this query is **[2, 4]** and number of distinct integers in range **[2, 4]** is **2**
For third query **L = 1** and **R = 7** , previous response was **2** , so effective **L** for this query is **1 xor 2 = 3,** and **R = R xor 2 = 5.** Count of distinct elements in range **[3, 5]** is **2.**